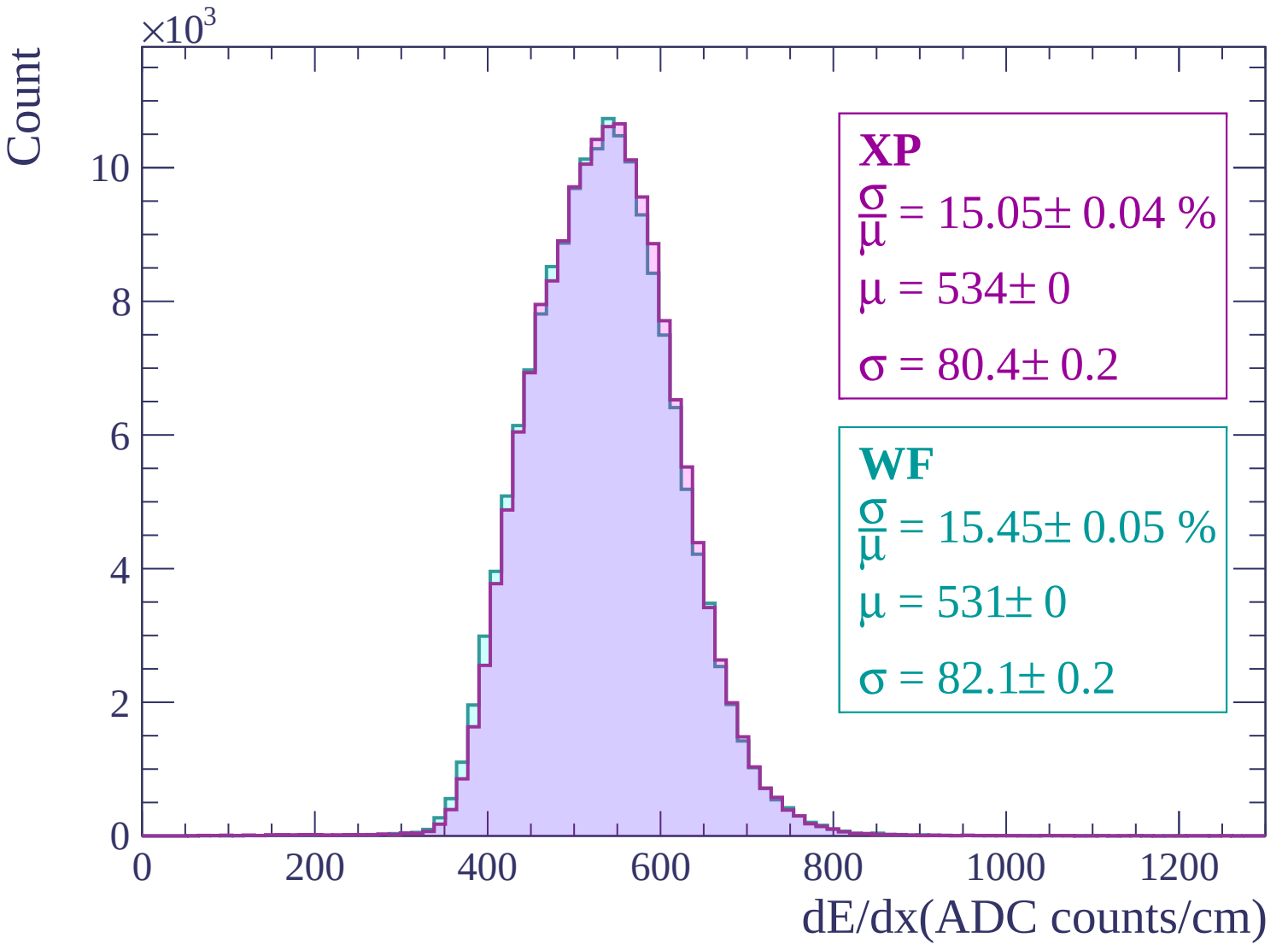
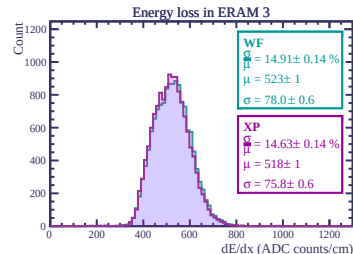
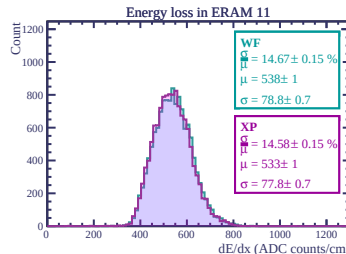
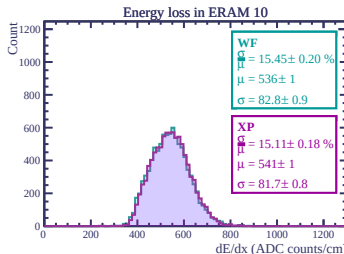
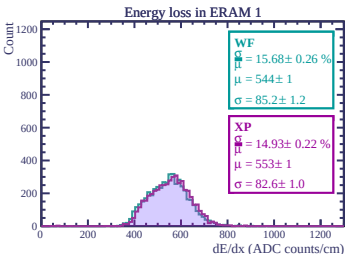
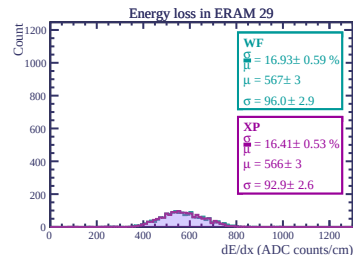
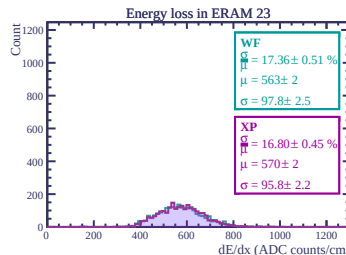
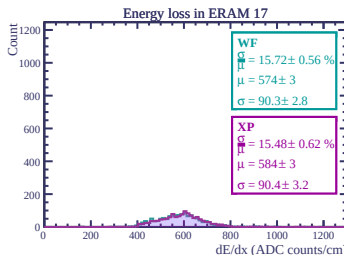
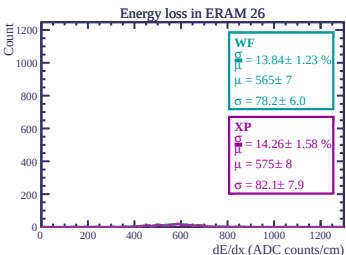
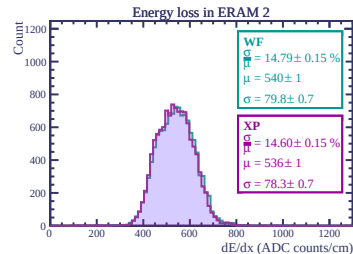
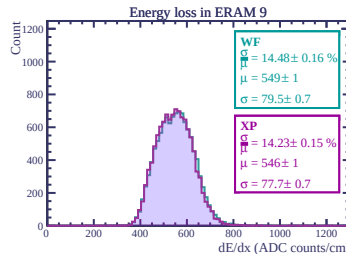
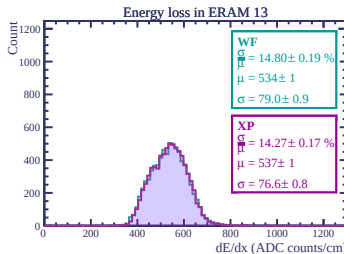
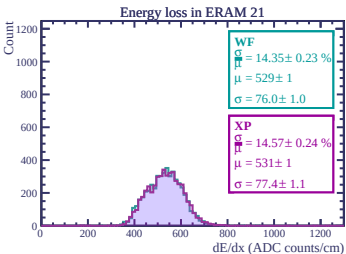
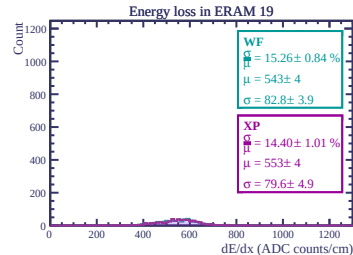
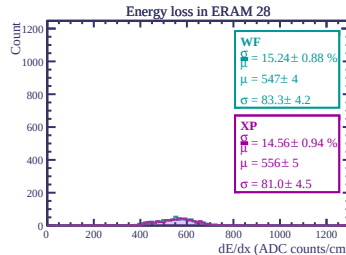
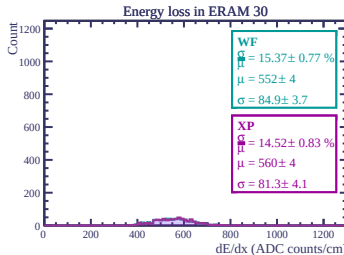
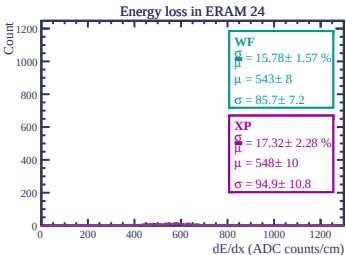
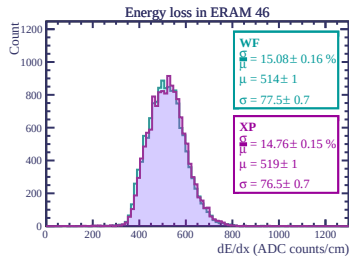
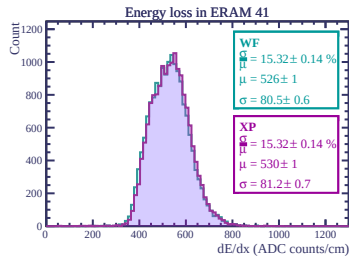
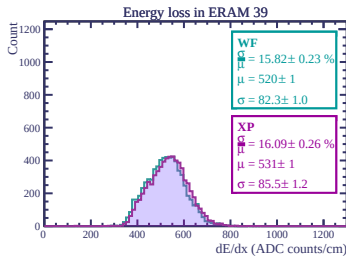
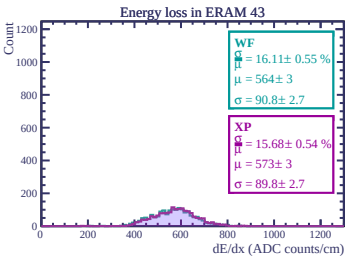
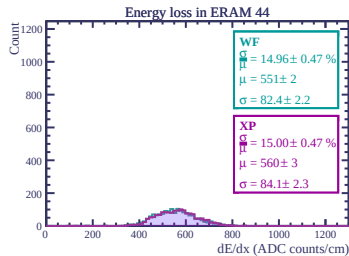
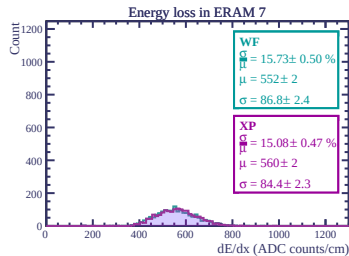
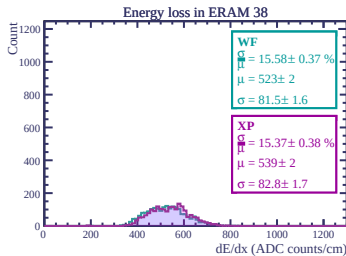
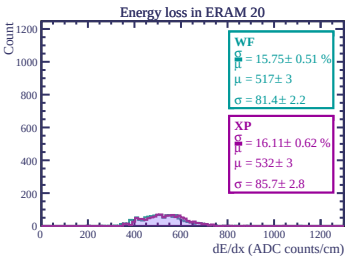
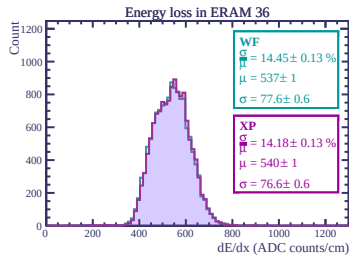
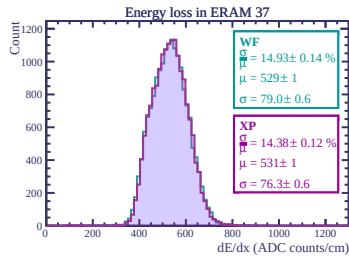
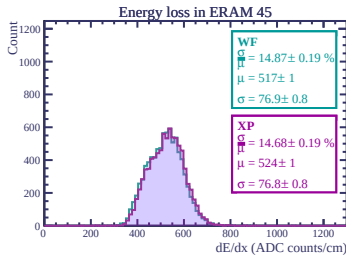
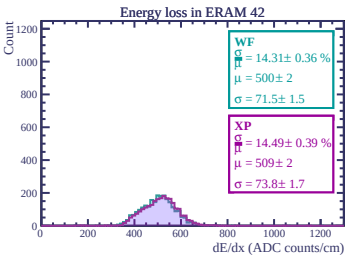
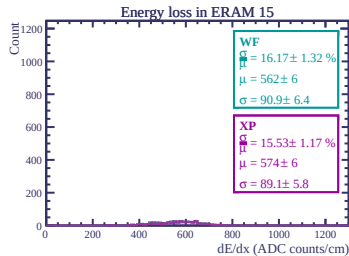
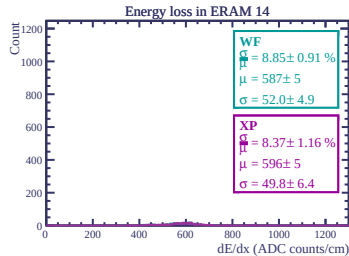
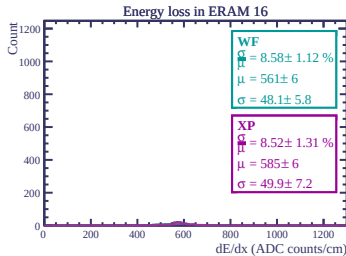
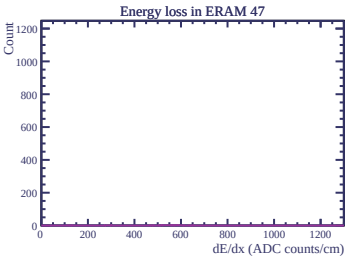
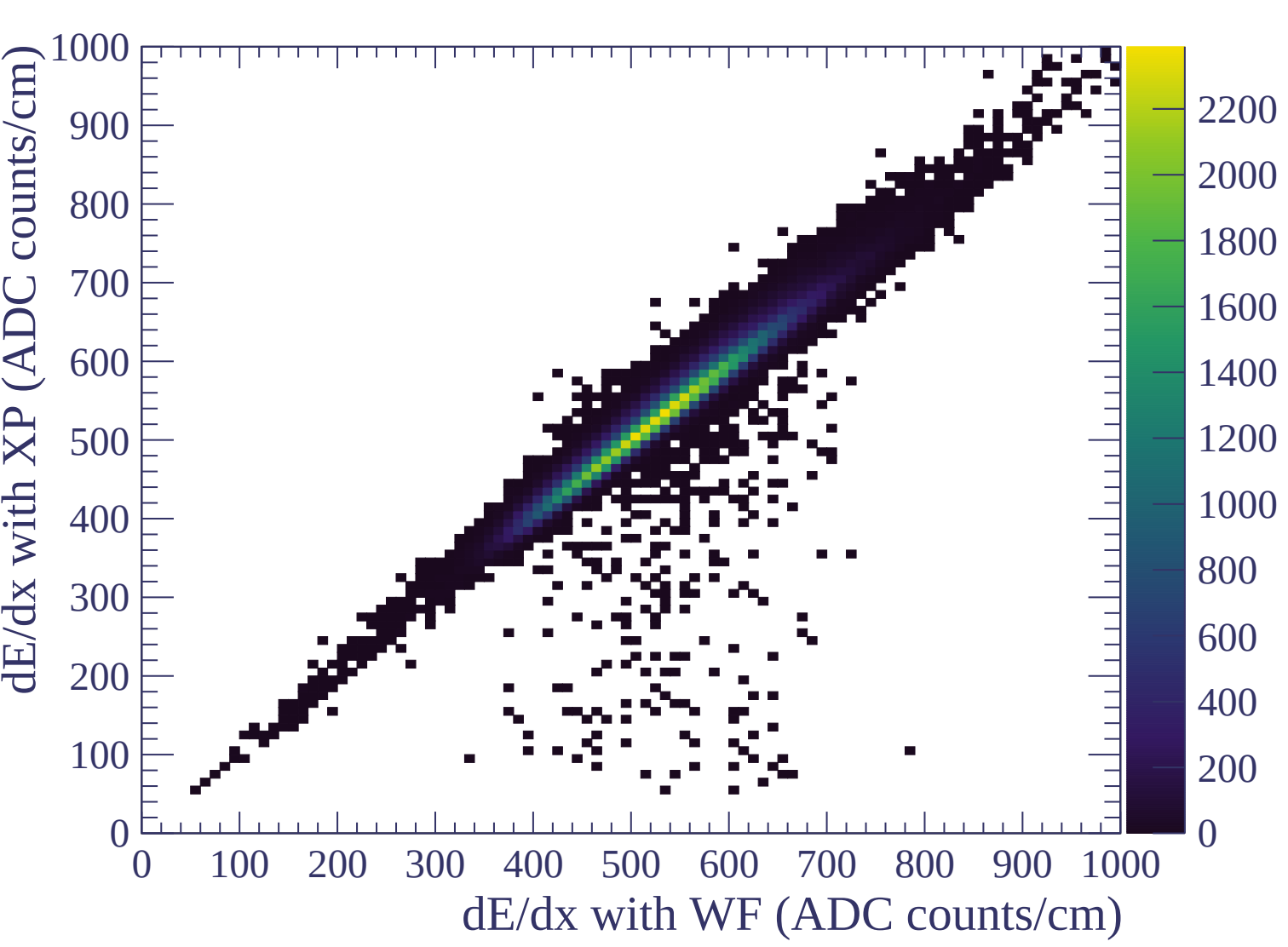
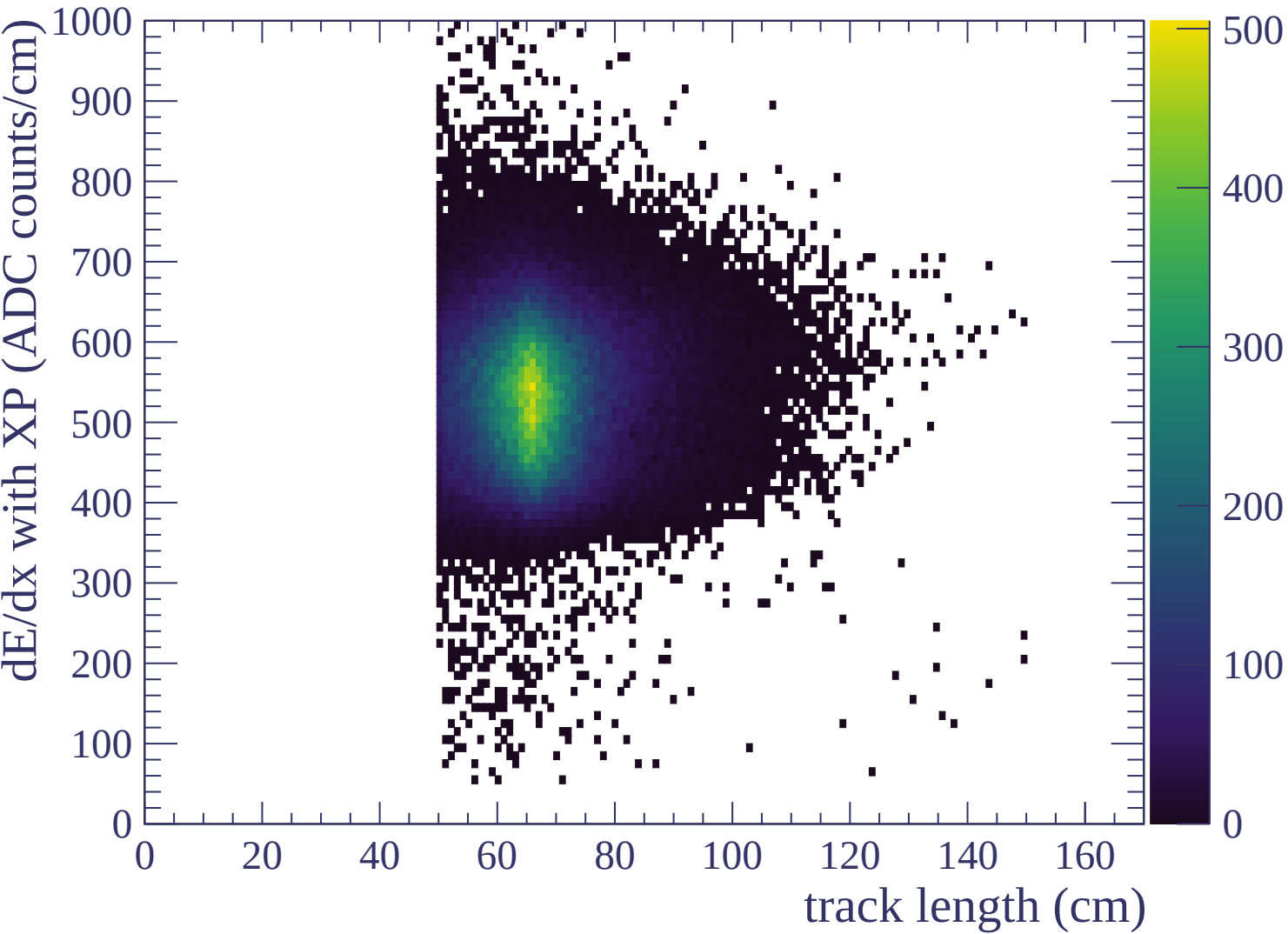


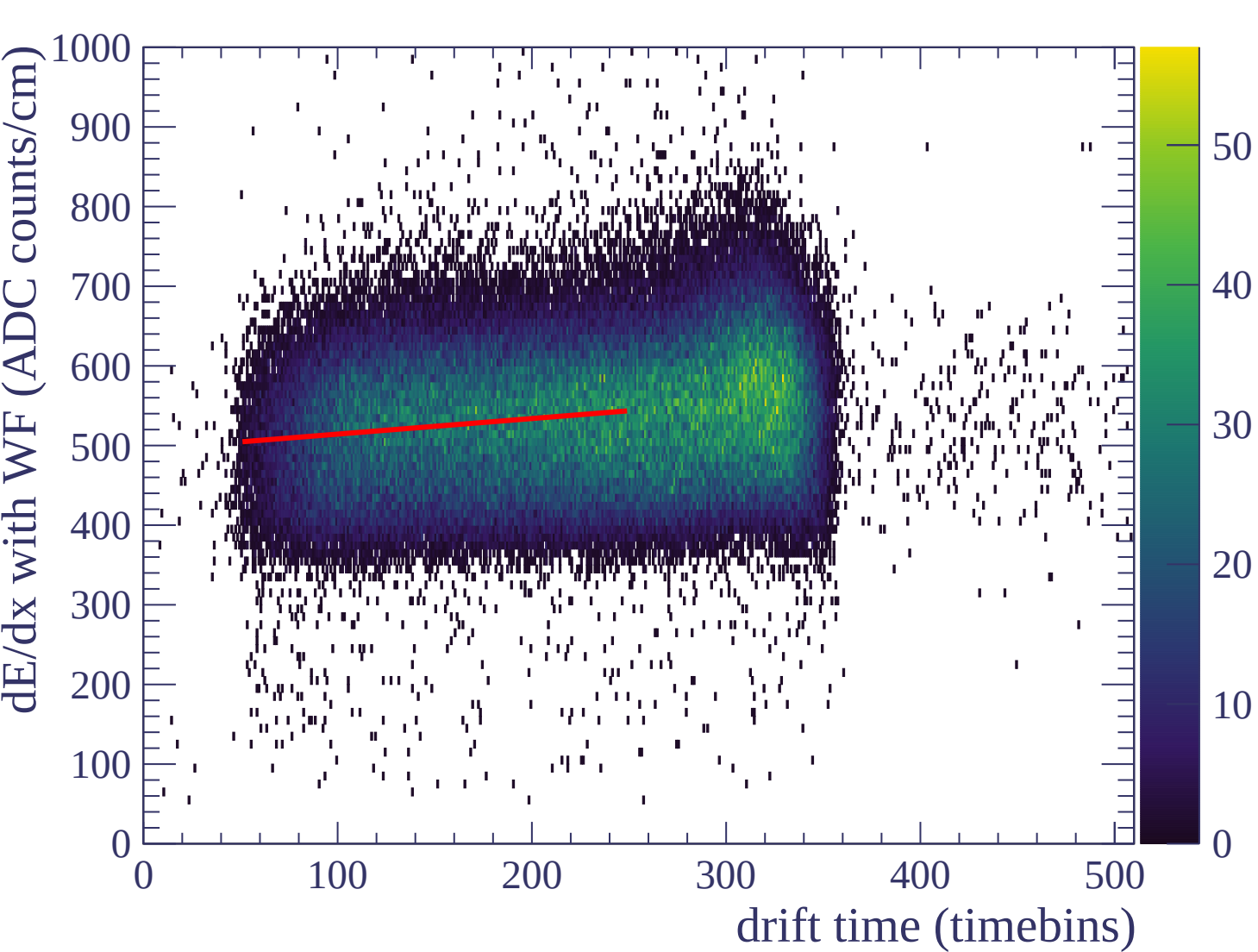


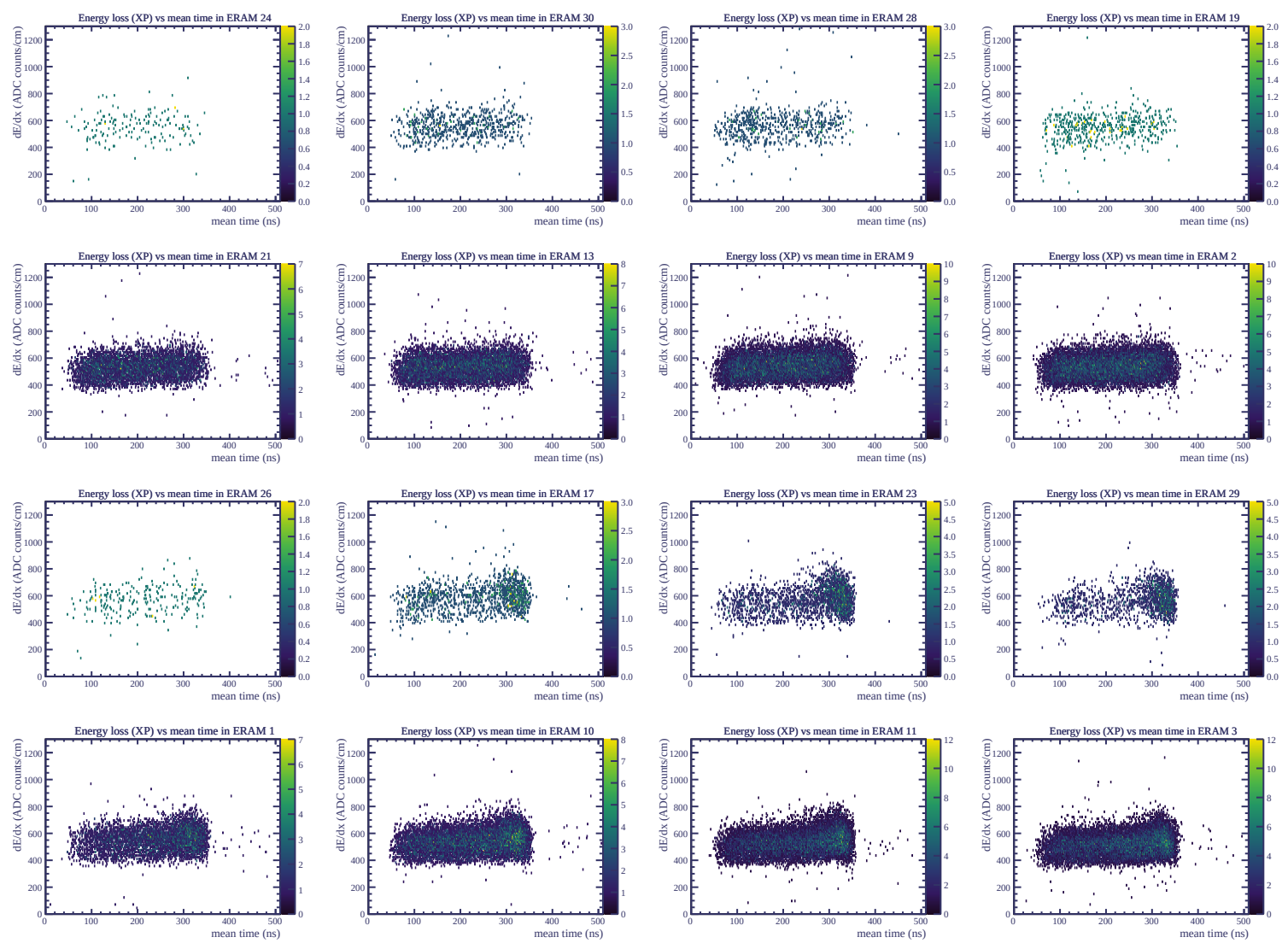


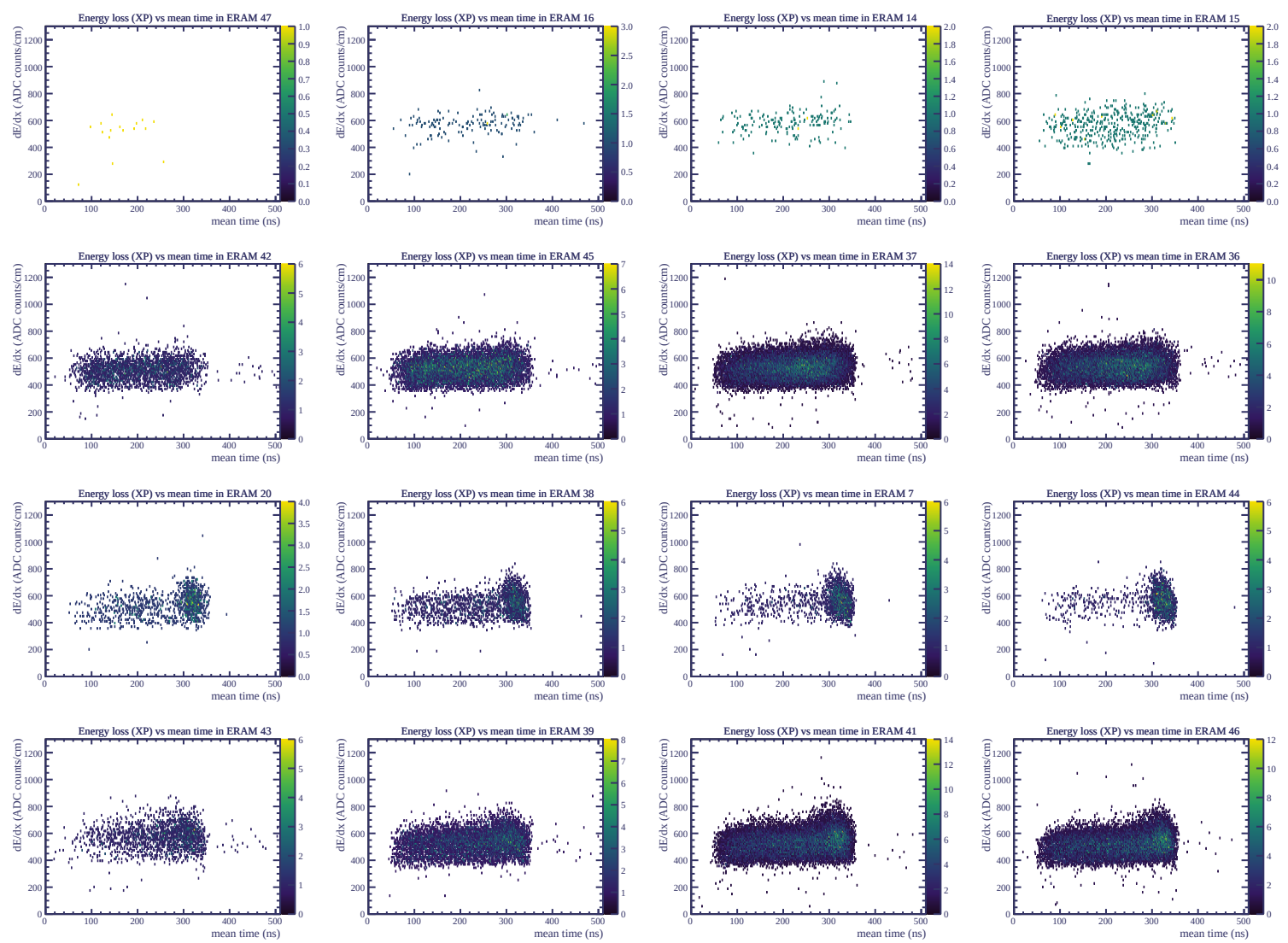




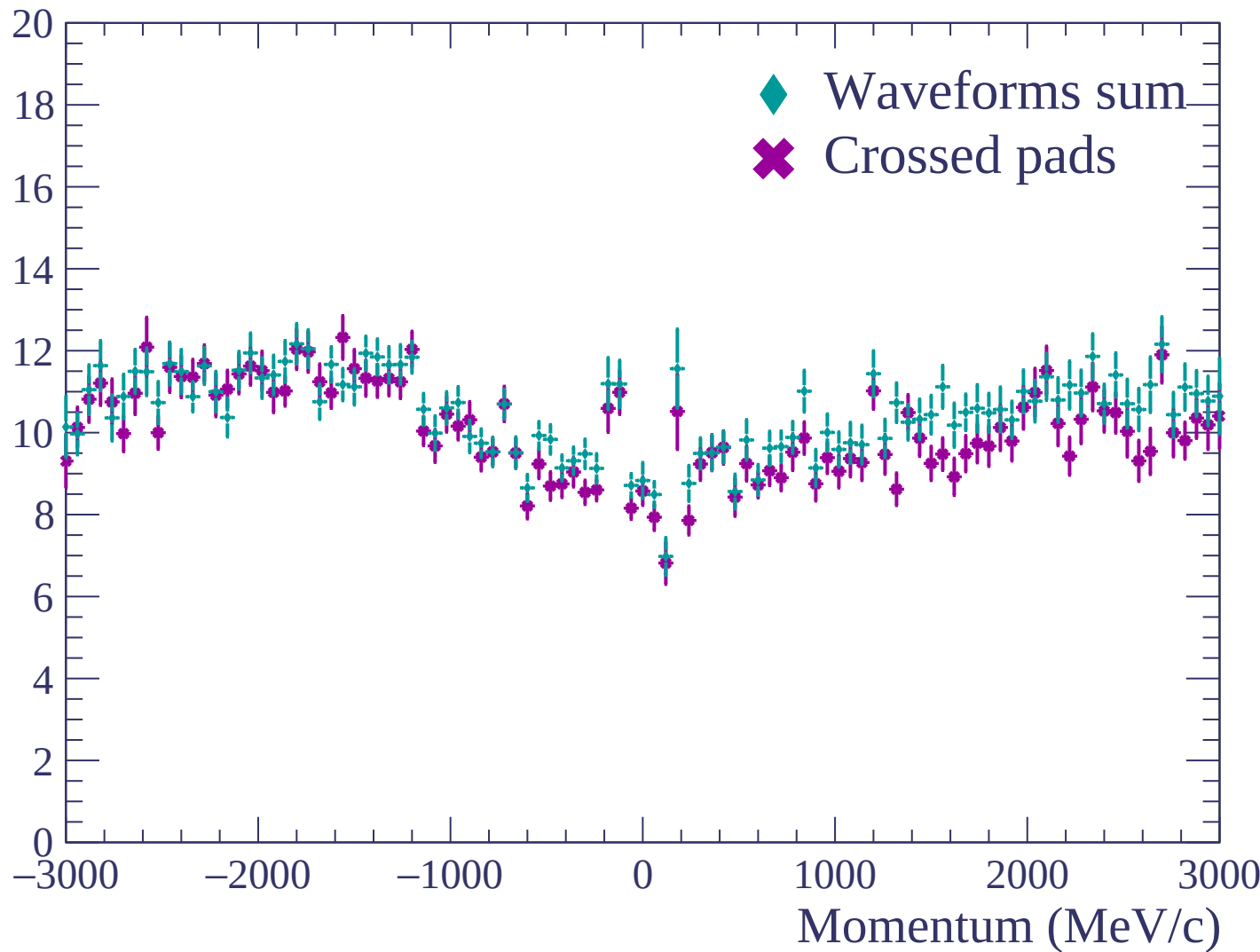


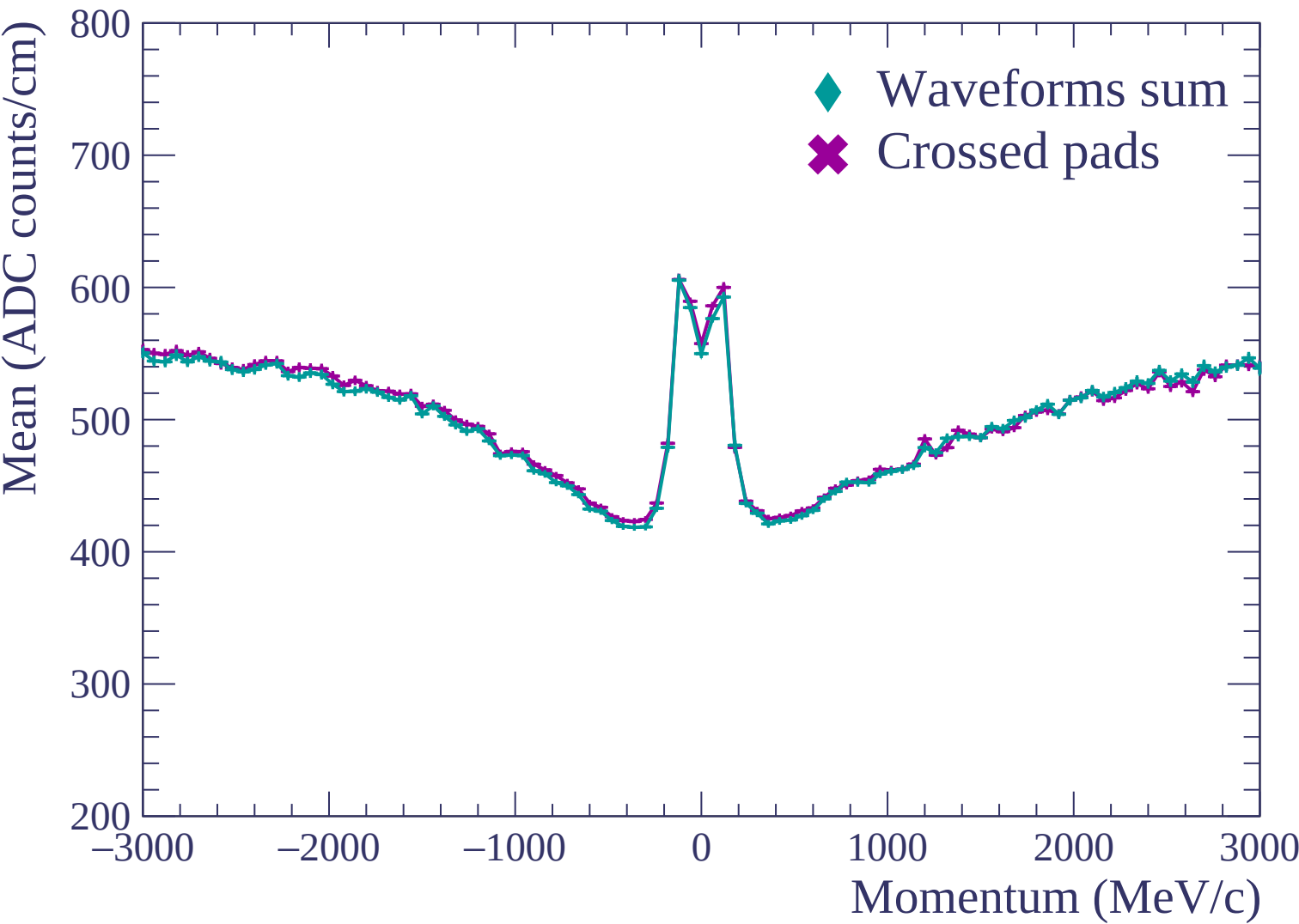


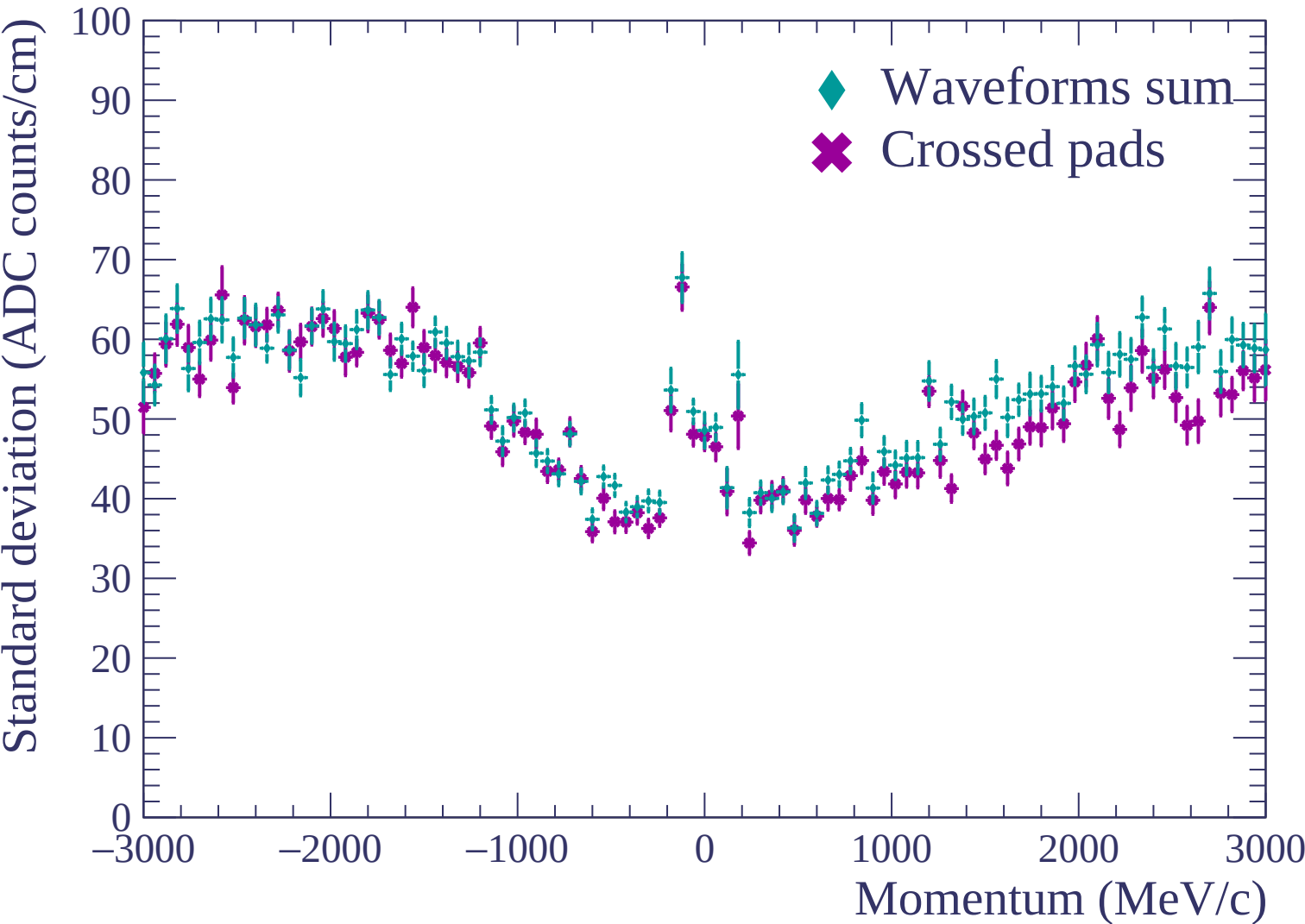


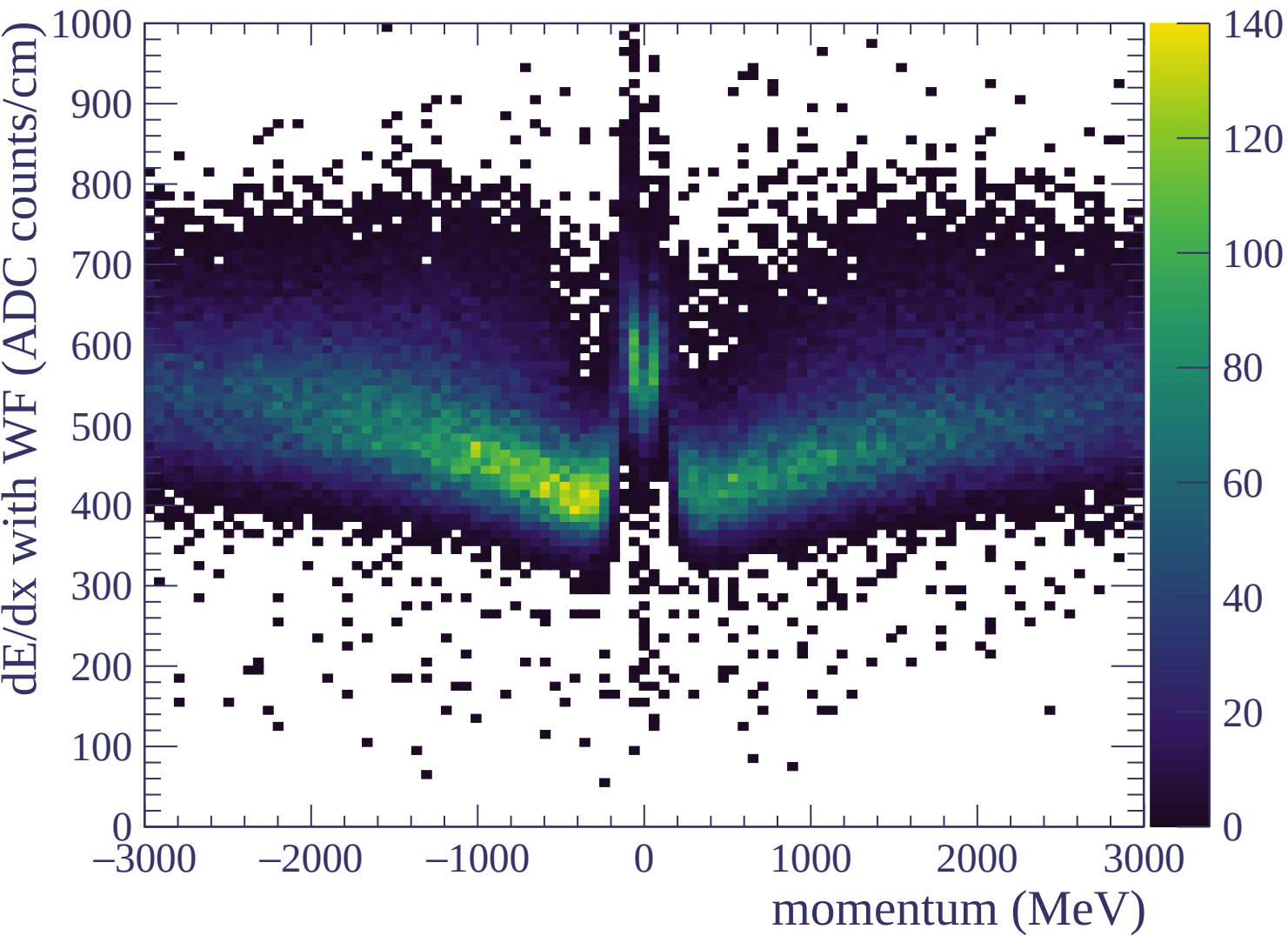


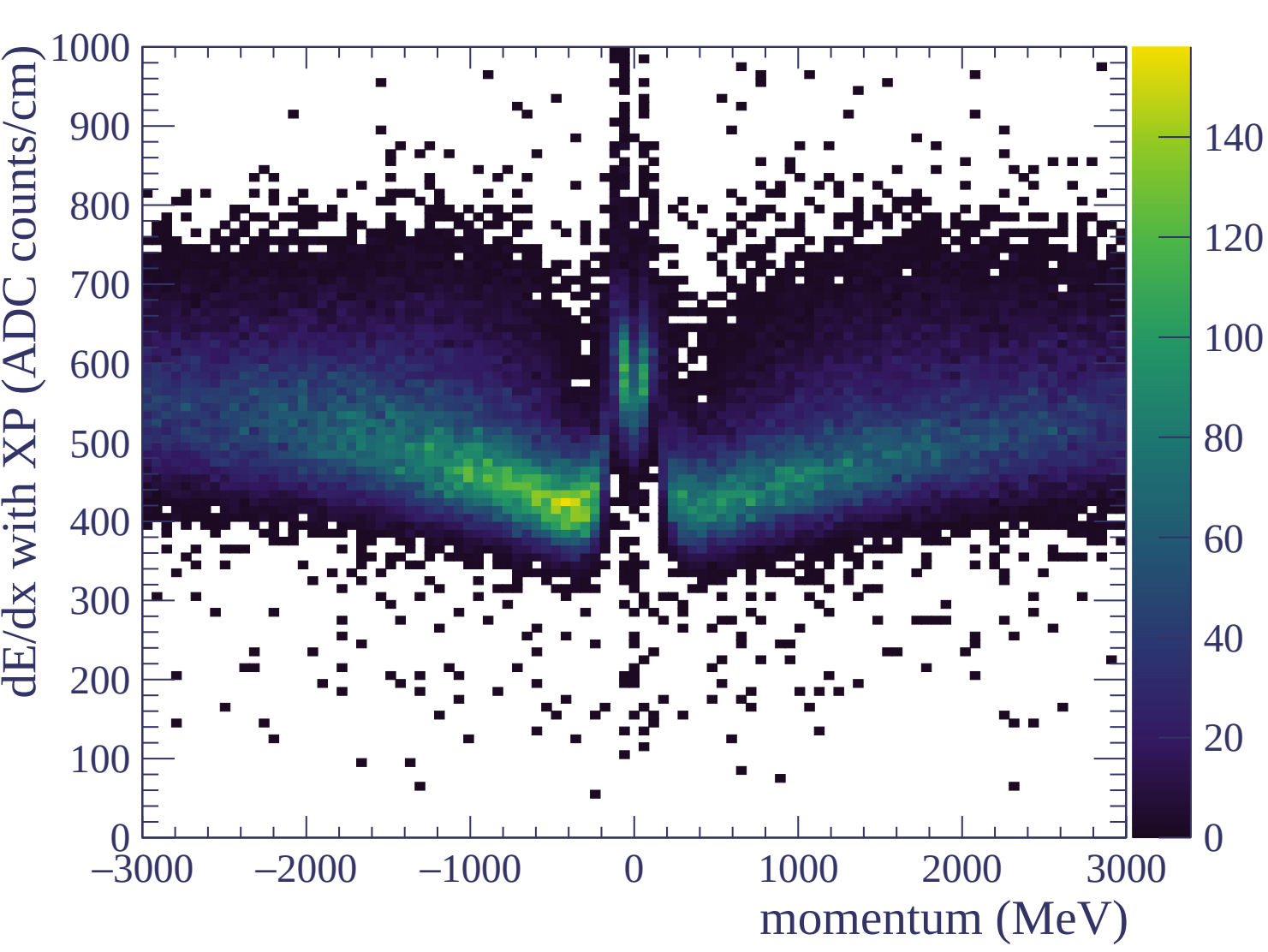
Resolution (%)

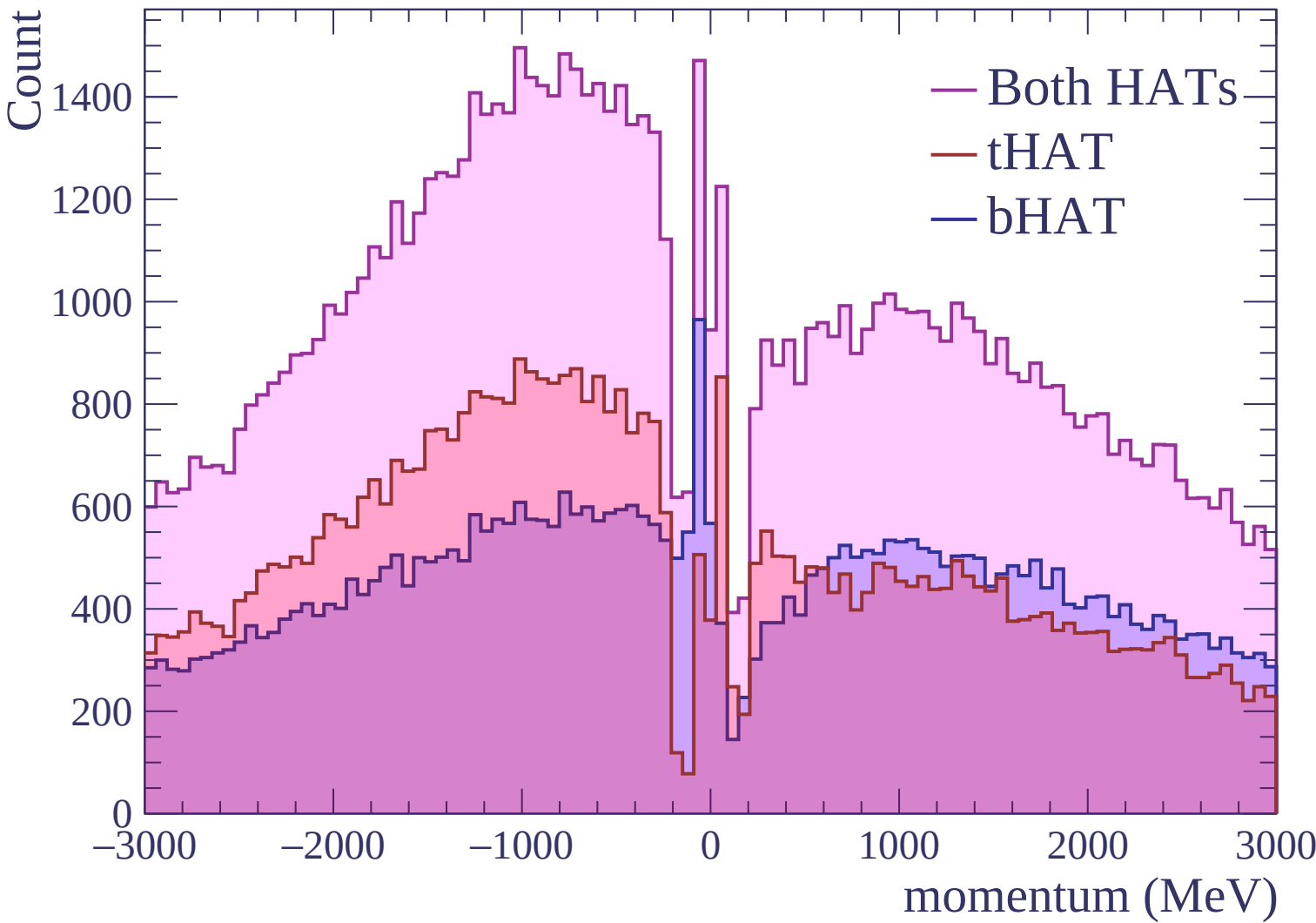






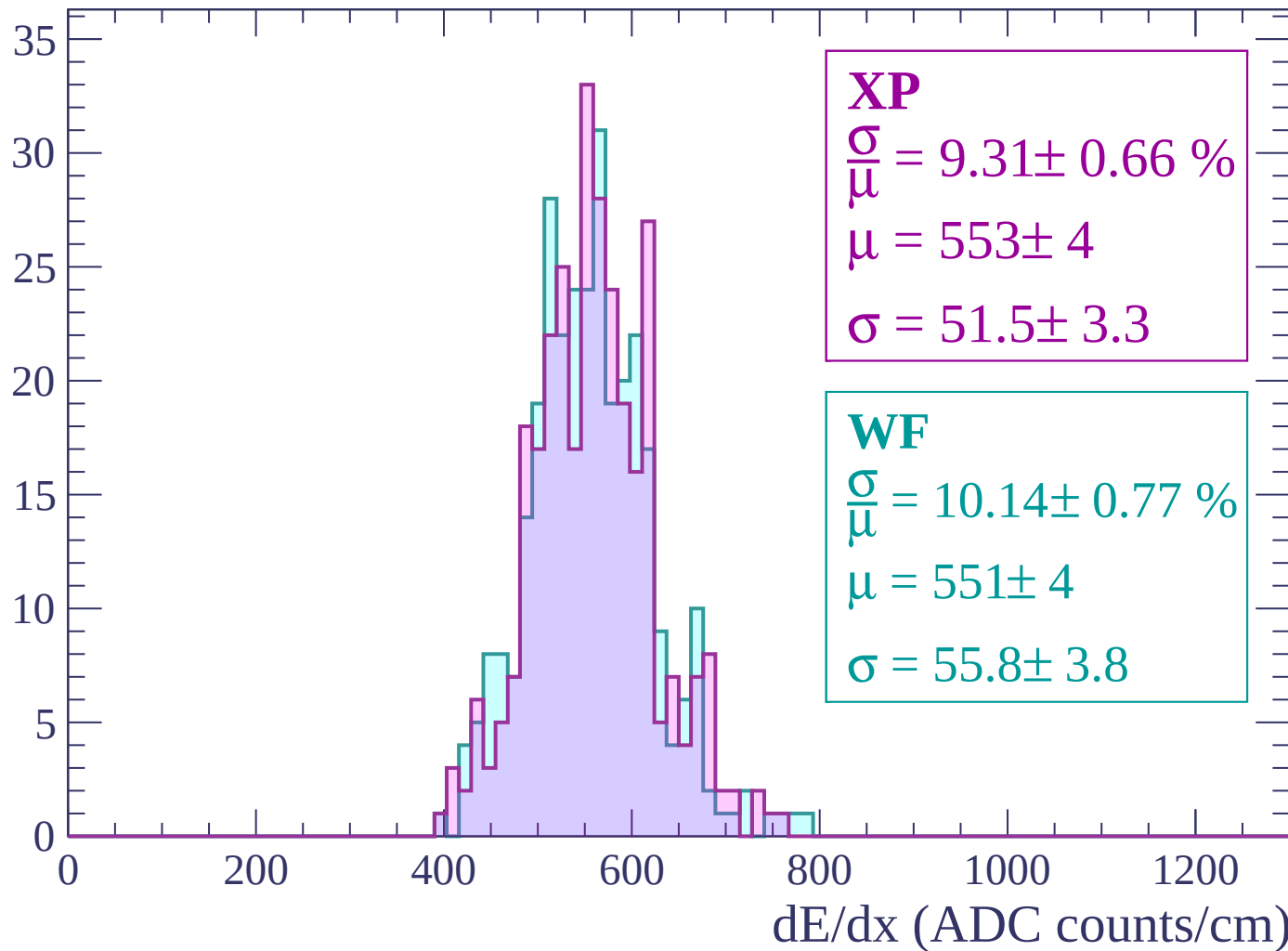






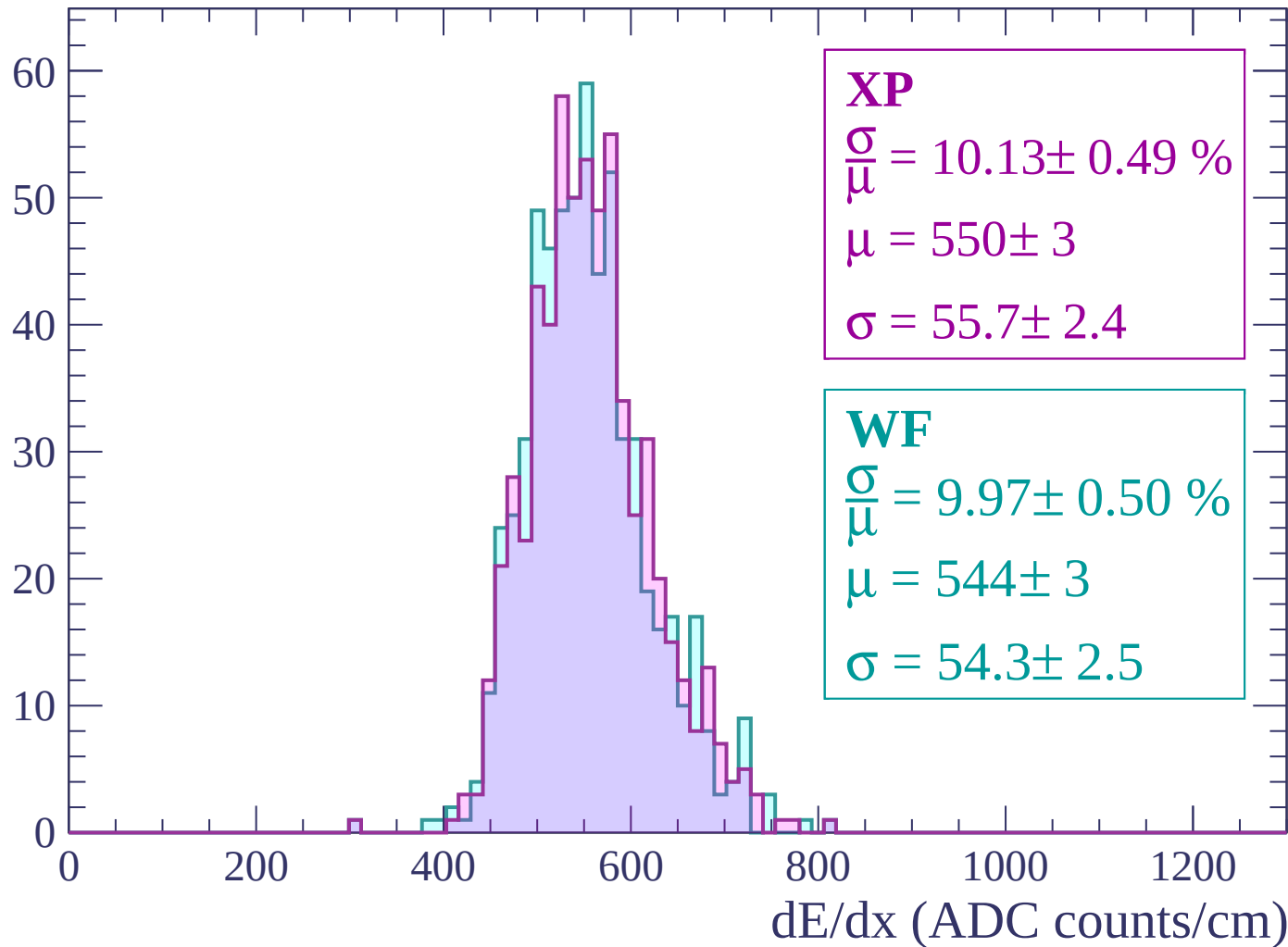
Energy loss | $-3000 < p < -2940$

Count



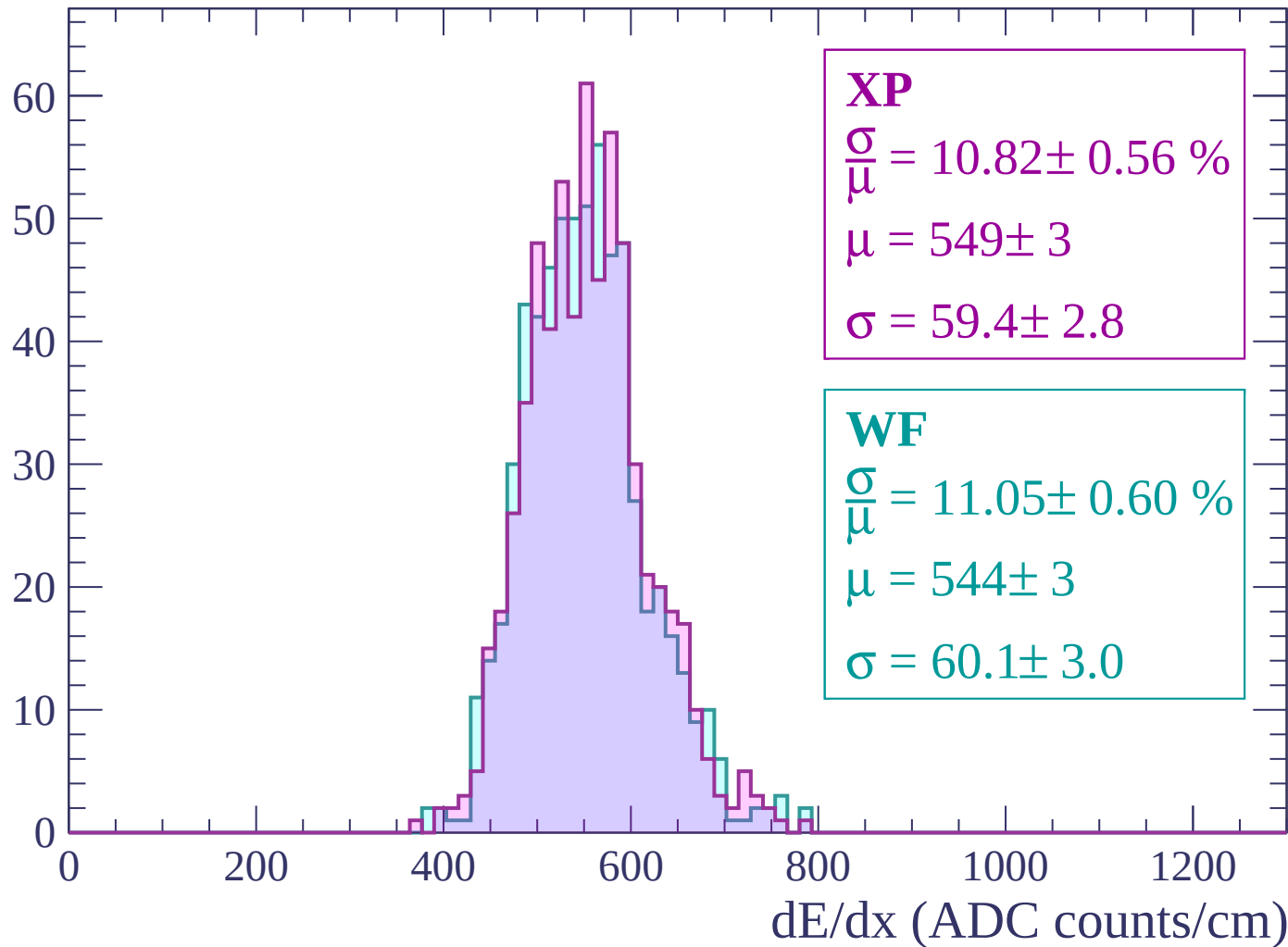
Energy loss | $-2940 < p < -2880$

Count



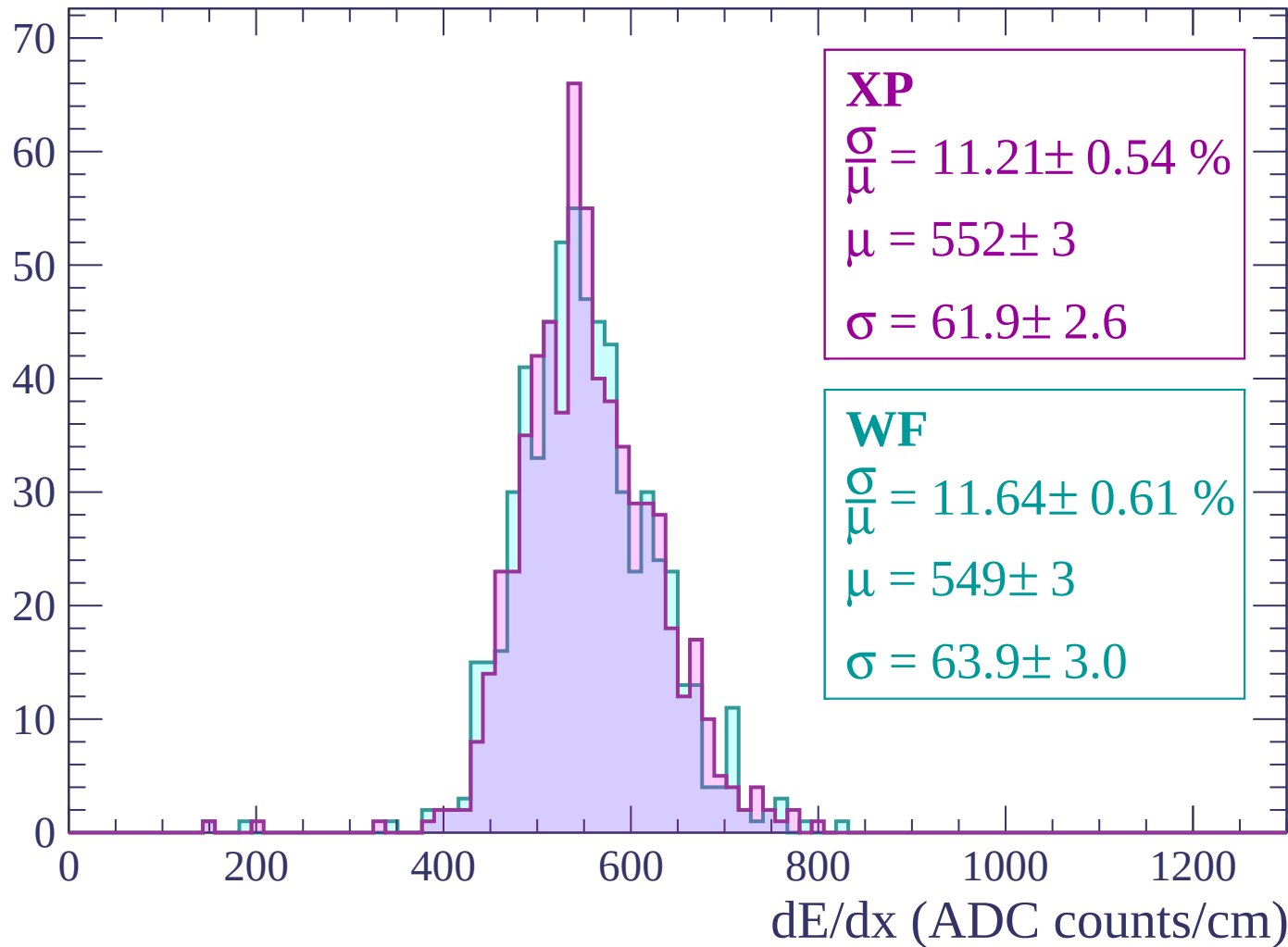
Energy loss | $-2880 < p < -2820$

Count



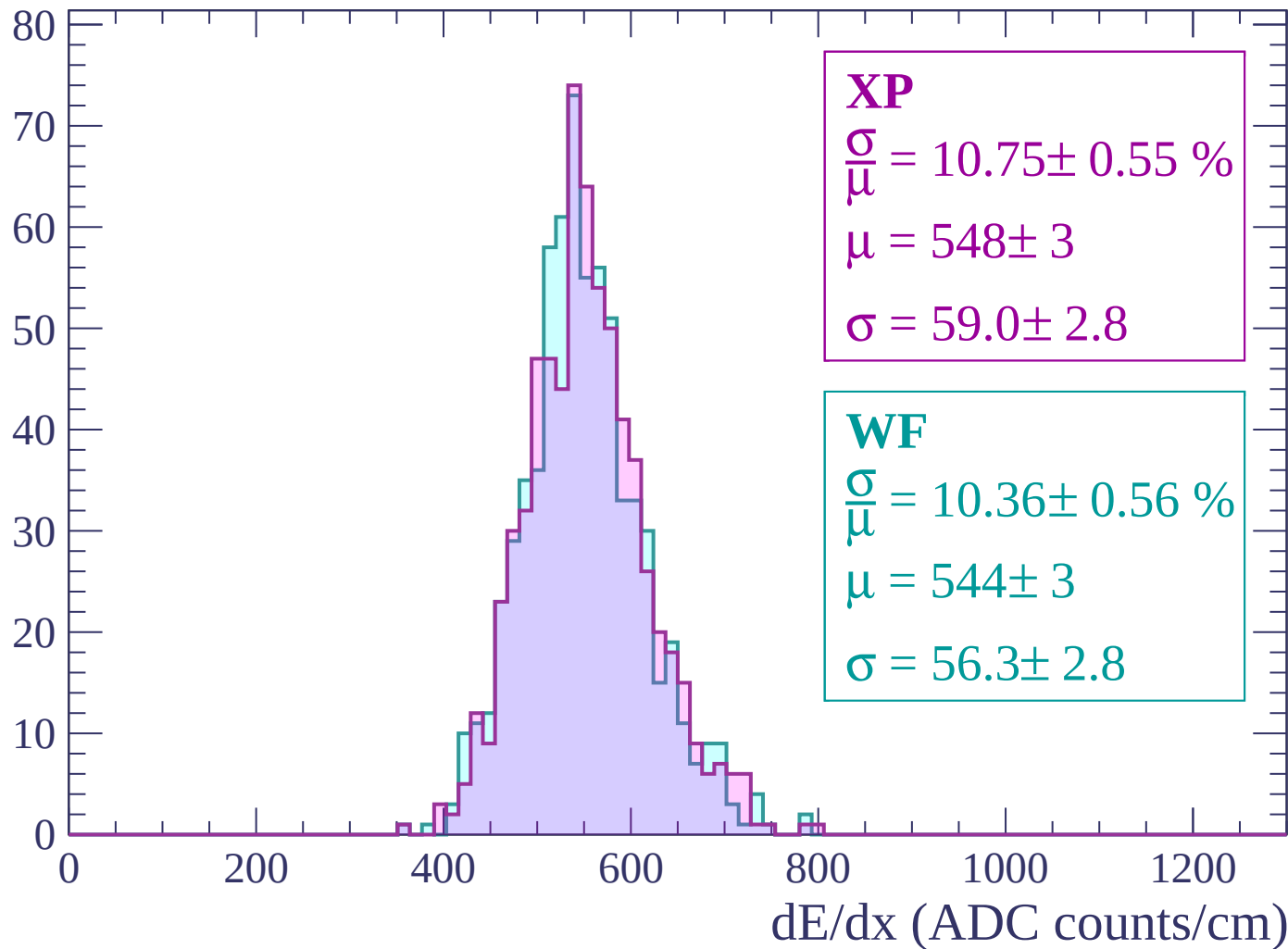
Energy loss | $-2820 < p < -2760$

Count



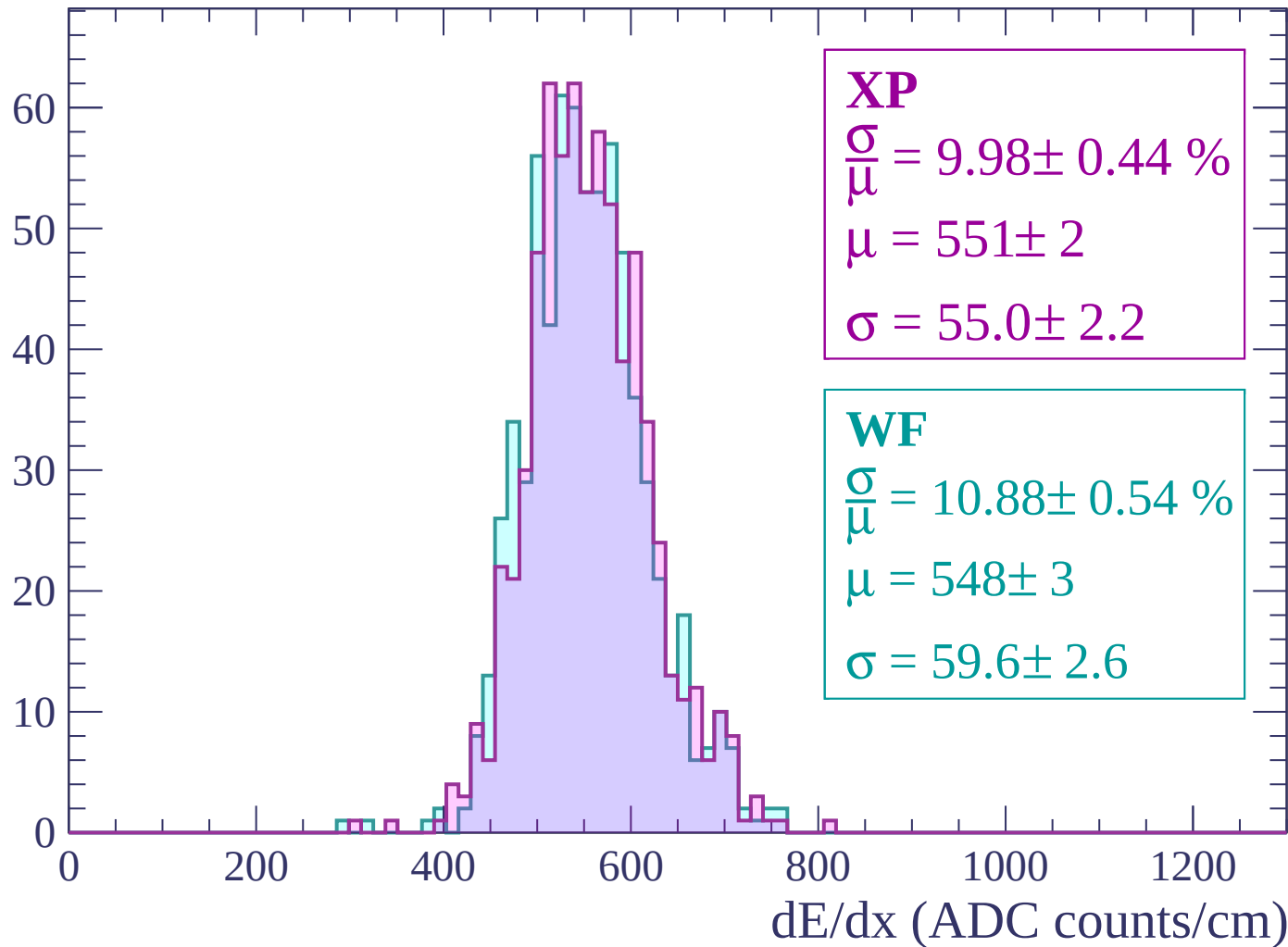
Energy loss | $-2760 < p < -2700$

Count



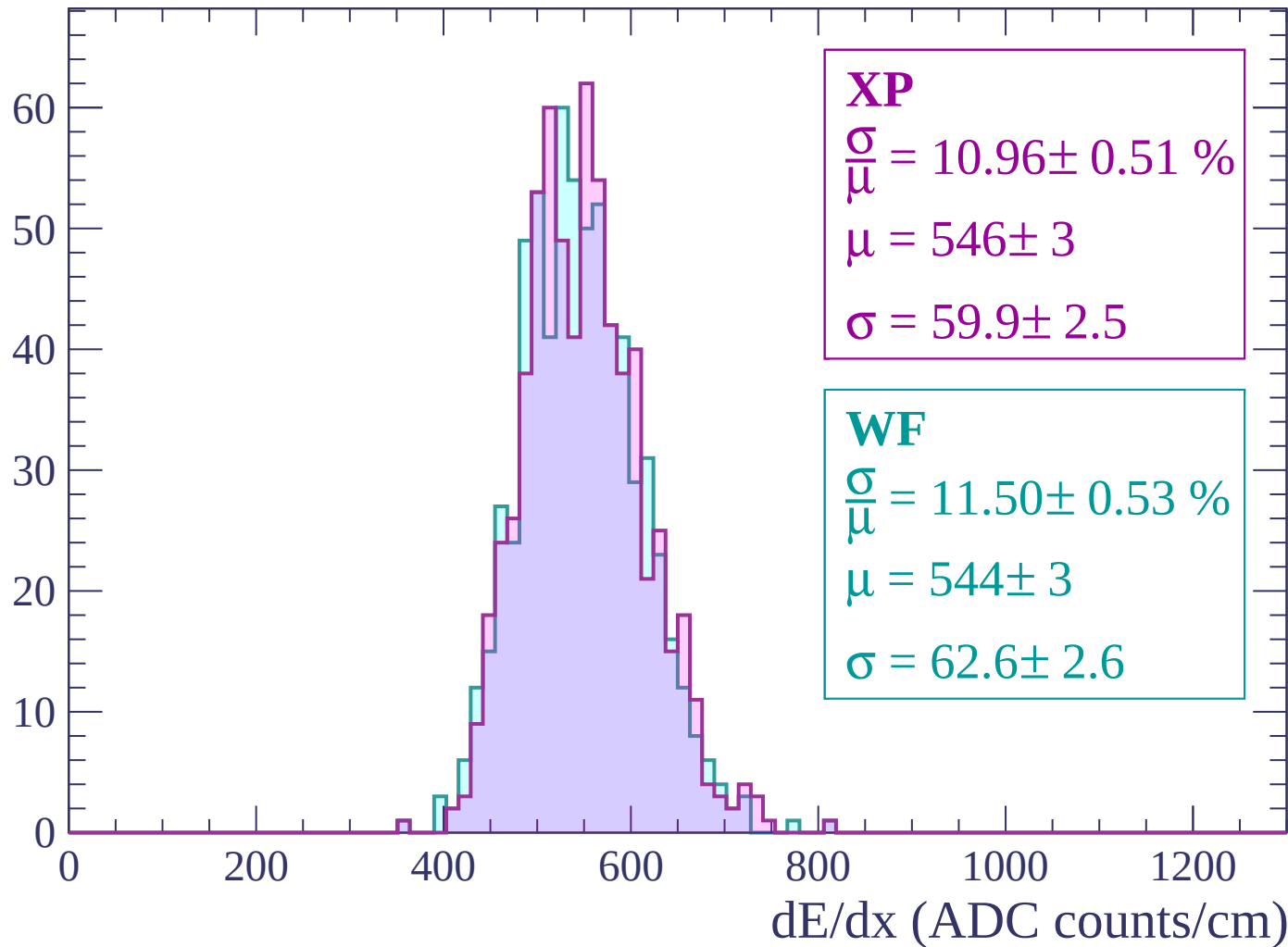
Energy loss | -2700 < p < -2640

Count



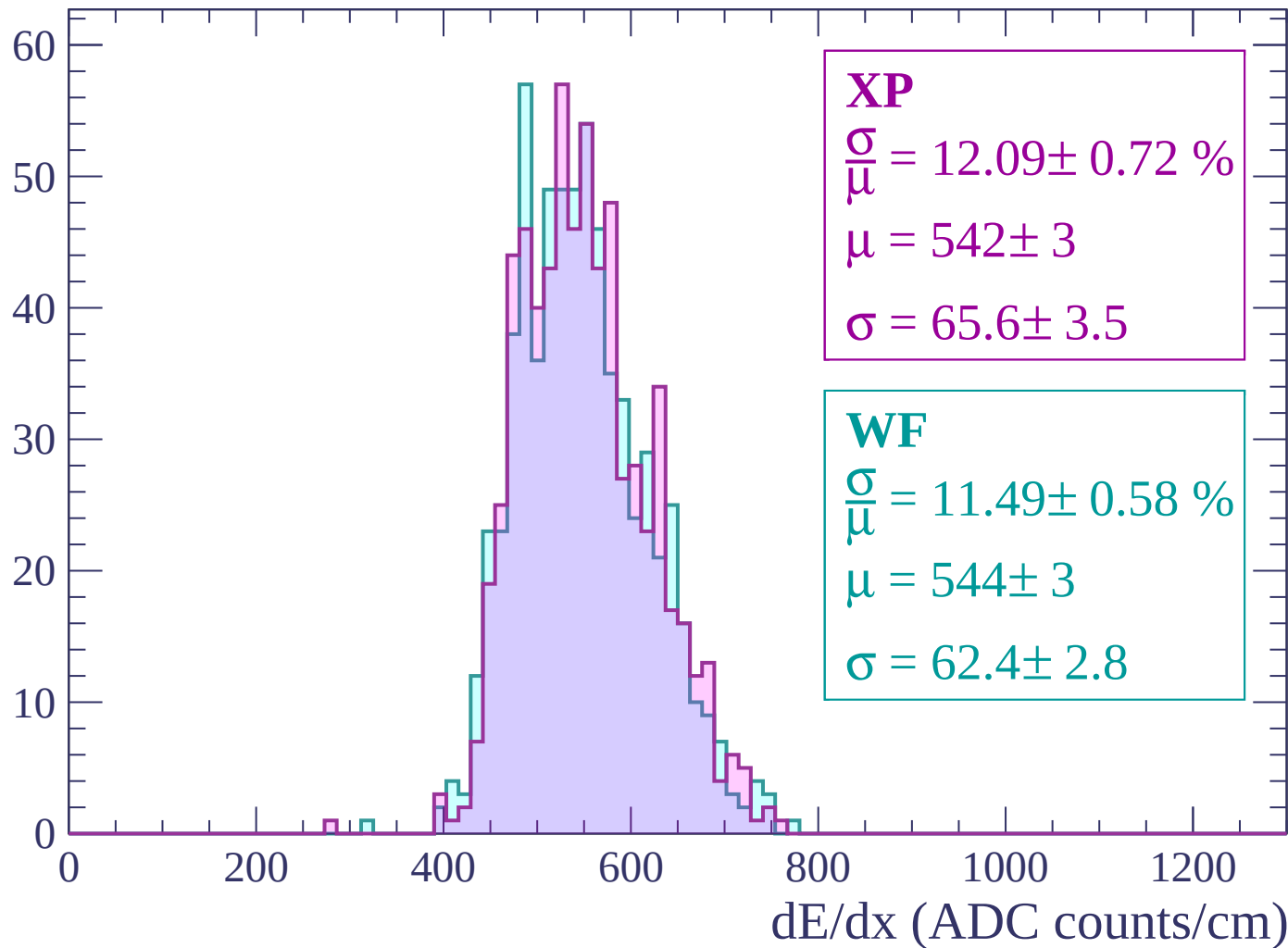
Energy loss | $-2640 < p < -2580$

Count



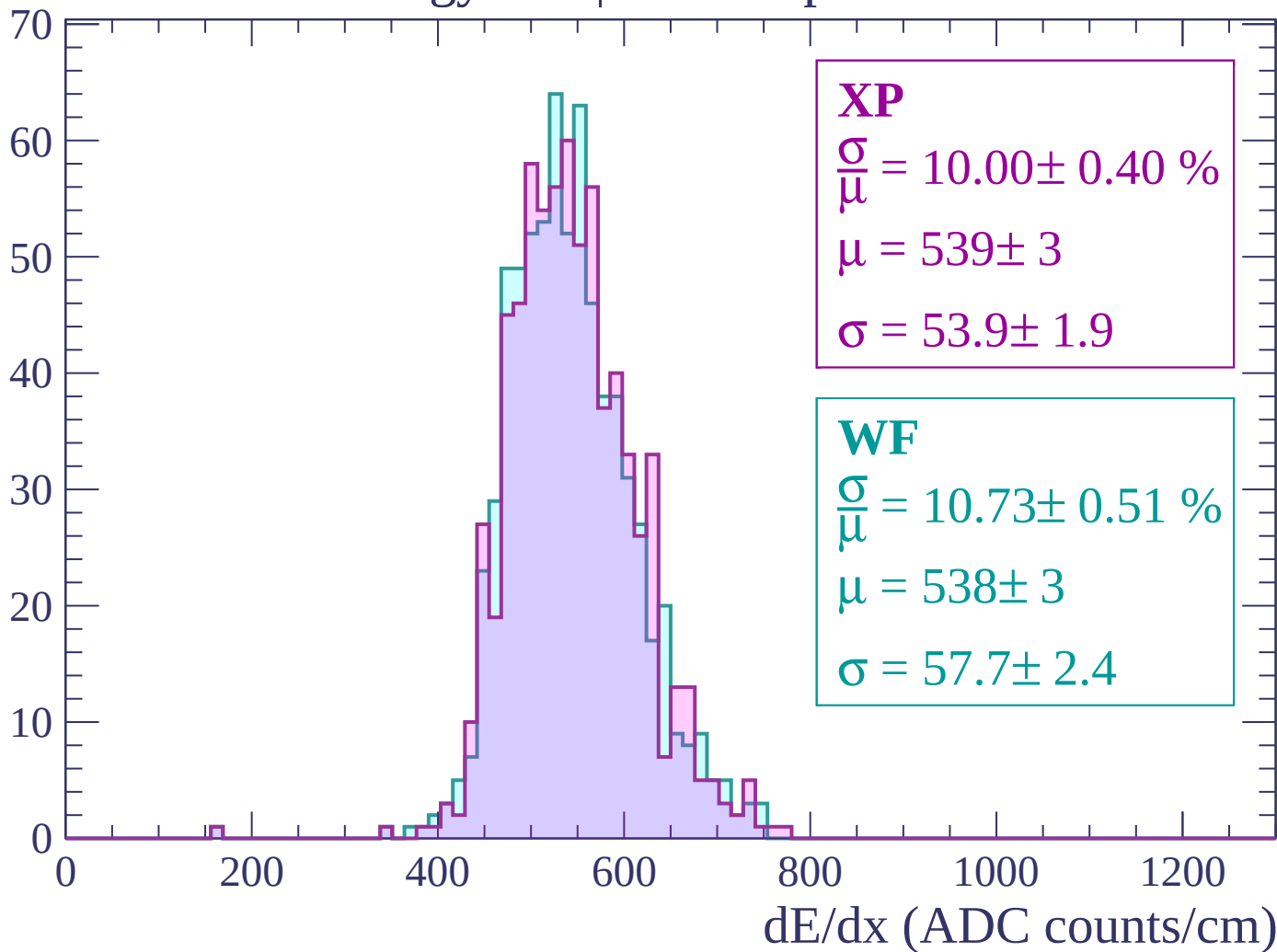
Energy loss | $-2580 < p < -2520$

Count



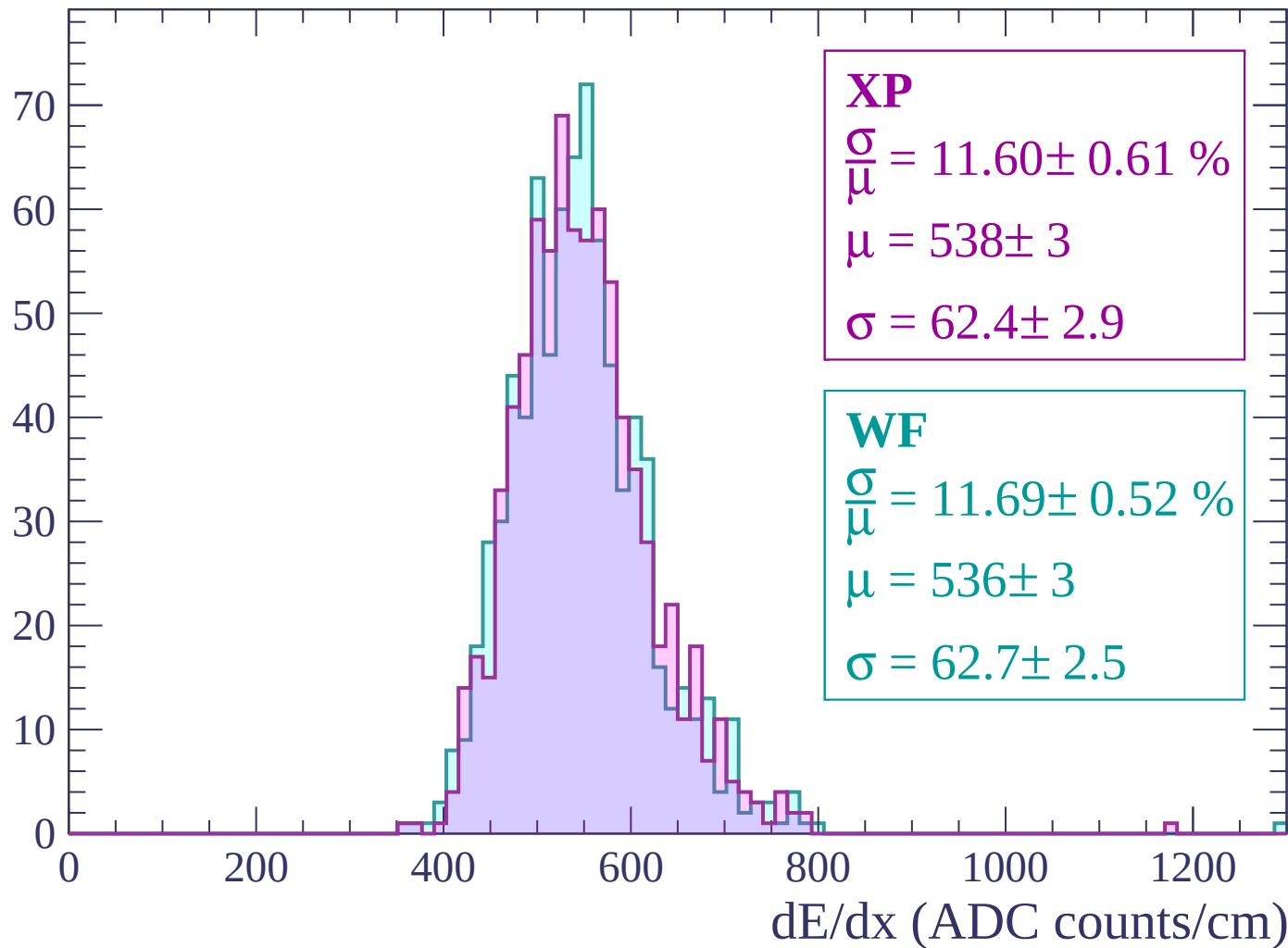
Energy loss | -2520 < p < -2460

Count



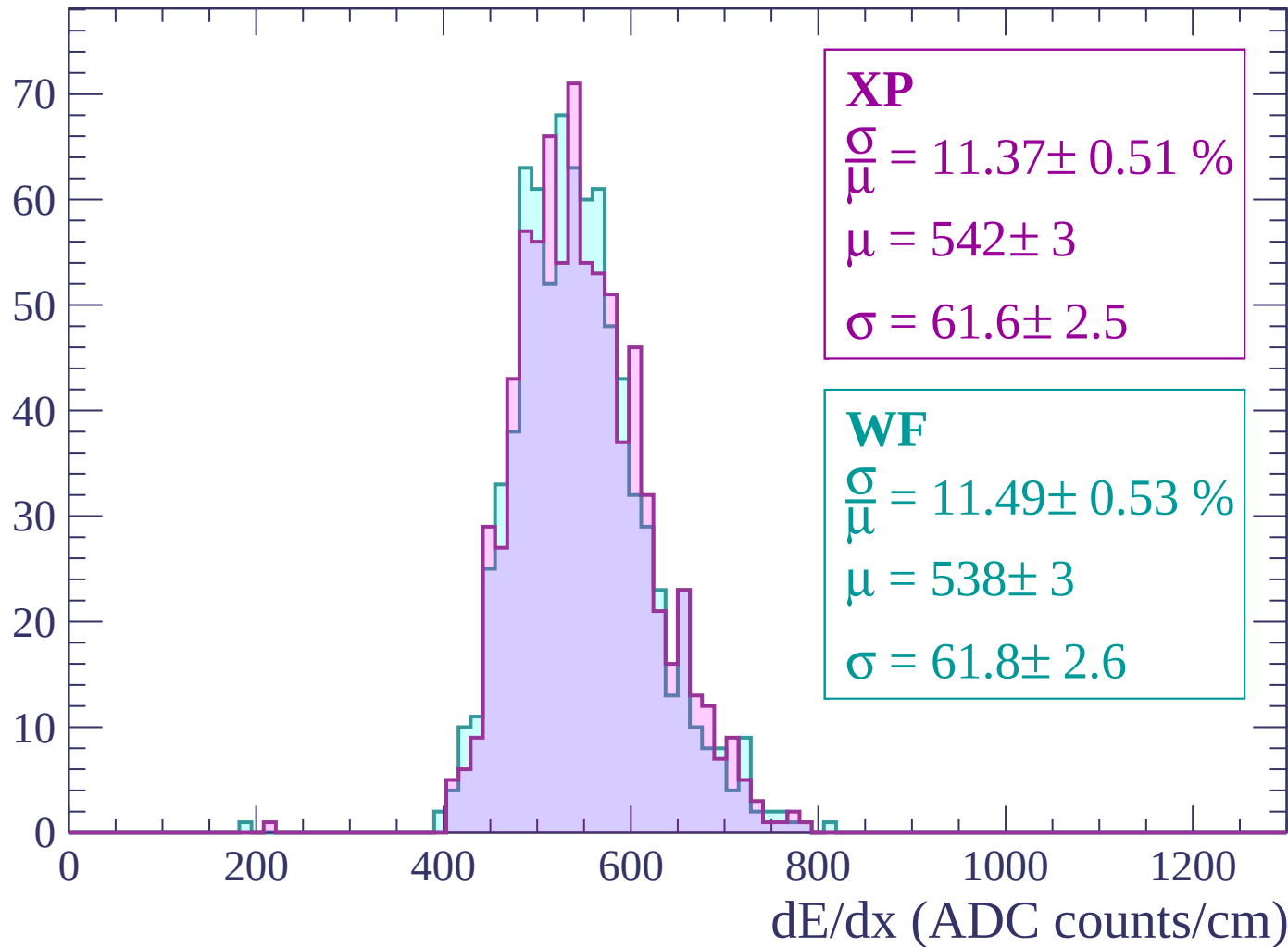
Energy loss | $-2460 < p < -2400$

Count



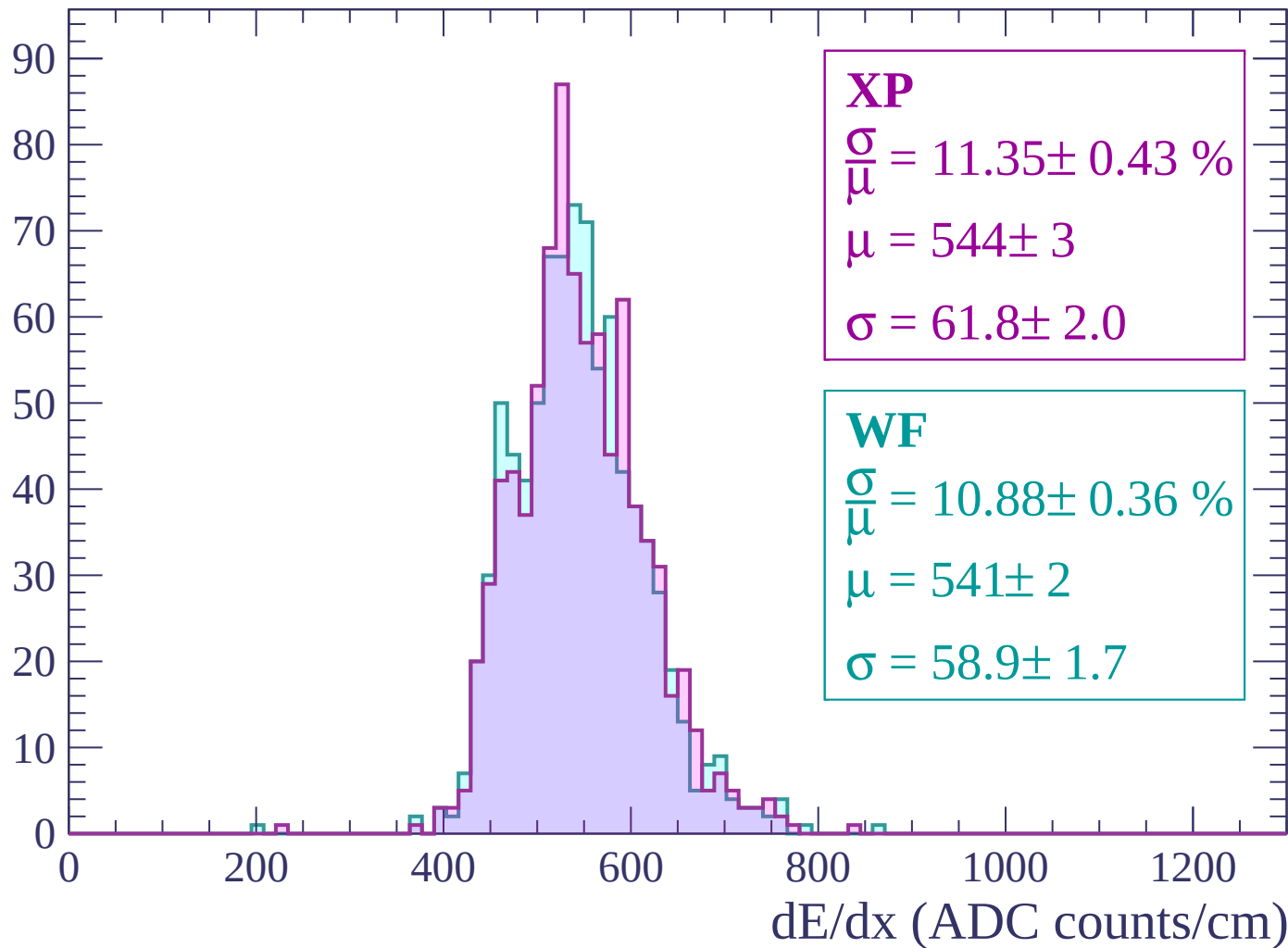
Energy loss | -2400 < p < -2340

Count



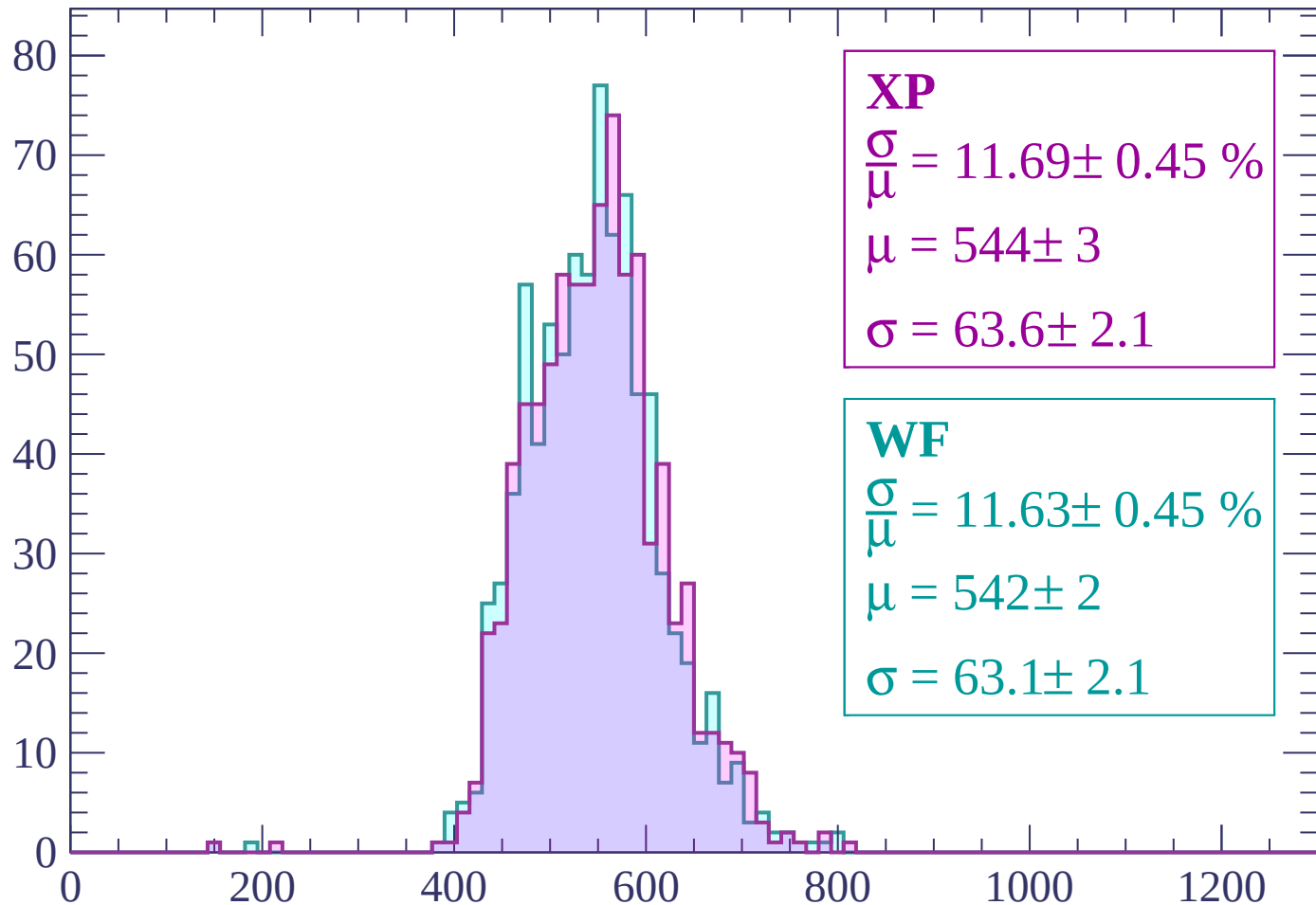
Energy loss | $-2340 < p < -2280$

Count



Energy loss | $-2280 < p < -2220$

Count



XP

$$\frac{\sigma}{\mu} = 11.69 \pm 0.45 \%$$

$$\mu = 544 \pm 3$$

$$\sigma = 63.6 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 11.63 \pm 0.45 \%$$

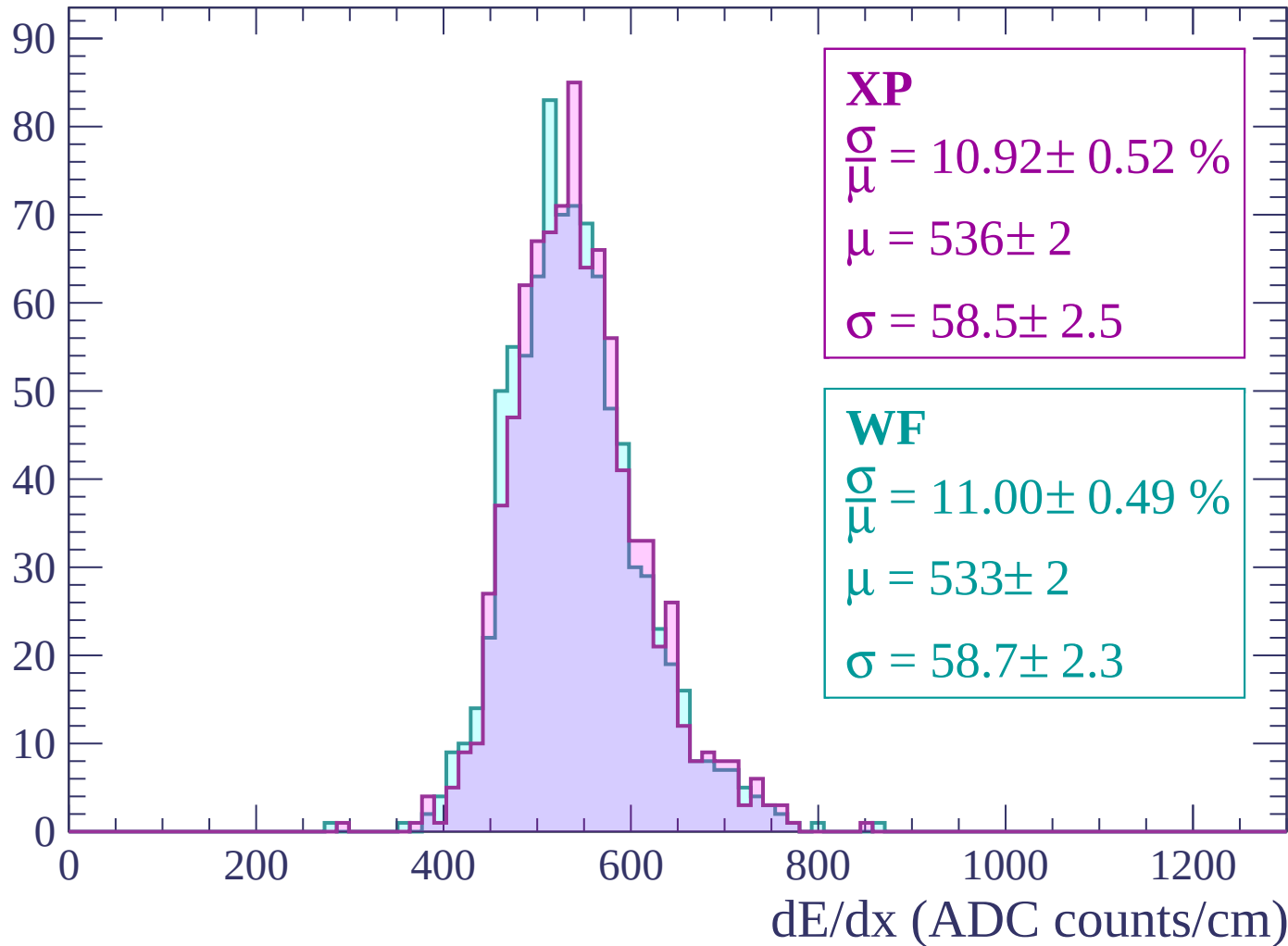
$$\mu = 542 \pm 2$$

$$\sigma = 63.1 \pm 2.1$$

dE/dx (ADC counts/cm)

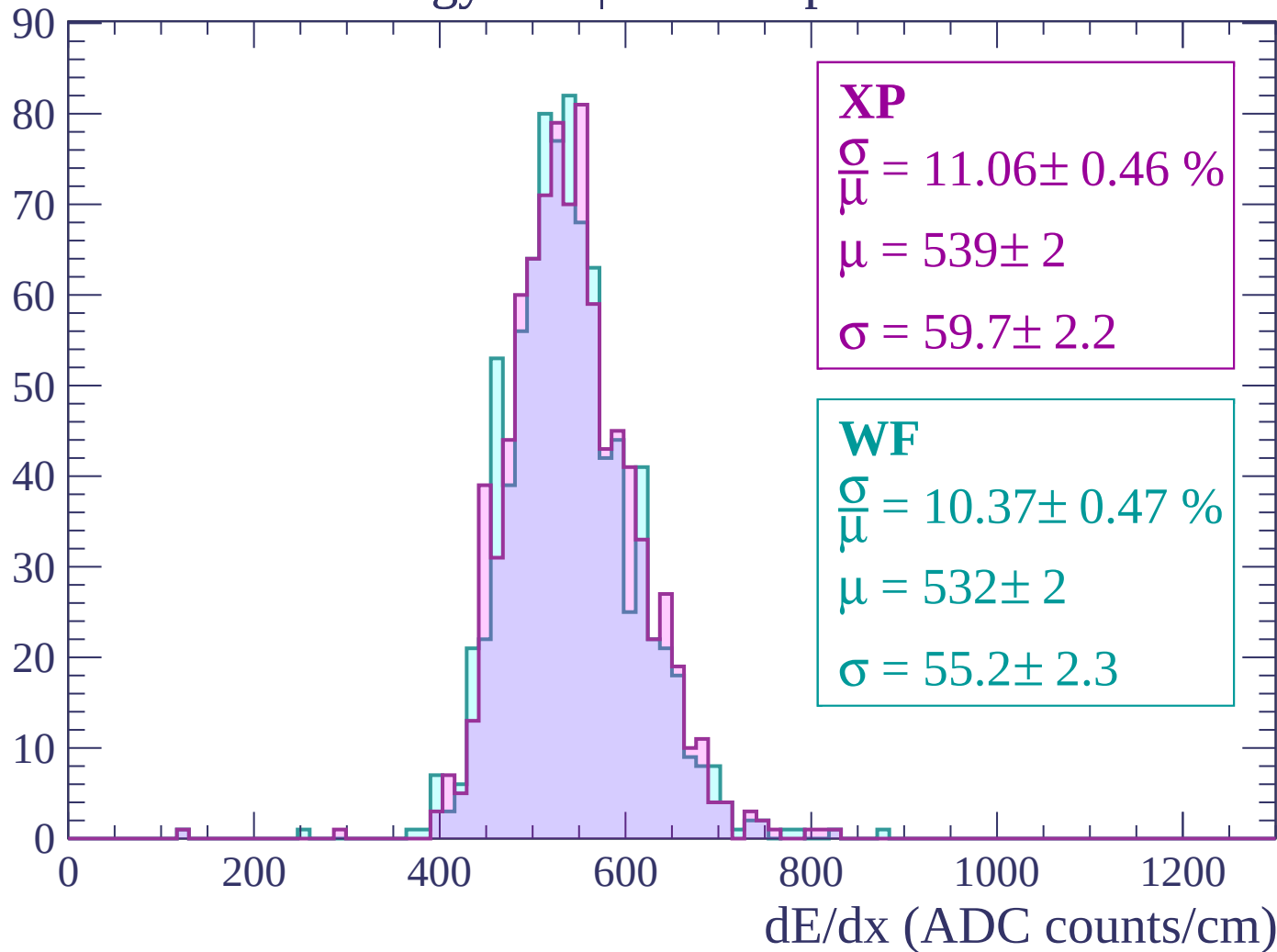
Energy loss | -2220 < p < -2160

Count



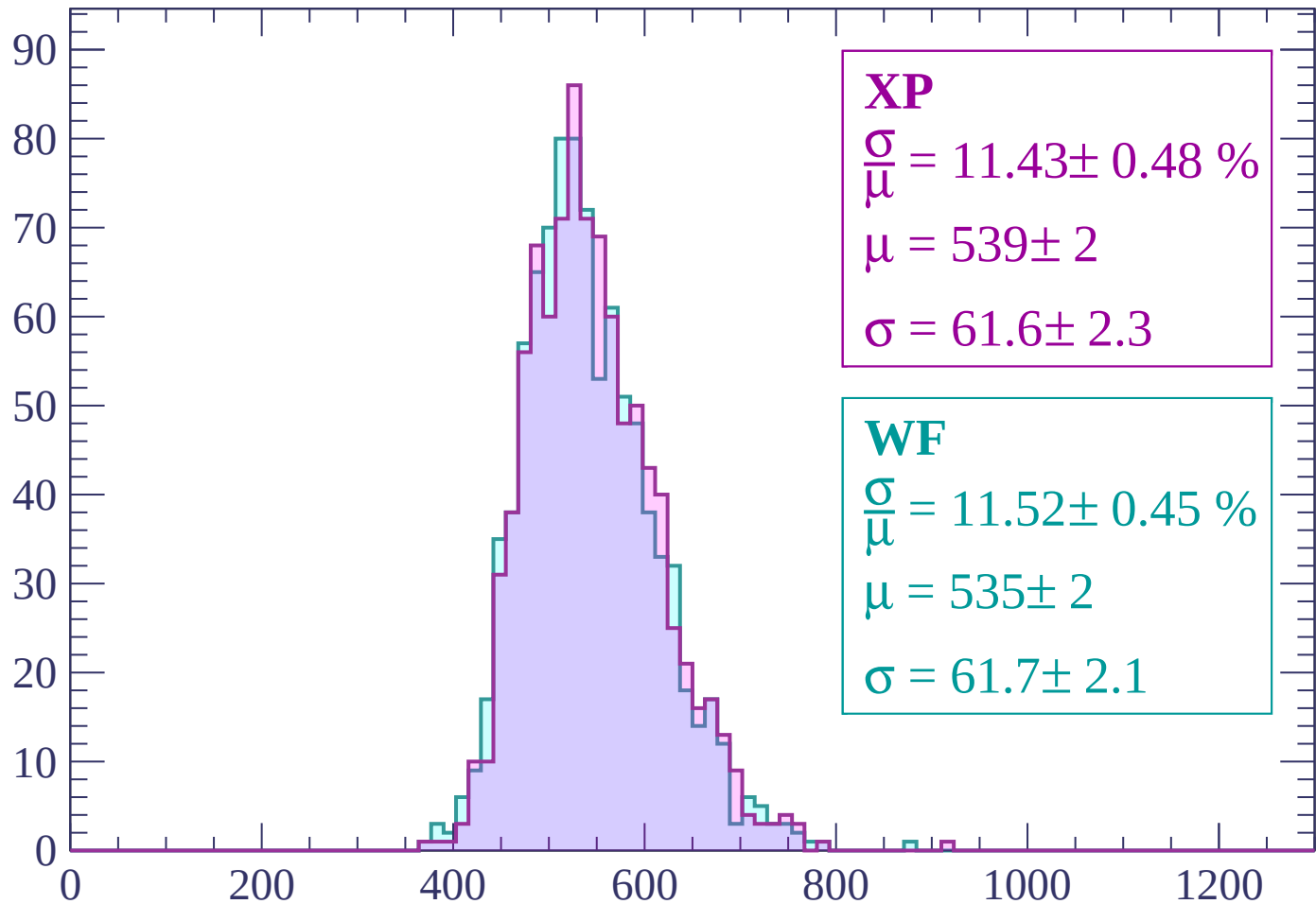
Energy loss | -2160 < p < -2100

Count



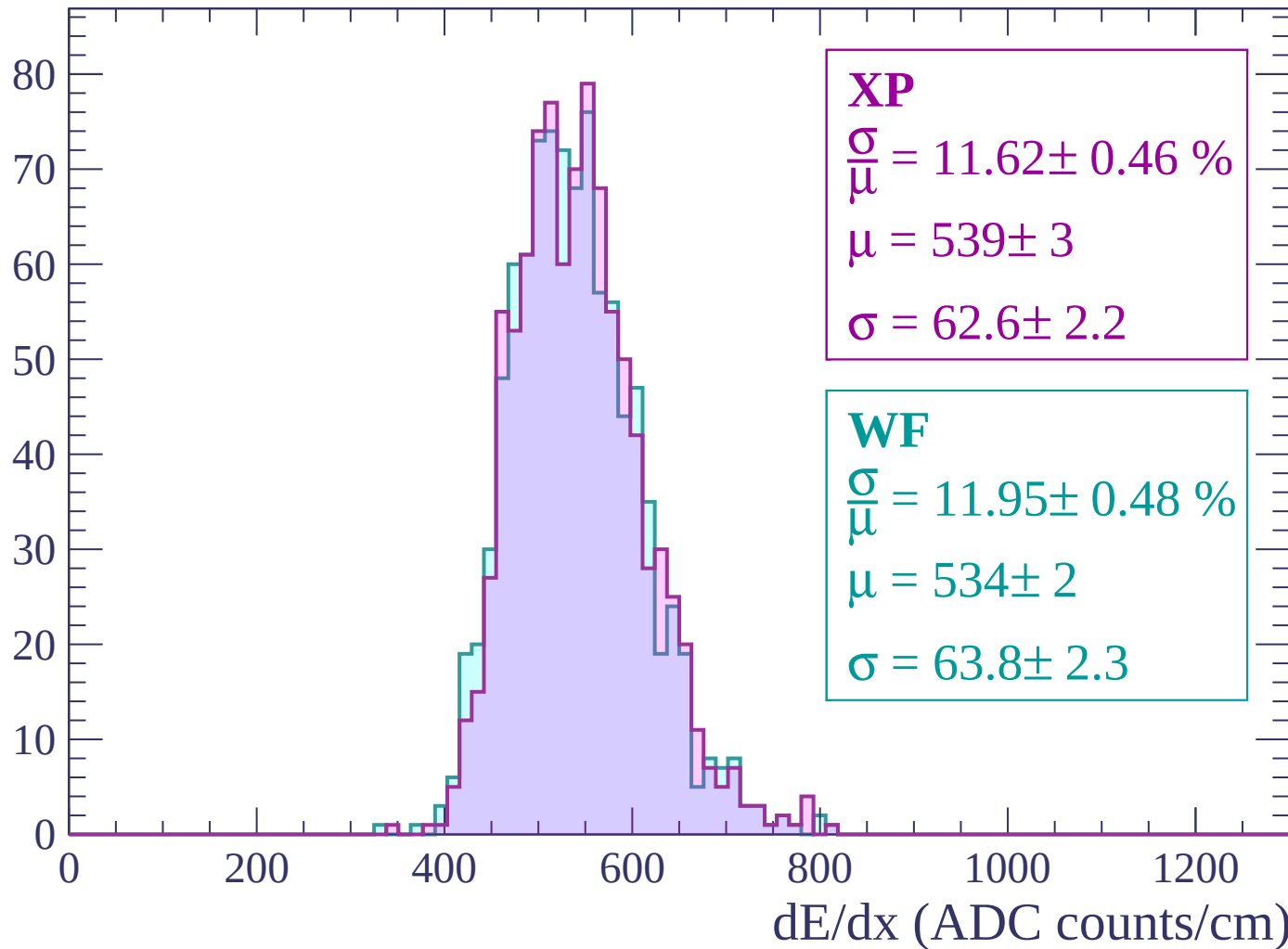
Energy loss | -2100 < p < -2040

Count



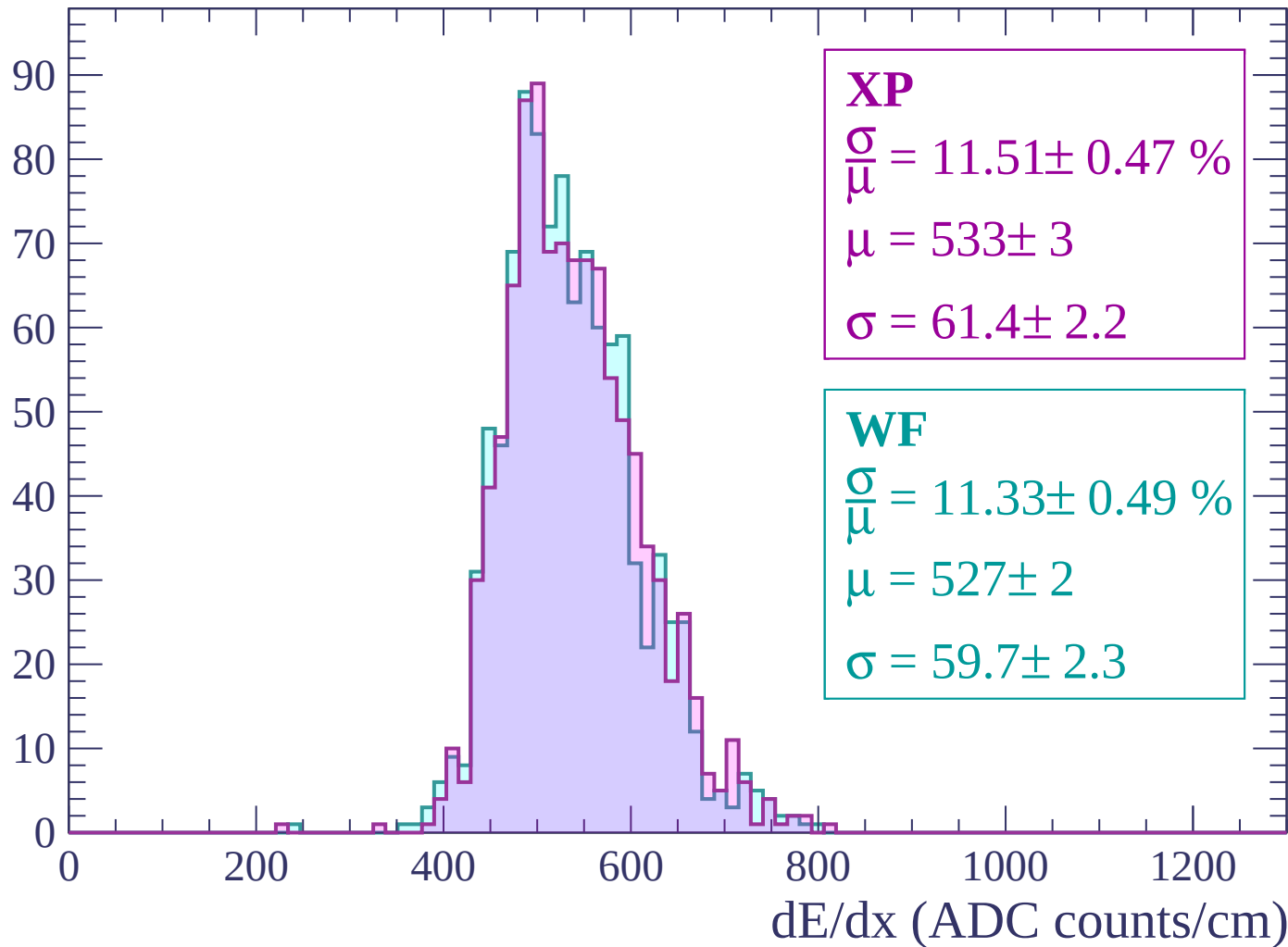
Energy loss | -2040 < p < -1980

Count



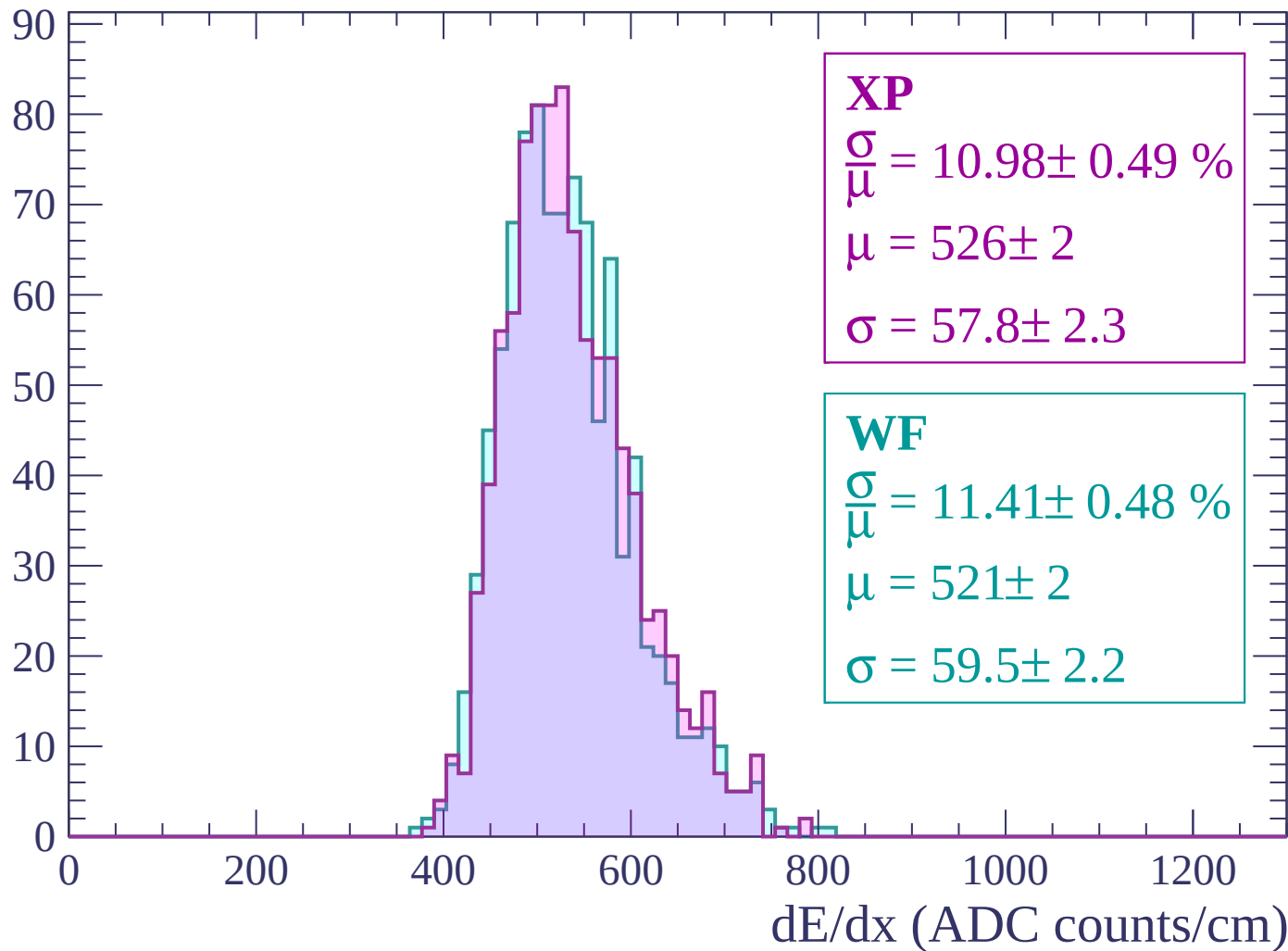
Energy loss | -1980 < p < -1920

Count



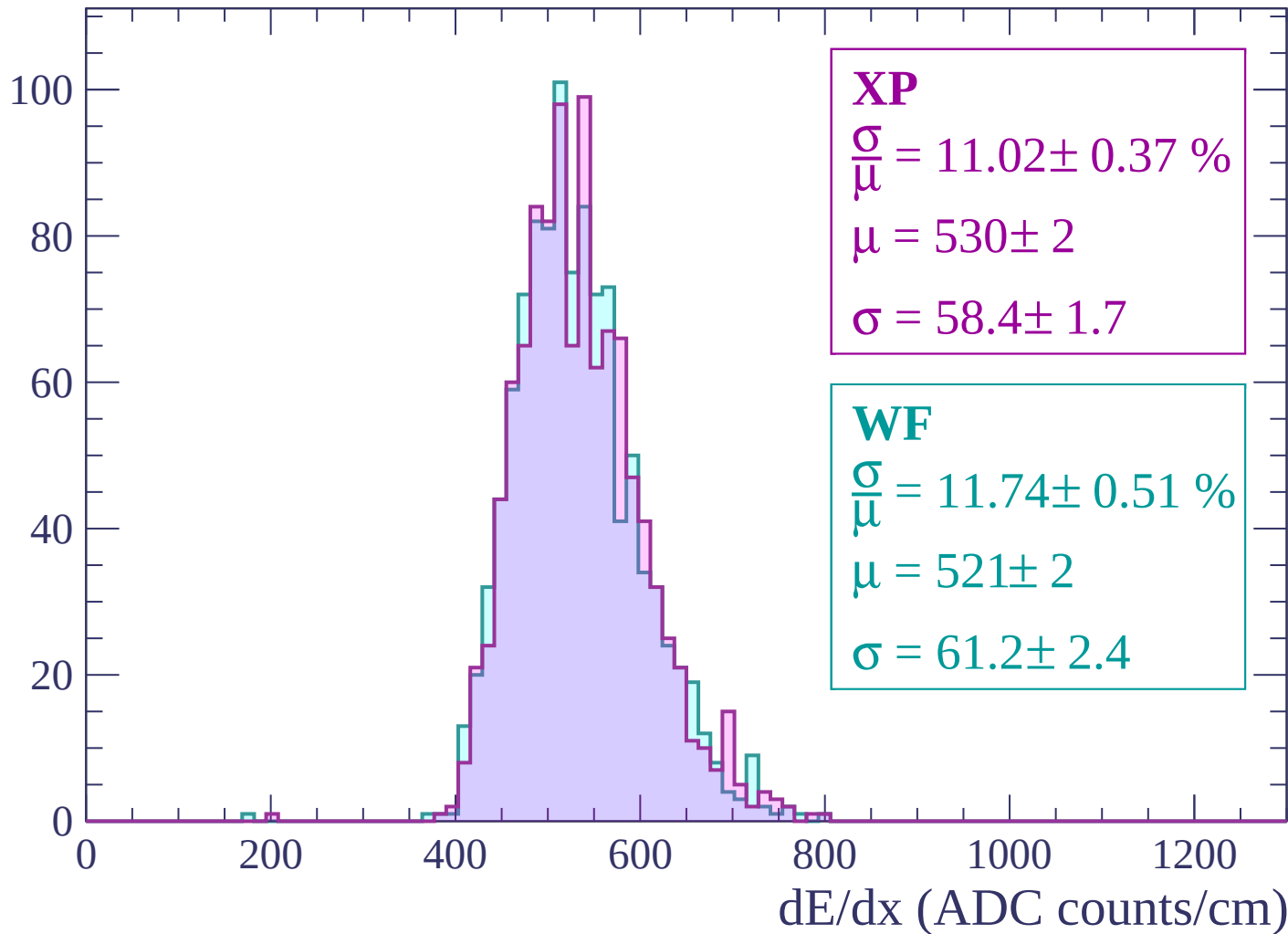
Energy loss | -1920 < p < -1860

Count



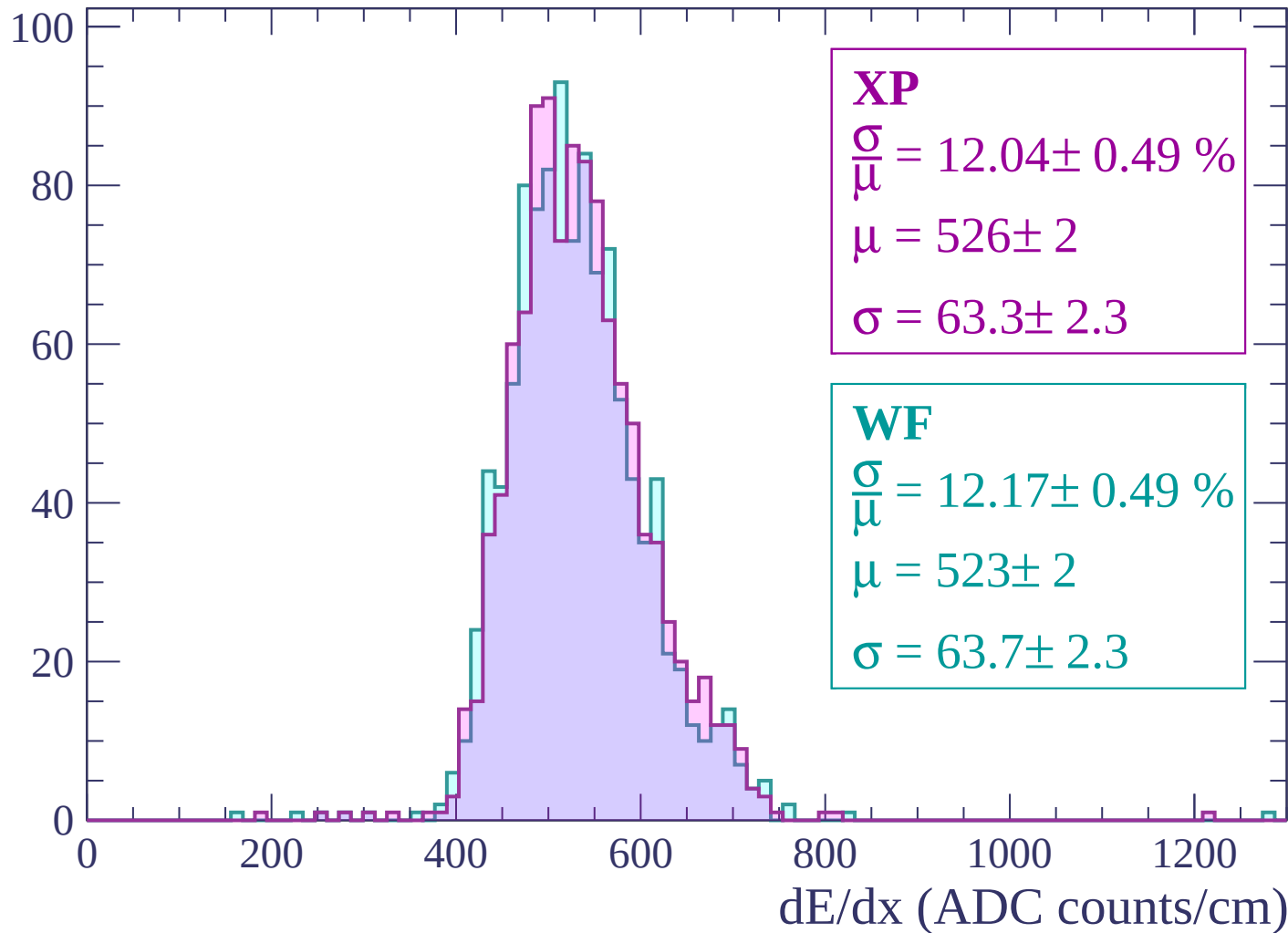
Energy loss | $-1860 < p < -1800$

Count



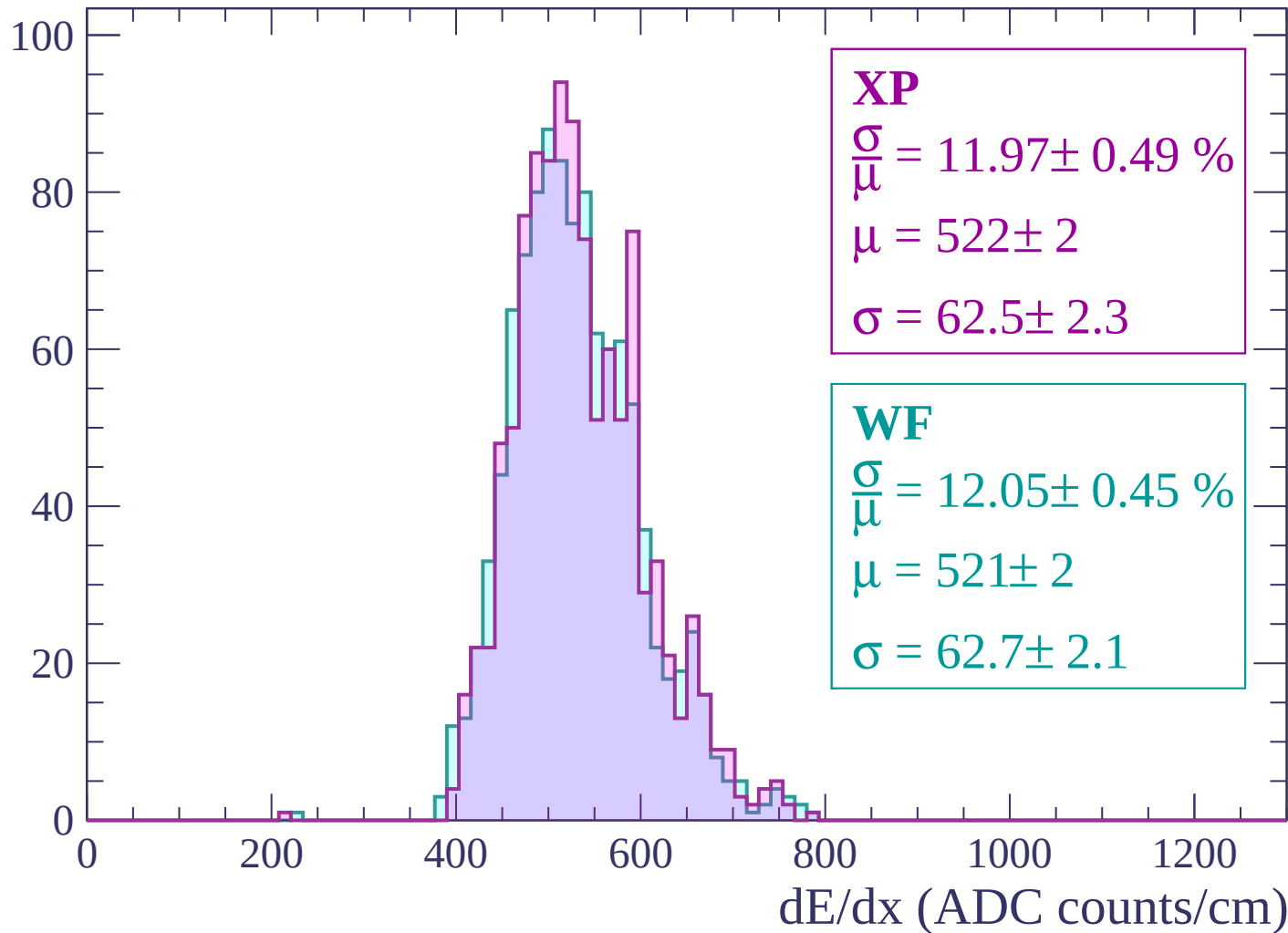
Energy loss | $-1800 < p < -1740$

Count



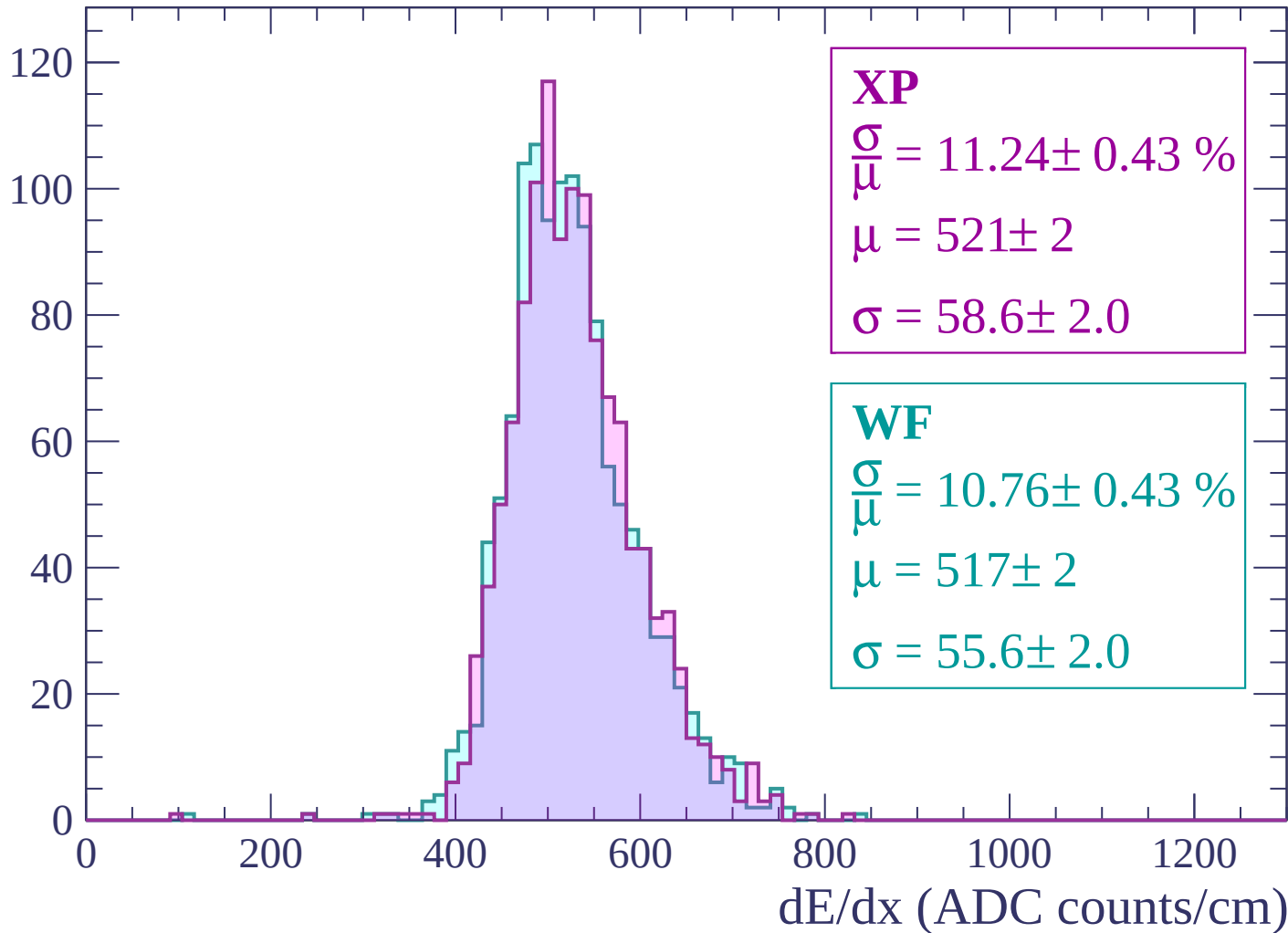
Energy loss | -1740 < p < -1680

Count



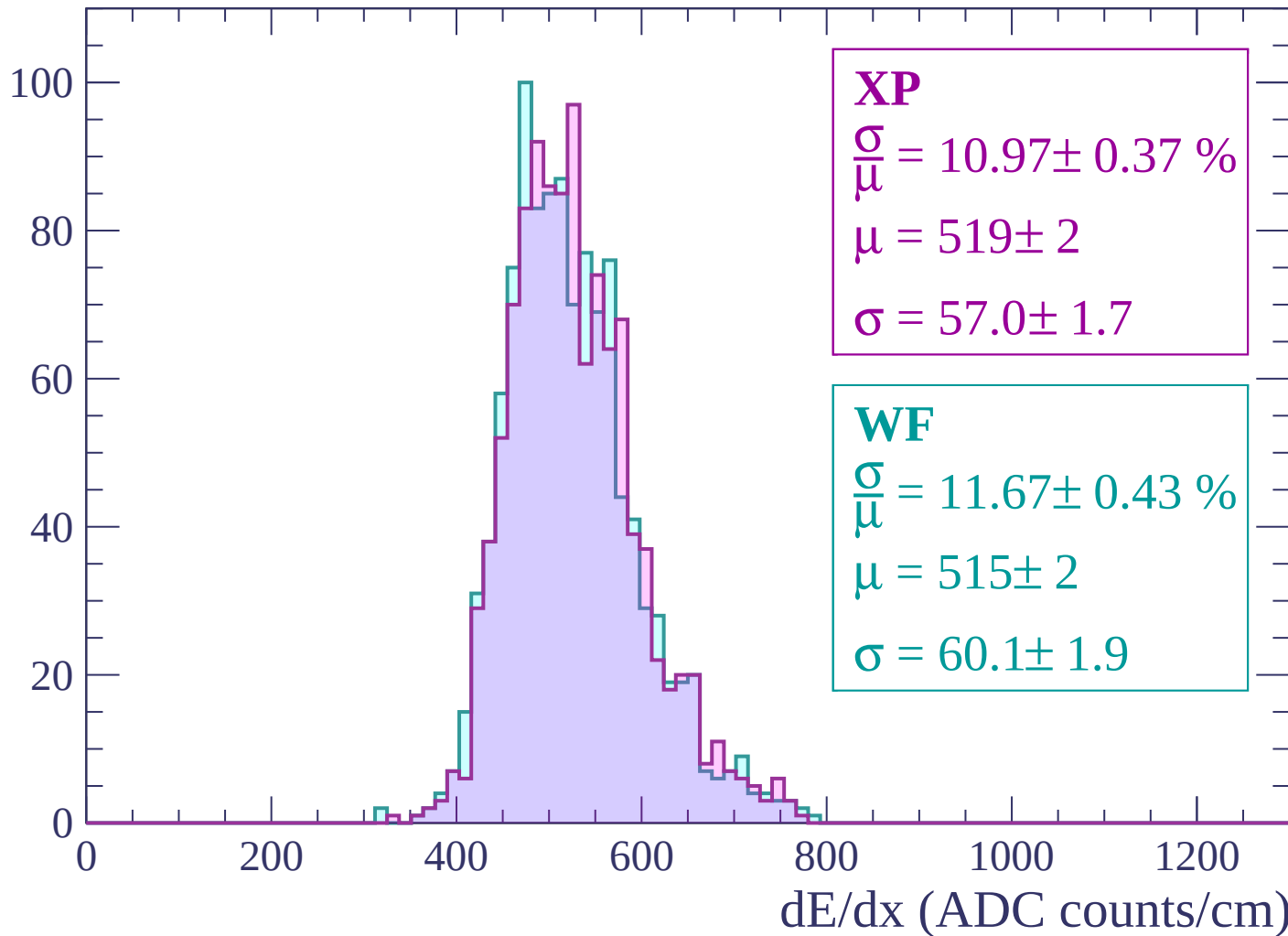
Energy loss | -1680 < p < -1620

Count



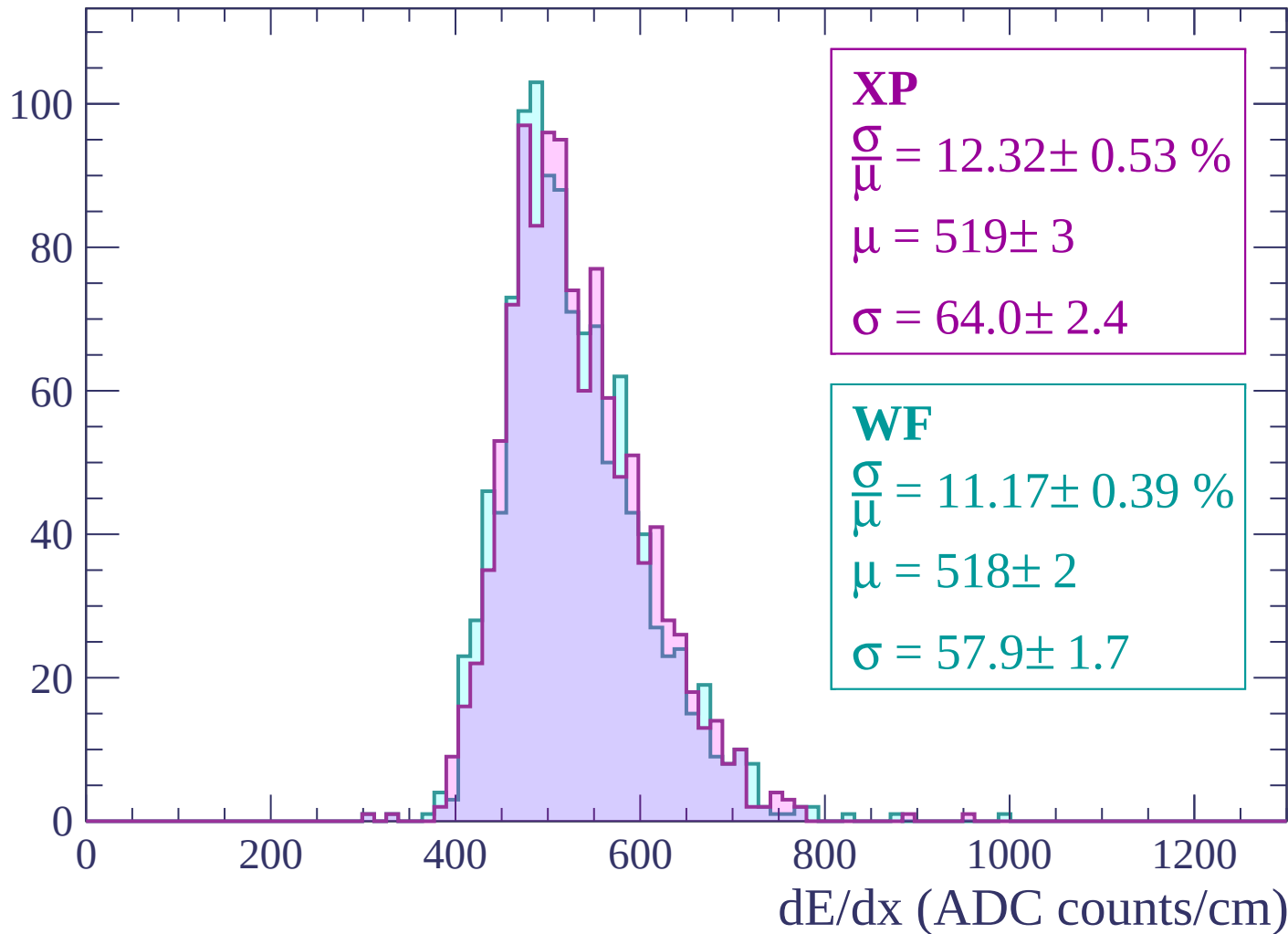
Energy loss | -1620 < p < -1560

Count



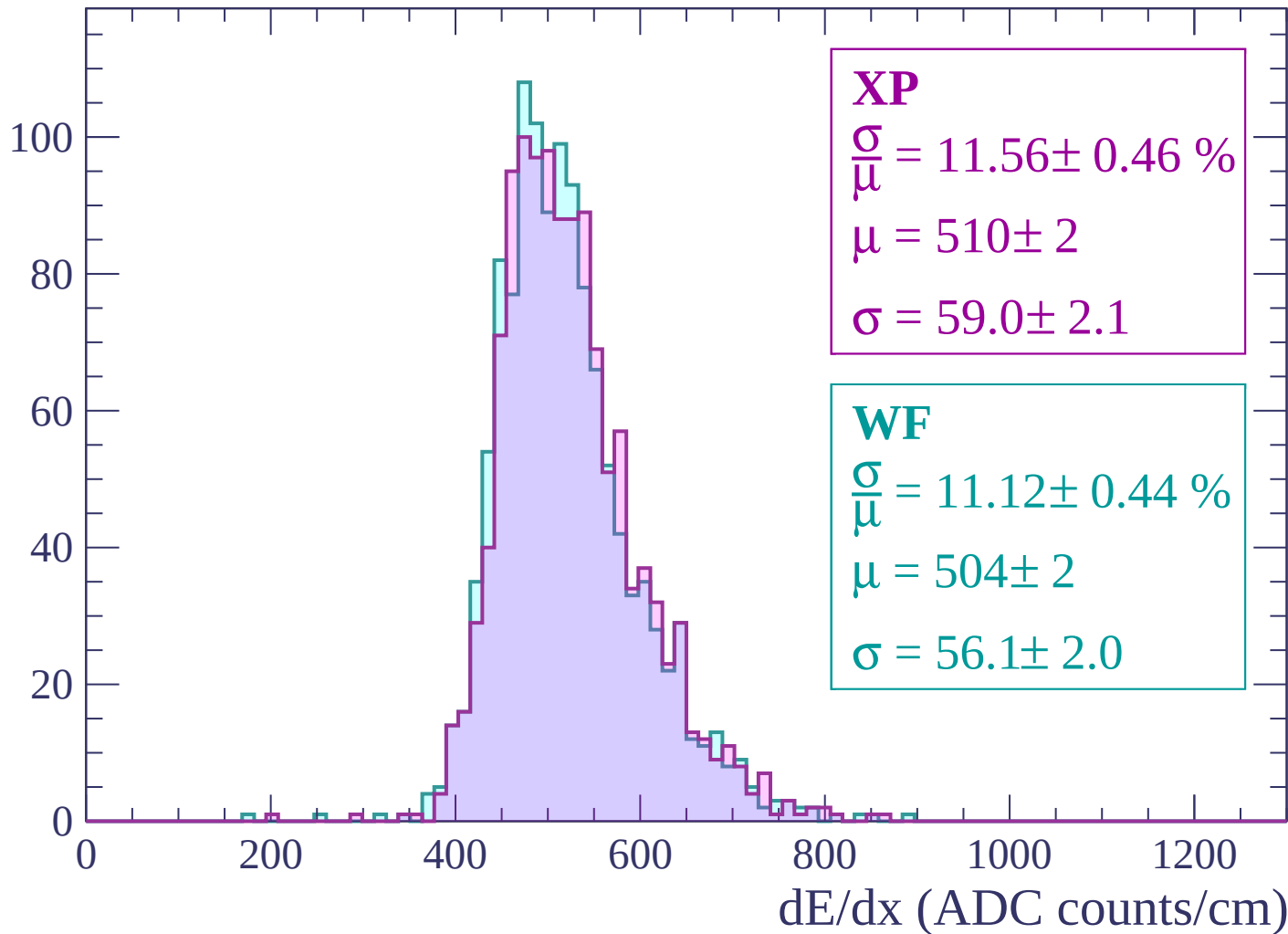
Energy loss | $-1560 < p < -1500$

Count



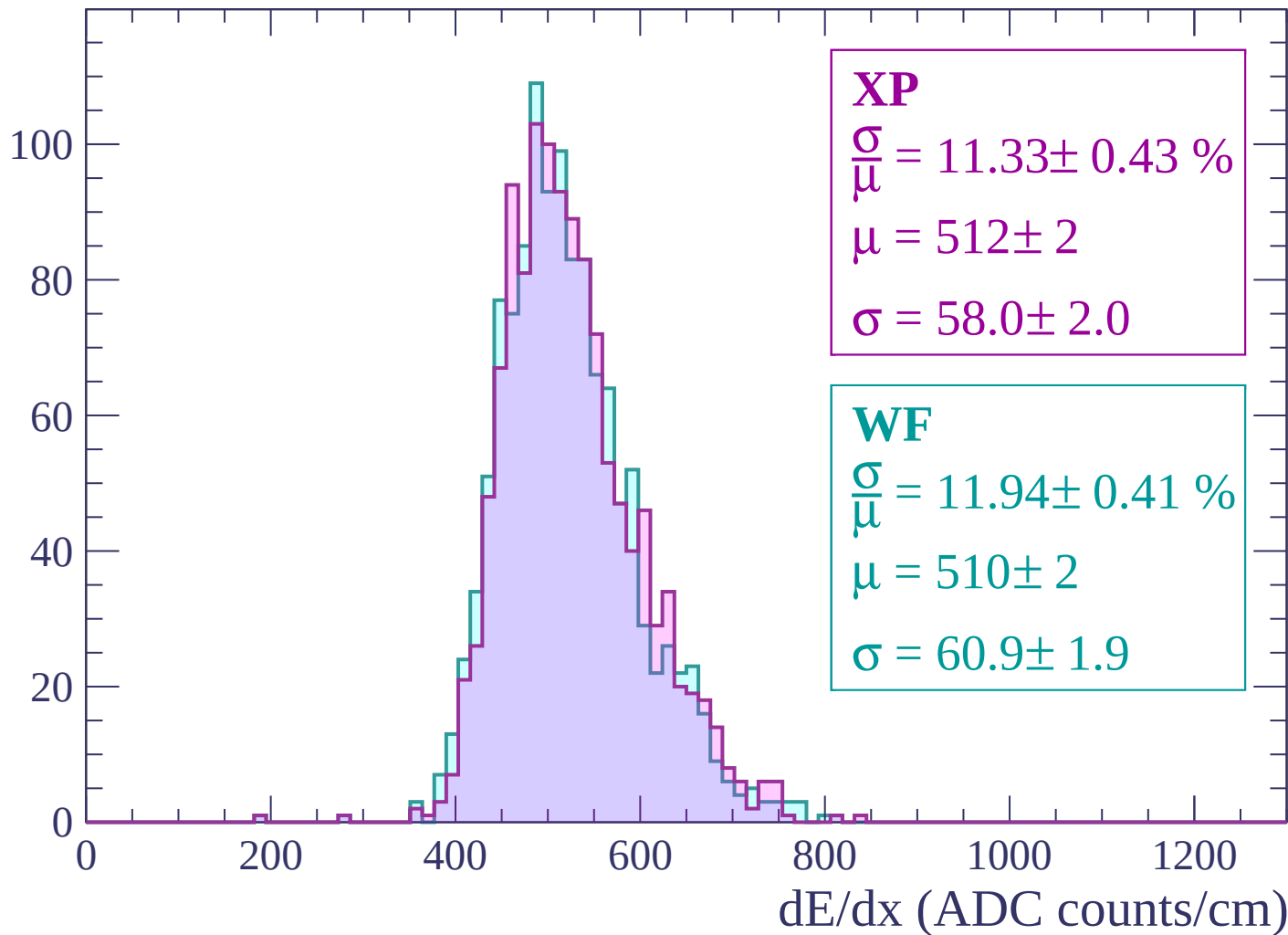
Energy loss | -1500 < p < -1440

Count



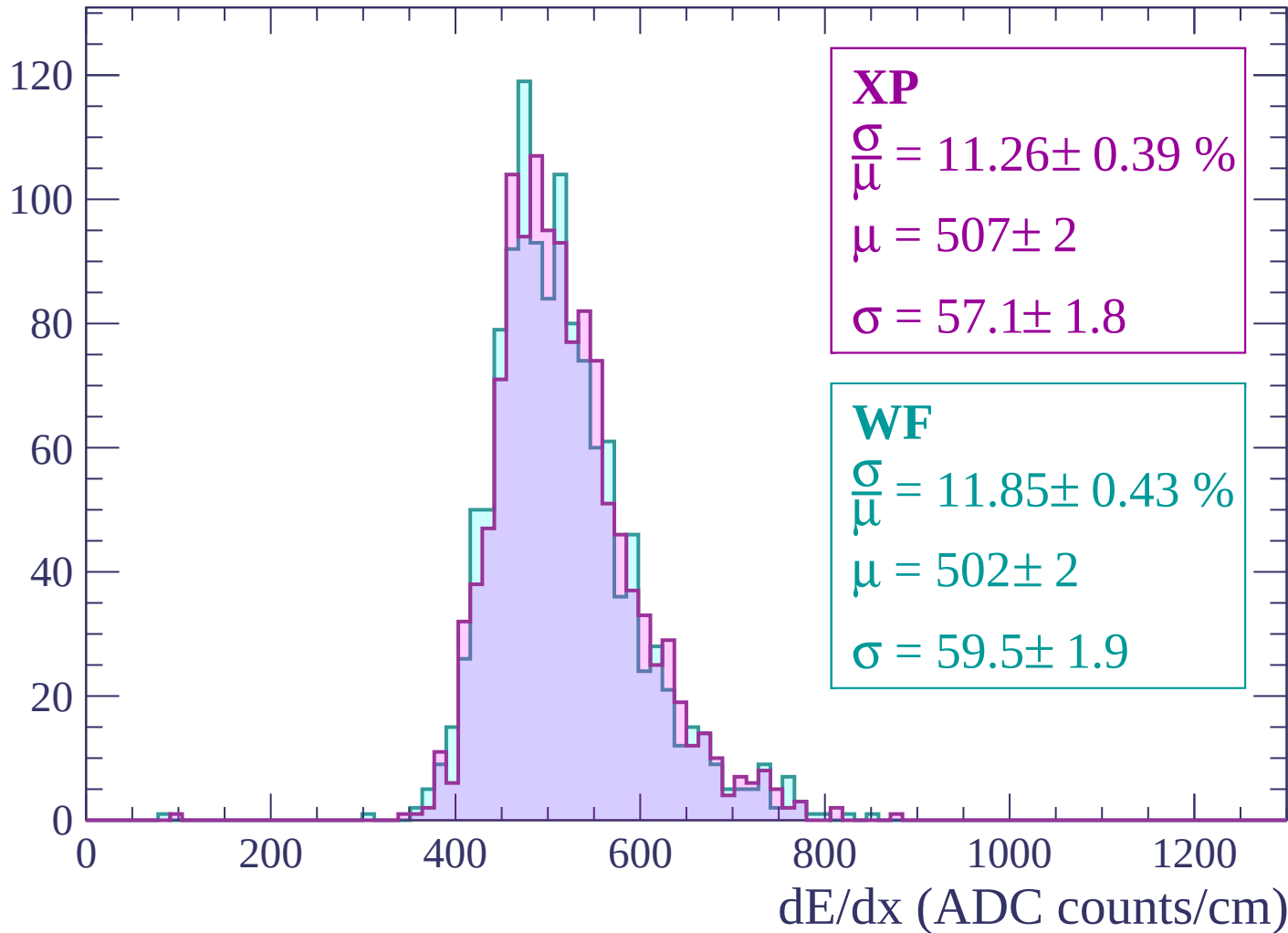
Energy loss | $-1440 < p < -1380$

Count



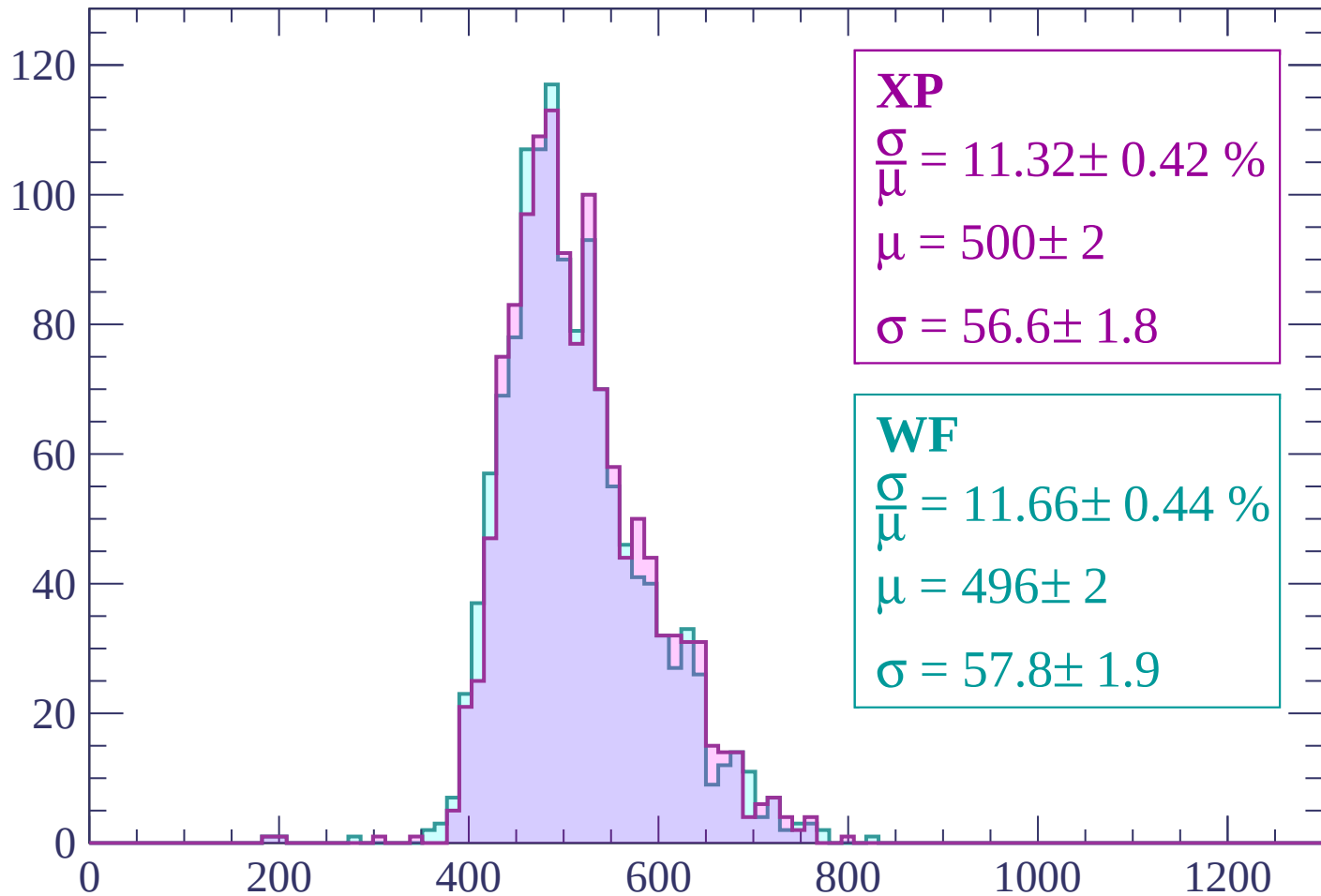
Energy loss | -1380 < p < -1320

Count



Energy loss | -1320 < p < -1260

Count



XP

$$\frac{\sigma}{\mu} = 11.32 \pm 0.42 \%$$

$$\mu = 500 \pm 2$$

$$\sigma = 56.6 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 11.66 \pm 0.44 \%$$

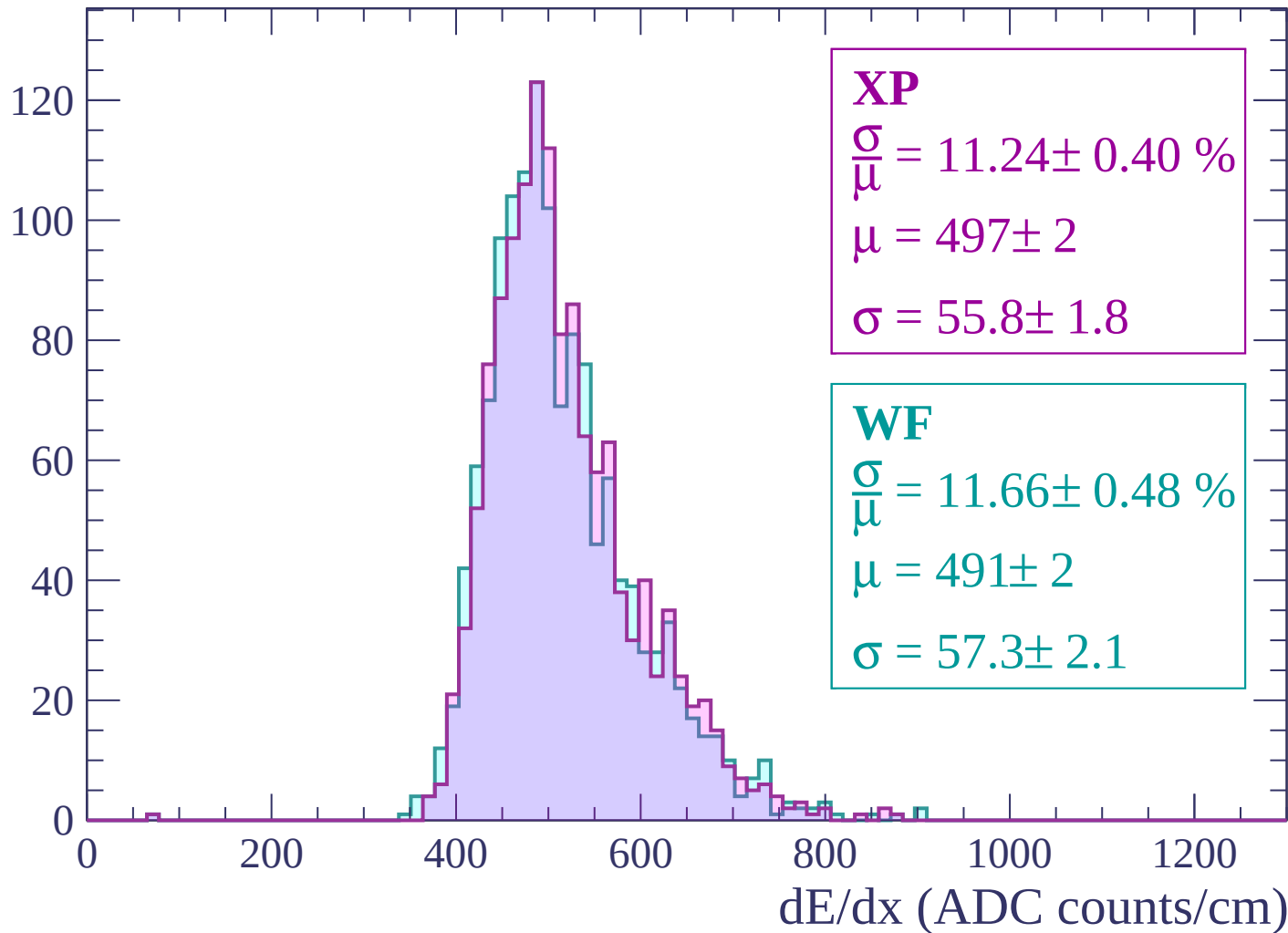
$$\mu = 496 \pm 2$$

$$\sigma = 57.8 \pm 1.9$$

dE/dx (ADC counts/cm)

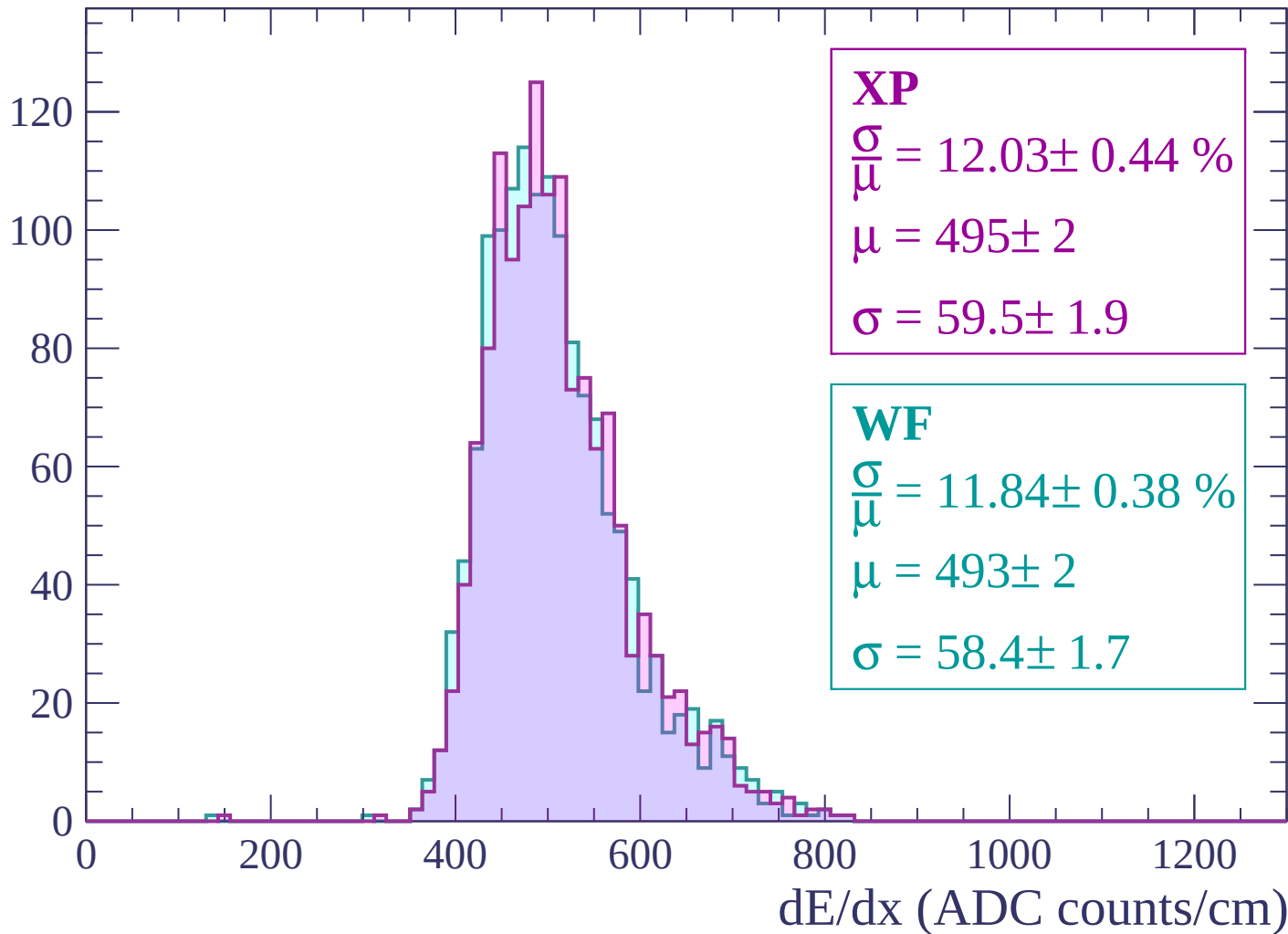
Energy loss | $-1260 < p < -1200$

Count



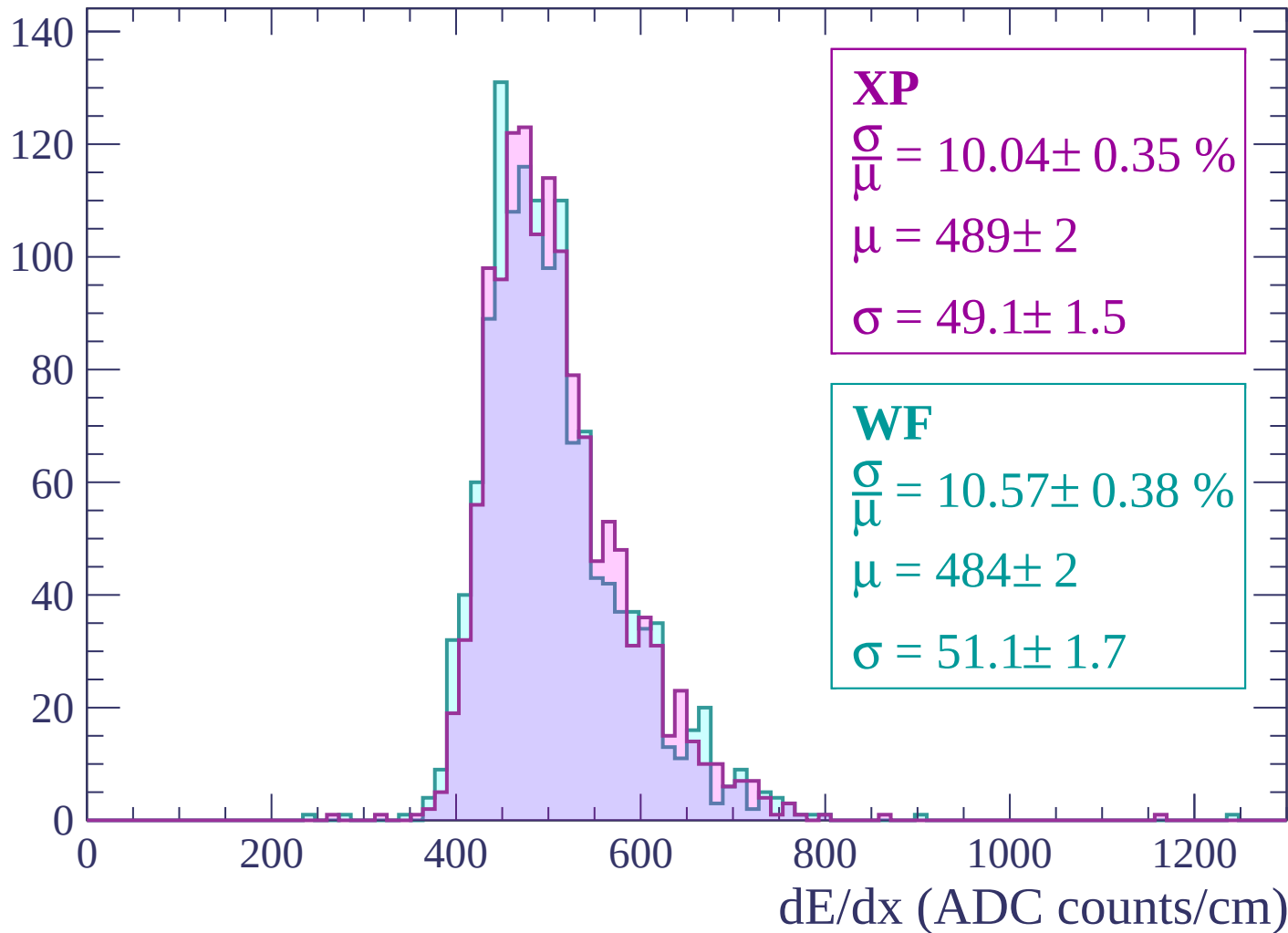
Energy loss | -1200 < p < -1140

Count



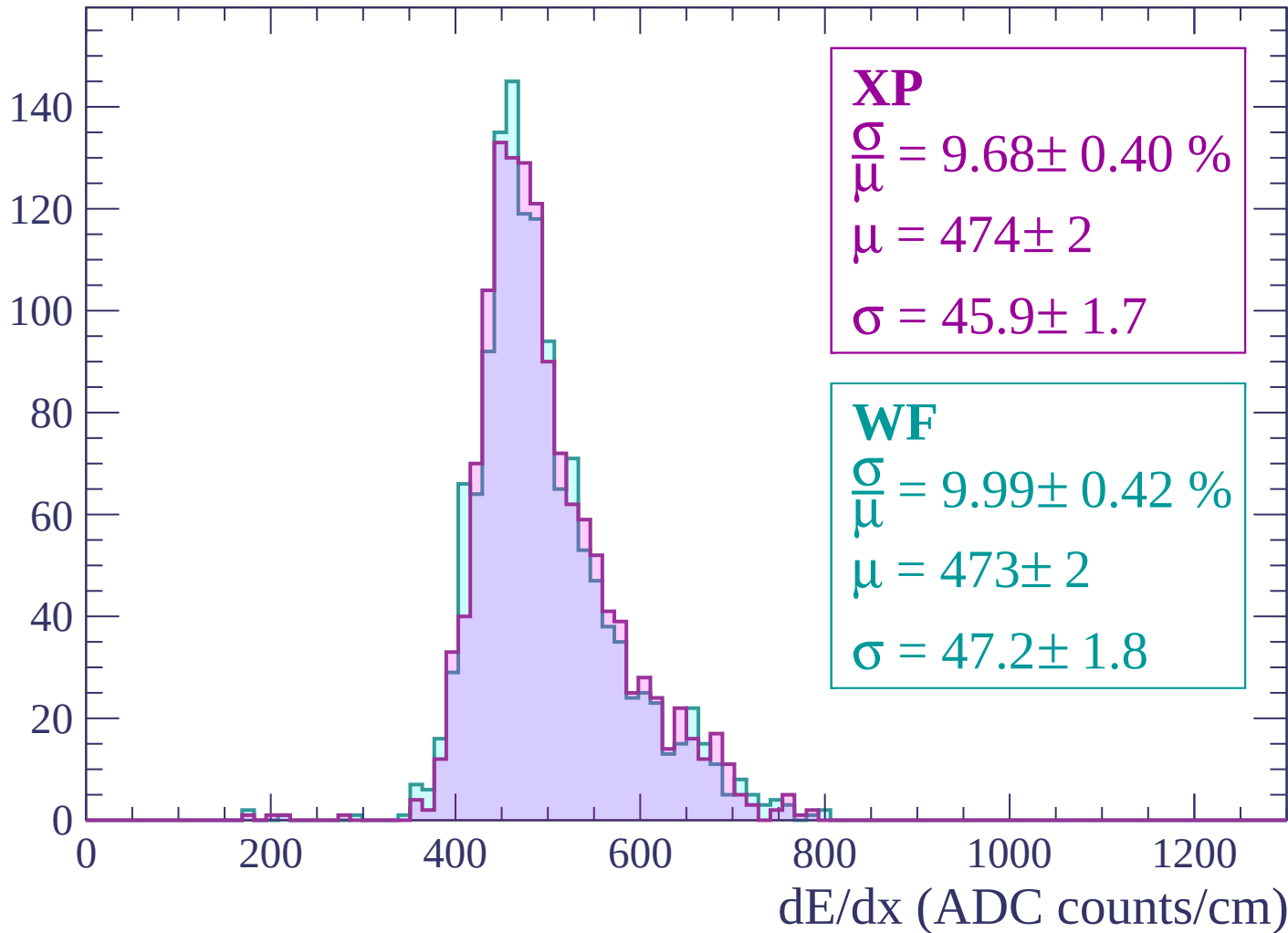
Energy loss | -1140 < p < -1080

Count



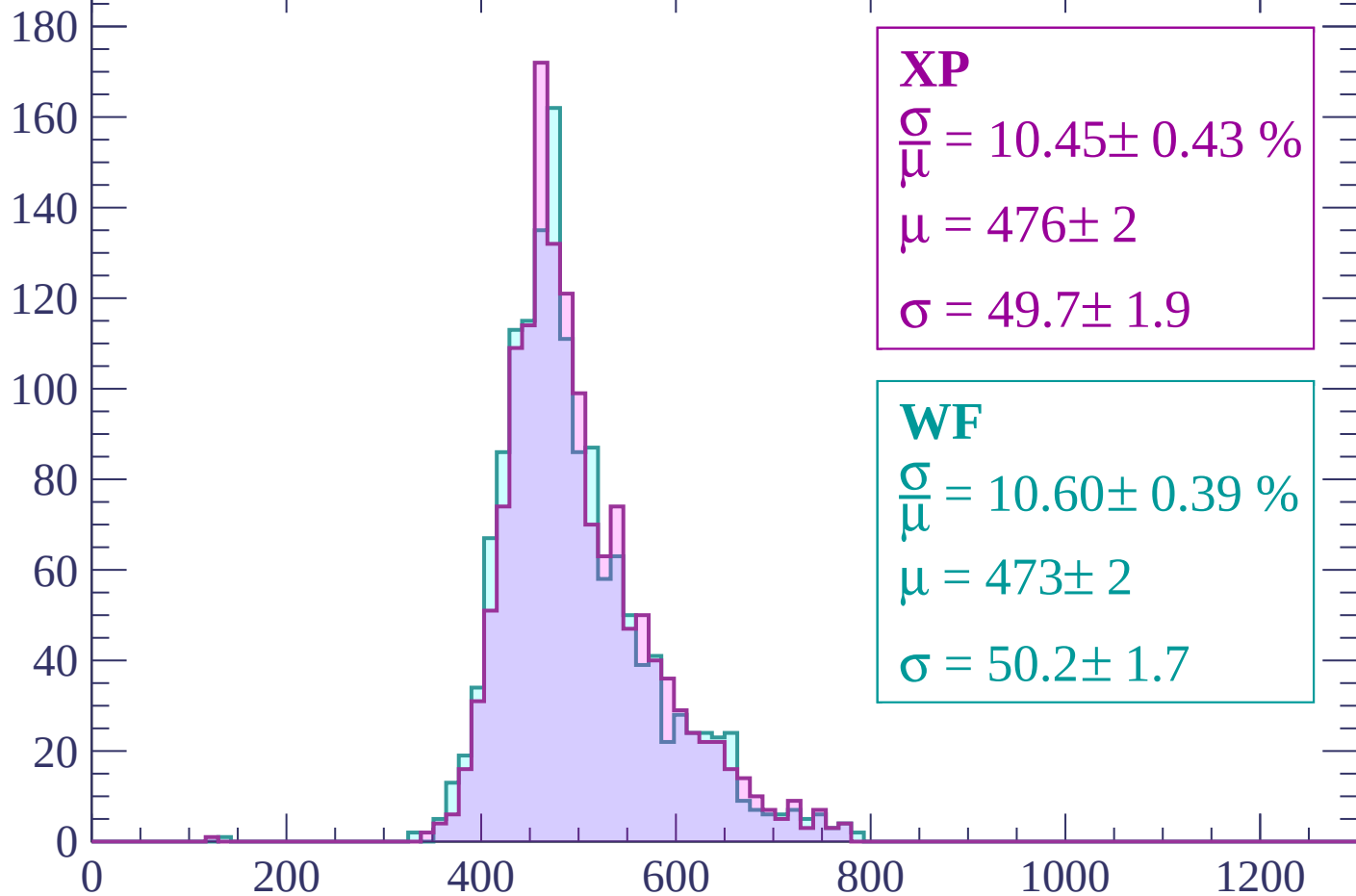
Energy loss | -1080 < p < -1020

Count



Energy loss | $-1020 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 10.45 \pm 0.43 \%$$

$$\mu = 476 \pm 2$$

$$\sigma = 49.7 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 10.60 \pm 0.39 \%$$

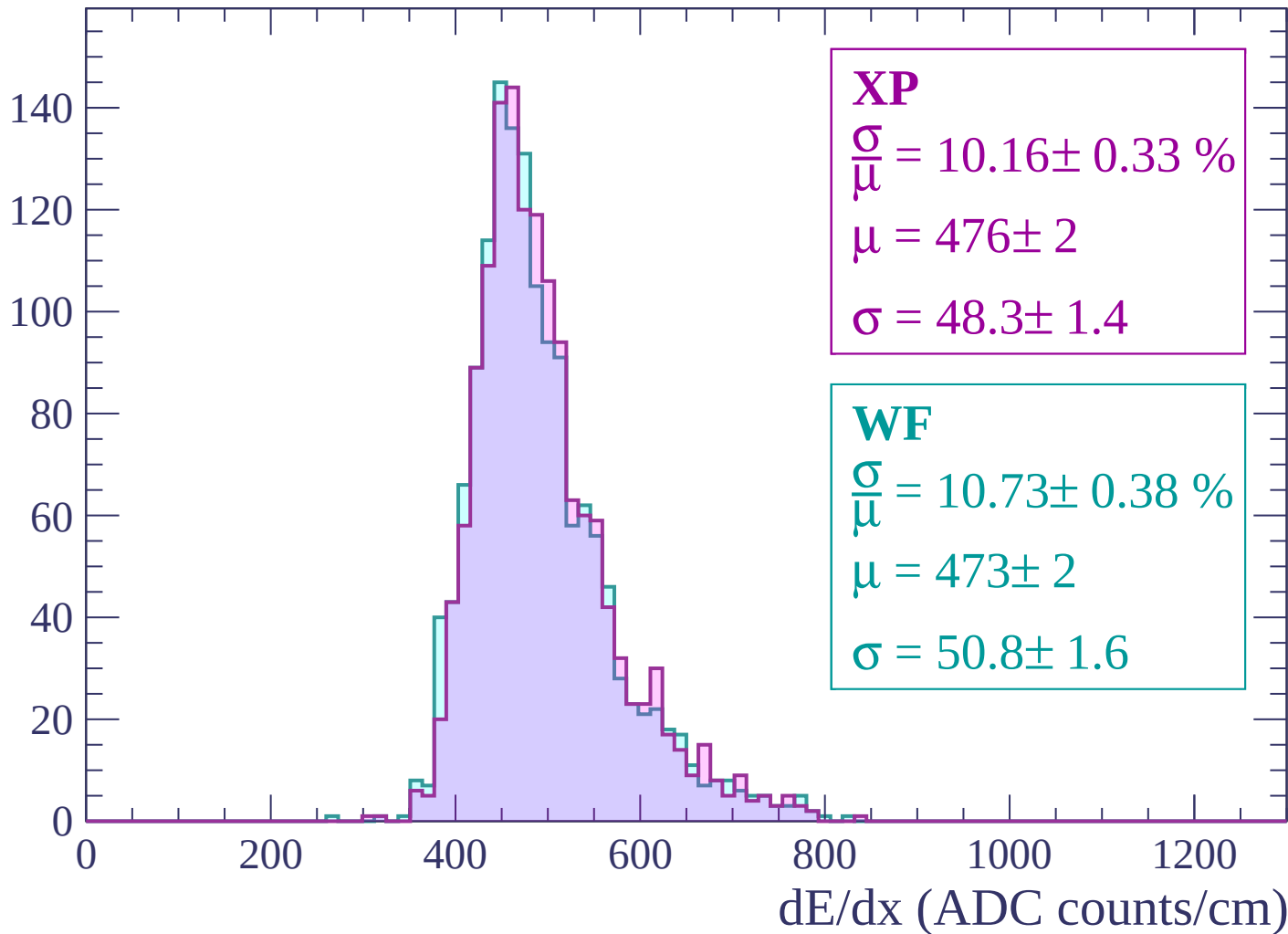
$$\mu = 473 \pm 2$$

$$\sigma = 50.2 \pm 1.7$$

dE/dx (ADC counts/cm)

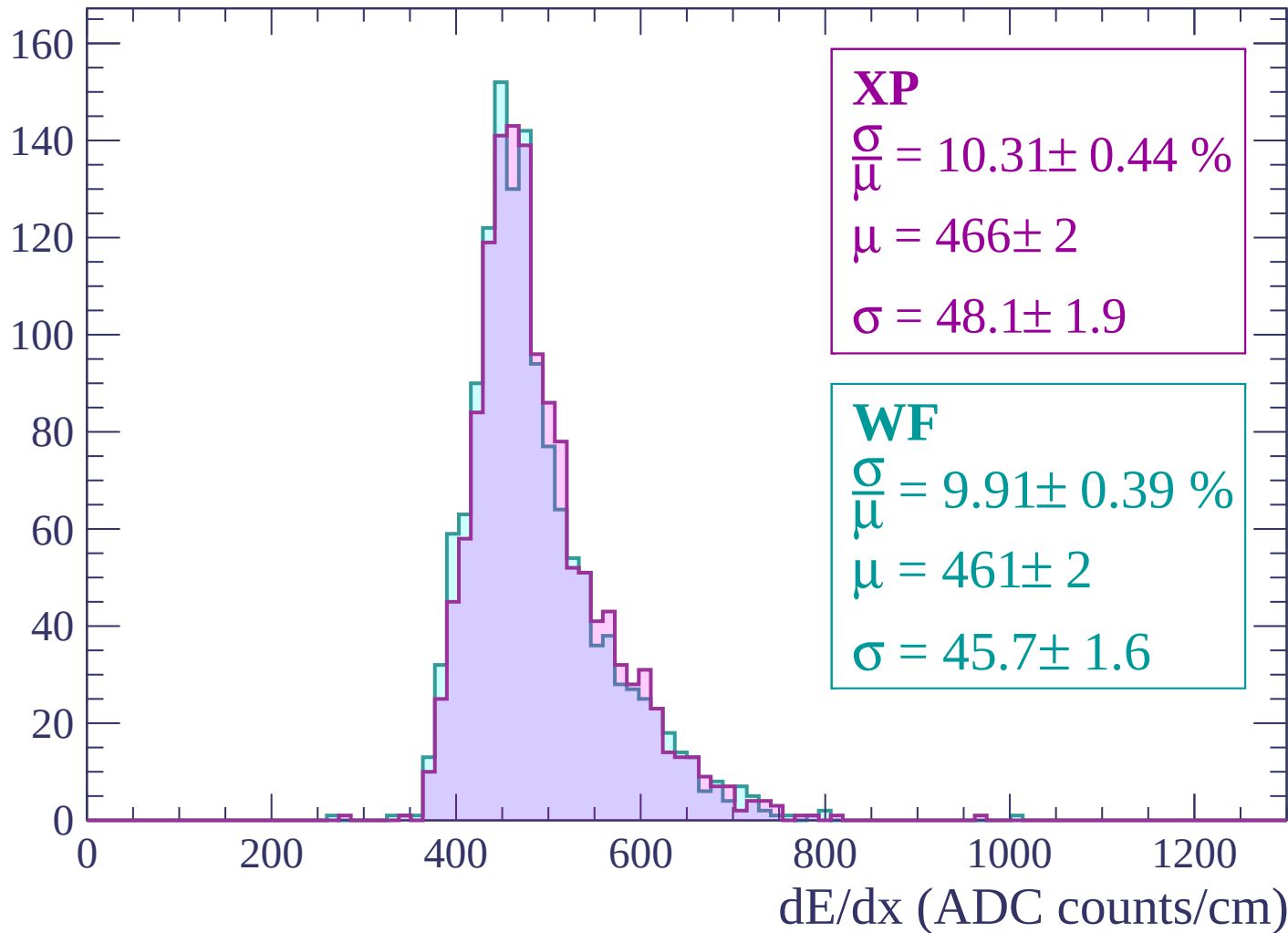
Energy loss | -960 < p < -900

Count



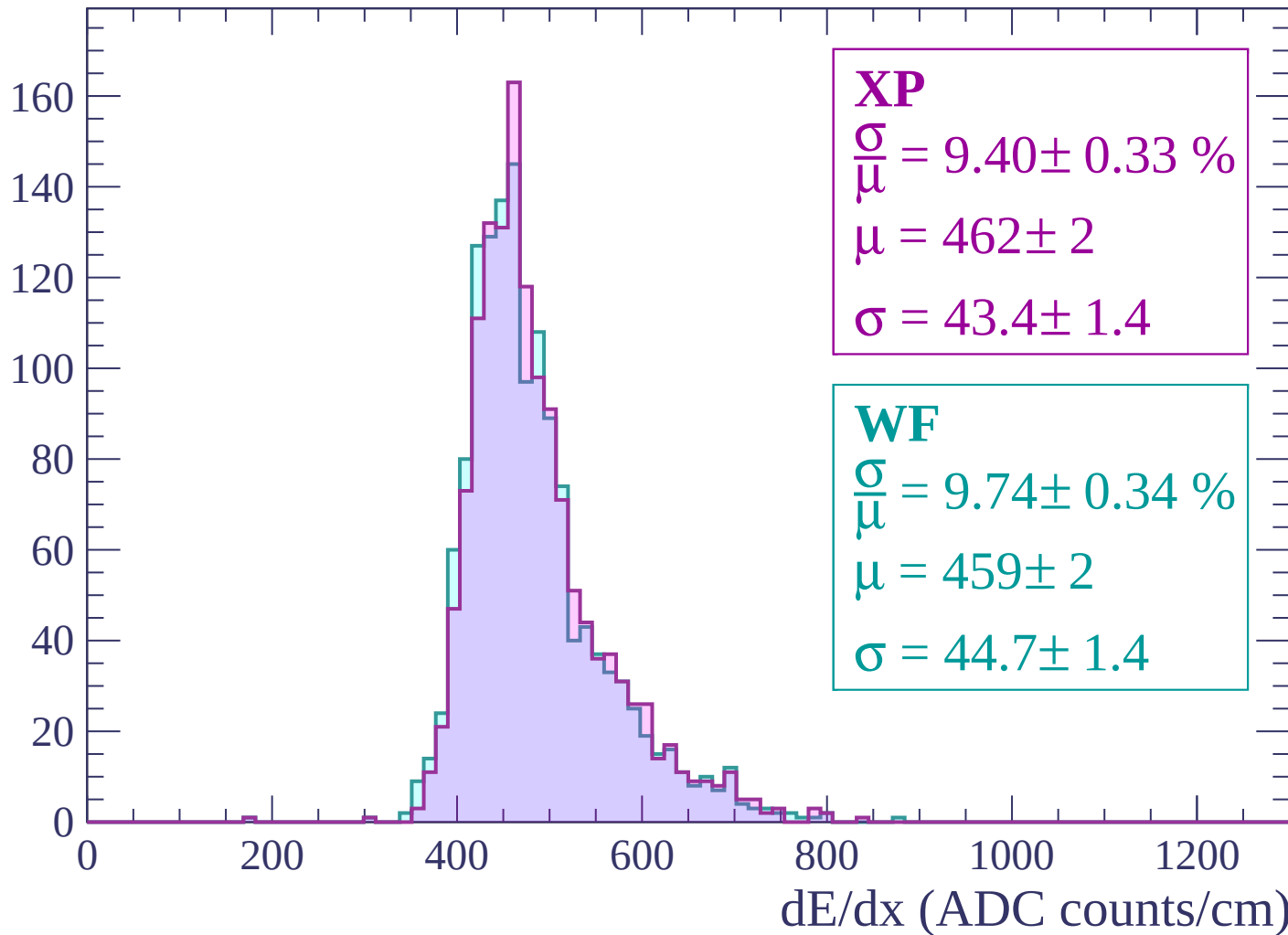
Energy loss | $-900 < p < -840$

Count



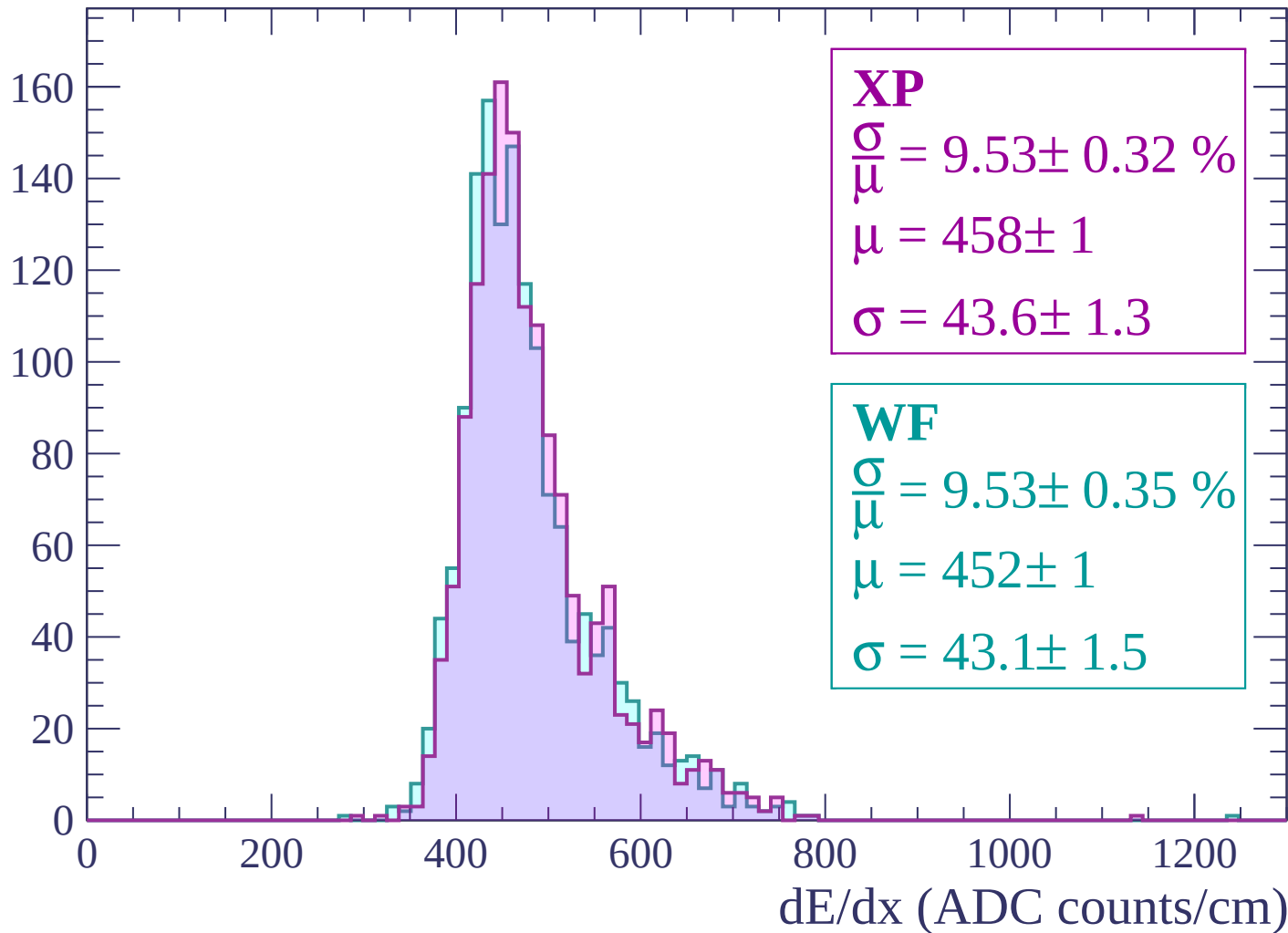
Energy loss | $-840 < p < -780$

Count



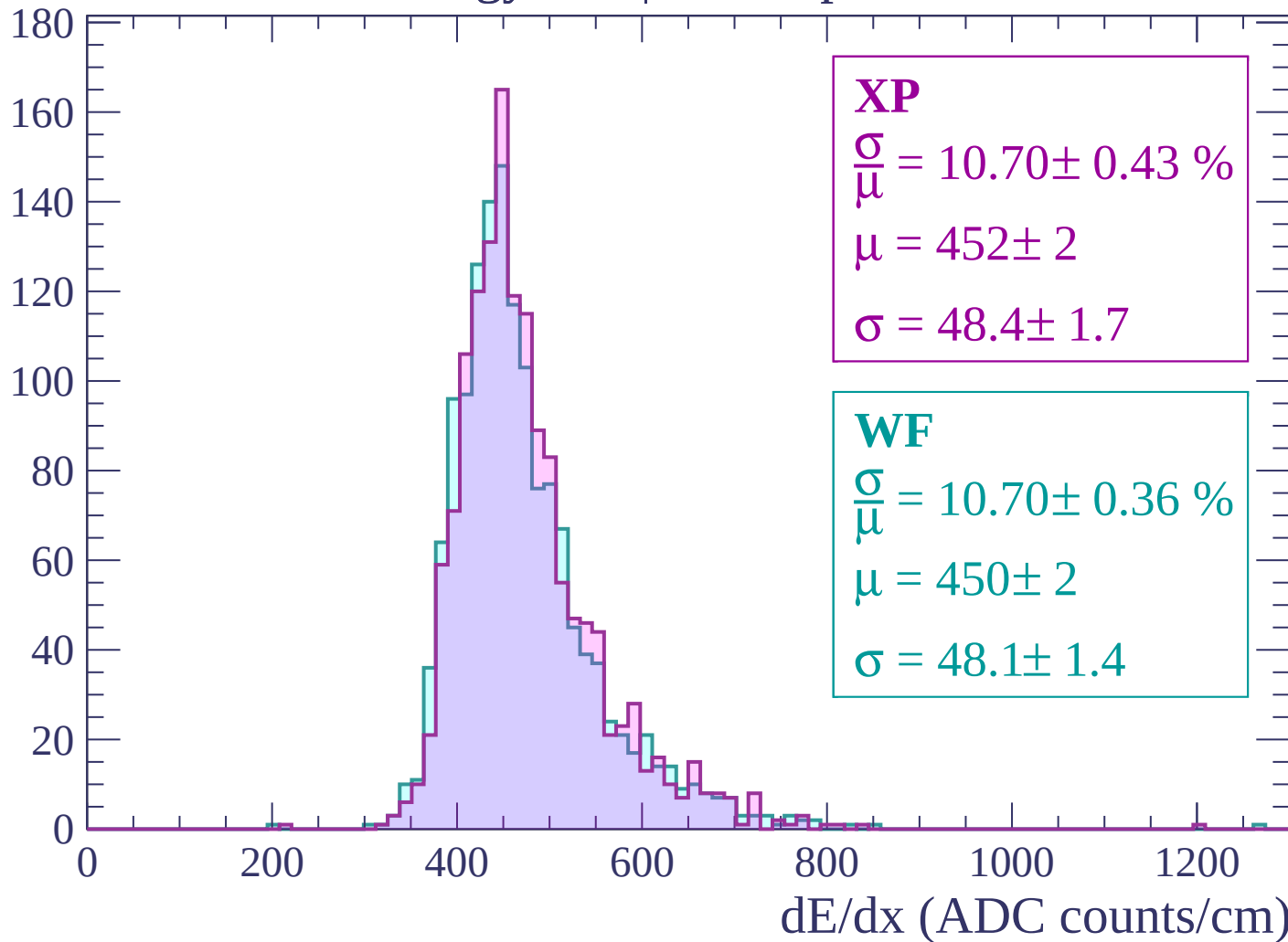
Energy loss | $-780 < p < -720$

Count



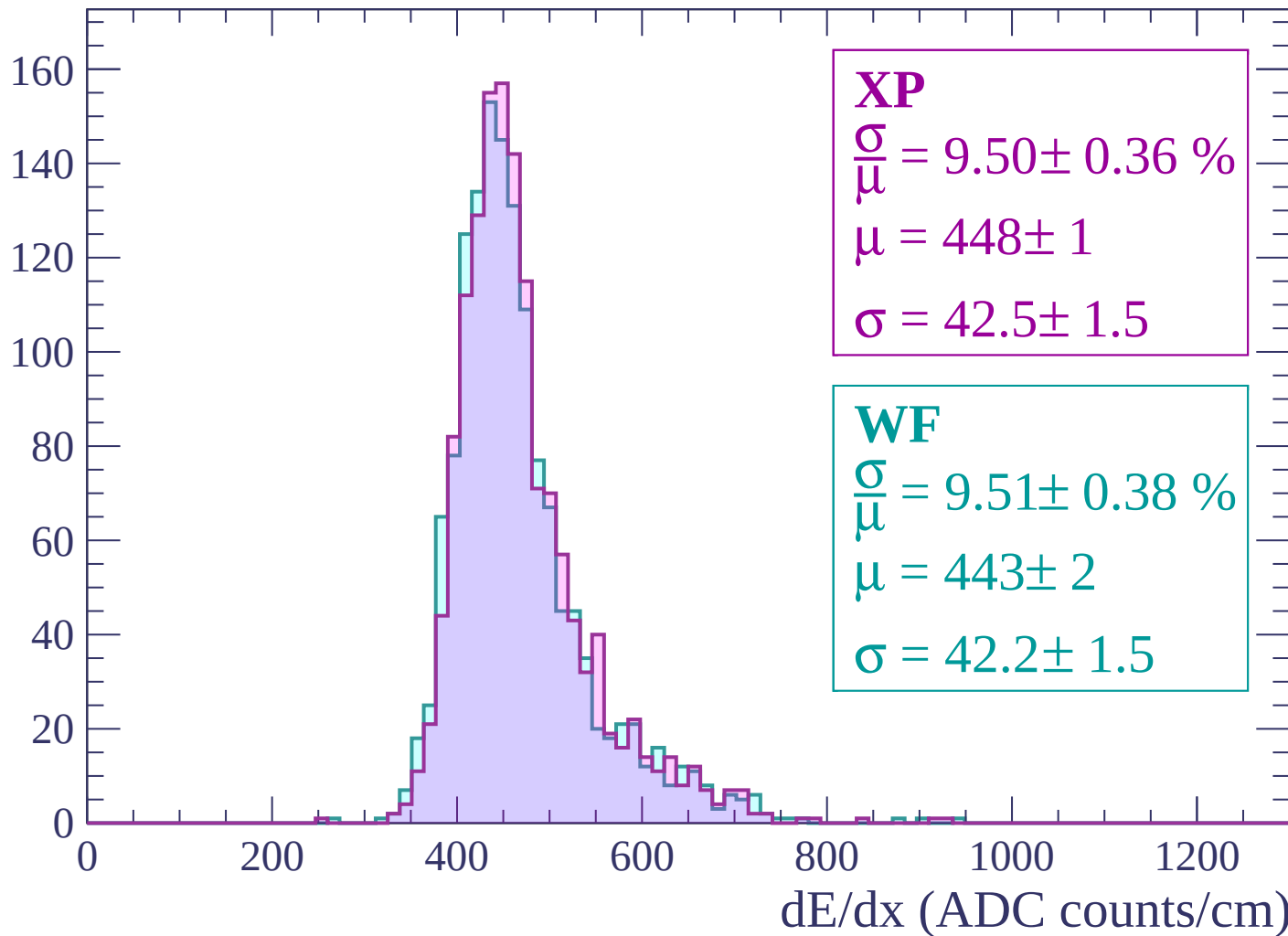
Energy loss | -720 < p < -660

Count



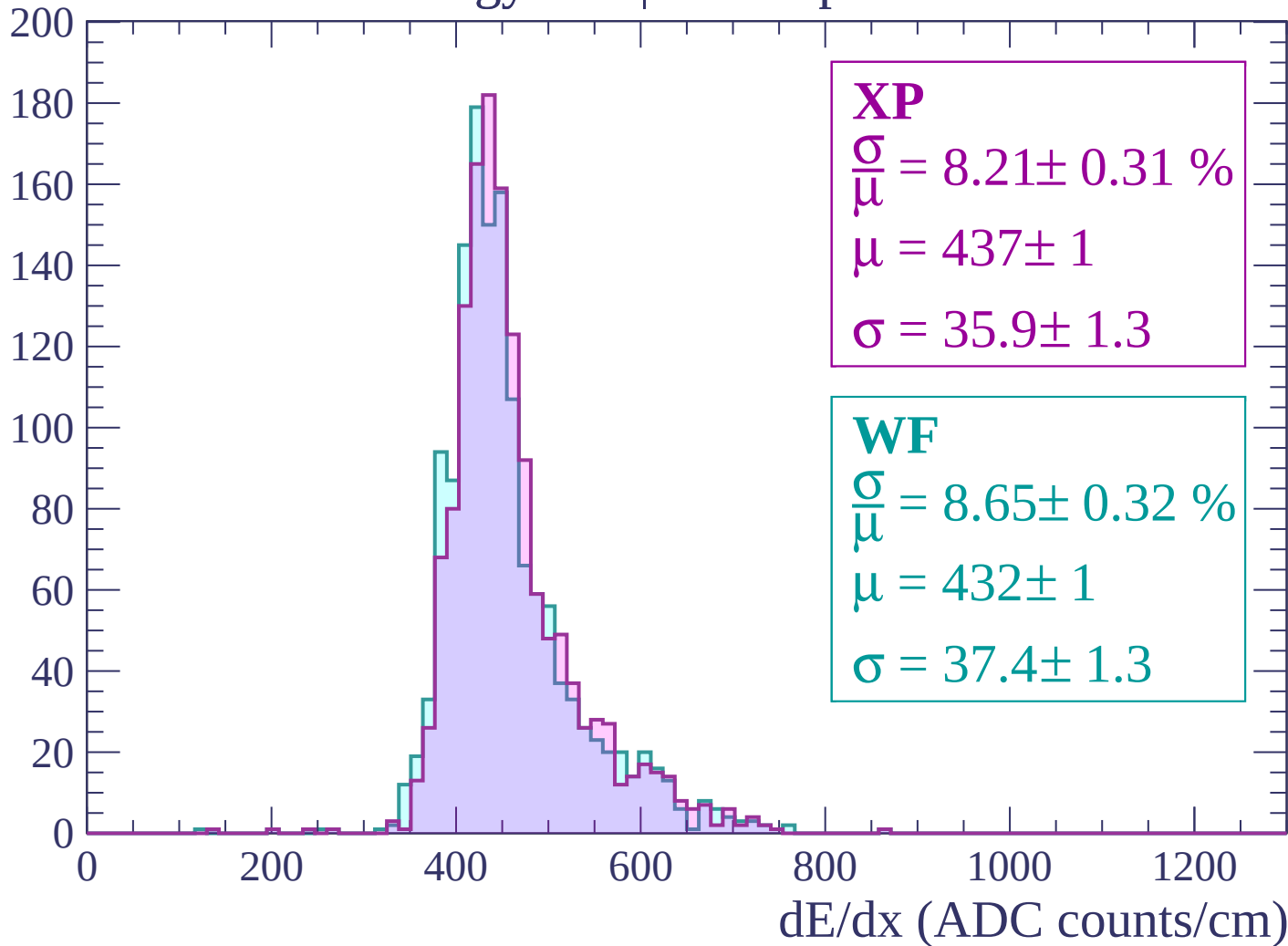
Energy loss | $-660 < p < -600$

Count



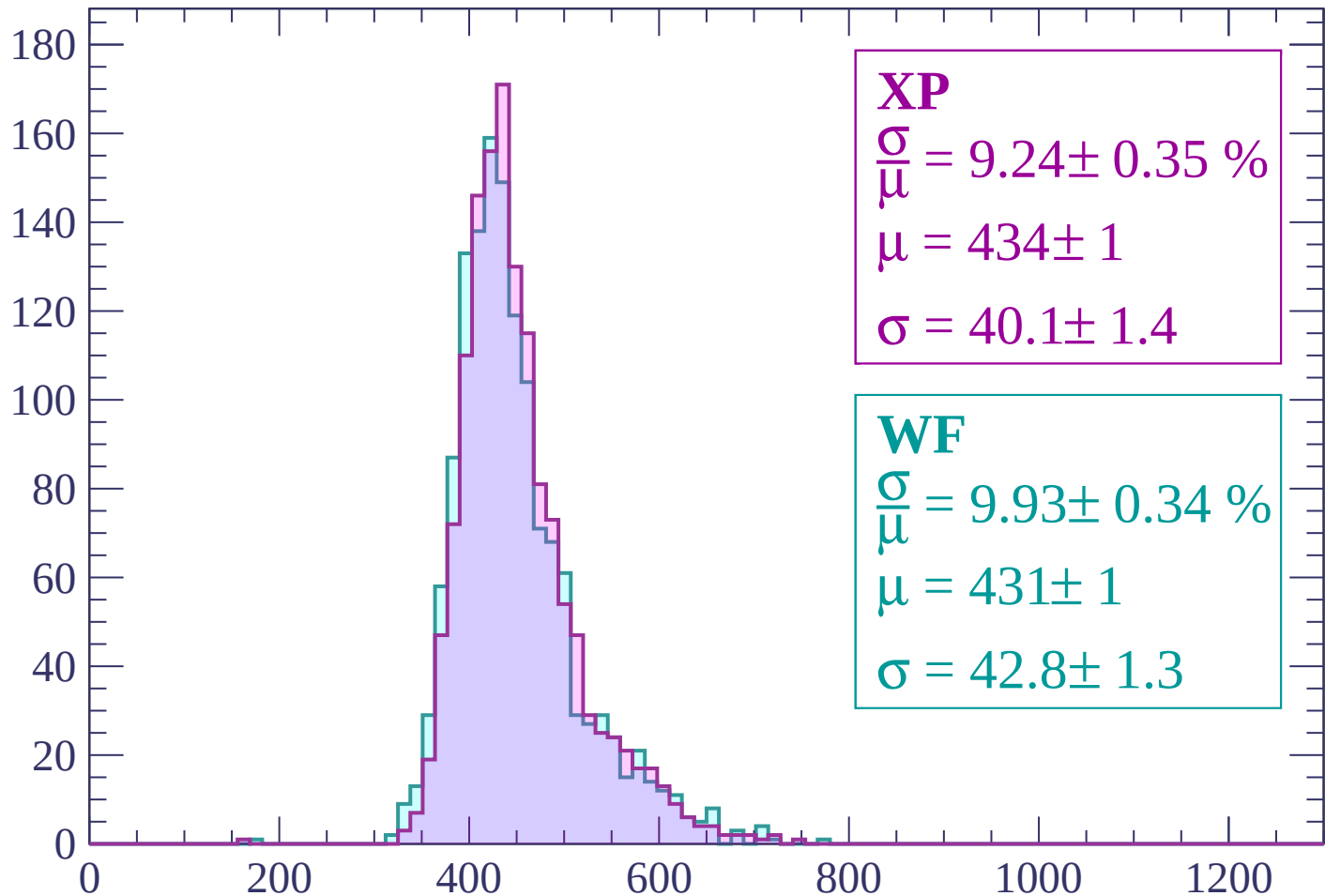
Energy loss | $-600 < p < -540$

Count



Energy loss | $-540 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 9.24 \pm 0.35 \%$$

$$\mu = 434 \pm 1$$

$$\sigma = 40.1 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 9.93 \pm 0.34 \%$$

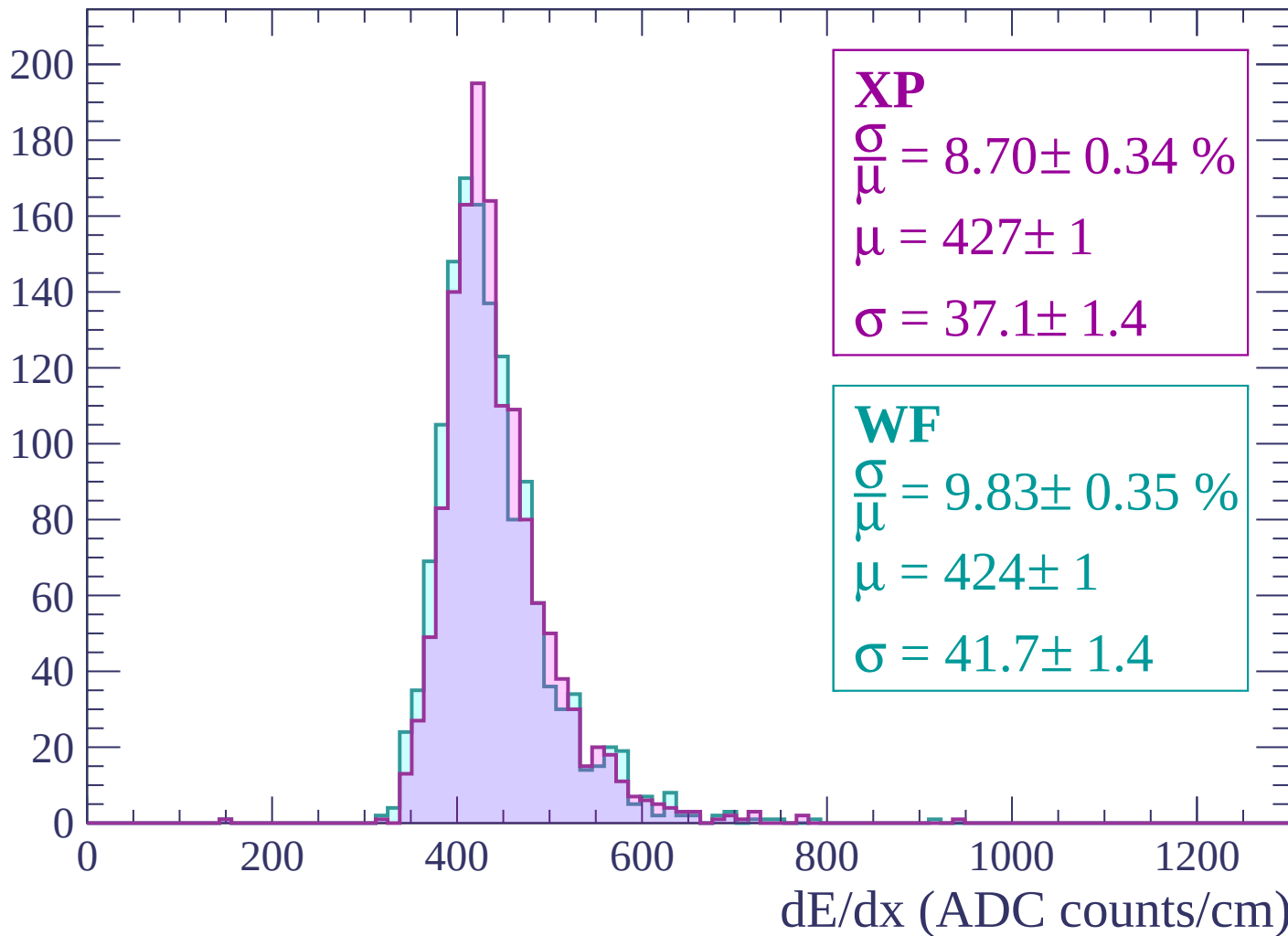
$$\mu = 431 \pm 1$$

$$\sigma = 42.8 \pm 1.3$$

dE/dx (ADC counts/cm)

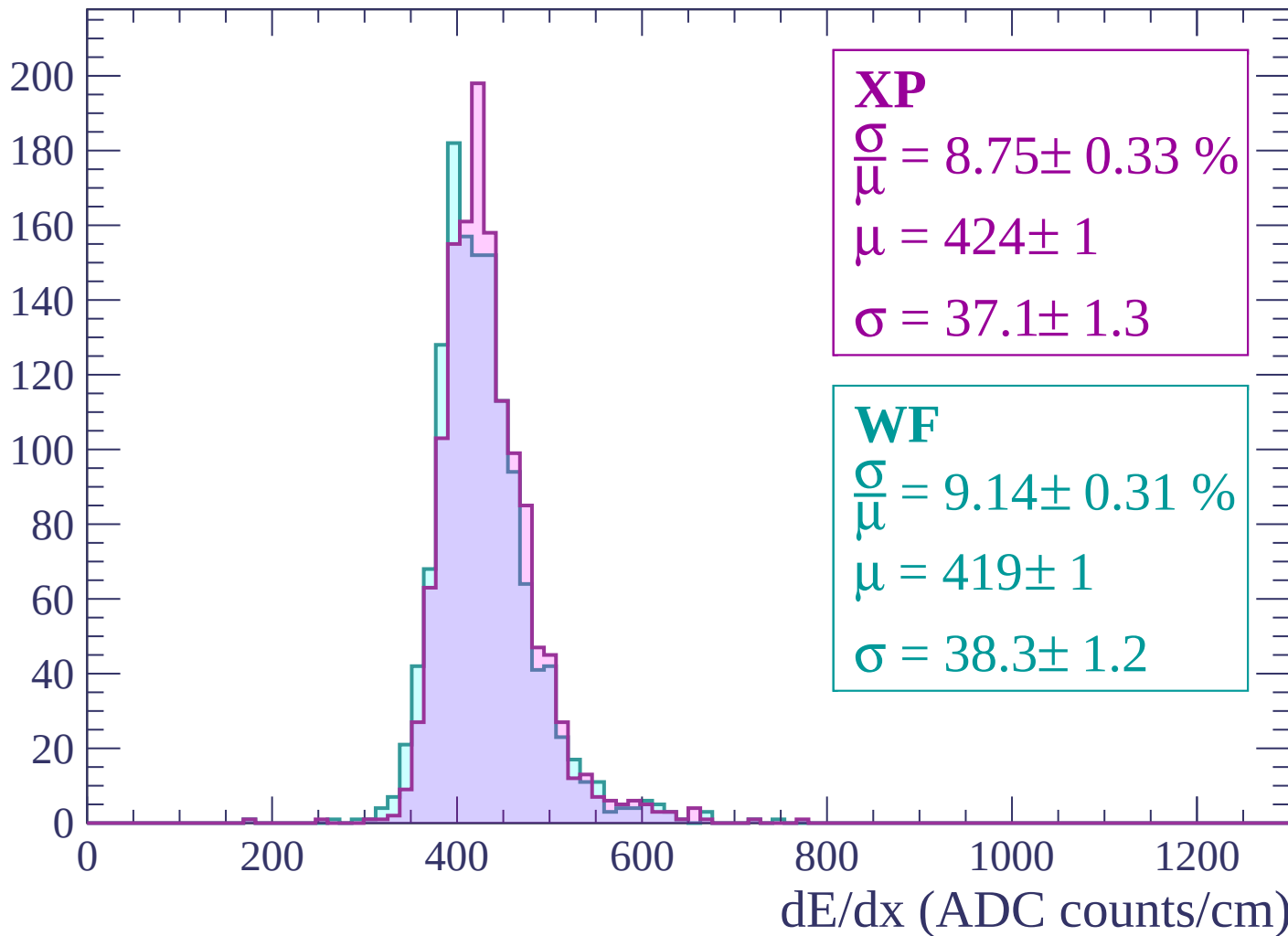
Energy loss | $-480 < p < -420$

Count

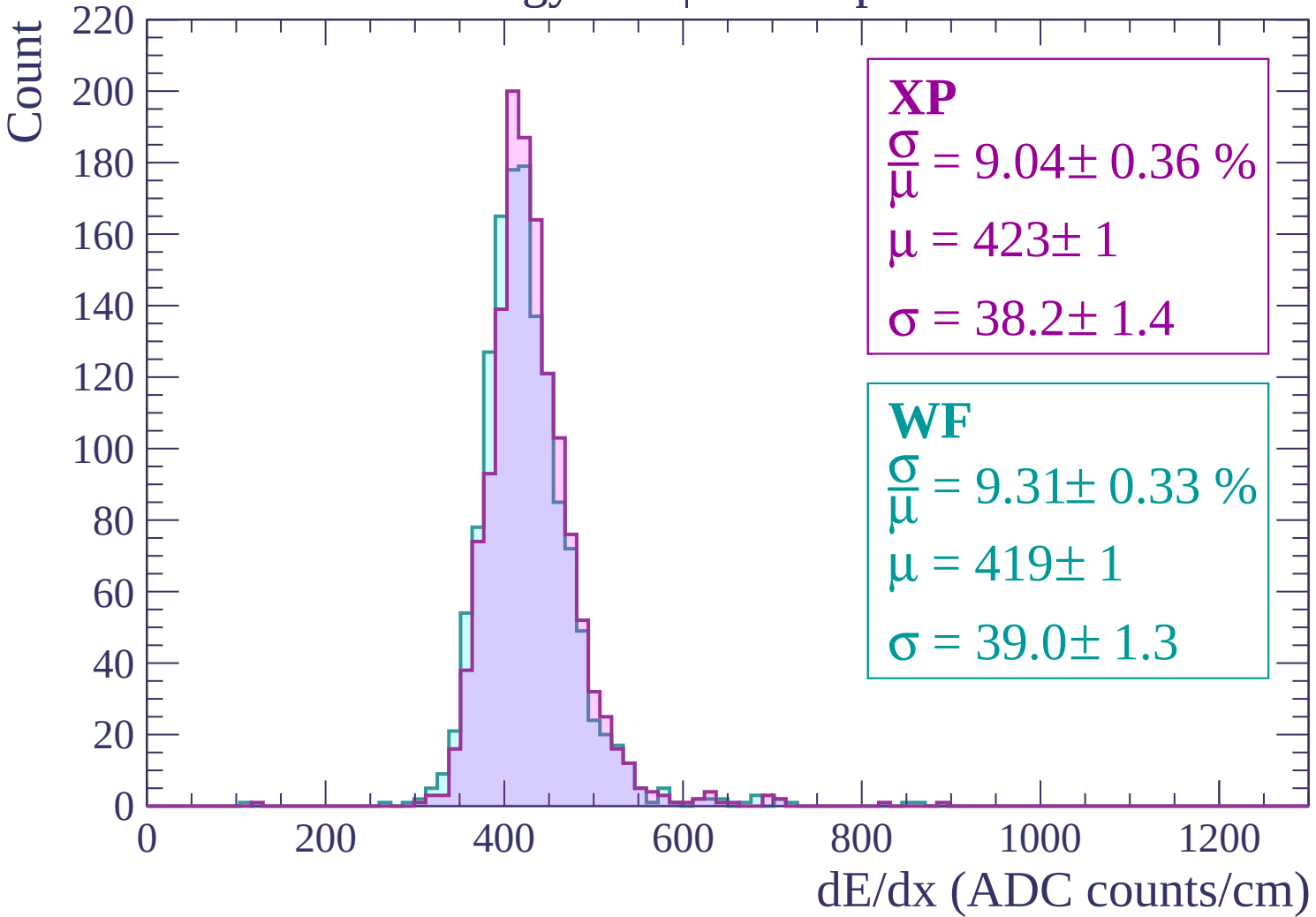


Energy loss | $-420 < p < -360$

Count

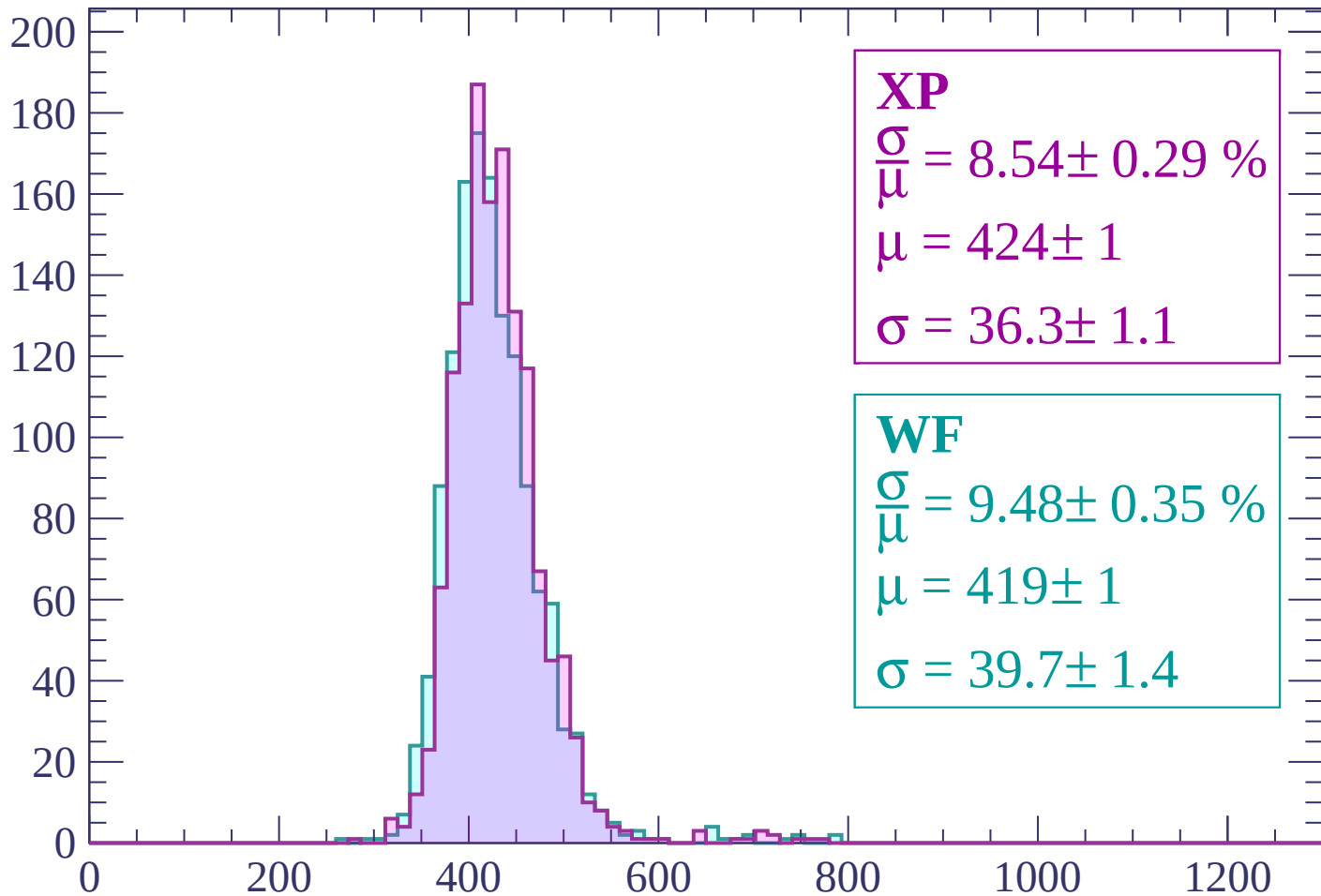


Energy loss | -360 < p < -300



Energy loss | $-300 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 8.54 \pm 0.29 \%$$

$$\mu = 424 \pm 1$$

$$\sigma = 36.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.48 \pm 0.35 \%$$

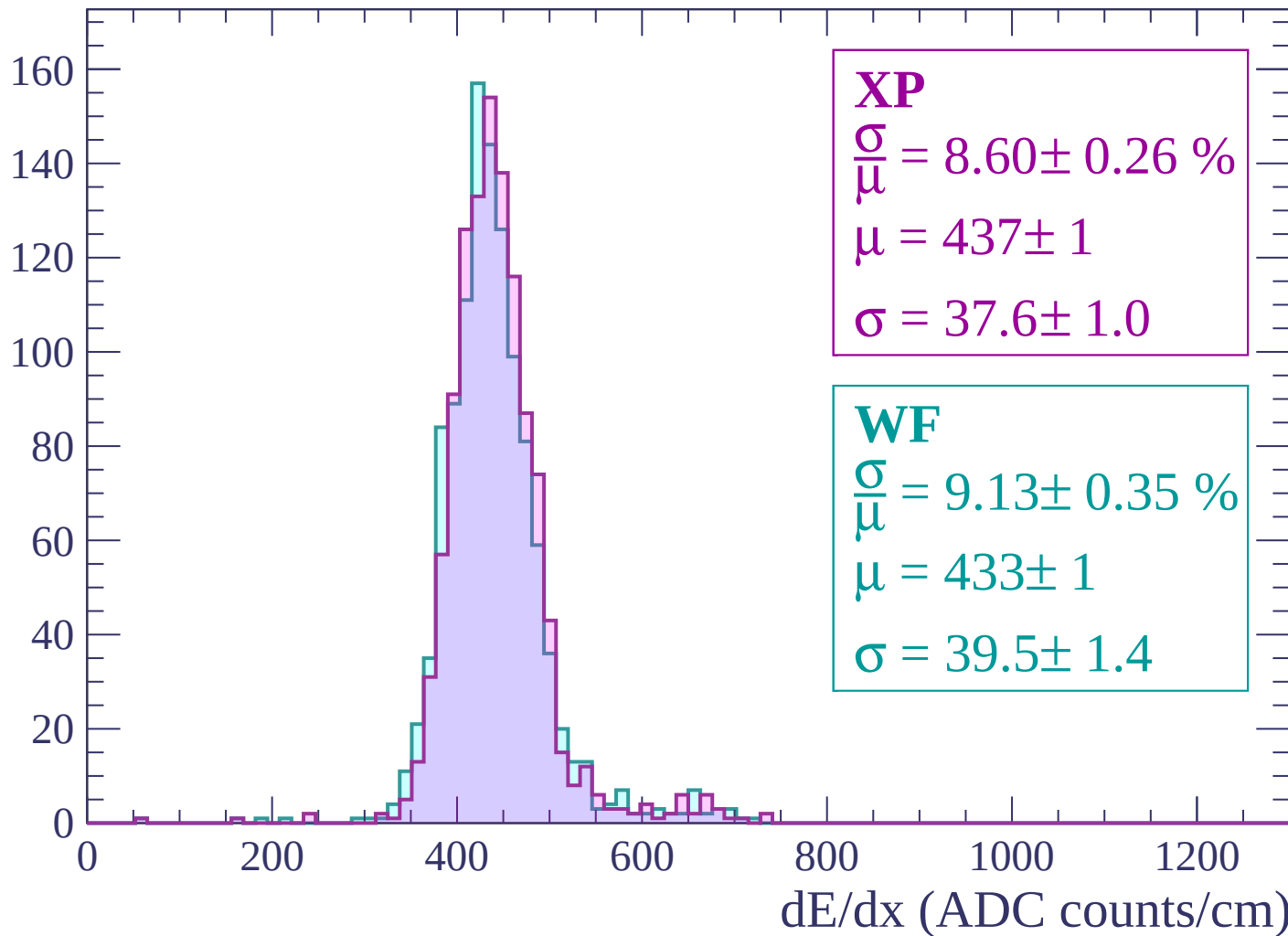
$$\mu = 419 \pm 1$$

$$\sigma = 39.7 \pm 1.4$$

dE/dx (ADC counts/cm)

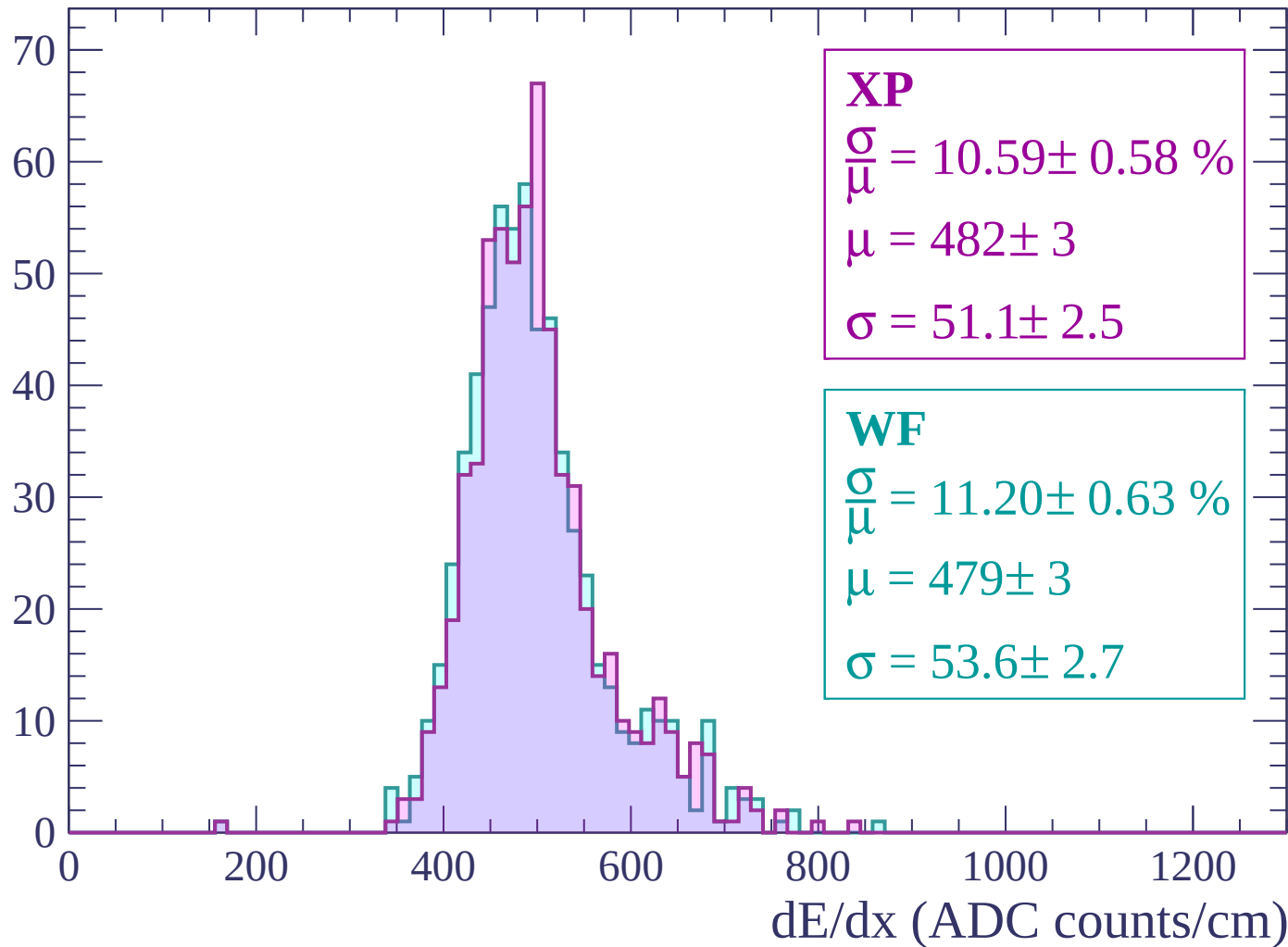
Energy loss | $-240 < p < -180$

Count



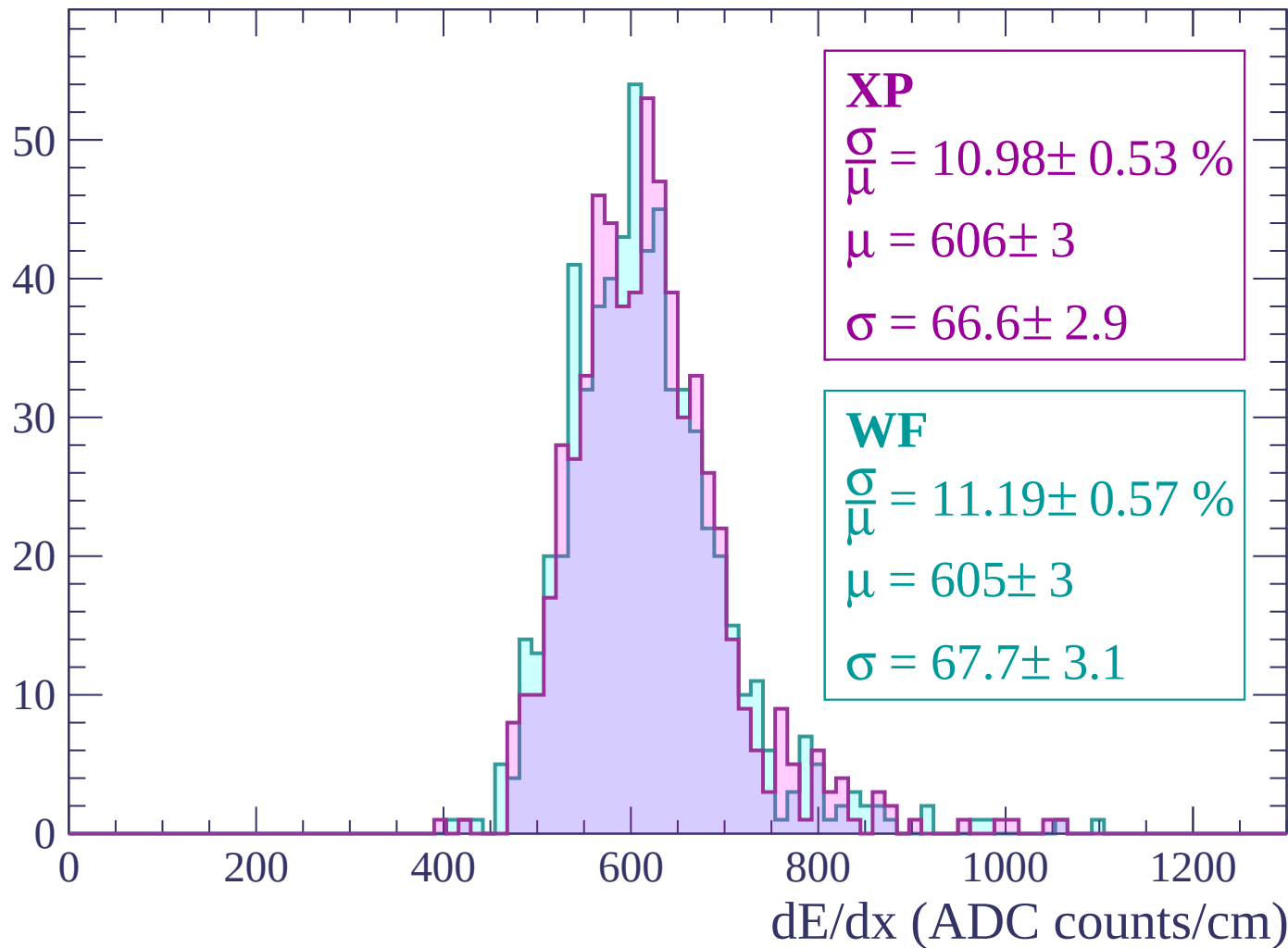
Energy loss | -180 < p < -120

Count



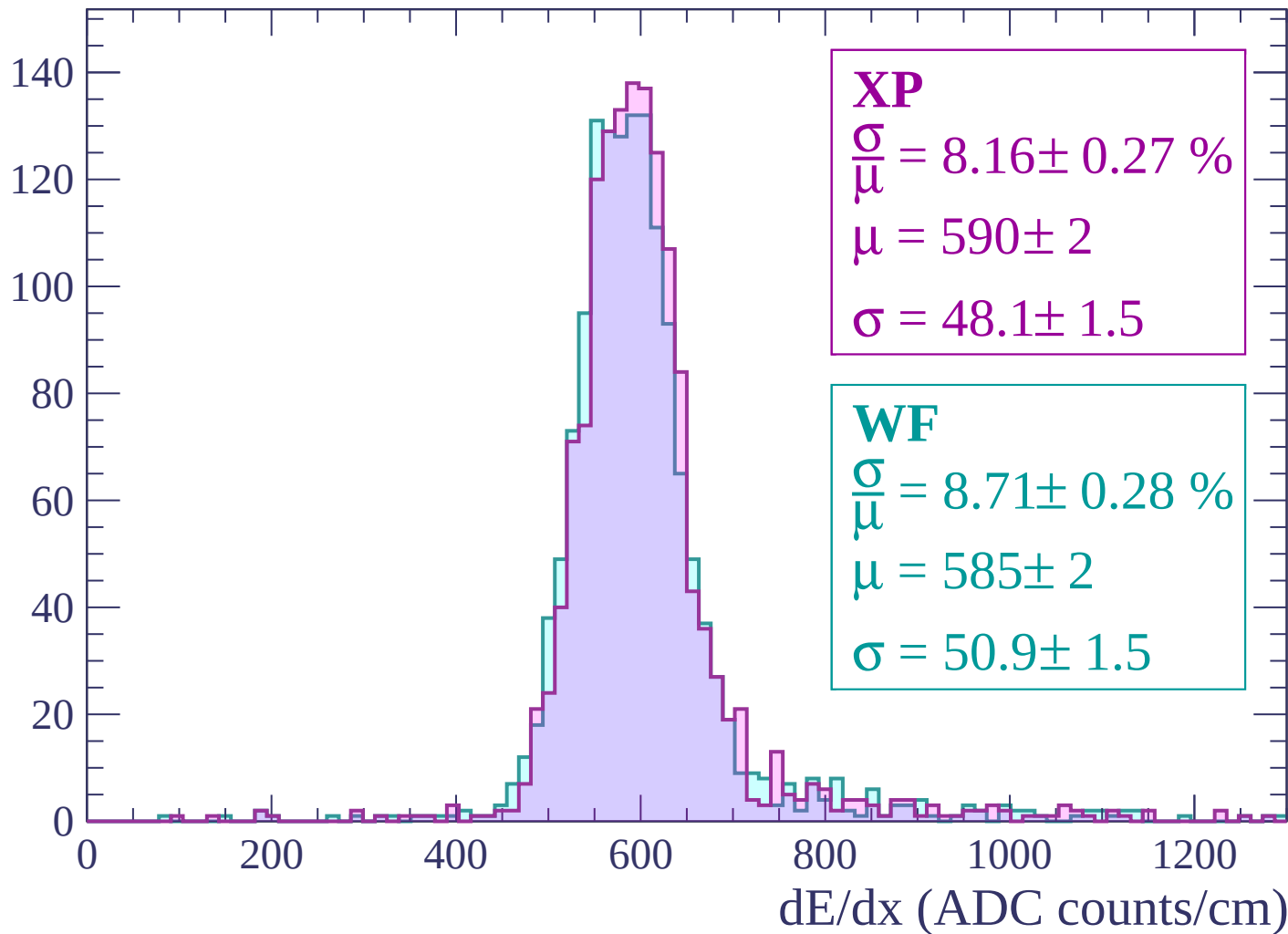
Energy loss | $-120 < p < -60$

Count



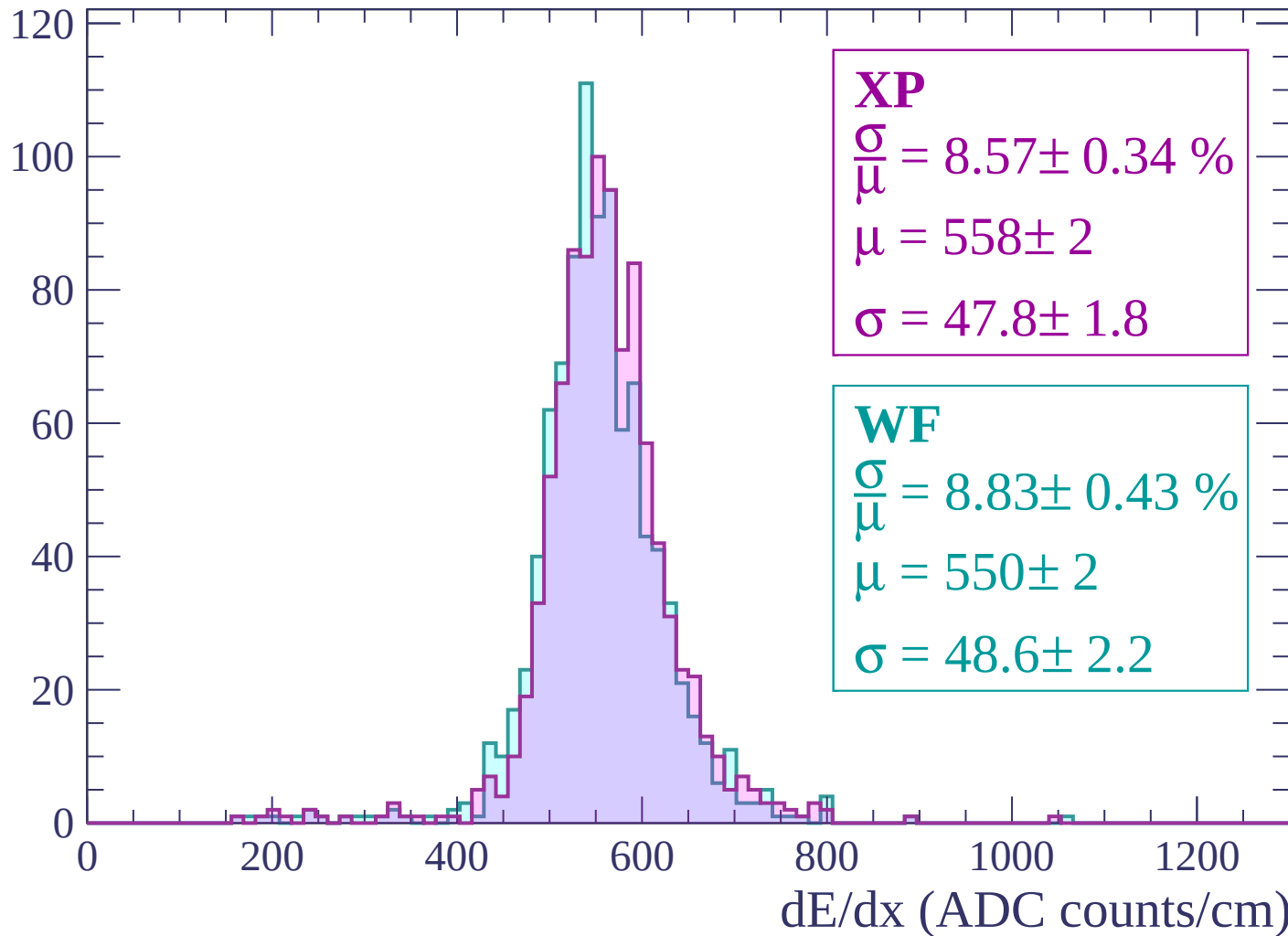
Energy loss | $-60 < p < 0$

Count



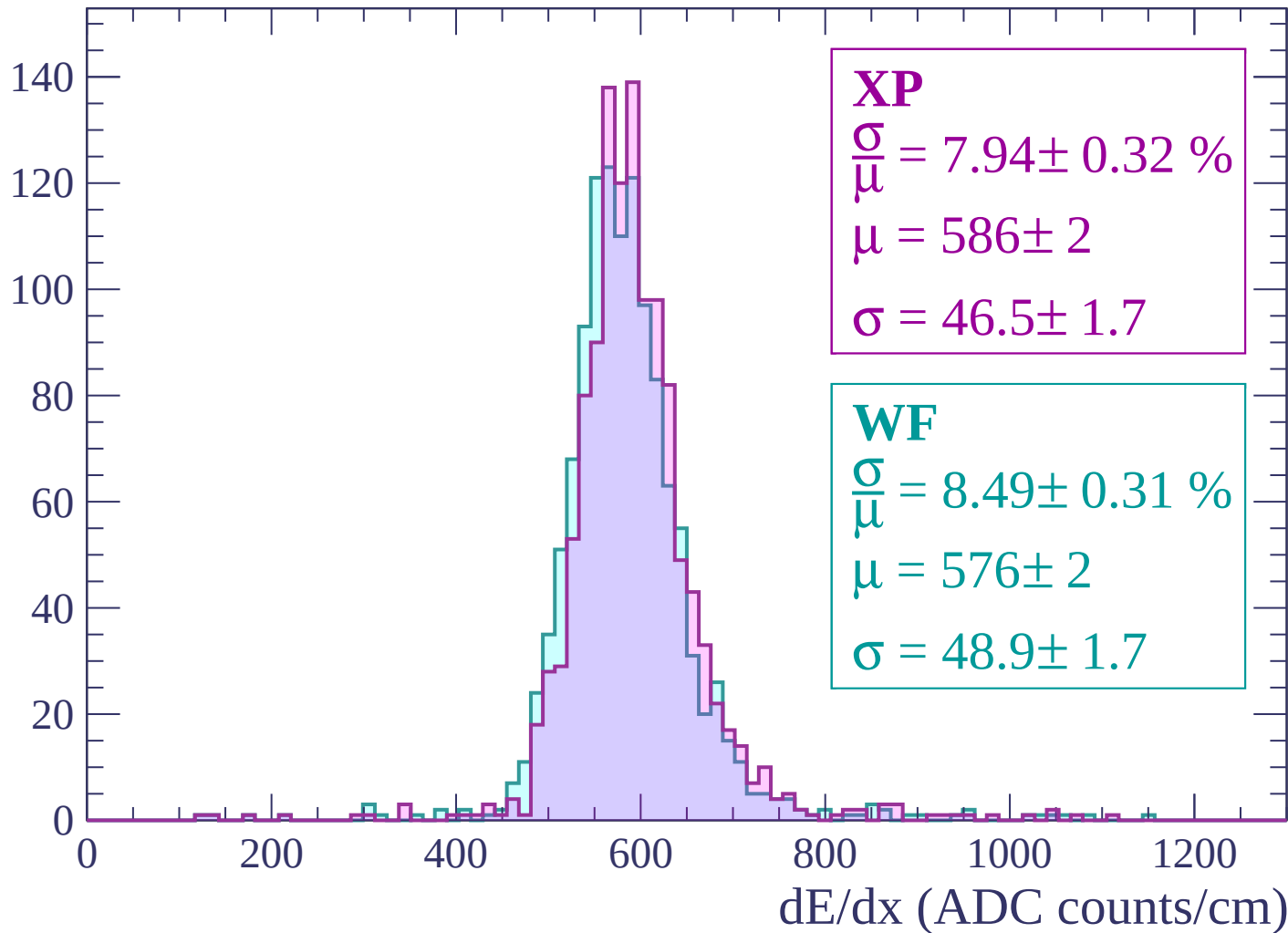
Energy loss | $0 < p < 60$

Count



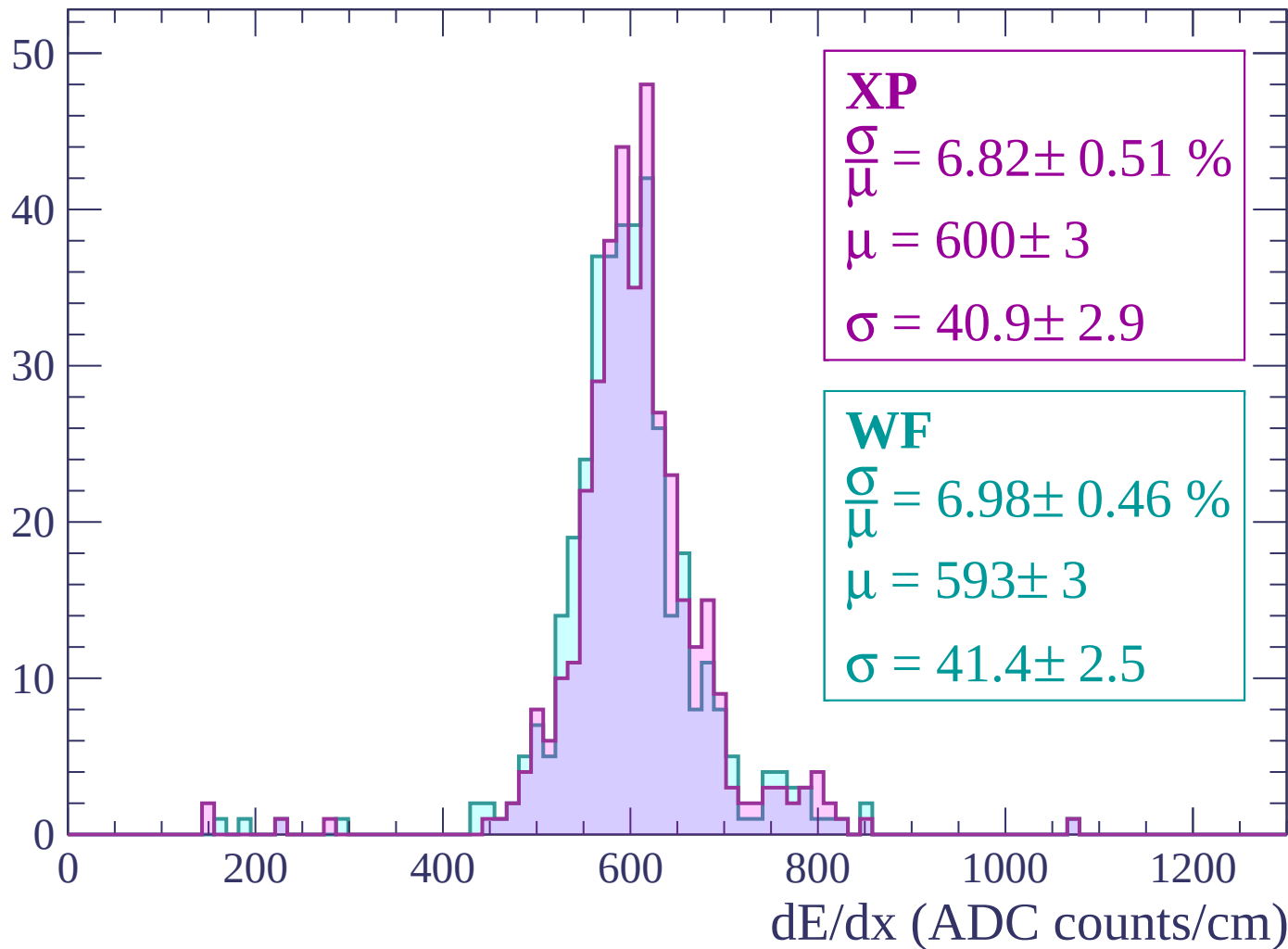
Energy loss | $60 < p < 120$

Count



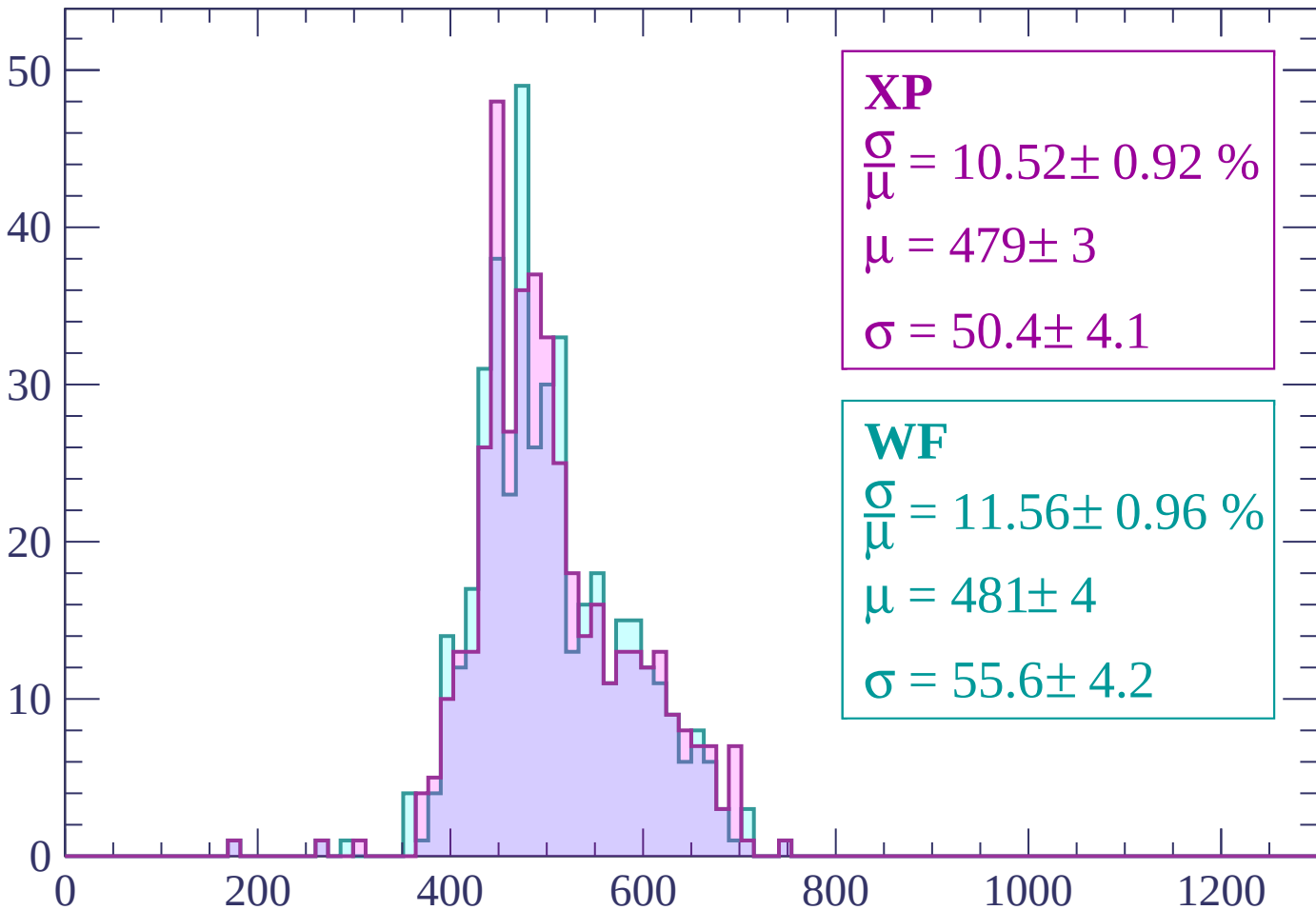
Energy loss | $120 < p < 180$

Count



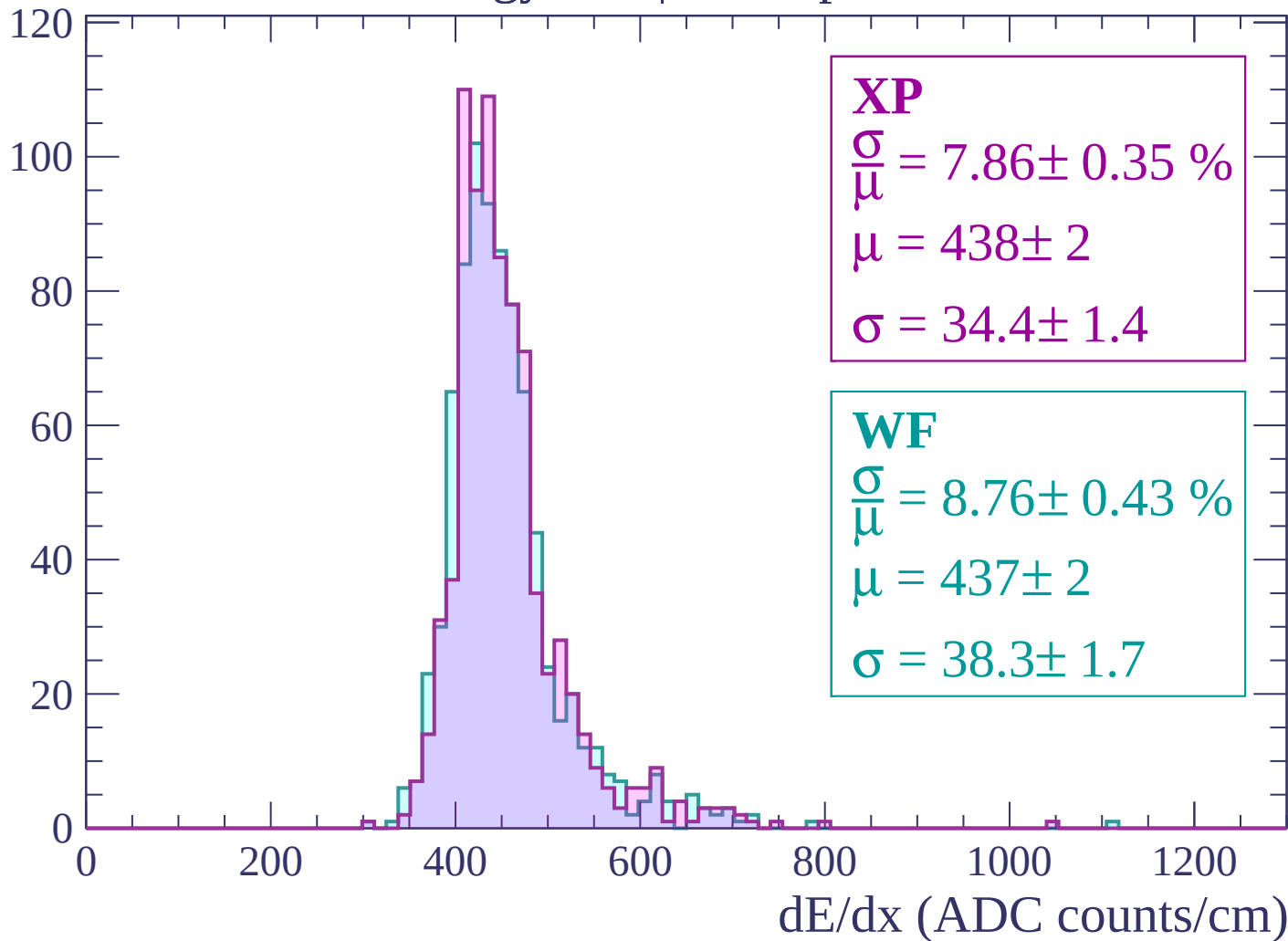
Energy loss | $180 < p < 240$

Count



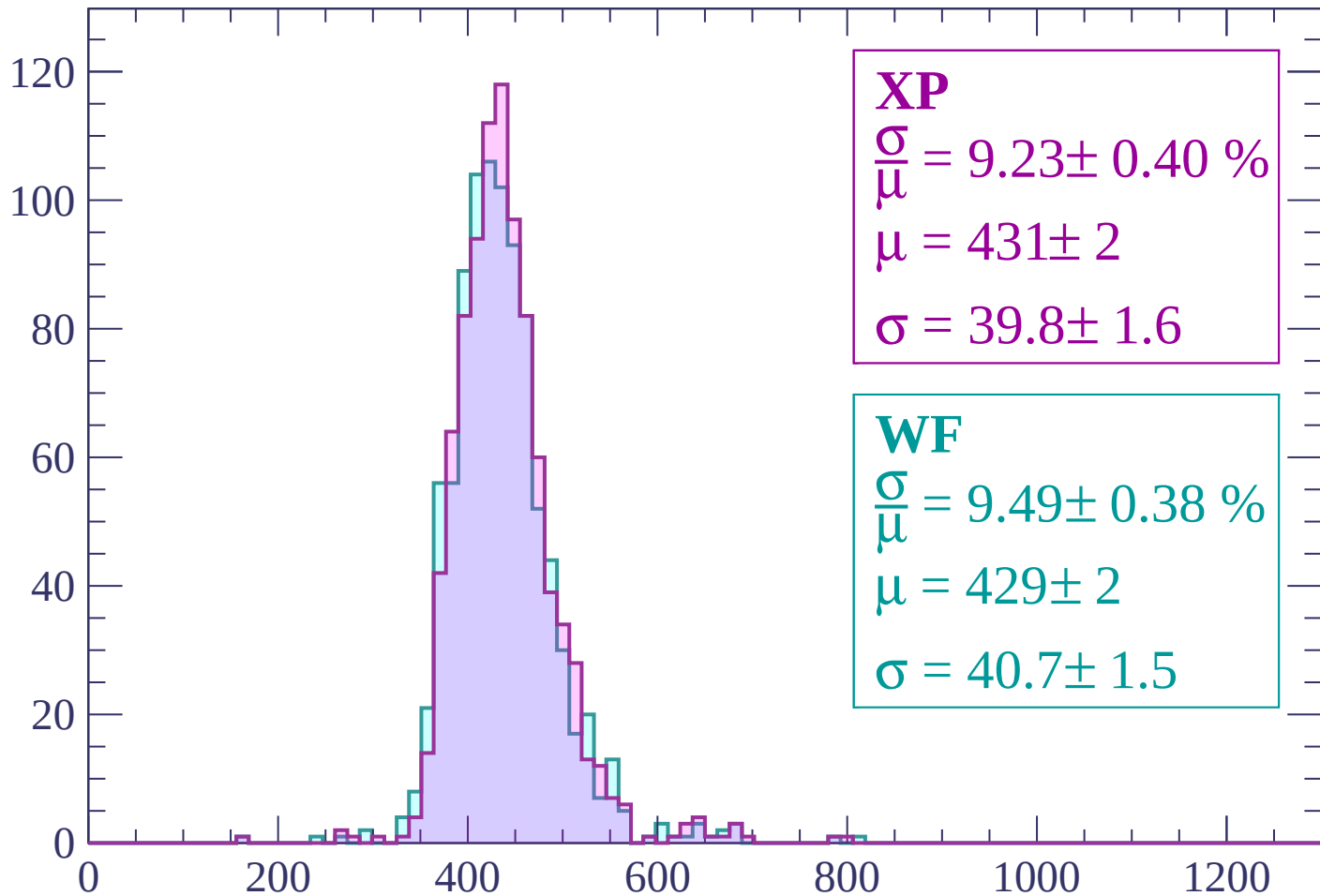
Energy loss | $240 < p < 300$

Count



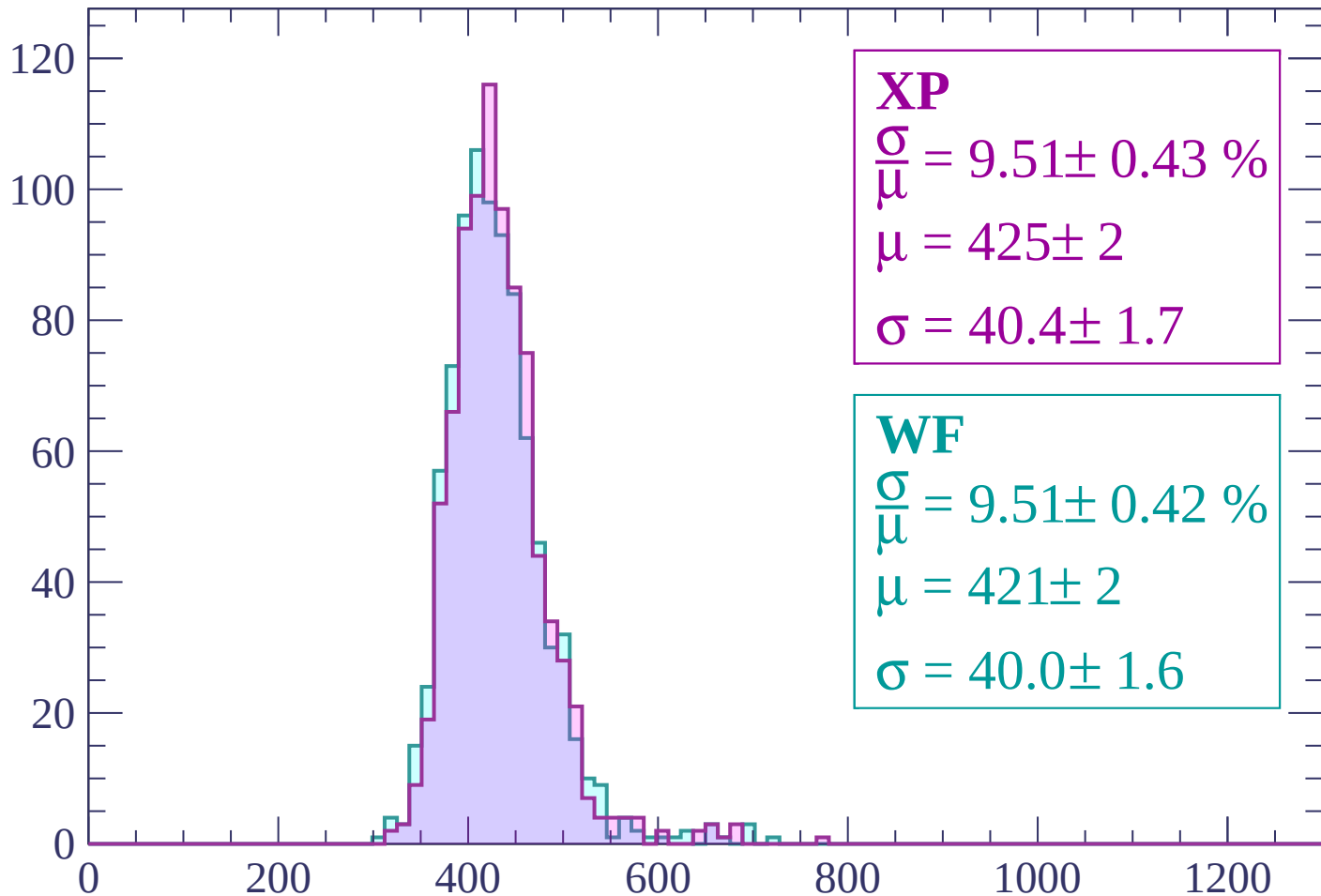
Energy loss | $300 < p < 360$

Count



Energy loss | $360 < p < 420$

Count



XP

$$\frac{\sigma}{\mu} = 9.51 \pm 0.43 \%$$

$$\mu = 425 \pm 2$$

$$\sigma = 40.4 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 9.51 \pm 0.42 \%$$

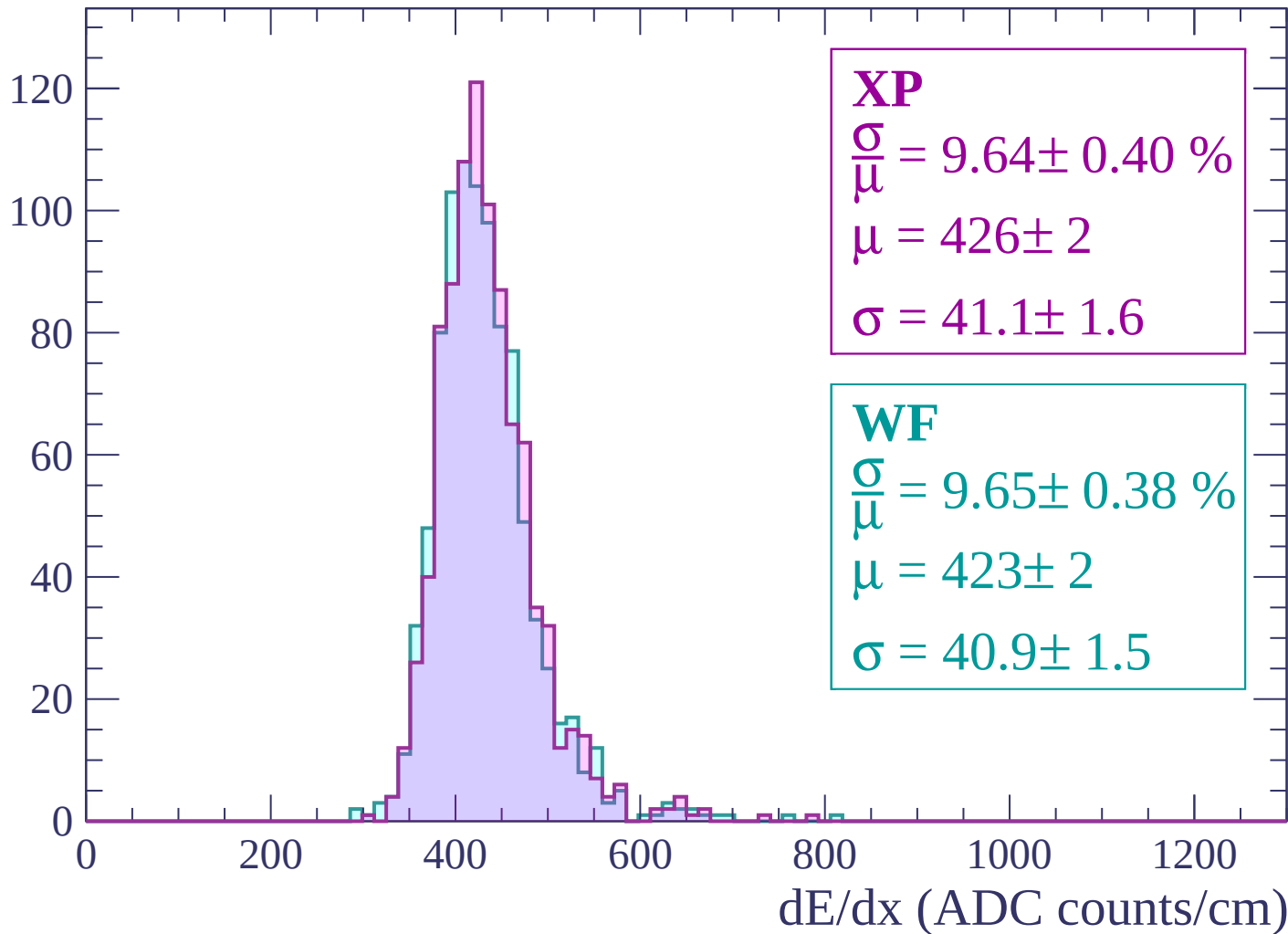
$$\mu = 421 \pm 2$$

$$\sigma = 40.0 \pm 1.6$$

dE/dx (ADC counts/cm)

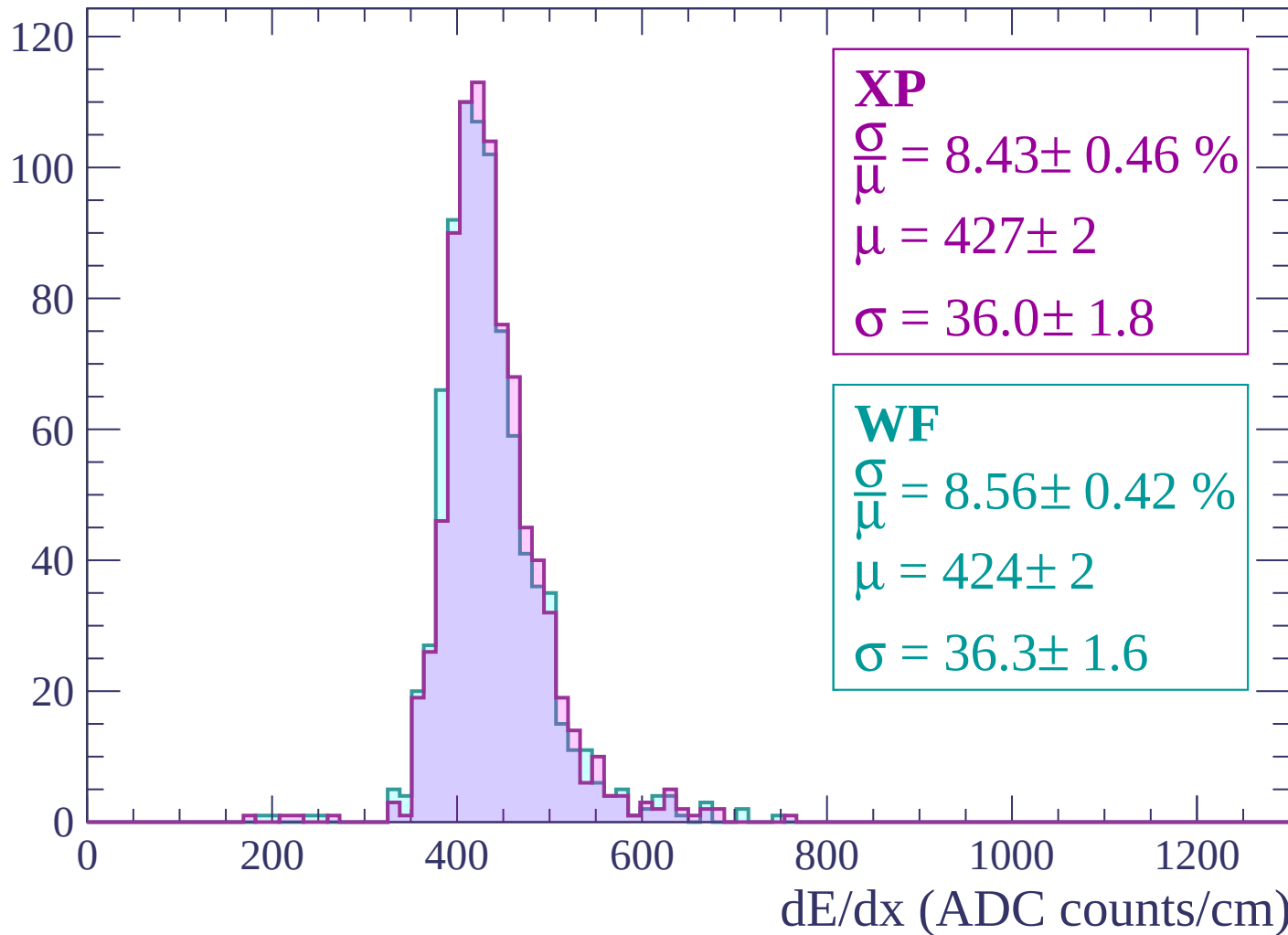
Energy loss | $420 < p < 480$

Count



Energy loss | $480 < p < 540$

Count



Energy loss | $540 < p < 600$

Count

140
120
100
80
60
40
20
0

XP

$$\frac{\sigma}{\mu} = 9.24 \pm 0.42 \%$$

$$\mu = 431 \pm 2$$

$$\sigma = 39.8 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 9.82 \pm 0.49 \%$$

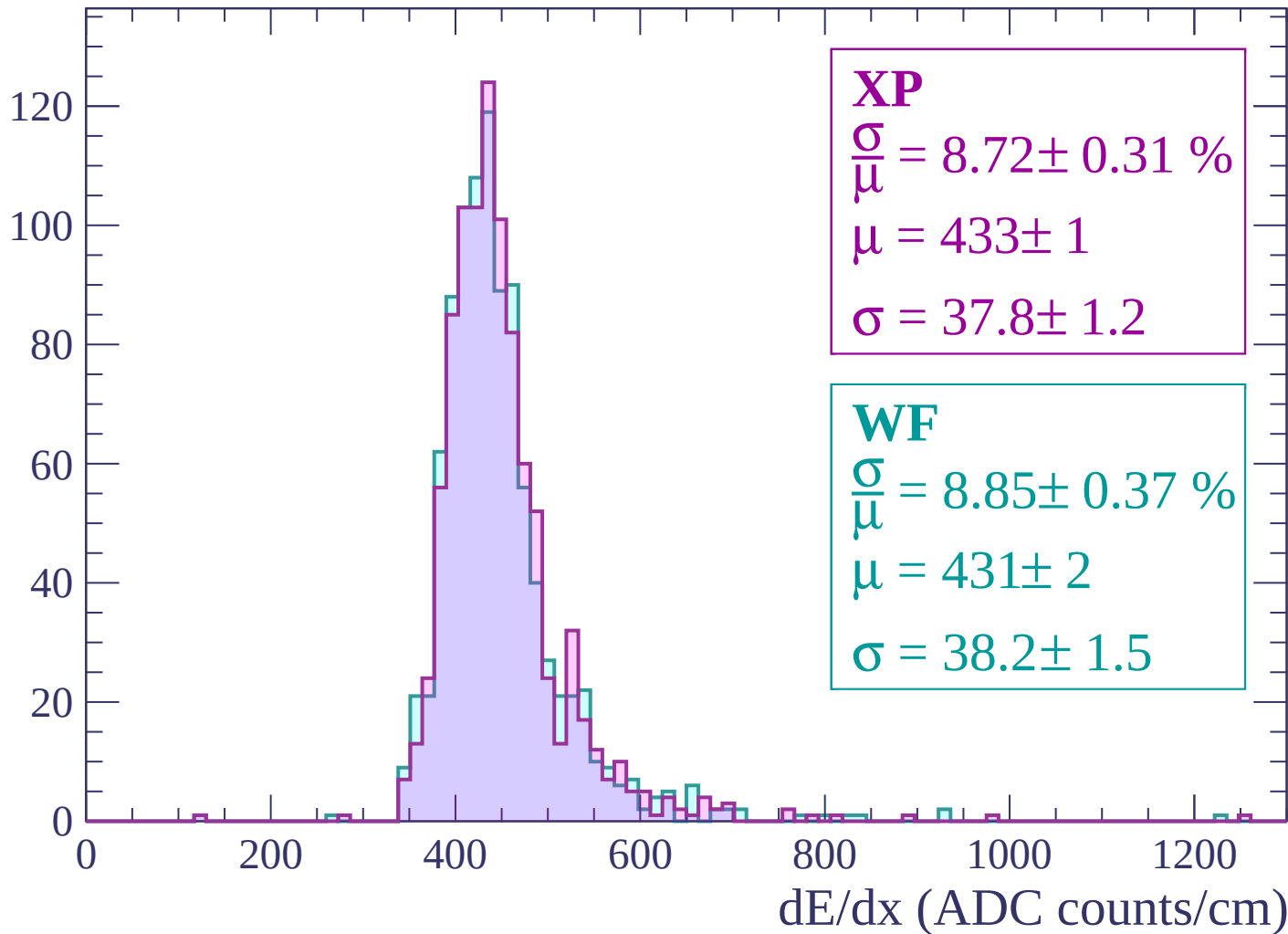
$$\mu = 427 \pm 2$$

$$\sigma = 42.0 \pm 1.9$$

0 200 400 600 800 1000 1200
dE/dx (ADC counts/cm)

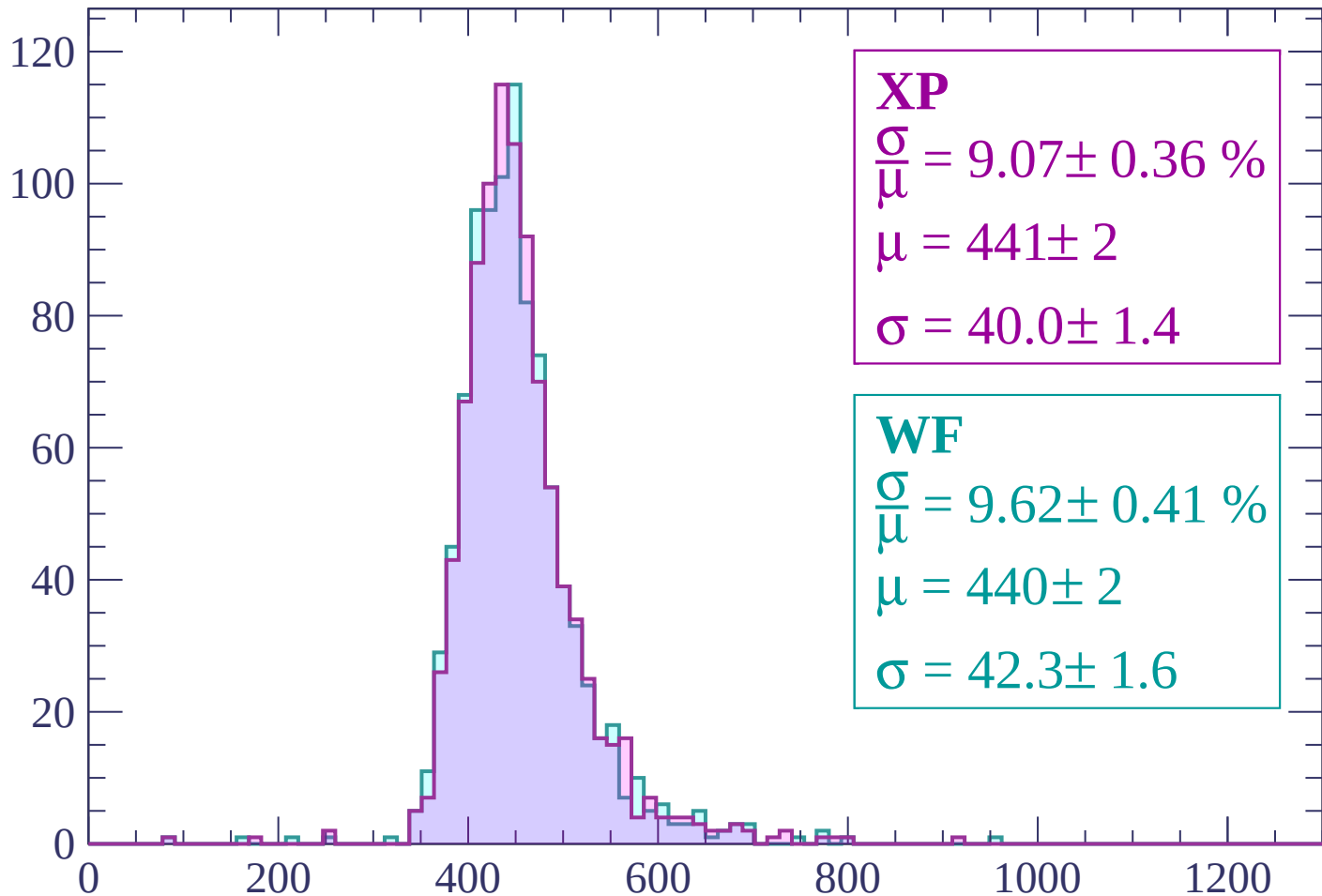
Energy loss | $600 < p < 660$

Count



Energy loss | $660 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 9.07 \pm 0.36 \%$$

$$\mu = 441 \pm 2$$

$$\sigma = 40.0 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 9.62 \pm 0.41 \%$$

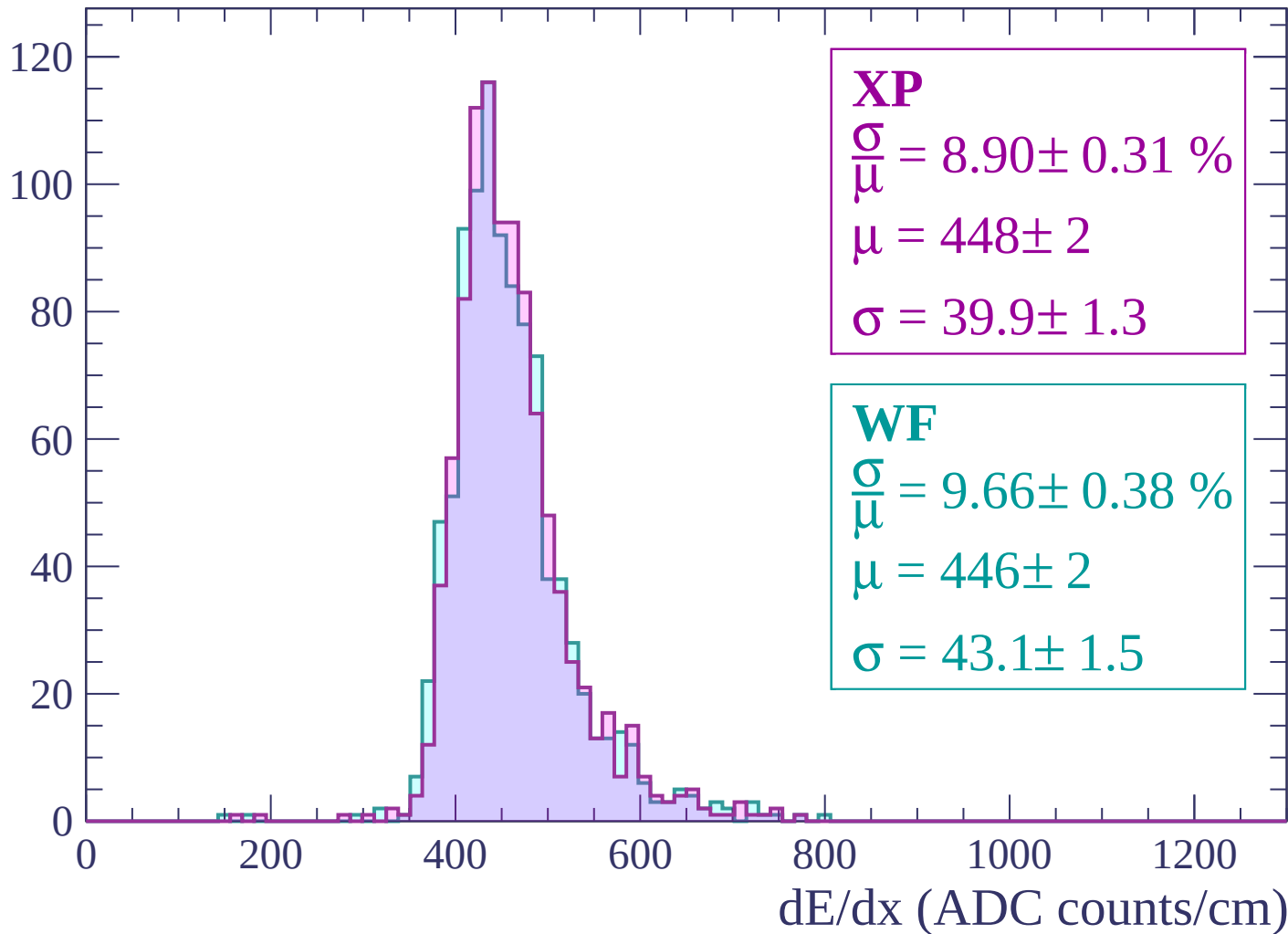
$$\mu = 440 \pm 2$$

$$\sigma = 42.3 \pm 1.6$$

dE/dx (ADC counts/cm)

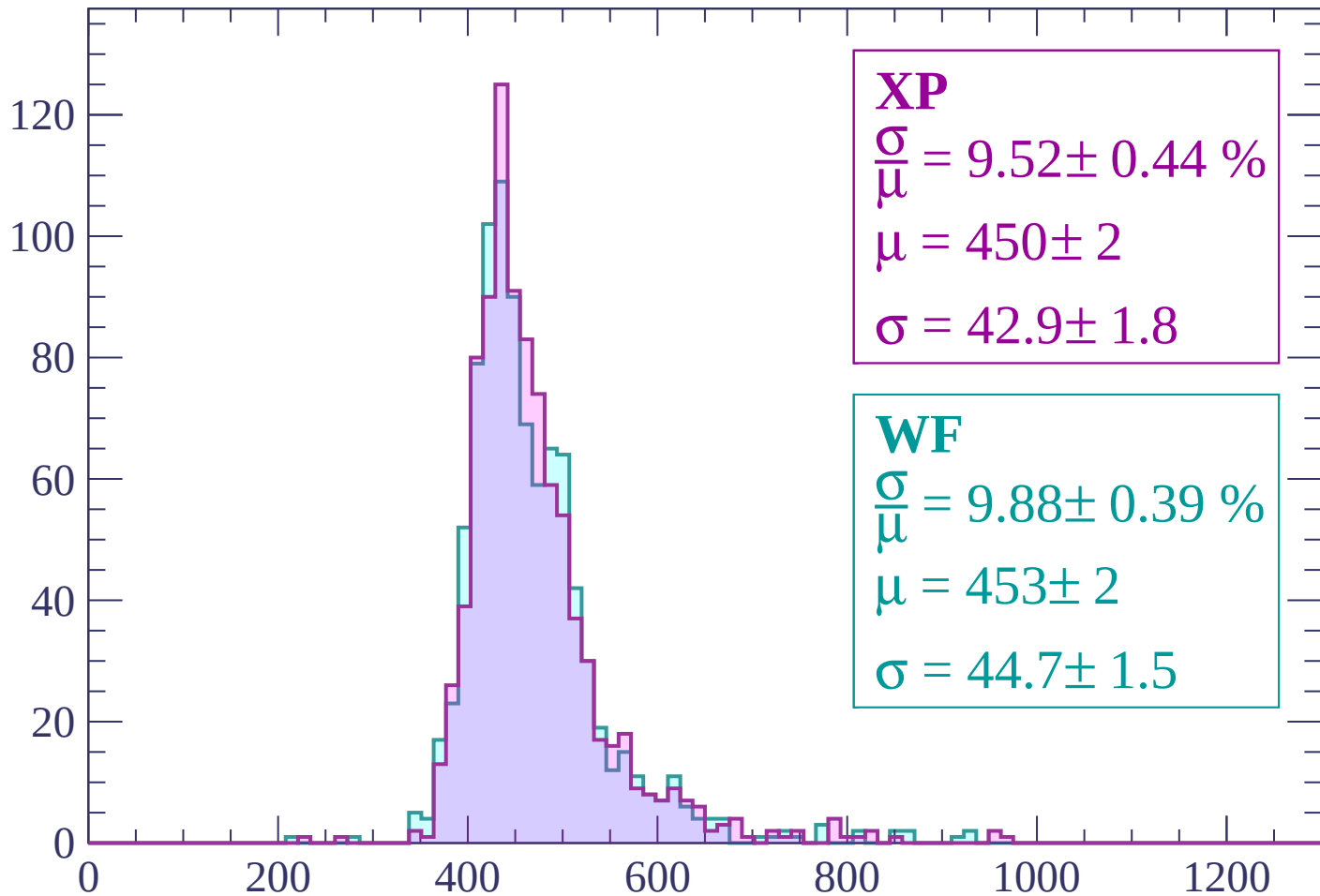
Energy loss | $720 < p < 780$

Count



Energy loss | $780 < p < 840$

Count



XP

$$\frac{\sigma}{\mu} = 9.52 \pm 0.44 \%$$

$$\mu = 450 \pm 2$$

$$\sigma = 42.9 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 9.88 \pm 0.39 \%$$

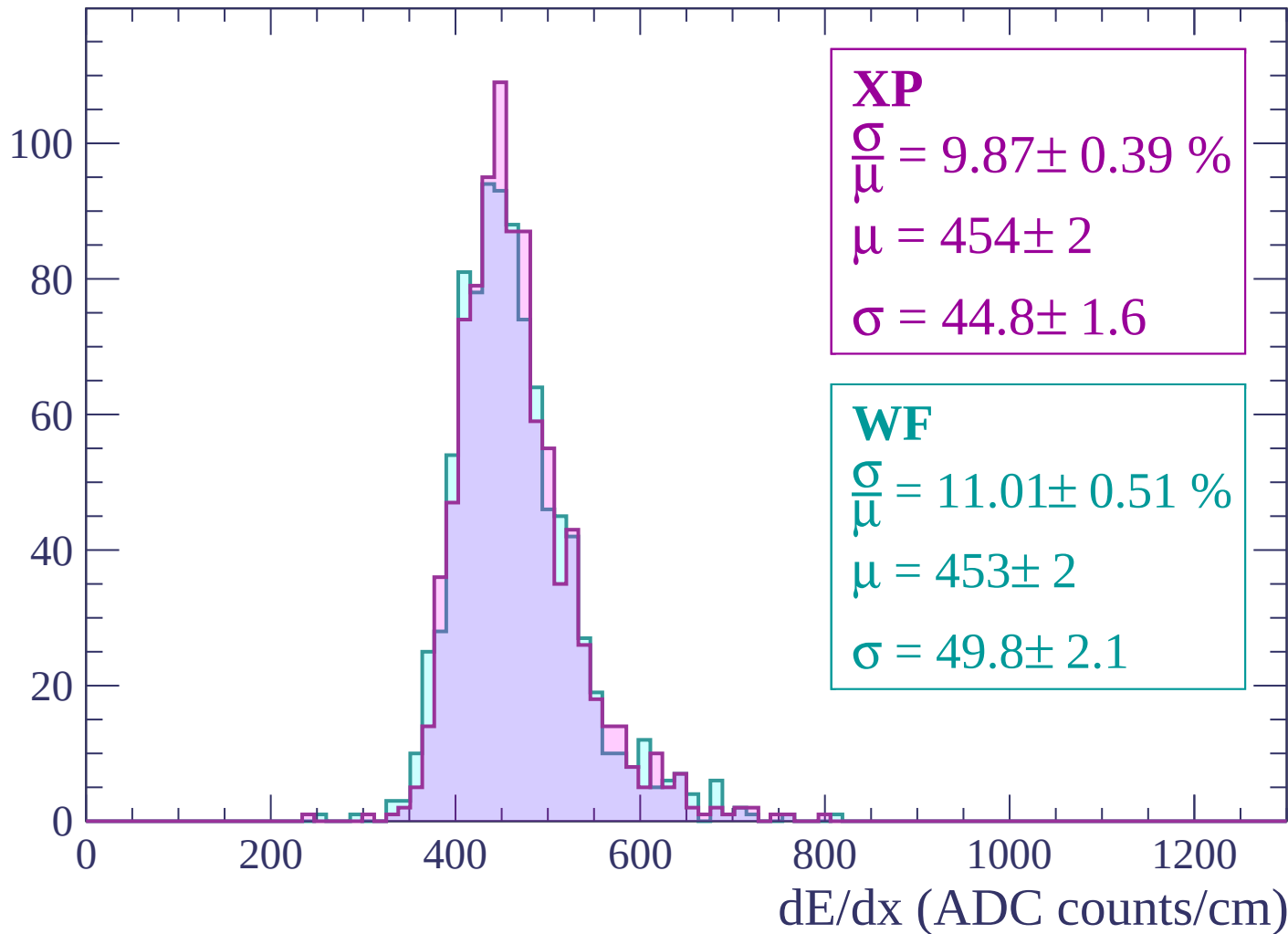
$$\mu = 453 \pm 2$$

$$\sigma = 44.7 \pm 1.5$$

dE/dx (ADC counts/cm)

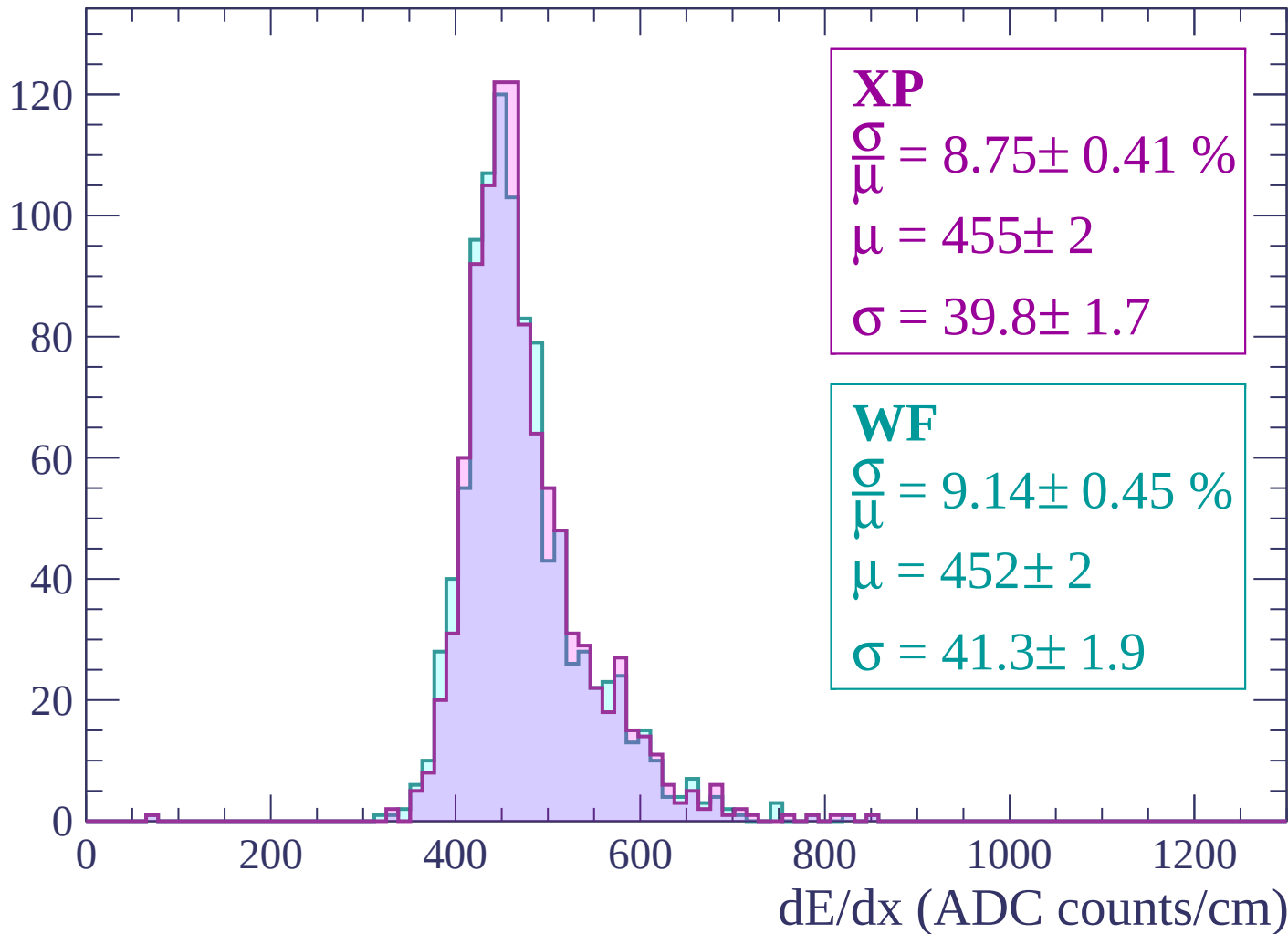
Energy loss | $840 < p < 900$

Count



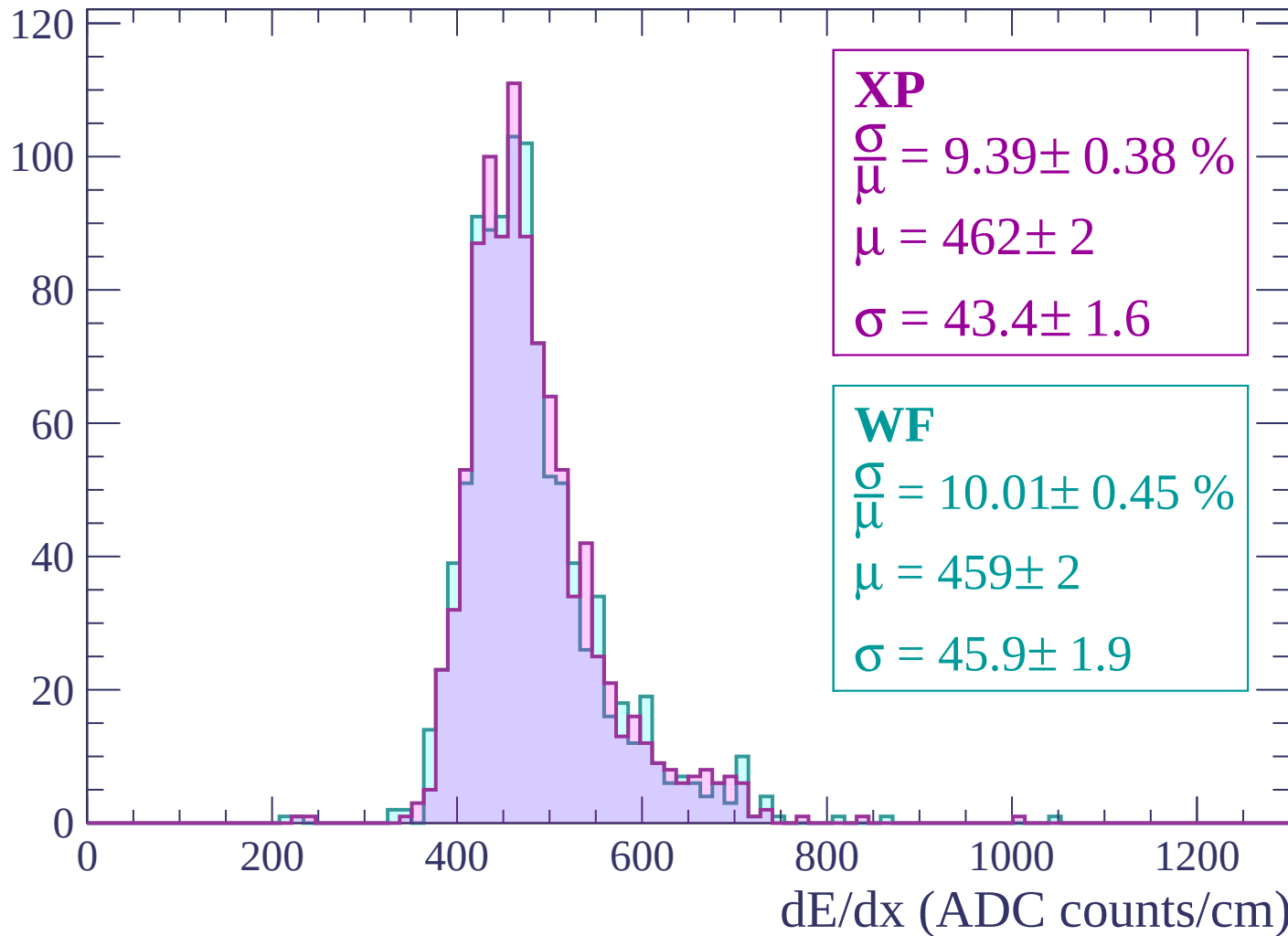
Energy loss | $900 < p < 960$

Count



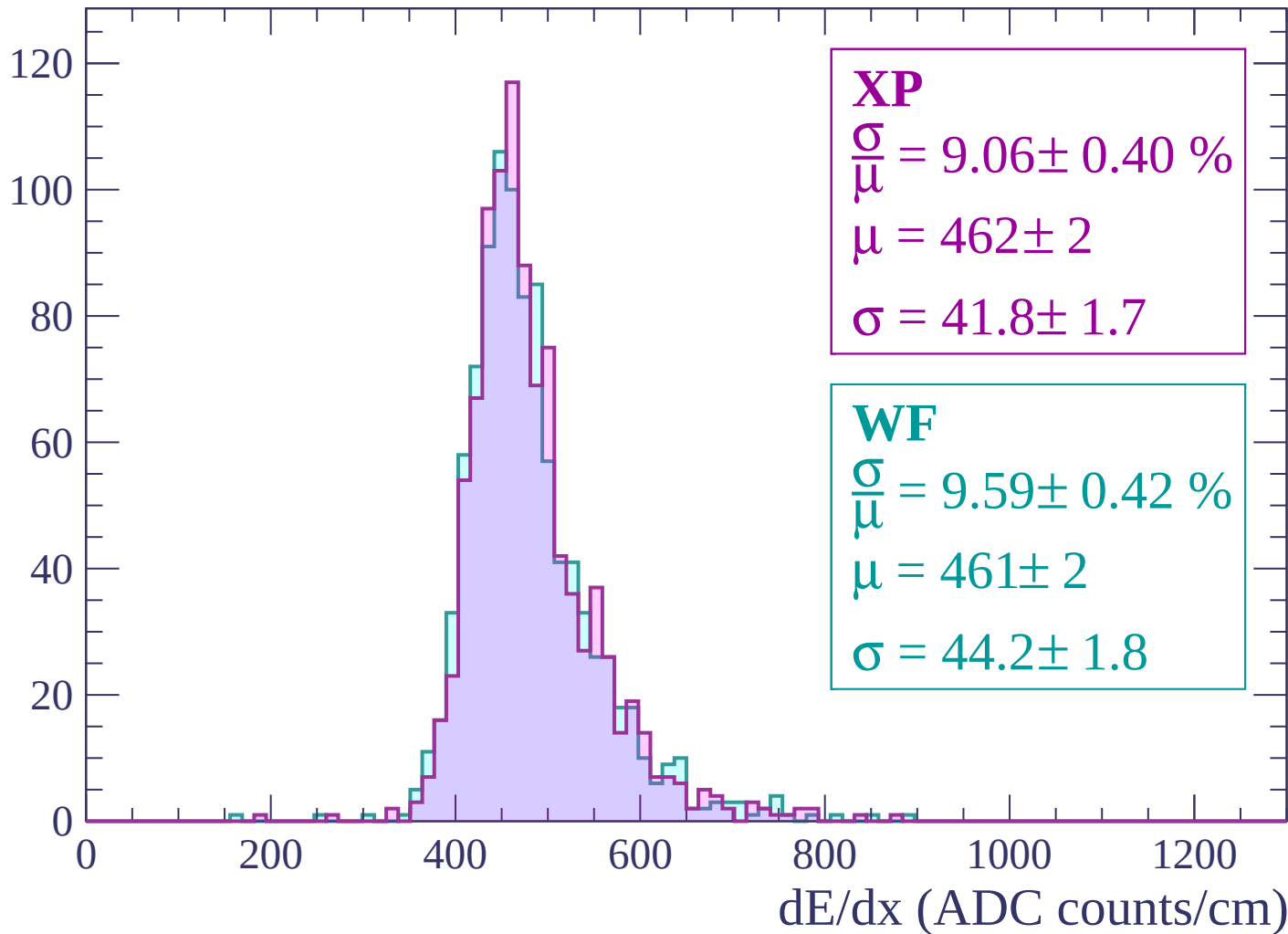
Energy loss | $960 < p < 1020$

Count



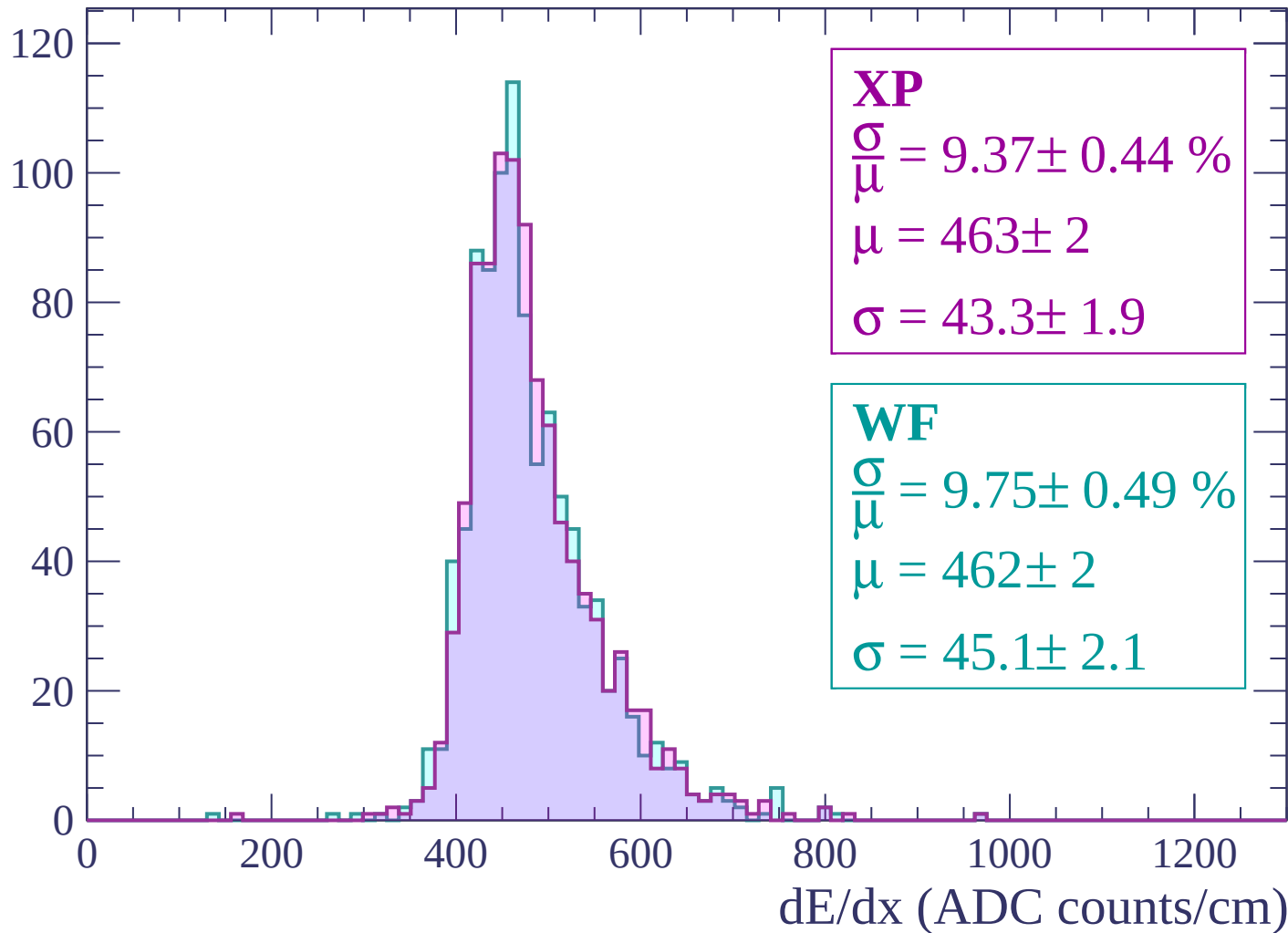
Energy loss | $1020 < p < 1080$

Count



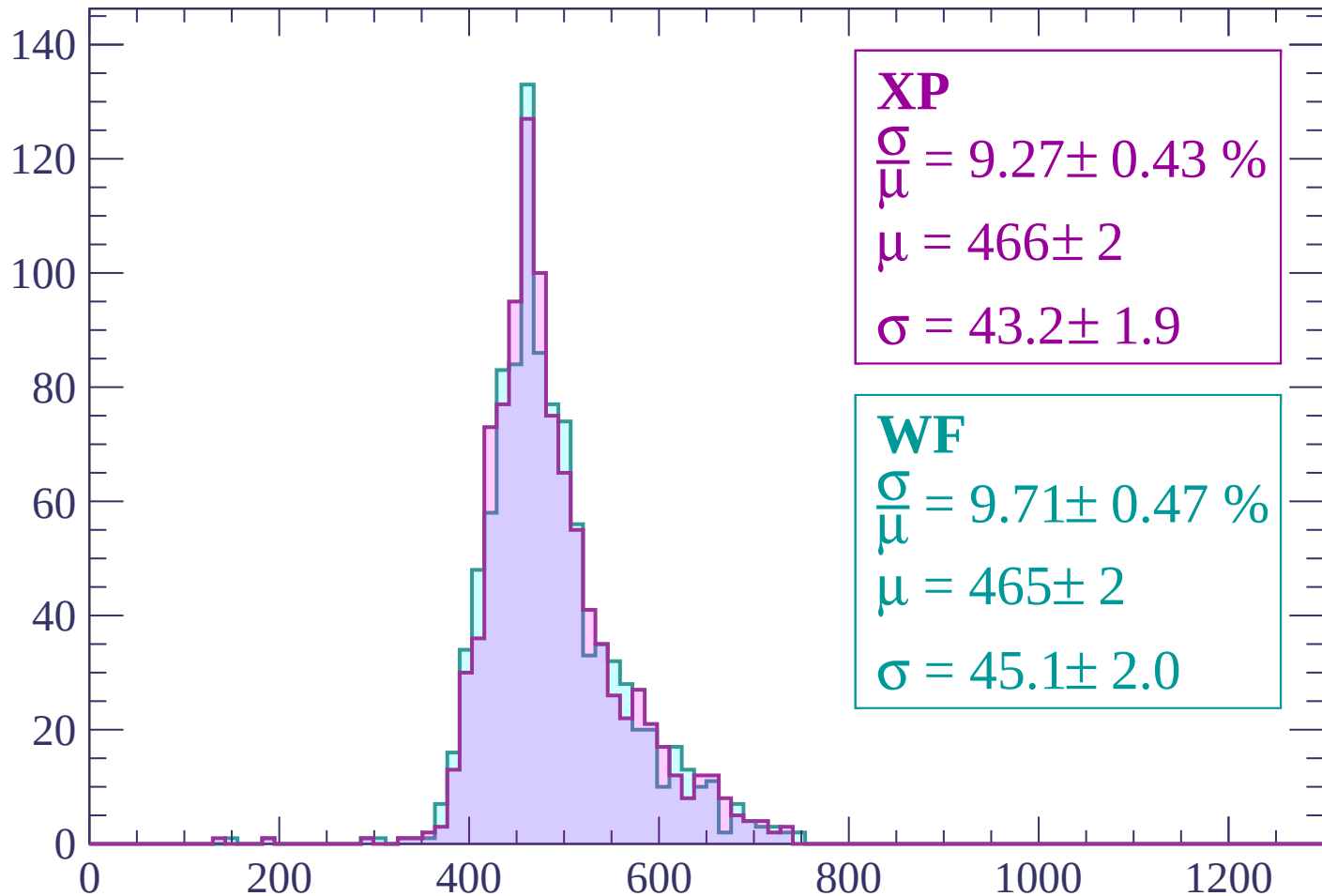
Energy loss | $1080 < p < 1140$

Count



Energy loss | $1140 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 9.27 \pm 0.43 \%$$

$$\mu = 466 \pm 2$$

$$\sigma = 43.2 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 9.71 \pm 0.47 \%$$

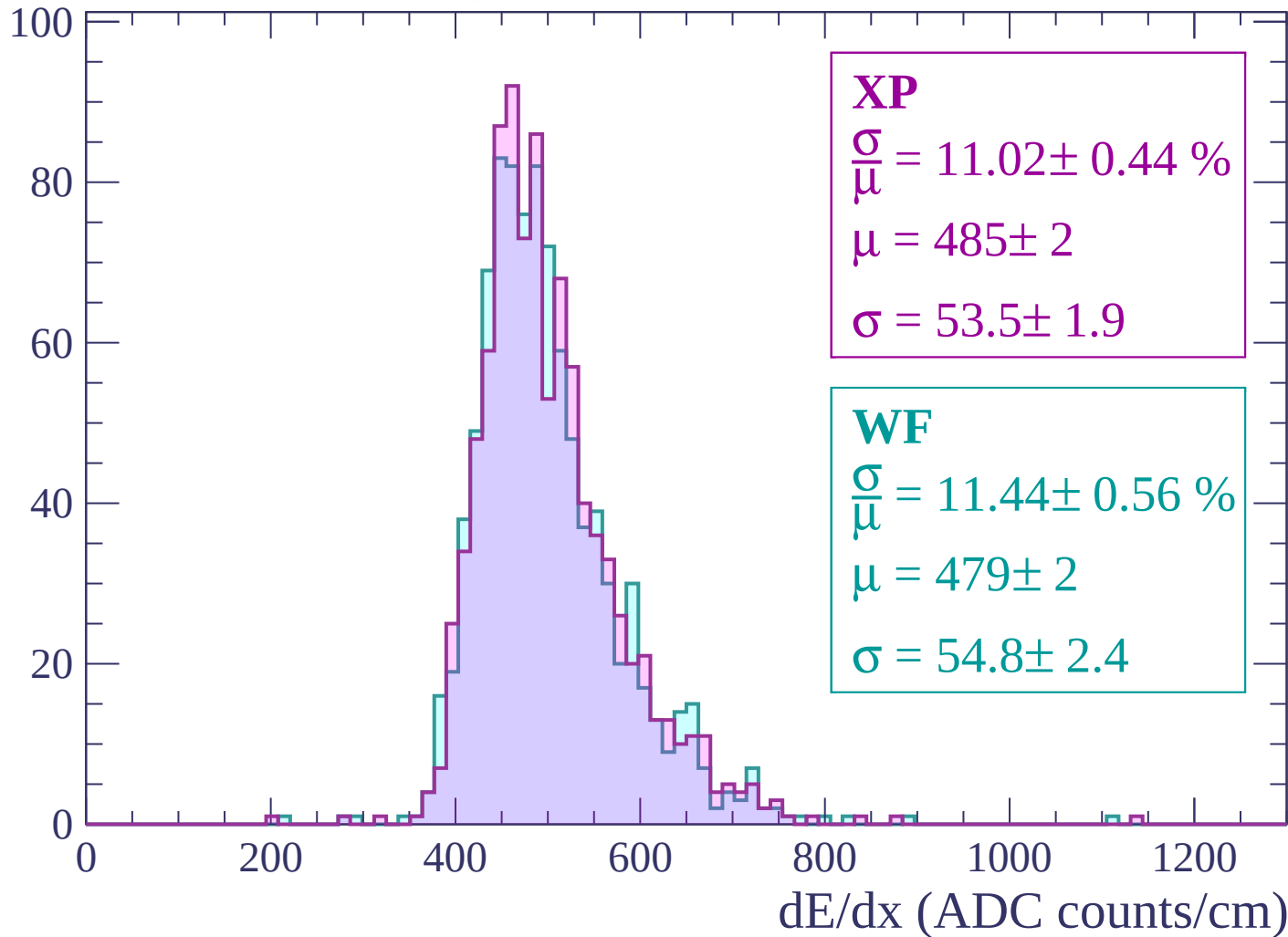
$$\mu = 465 \pm 2$$

$$\sigma = 45.1 \pm 2.0$$

dE/dx (ADC counts/cm)

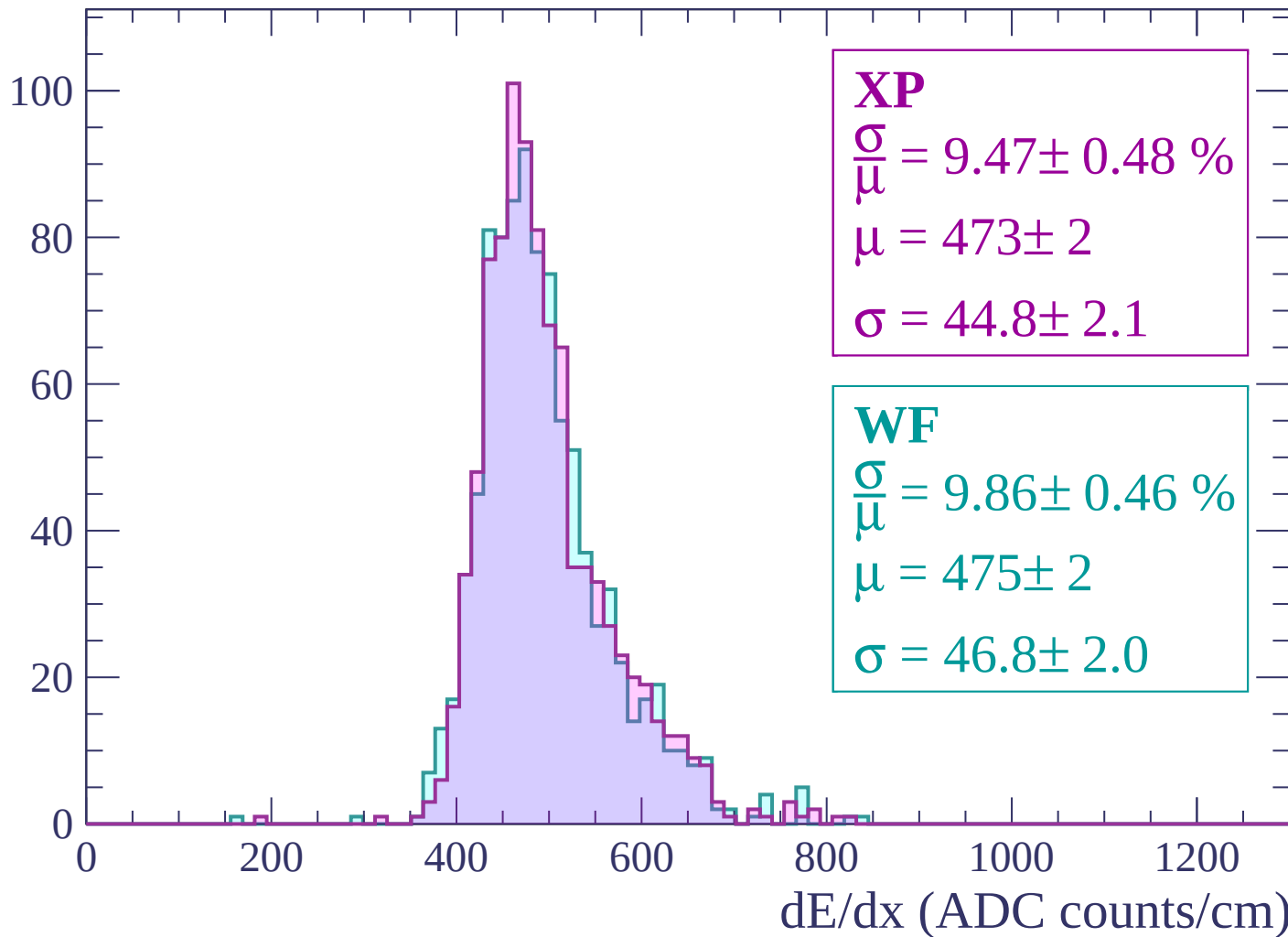
Energy loss | $1200 < p < 1260$

Count



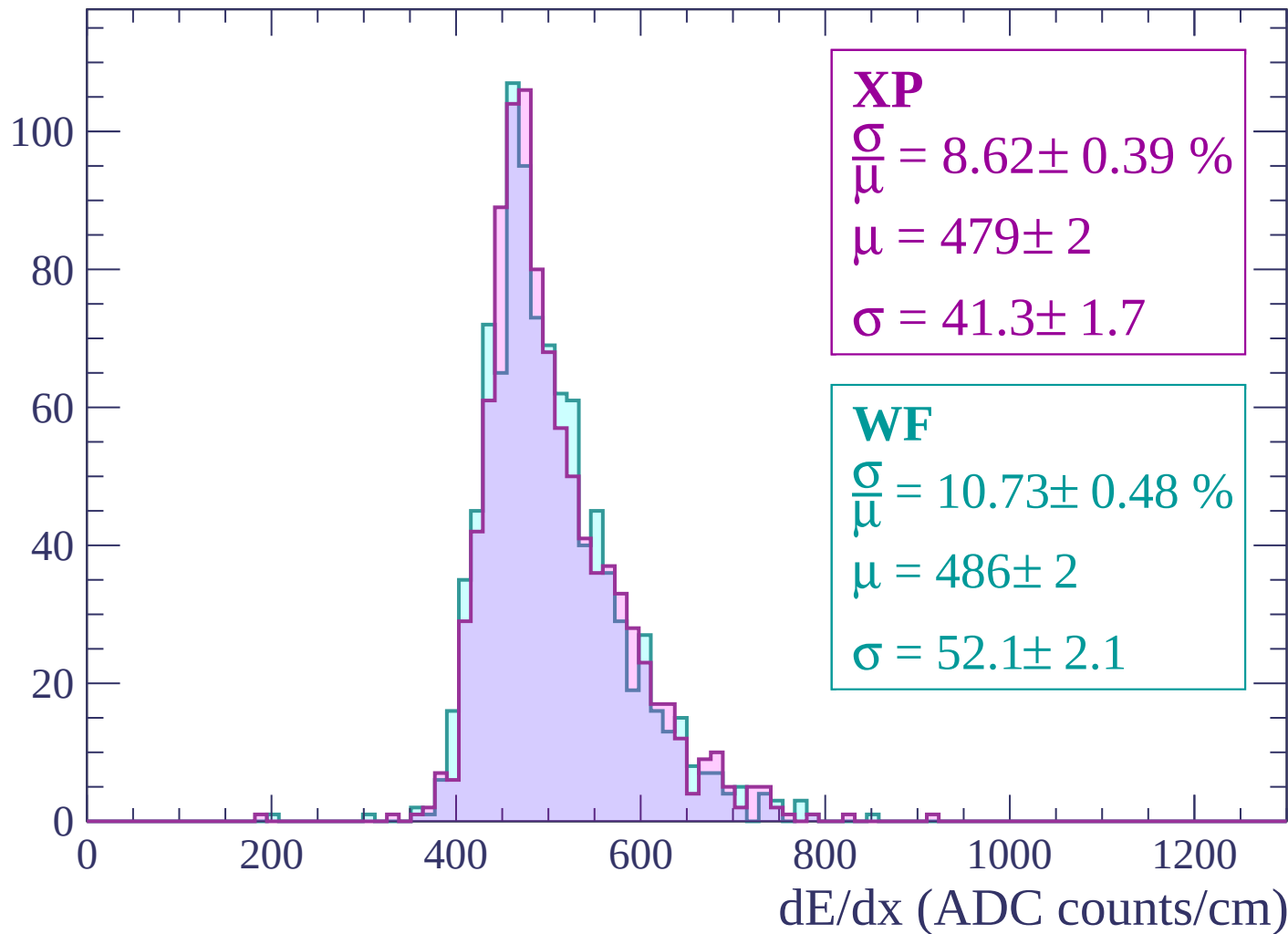
Energy loss | $1260 < p < 1320$

Count



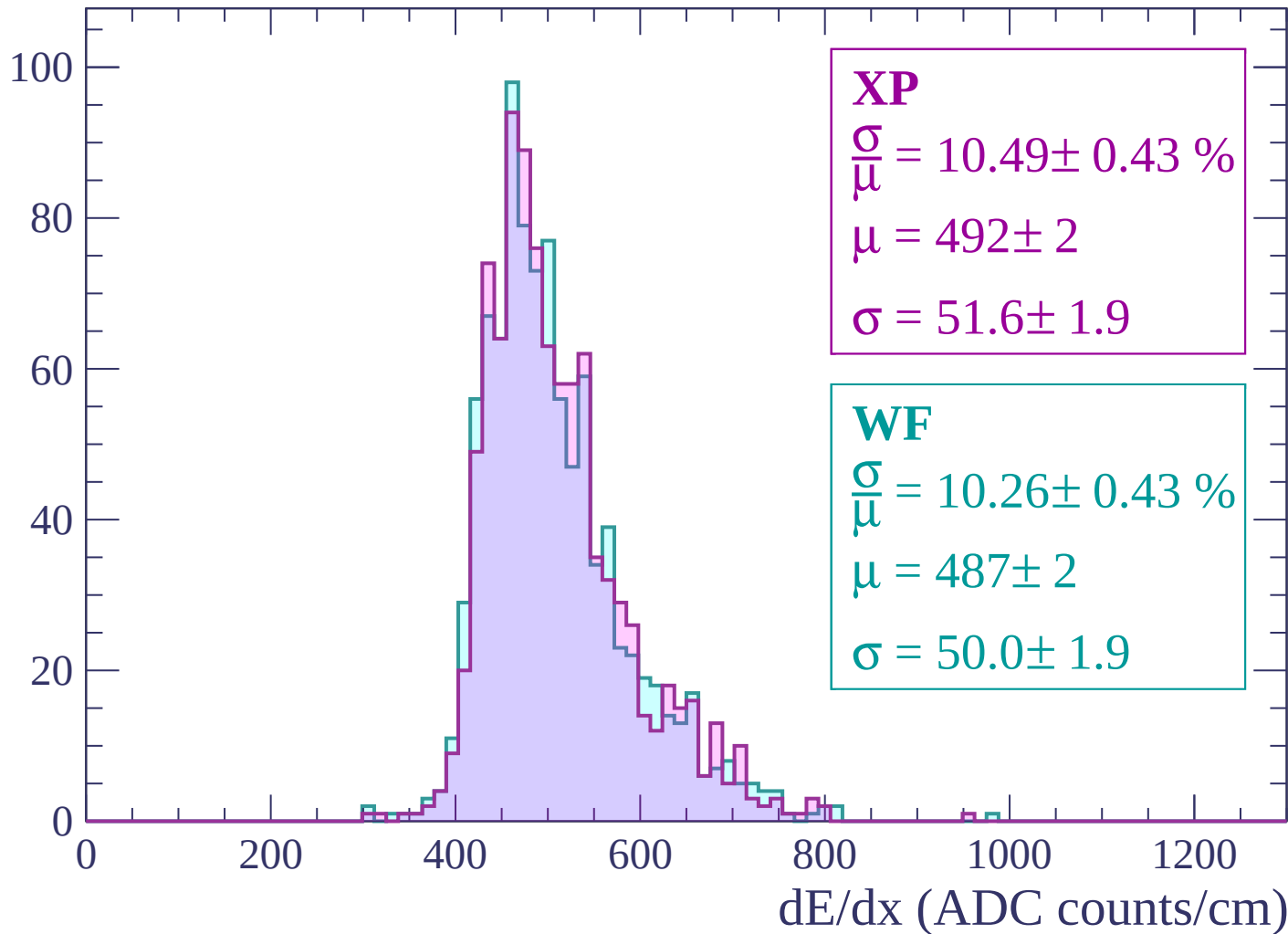
Energy loss | $1320 < p < 1380$

Count



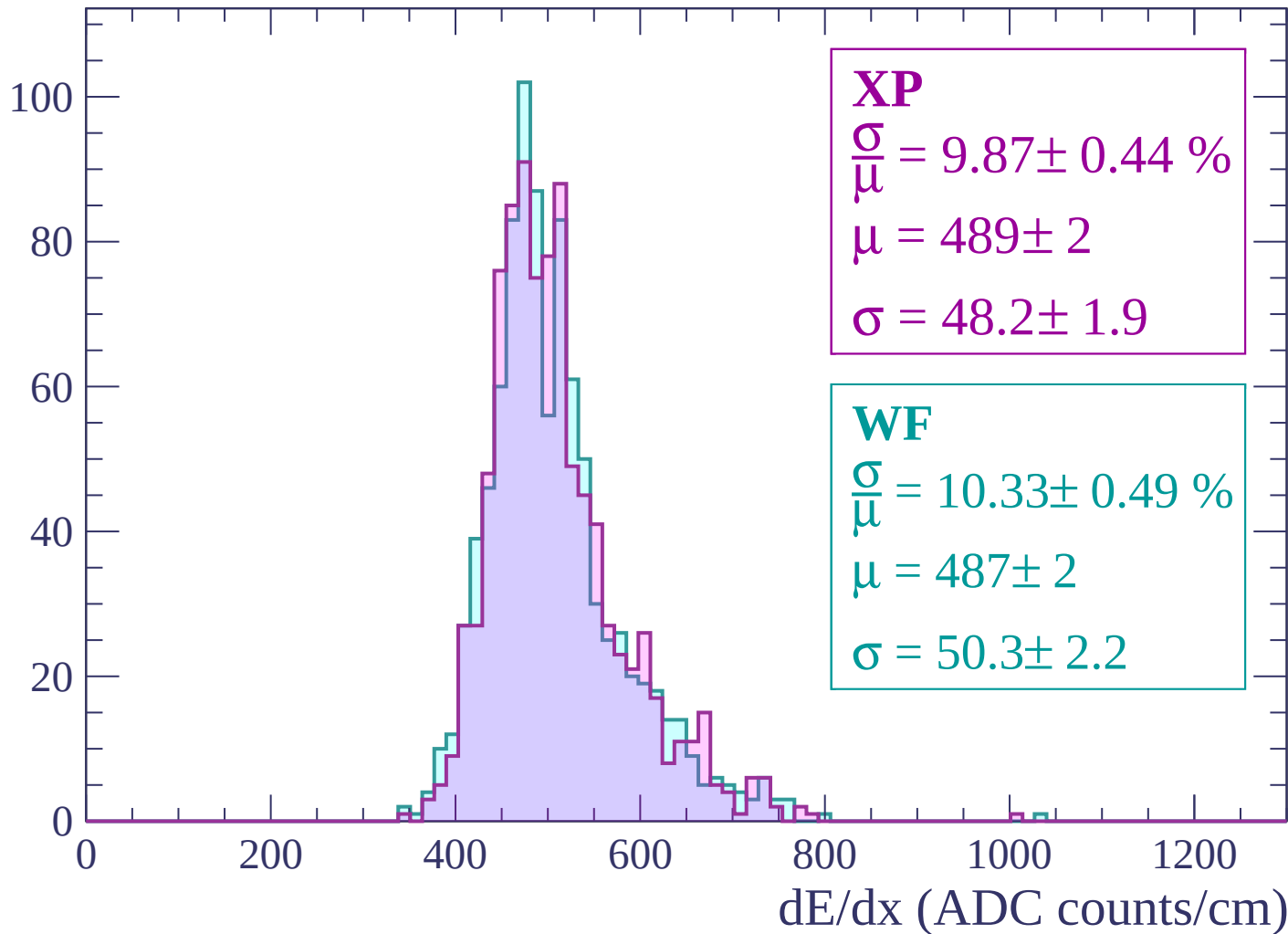
Energy loss | $1380 < p < 1440$

Count



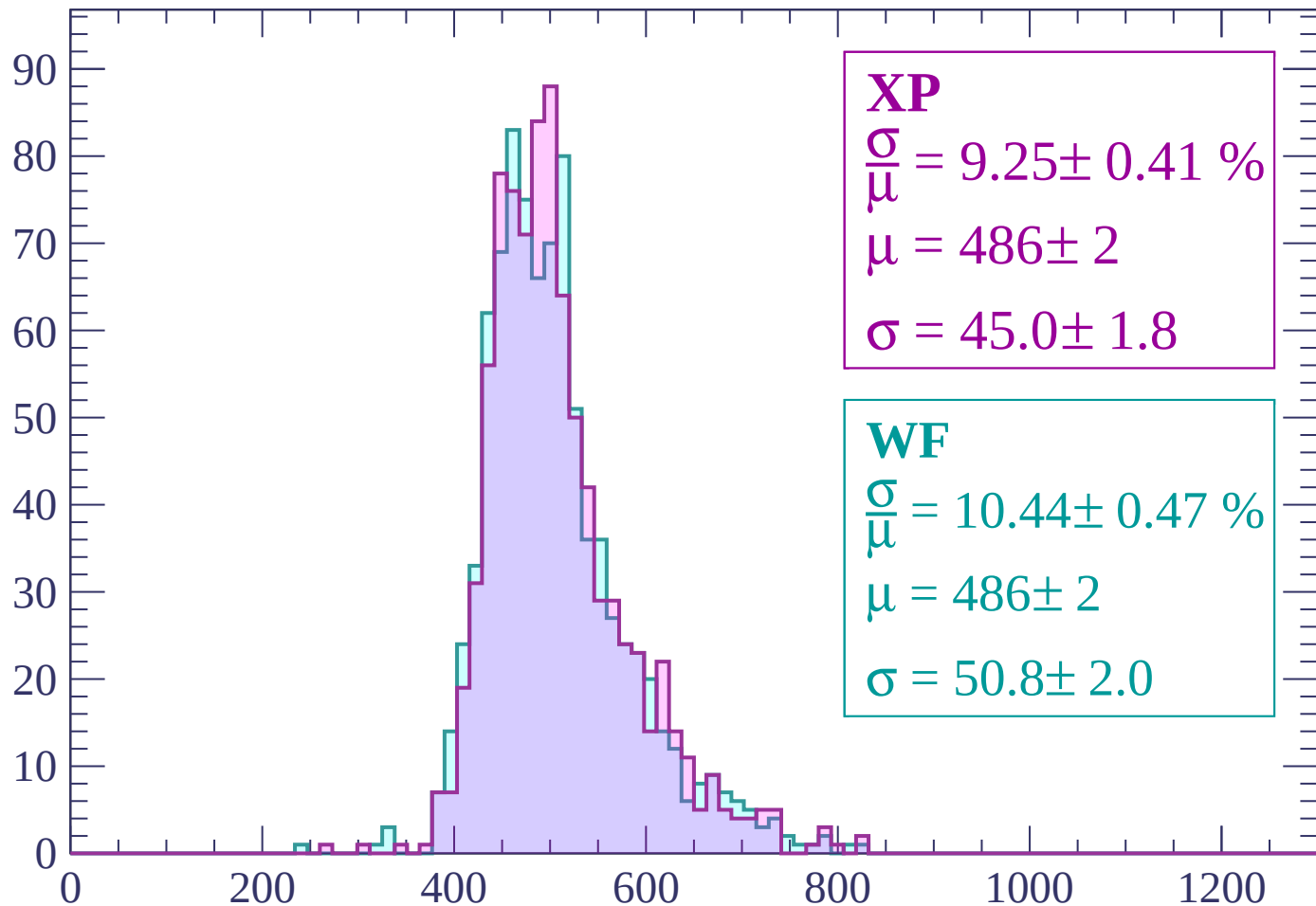
Energy loss | $1440 < p < 1500$

Count



Energy loss | $1500 < p < 1560$

Count



XP

$$\frac{\sigma}{\mu} = 9.25 \pm 0.41 \%$$

$$\mu = 486 \pm 2$$

$$\sigma = 45.0 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 10.44 \pm 0.47 \%$$

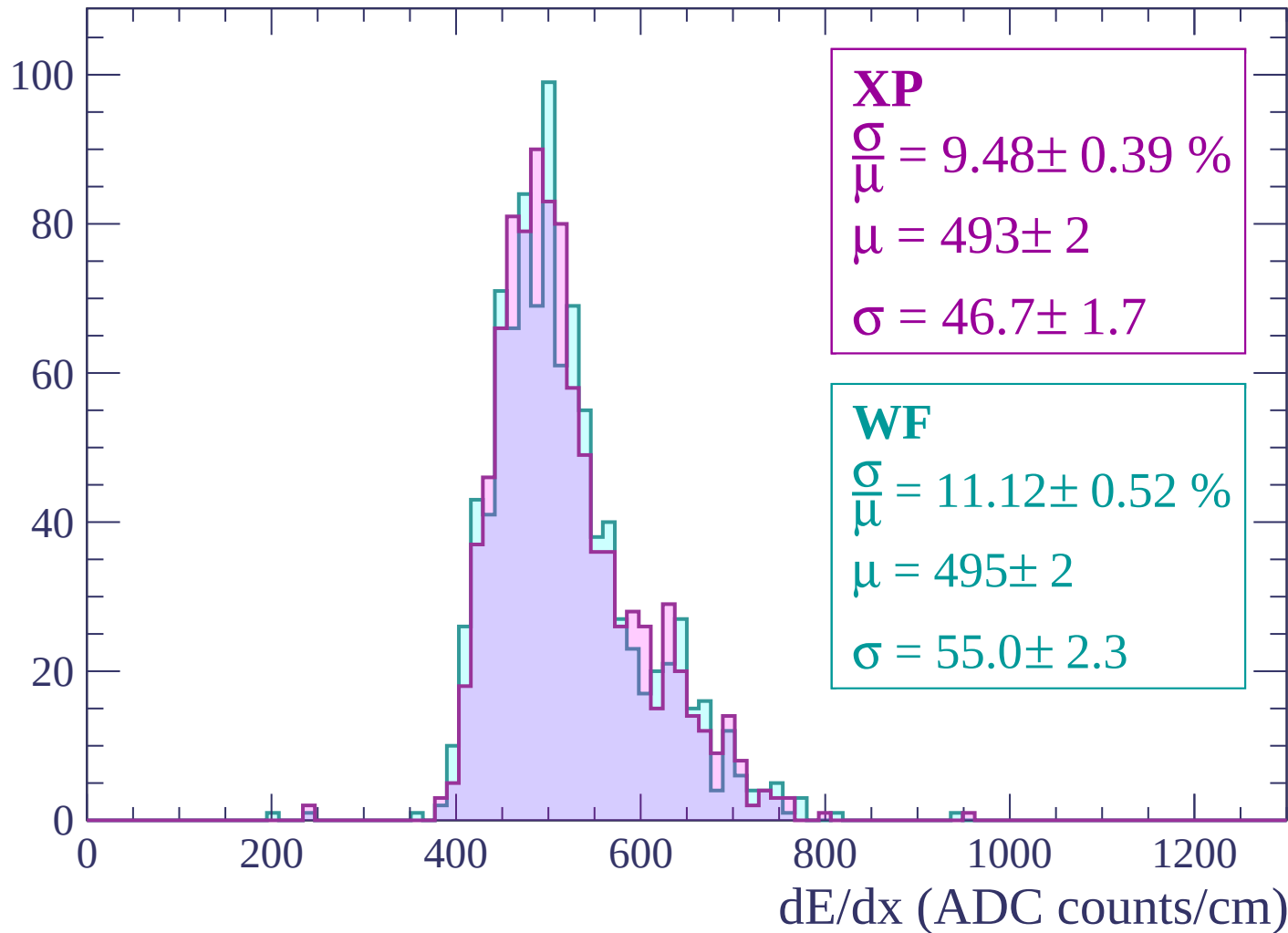
$$\mu = 486 \pm 2$$

$$\sigma = 50.8 \pm 2.0$$

dE/dx (ADC counts/cm)

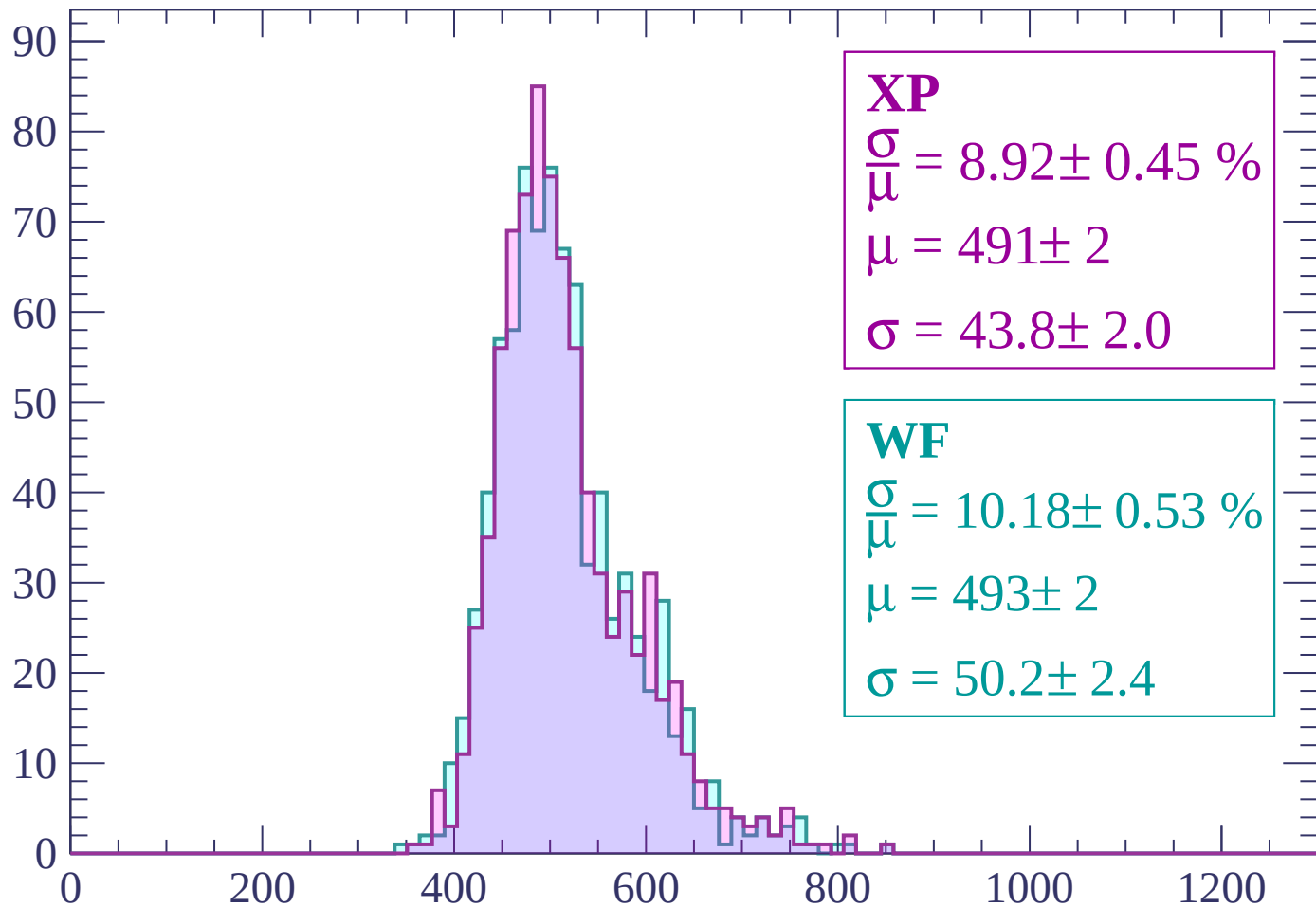
Energy loss | $1560 < p < 1620$

Count



Energy loss | $1620 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 8.92 \pm 0.45 \%$$

$$\mu = 491 \pm 2$$

$$\sigma = 43.8 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 10.18 \pm 0.53 \%$$

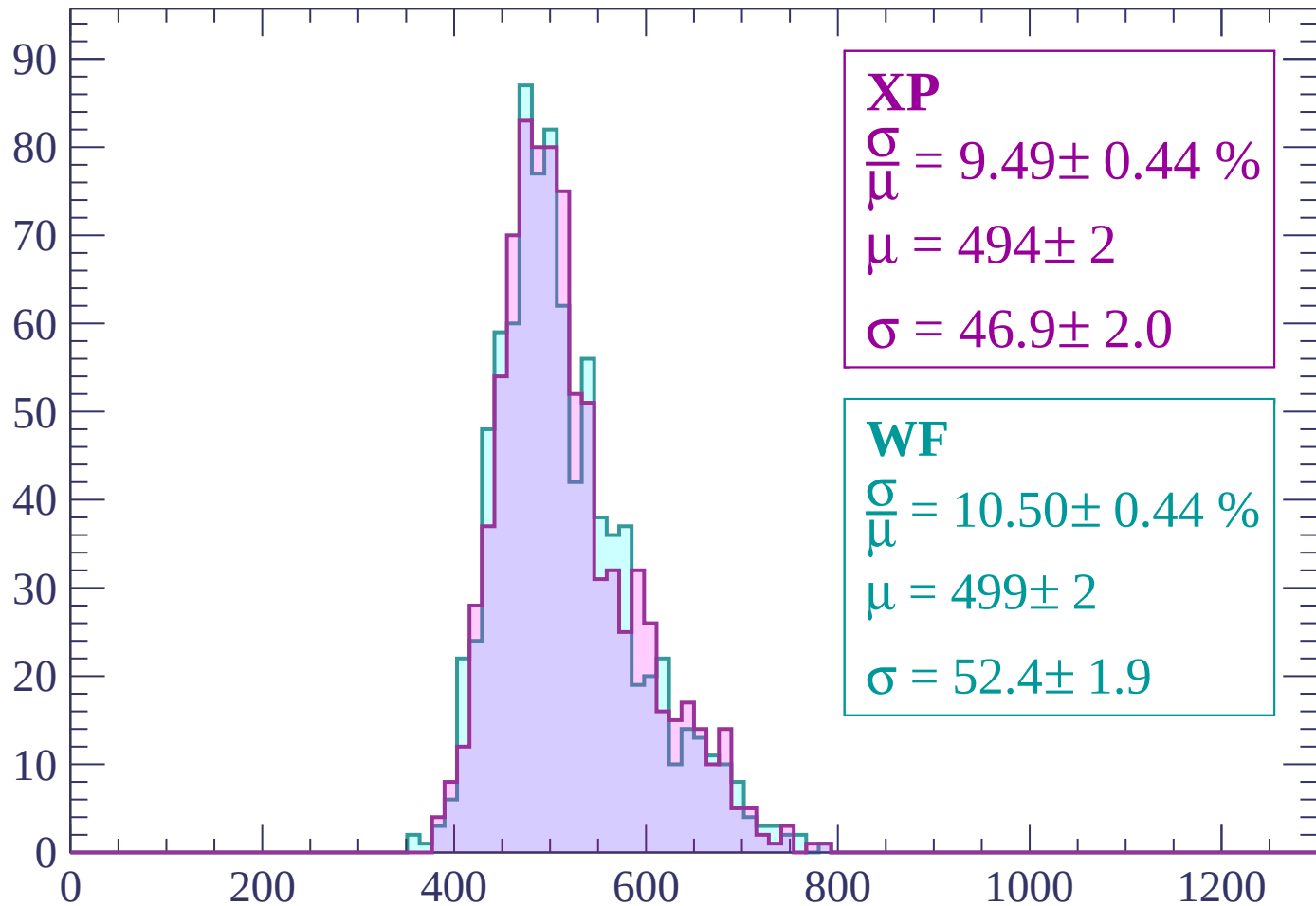
$$\mu = 493 \pm 2$$

$$\sigma = 50.2 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP

$$\frac{\sigma}{\mu} = 9.49 \pm 0.44 \%$$

$$\mu = 494 \pm 2$$

$$\sigma = 46.9 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 10.50 \pm 0.44 \%$$

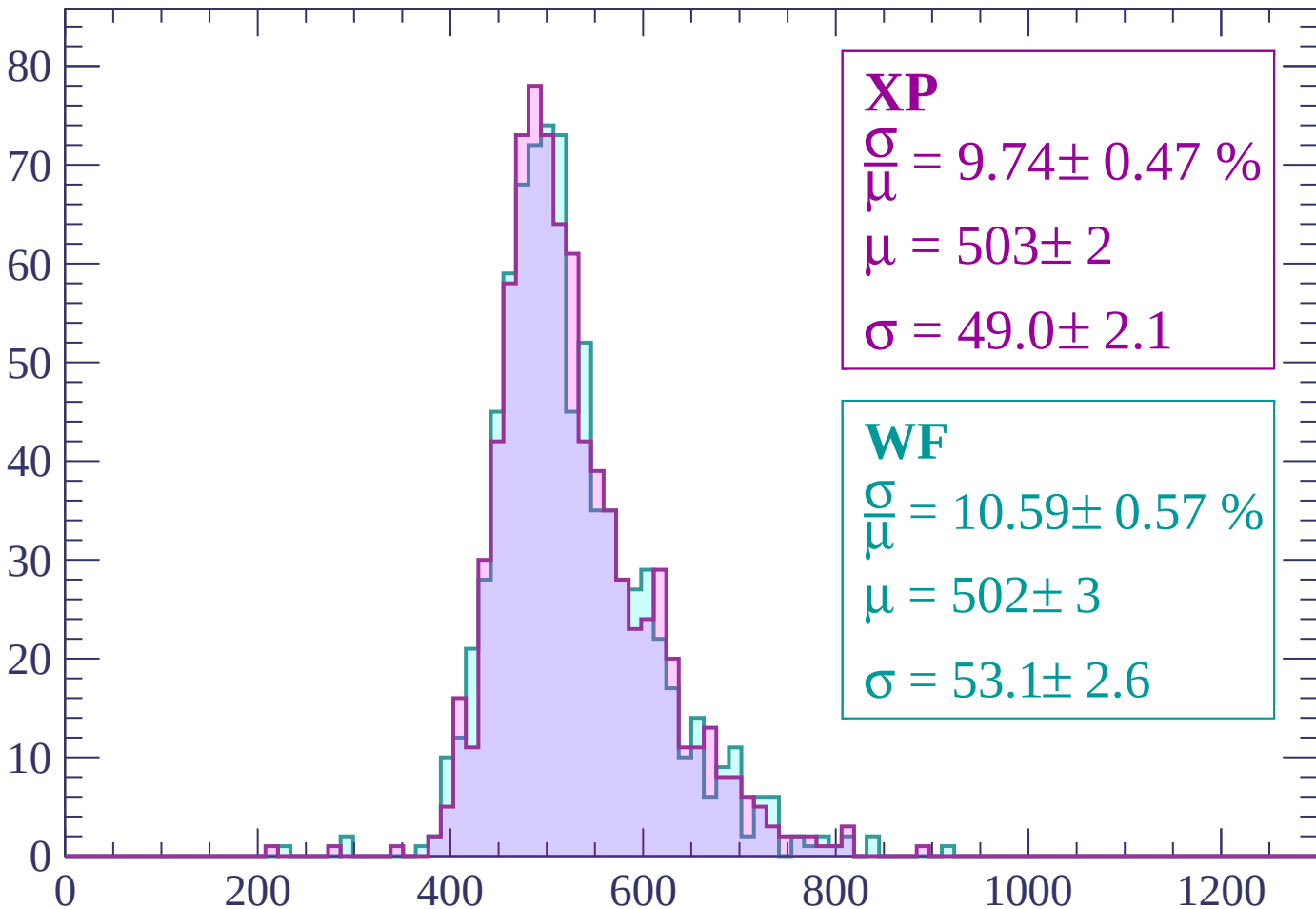
$$\mu = 499 \pm 2$$

$$\sigma = 52.4 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP

$$\frac{\sigma}{\mu} = 9.74 \pm 0.47 \%$$

$$\mu = 503 \pm 2$$

$$\sigma = 49.0 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 10.59 \pm 0.57 \%$$

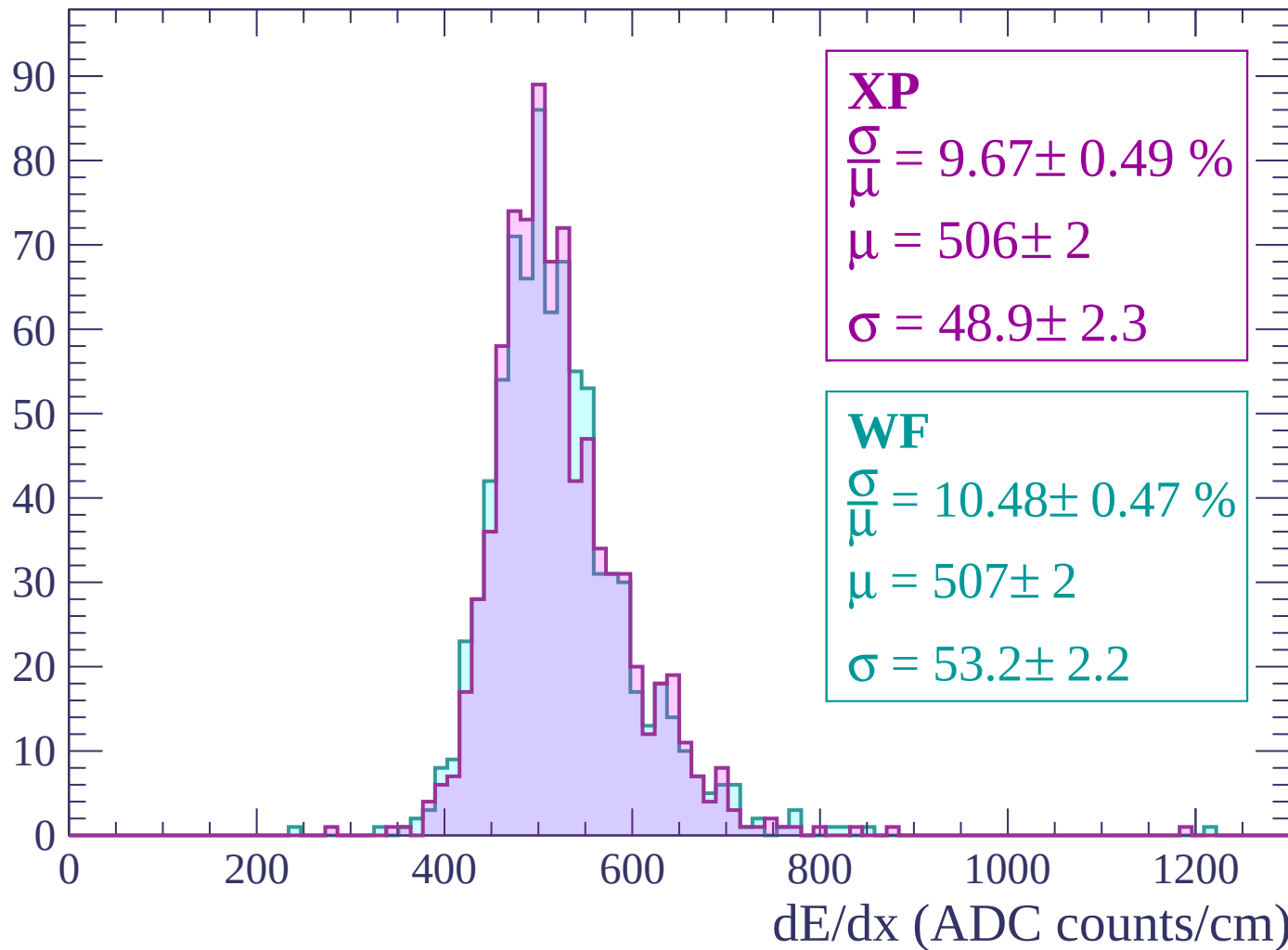
$$\mu = 502 \pm 3$$

$$\sigma = 53.1 \pm 2.6$$

dE/dx (ADC counts/cm)

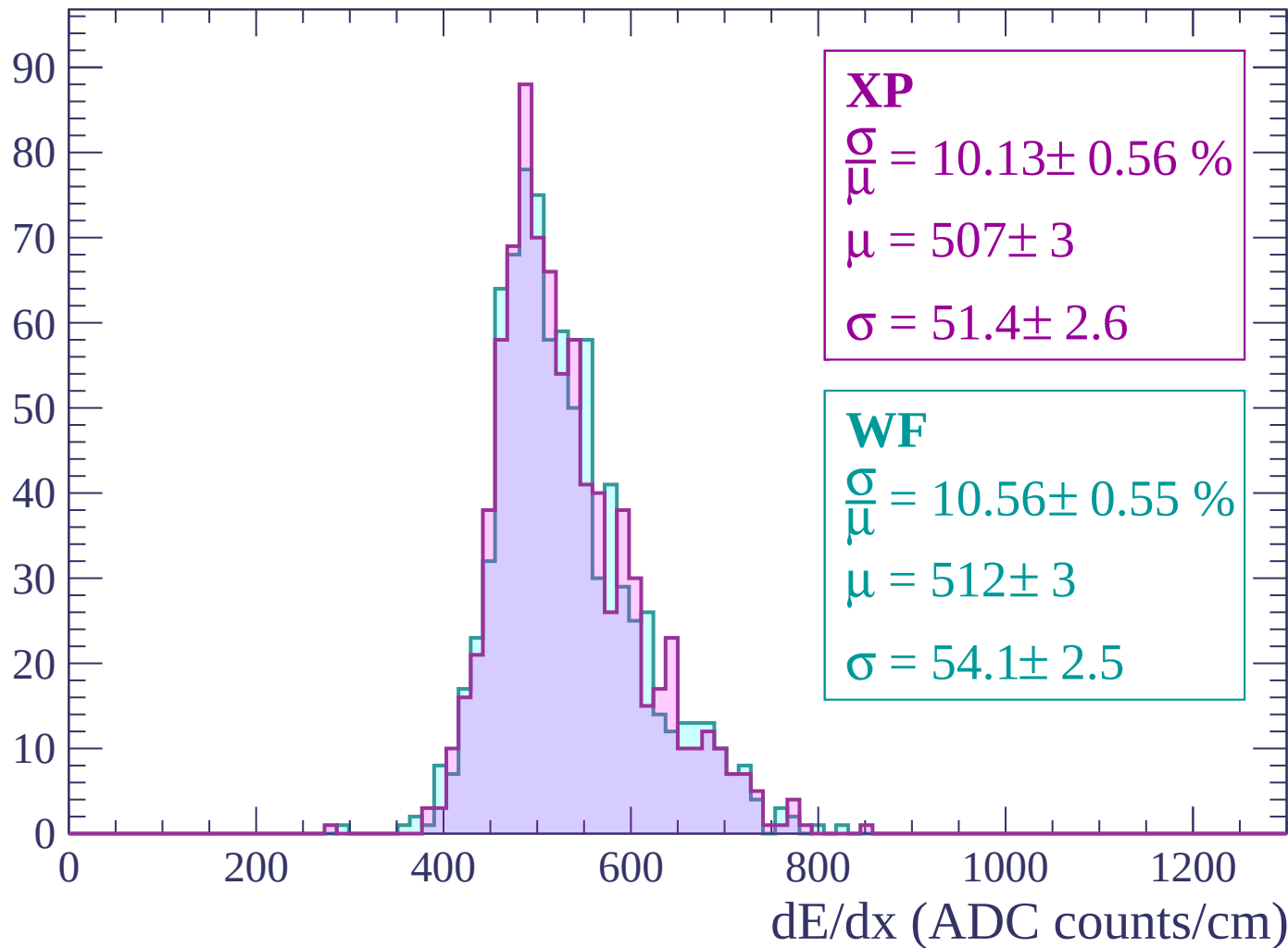
Energy loss | $1800 < p < 1860$

Count



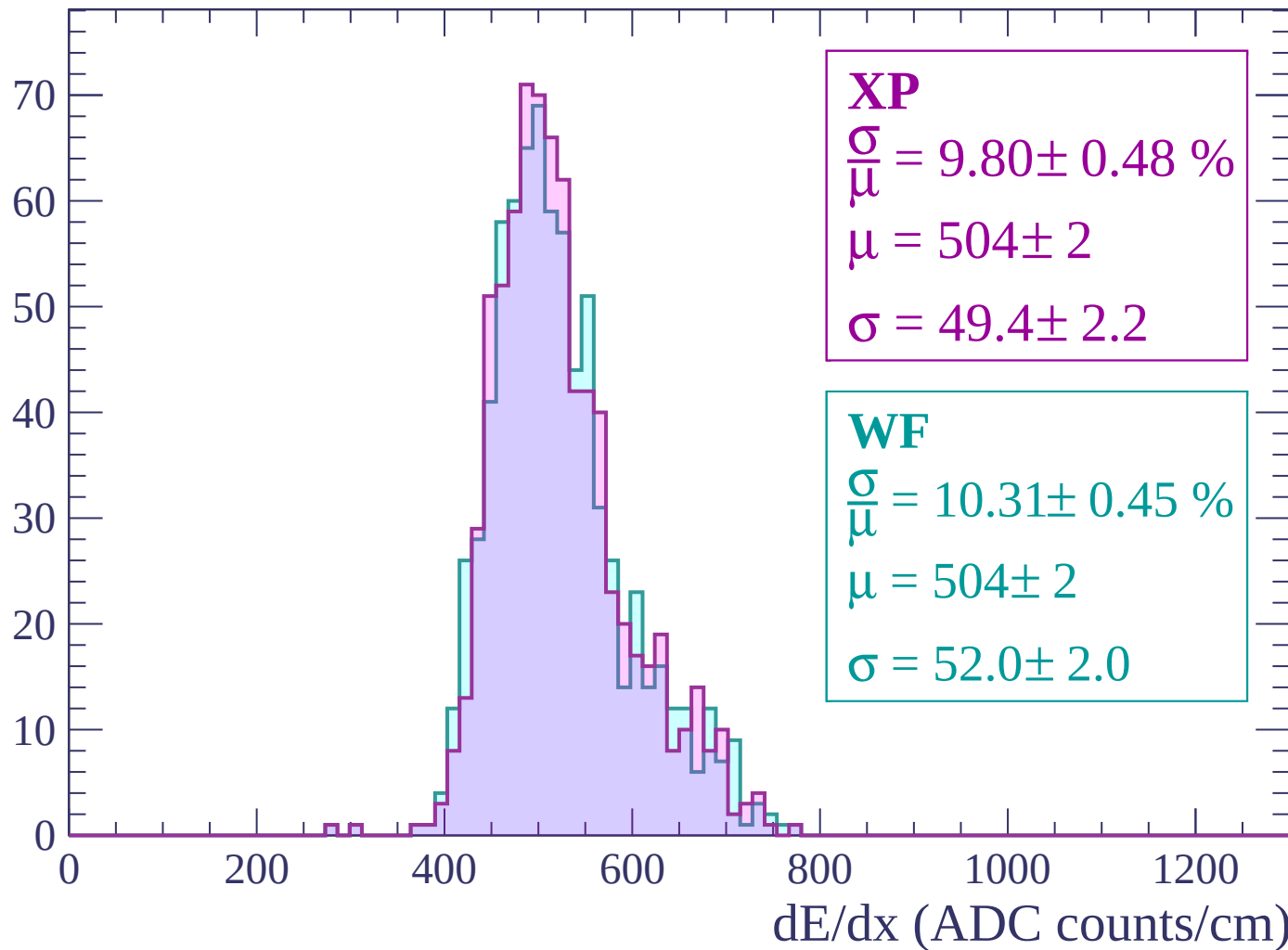
Energy loss | $1860 < p < 1920$

Count



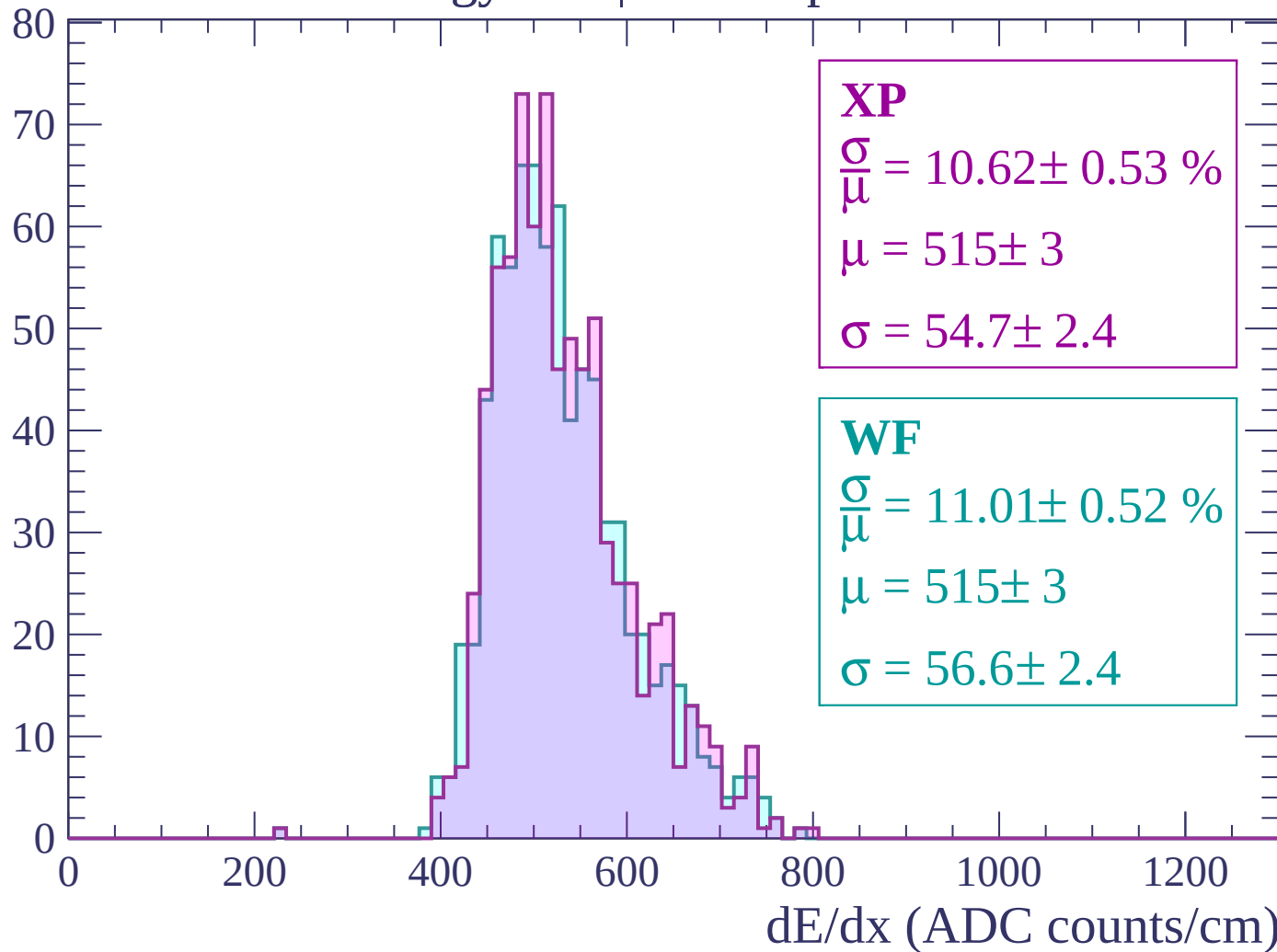
Energy loss | $1920 < p < 1980$

Count



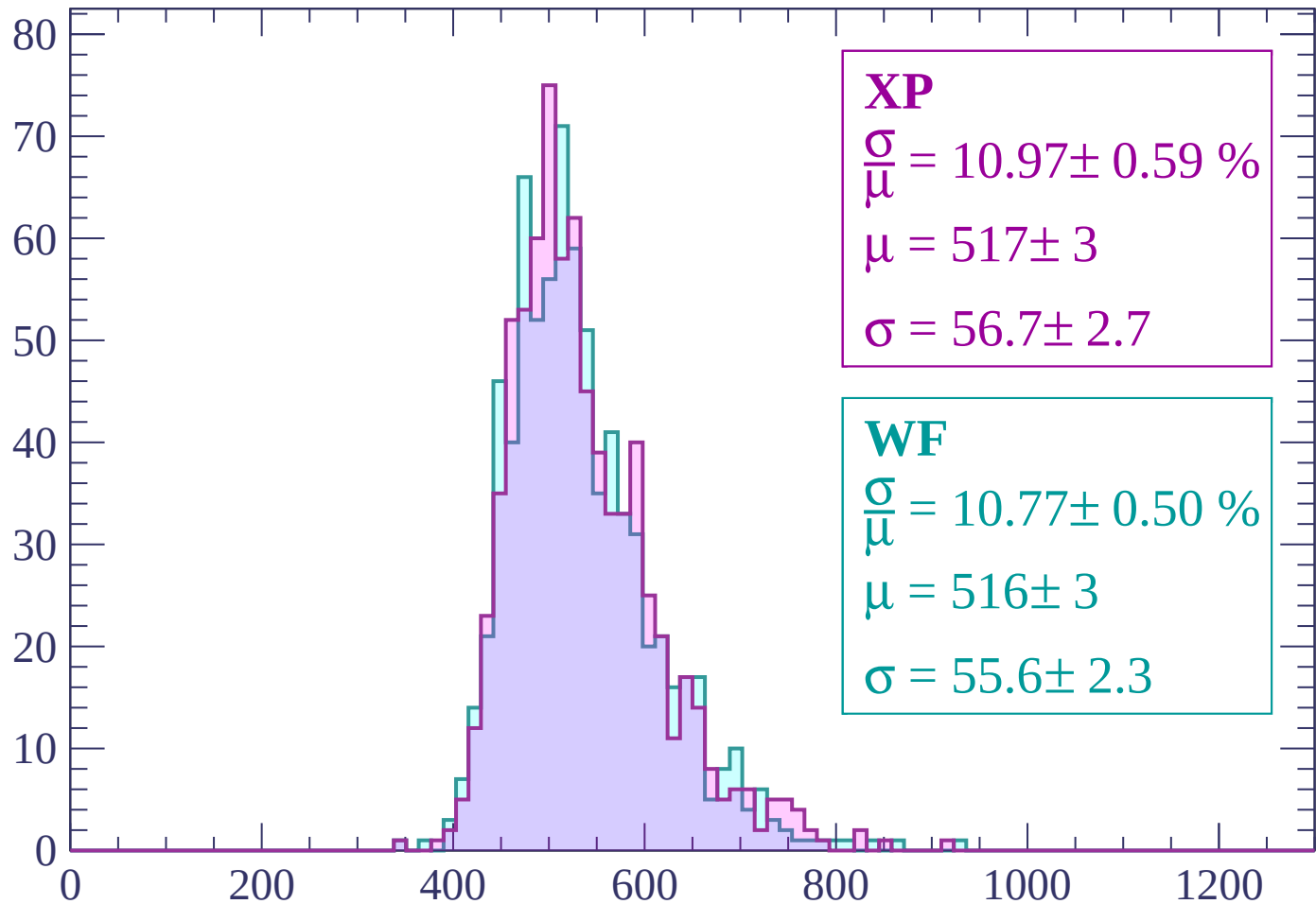
Energy loss | $1980 < p < 2040$

Count



Energy loss | $2040 < p < 2100$

Count



XP

$$\frac{\sigma}{\mu} = 10.97 \pm 0.59 \%$$

$$\mu = 517 \pm 3$$

$$\sigma = 56.7 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 10.77 \pm 0.50 \%$$

$$\mu = 516 \pm 3$$

$$\sigma = 55.6 \pm 2.3$$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count

XP

$$\frac{\sigma}{\mu} = 11.52 \pm 0.59 \%$$

$$\mu = 522 \pm 3$$

$$\sigma = 60.1 \pm 2.7$$

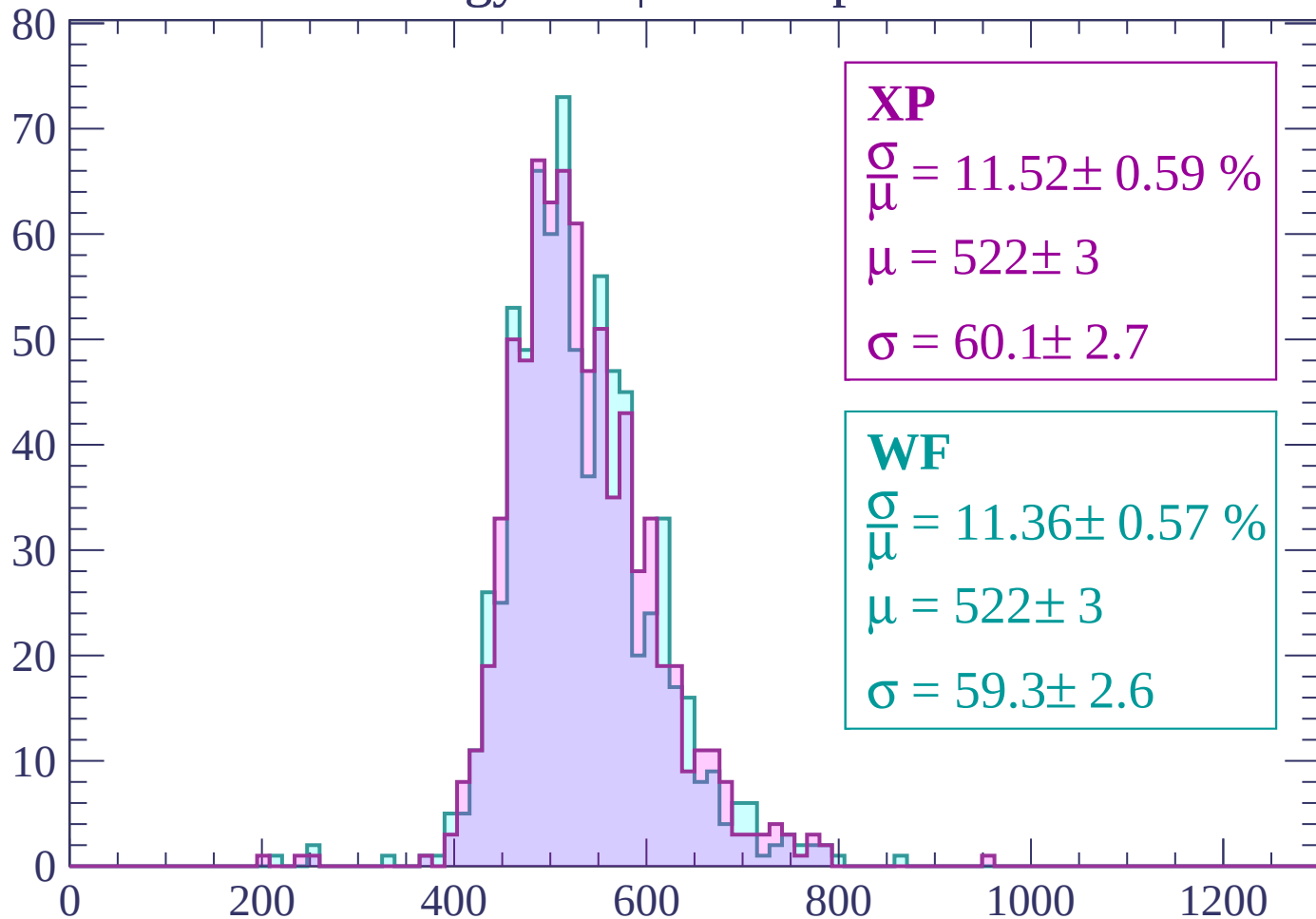
WF

$$\frac{\sigma}{\mu} = 11.36 \pm 0.57 \%$$

$$\mu = 522 \pm 3$$

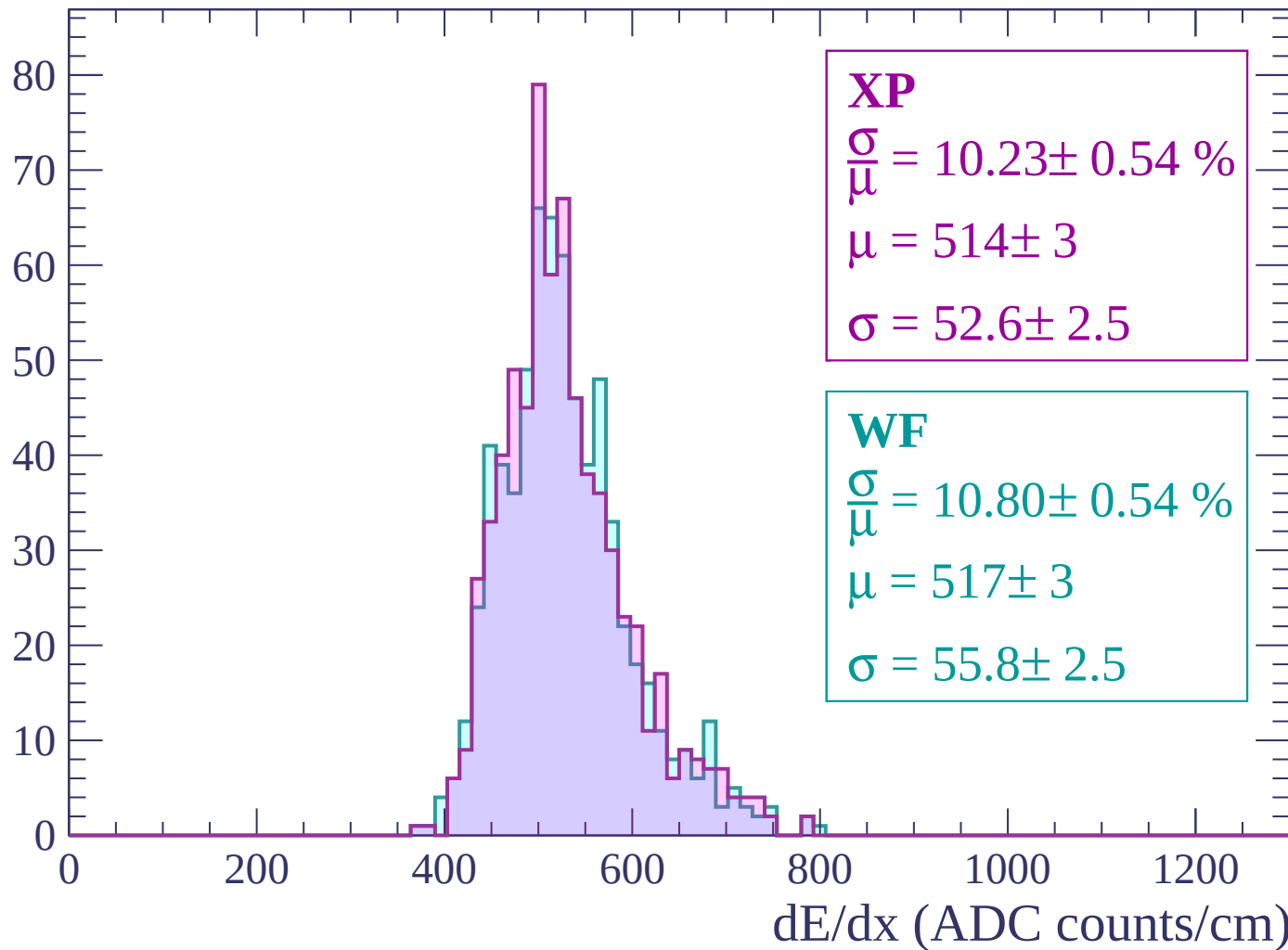
$$\sigma = 59.3 \pm 2.6$$

dE/dx (ADC counts/cm)



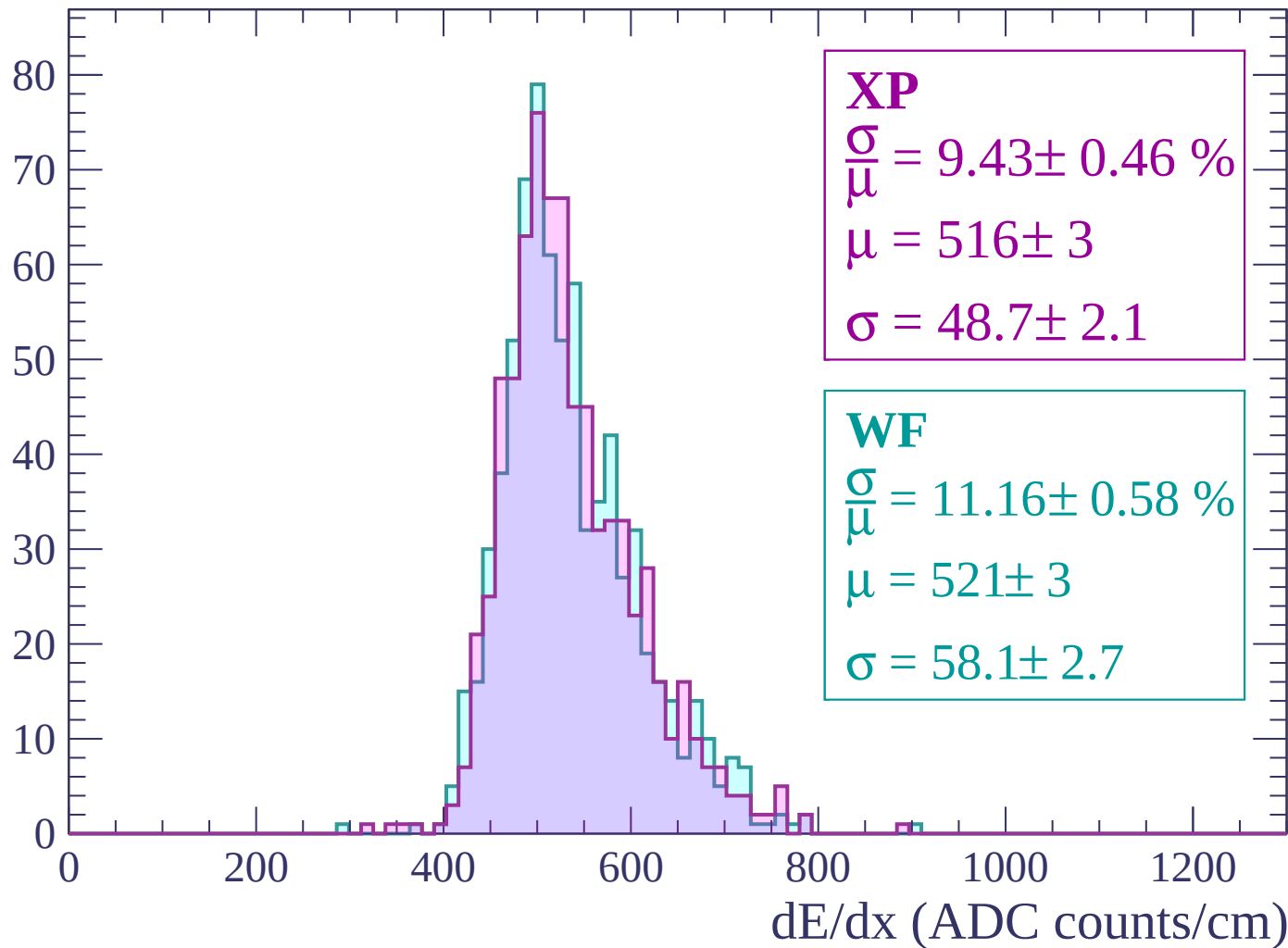
Energy loss | $2160 < p < 2220$

Count



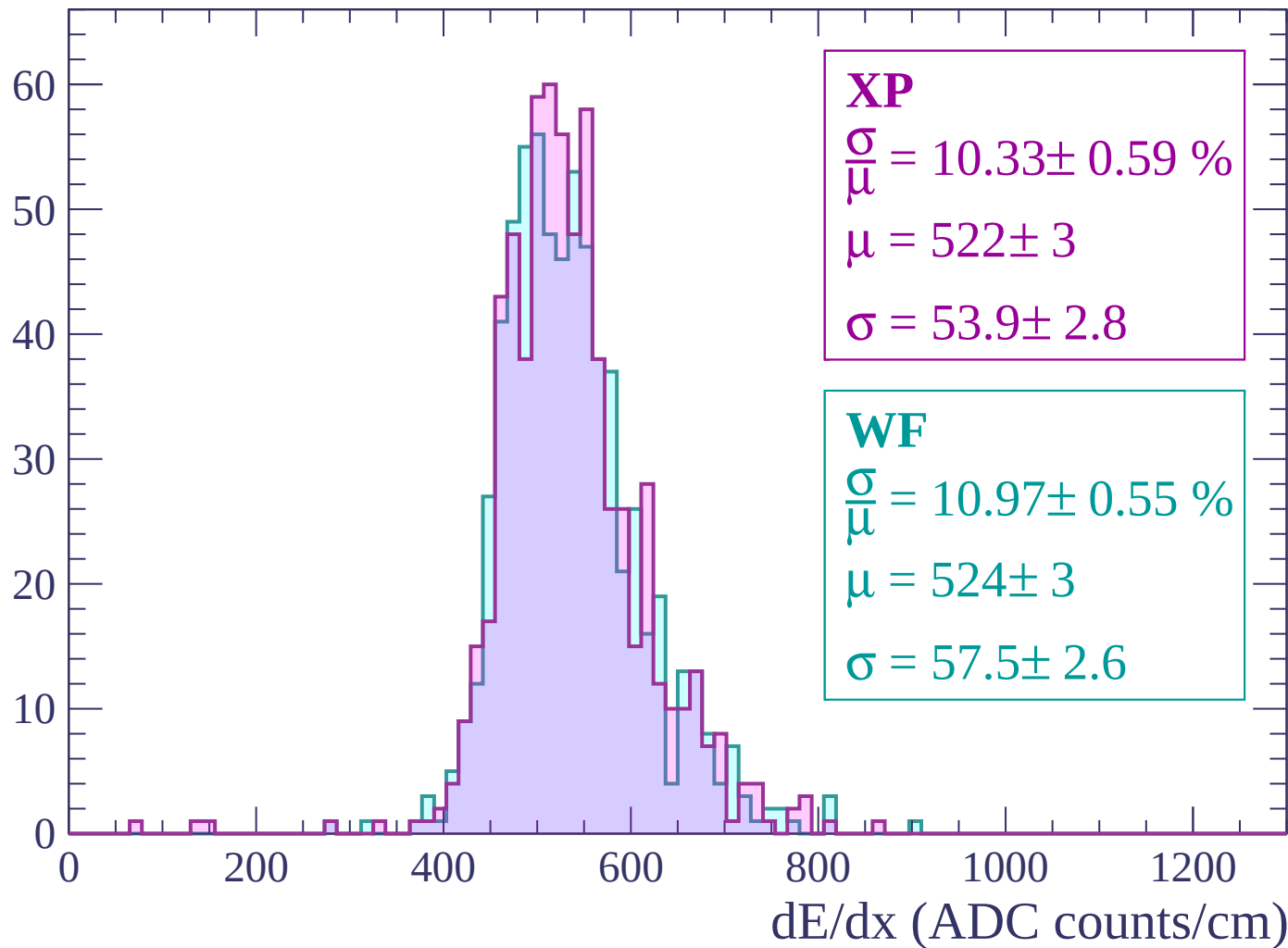
Energy loss | $2220 < p < 2280$

Count



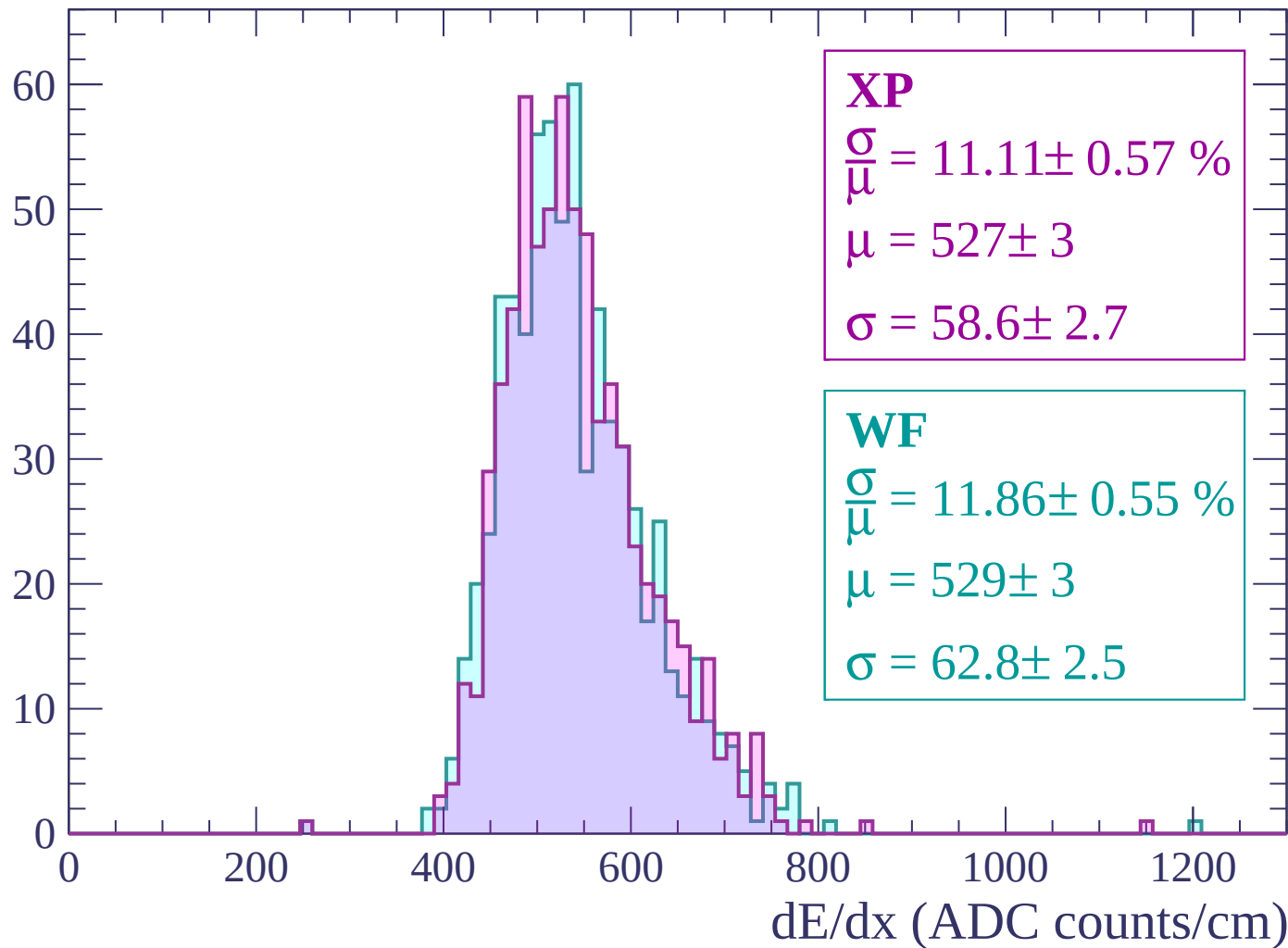
Energy loss | $2280 < p < 2340$

Count



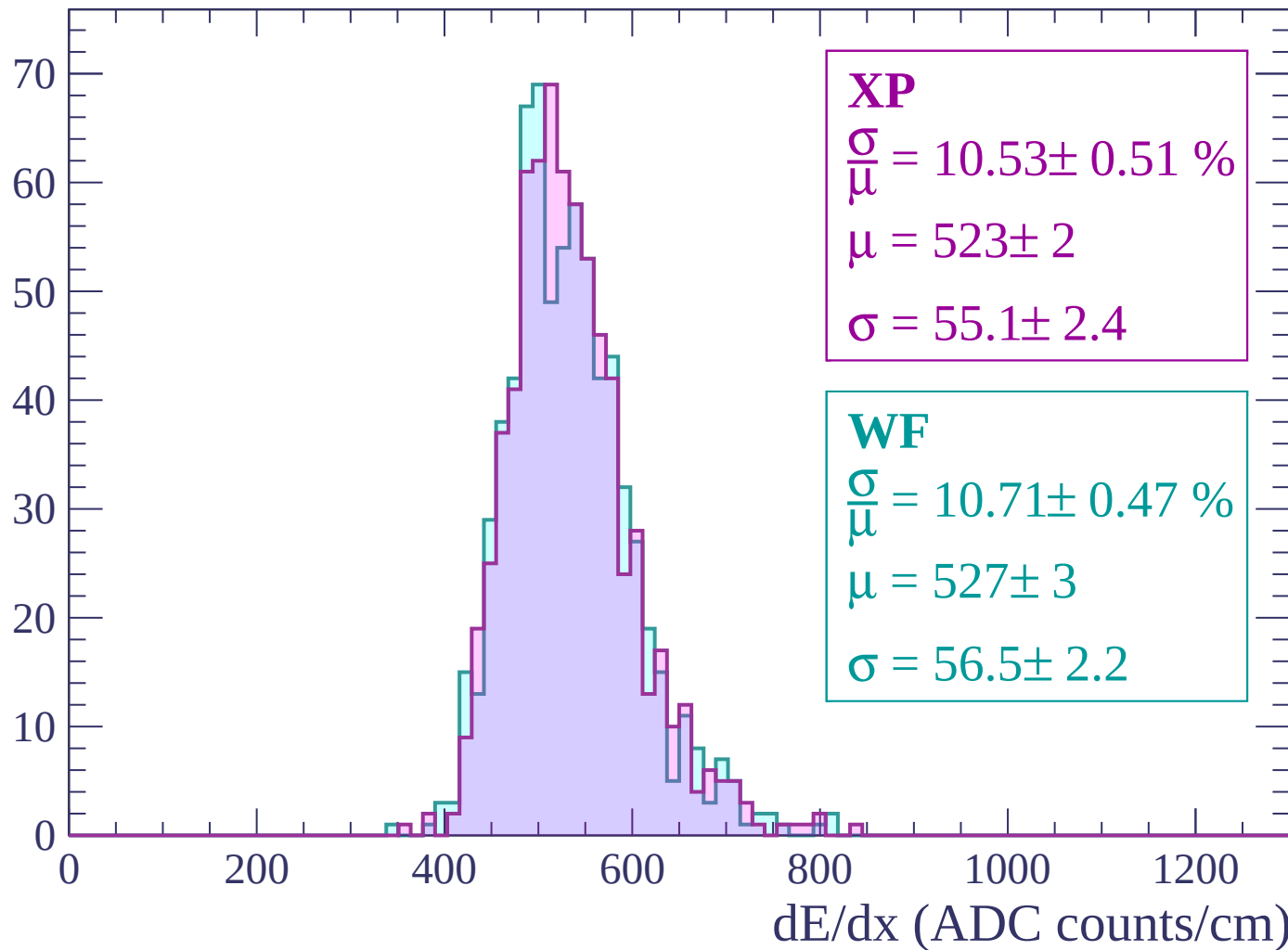
Energy loss | $2340 < p < 2400$

Count



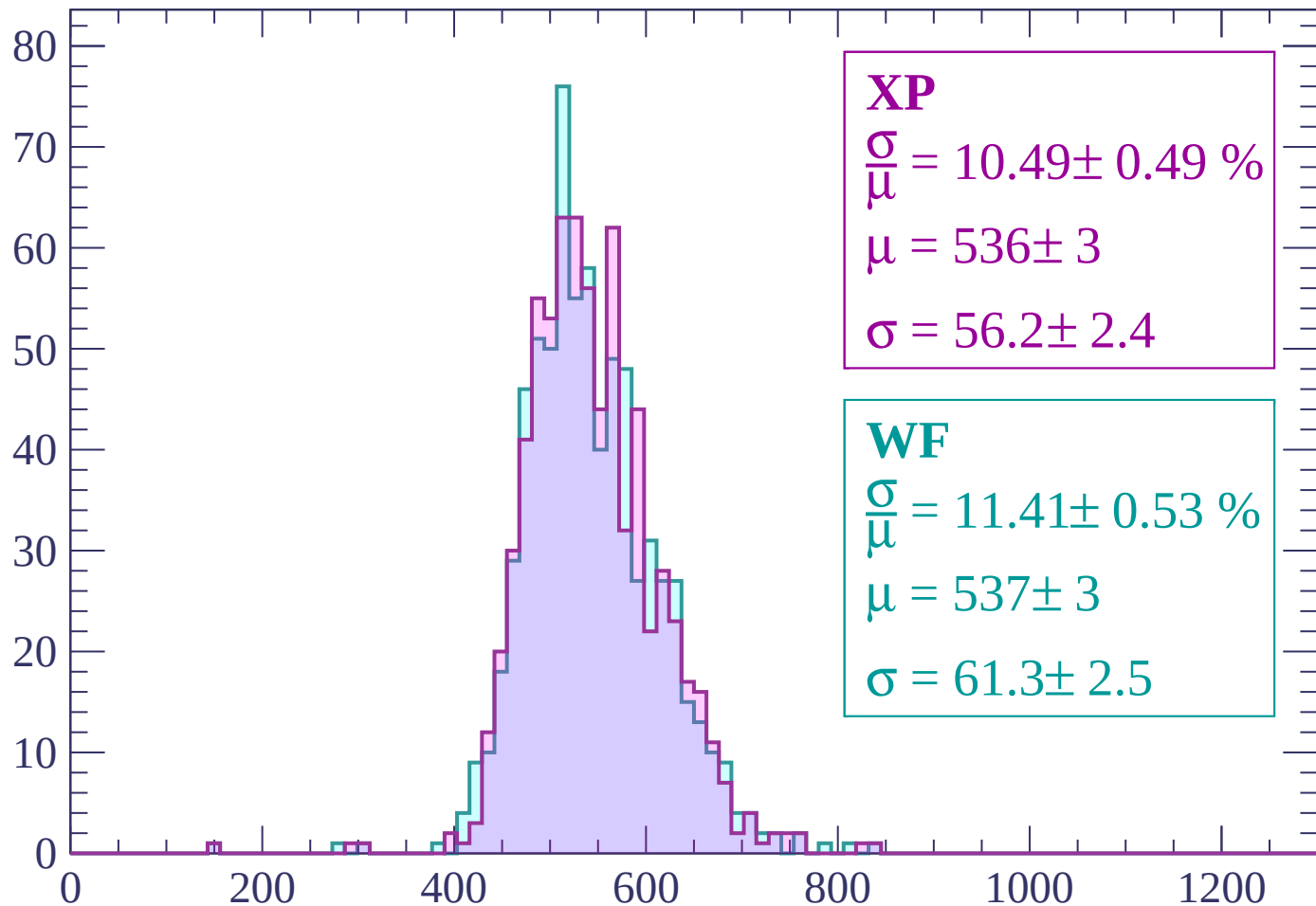
Energy loss | $2400 < p < 2460$

Count



Energy loss | $2460 < p < 2520$

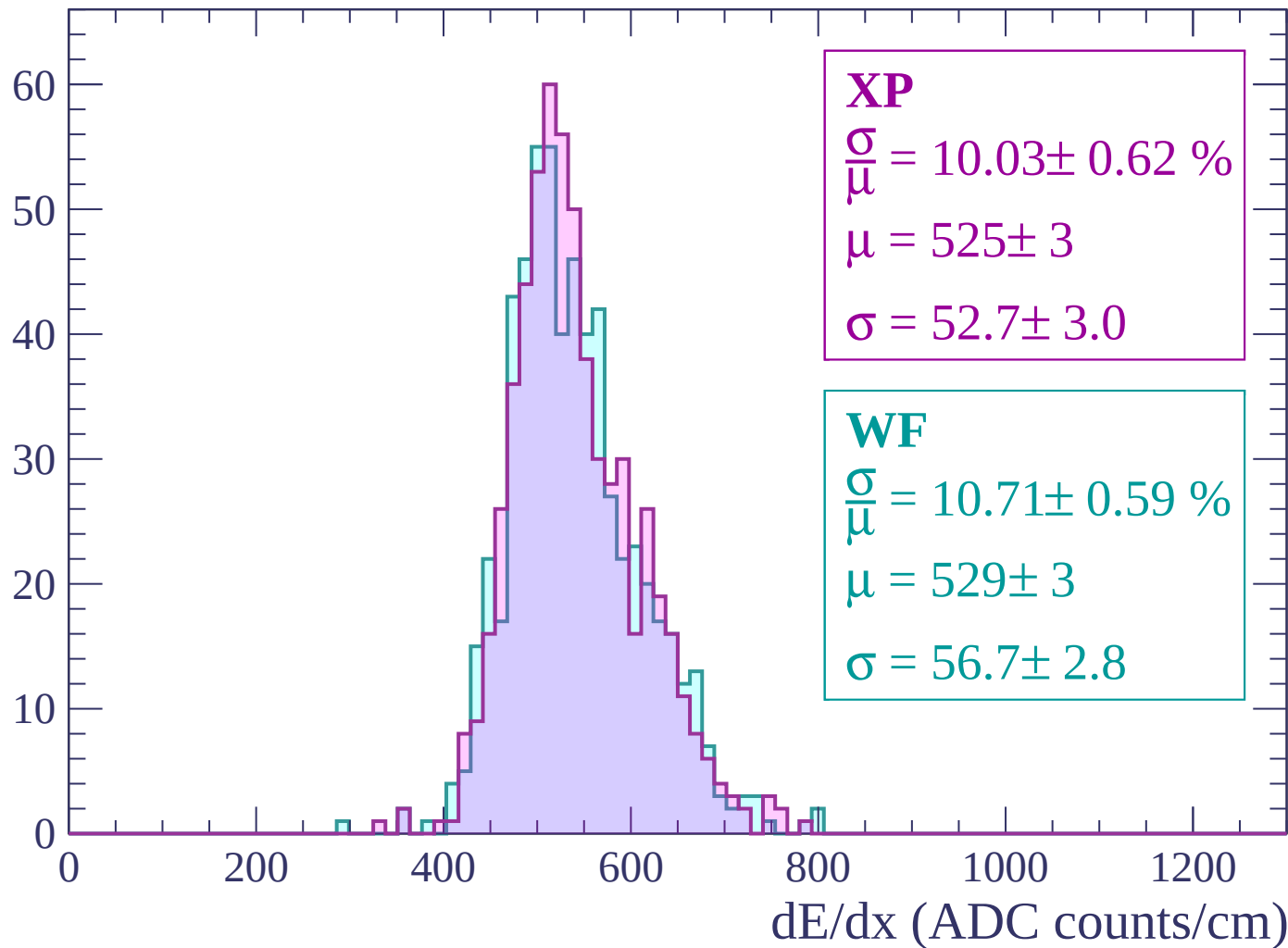
Count



dE/dx (ADC counts/cm)

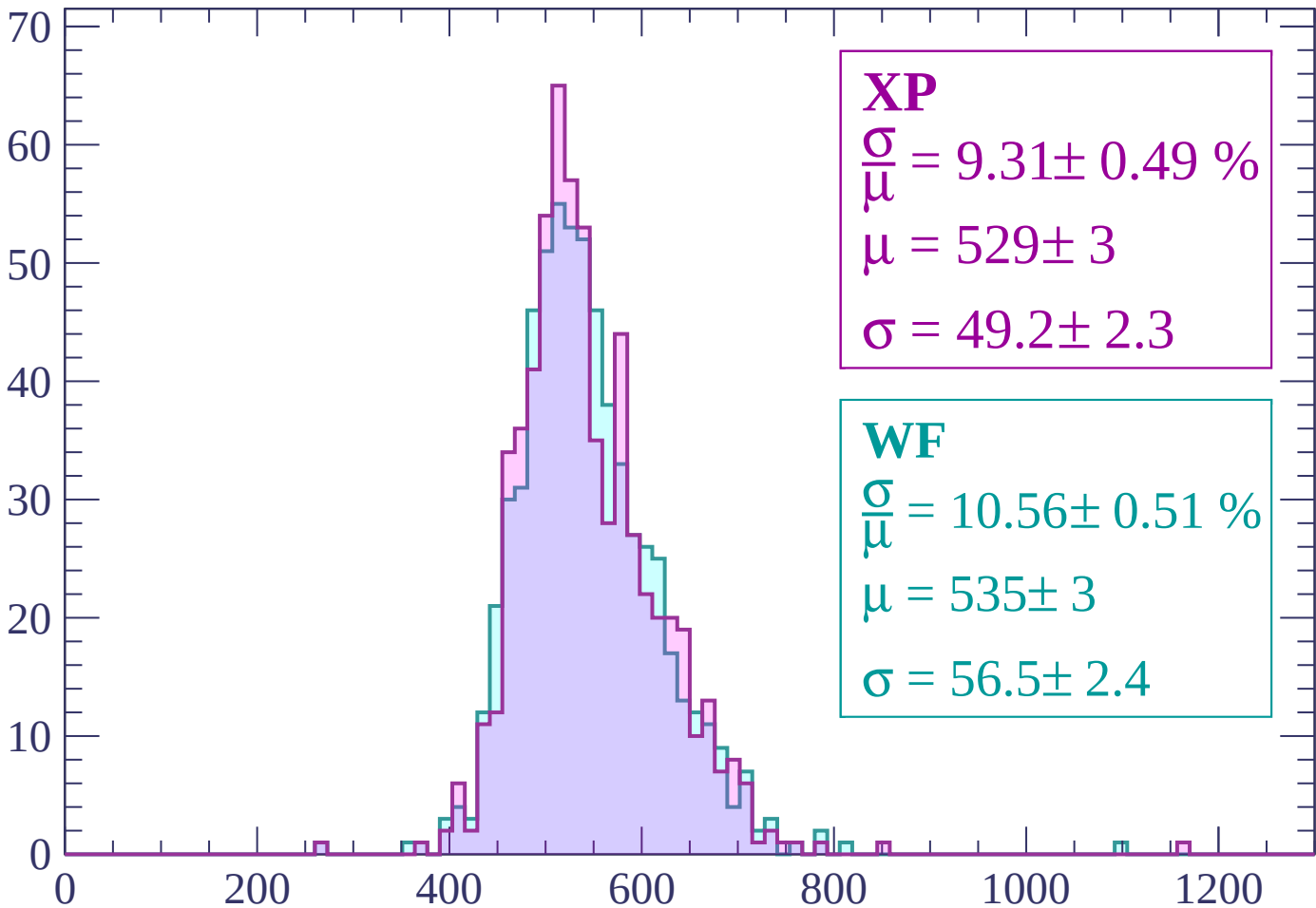
Energy loss | $2520 < p < 2580$

Count



Energy loss | $2580 < p < 2640$

Count



XP

$$\frac{\sigma}{\mu} = 9.31 \pm 0.49 \%$$

$$\mu = 529 \pm 3$$

$$\sigma = 49.2 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 10.56 \pm 0.51 \%$$

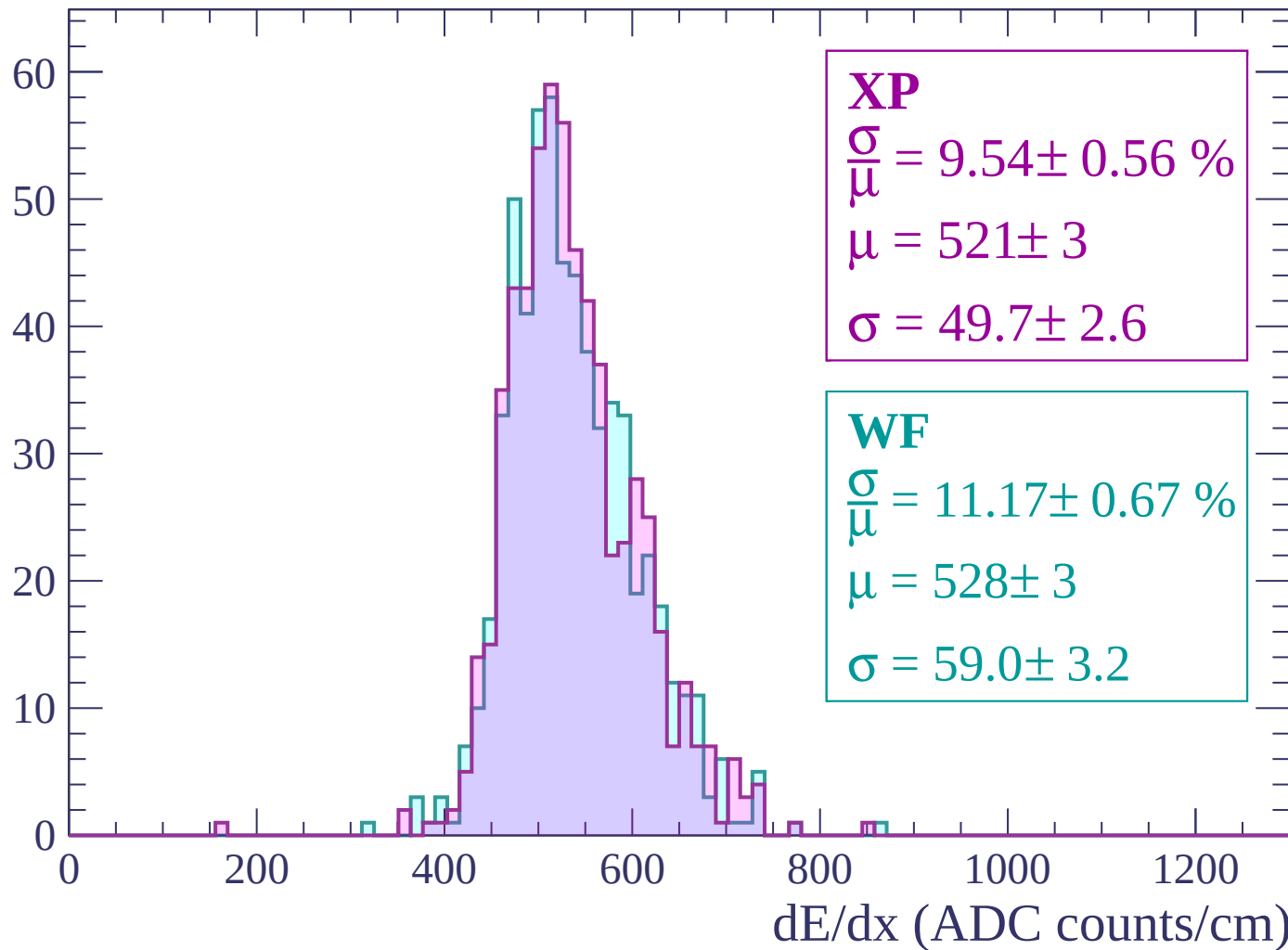
$$\mu = 535 \pm 3$$

$$\sigma = 56.5 \pm 2.4$$

dE/dx (ADC counts/cm)

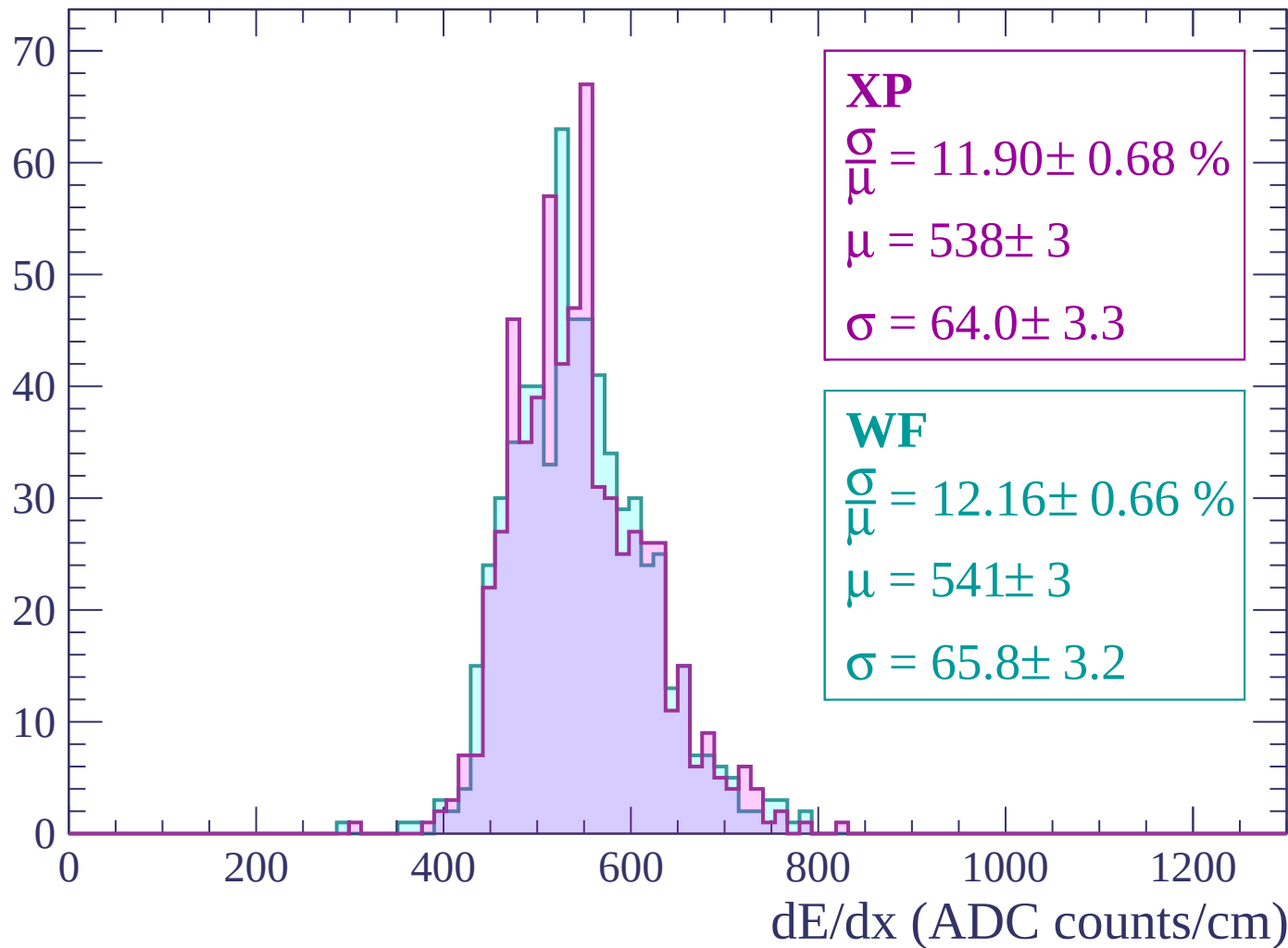
Energy loss | $2640 < p < 2700$

Count



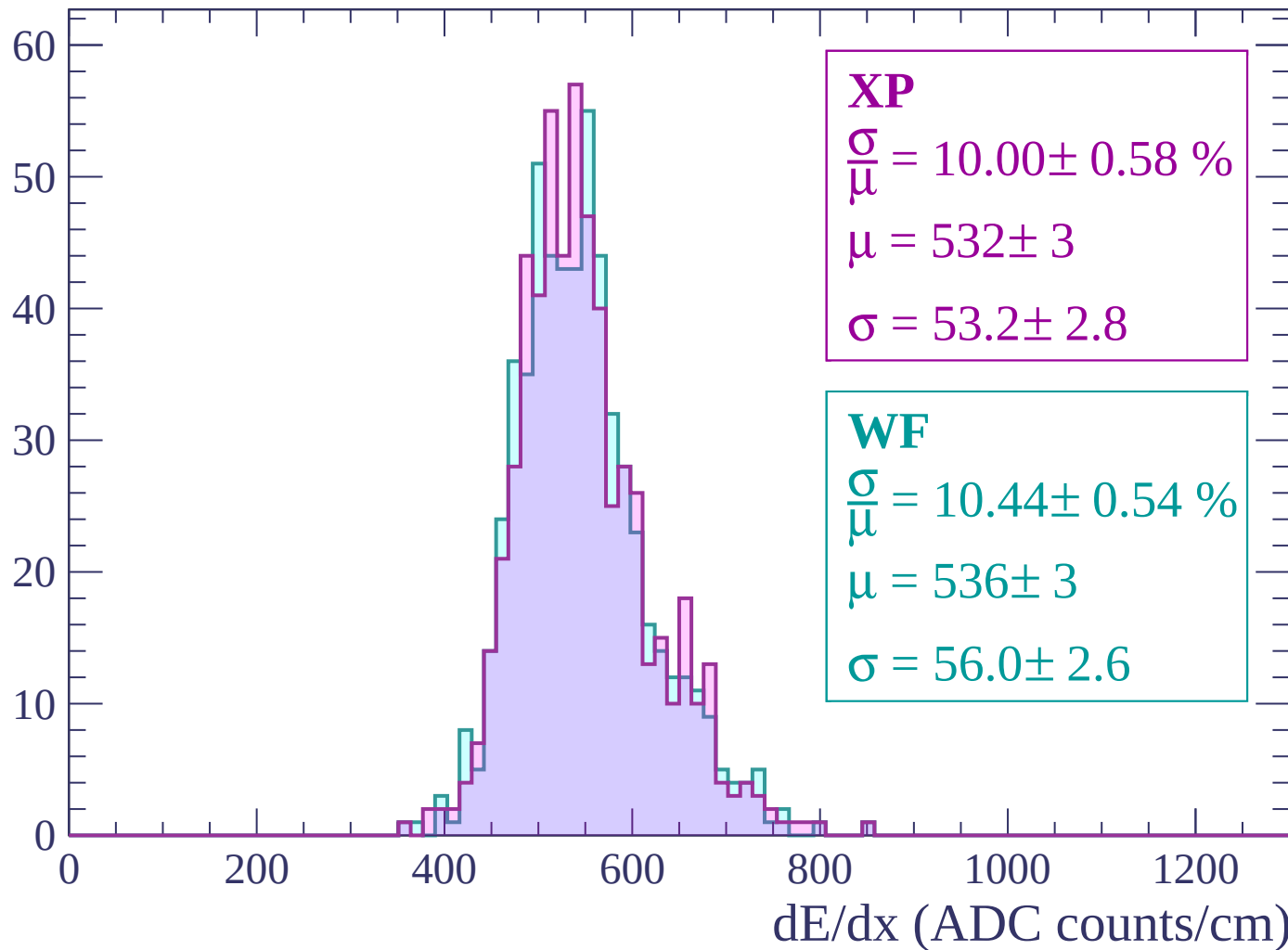
Energy loss | $2700 < p < 2760$

Count



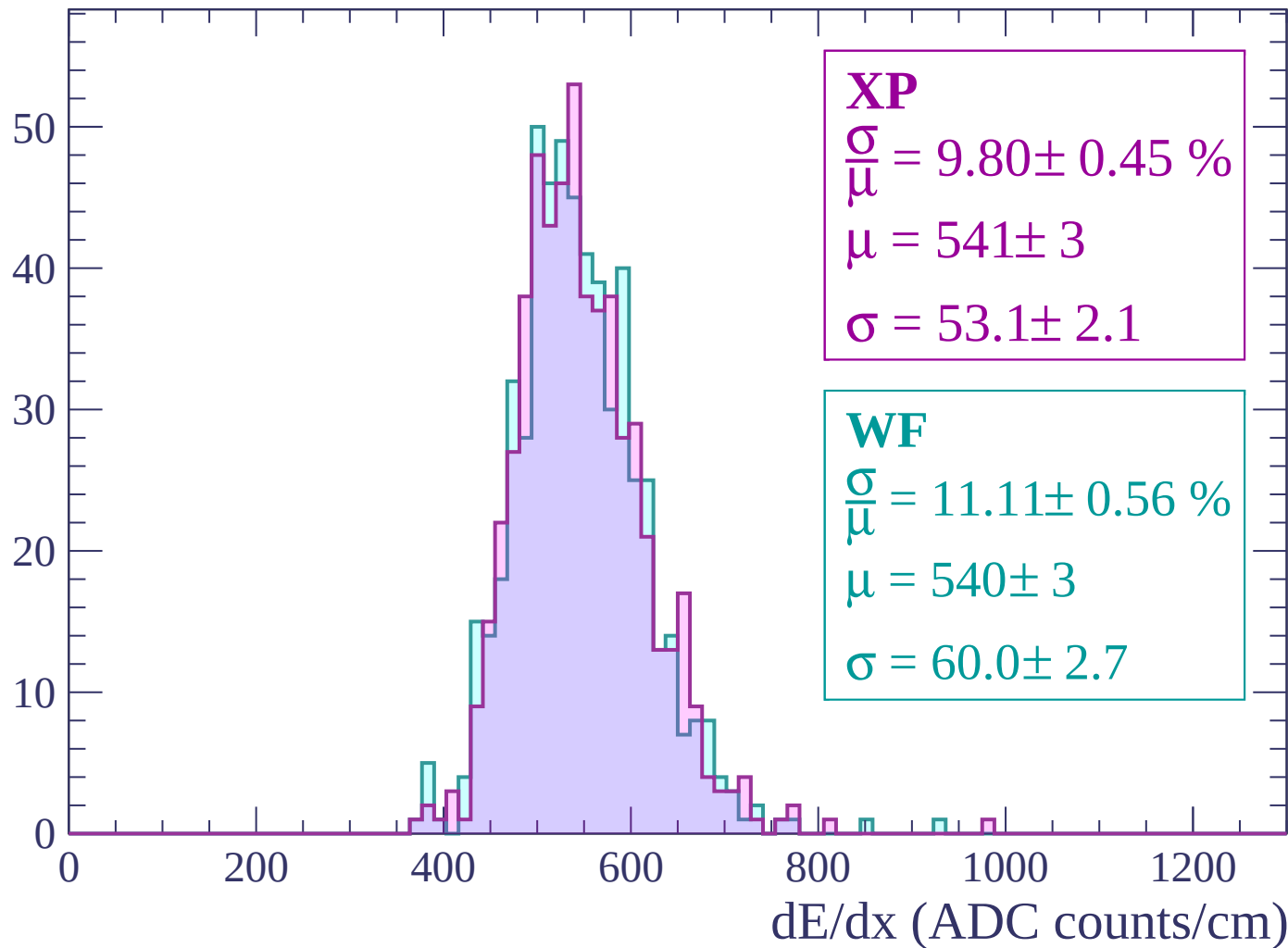
Energy loss | $2760 < p < 2820$

Count



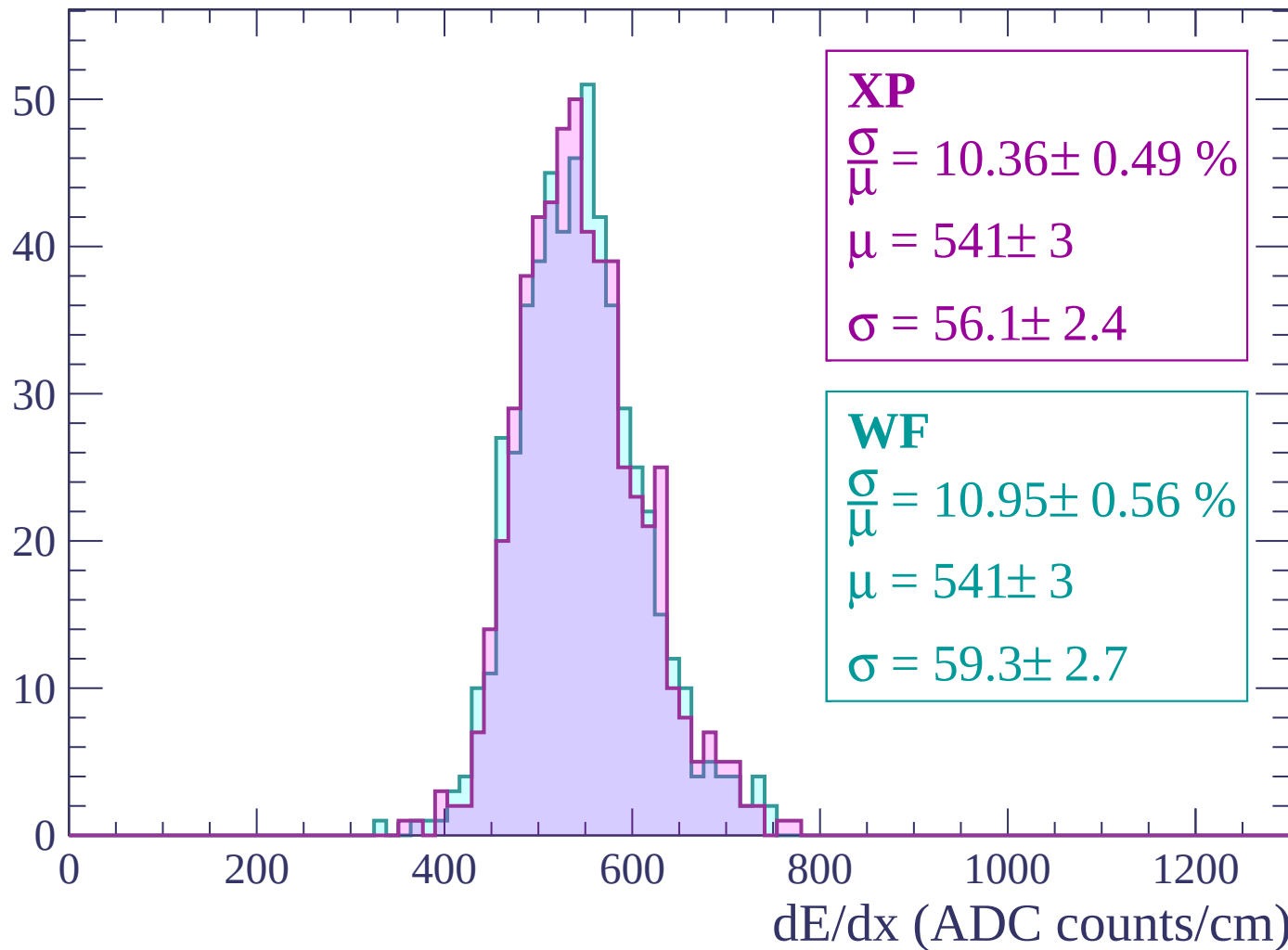
Energy loss | $2820 < p < 2880$

Count



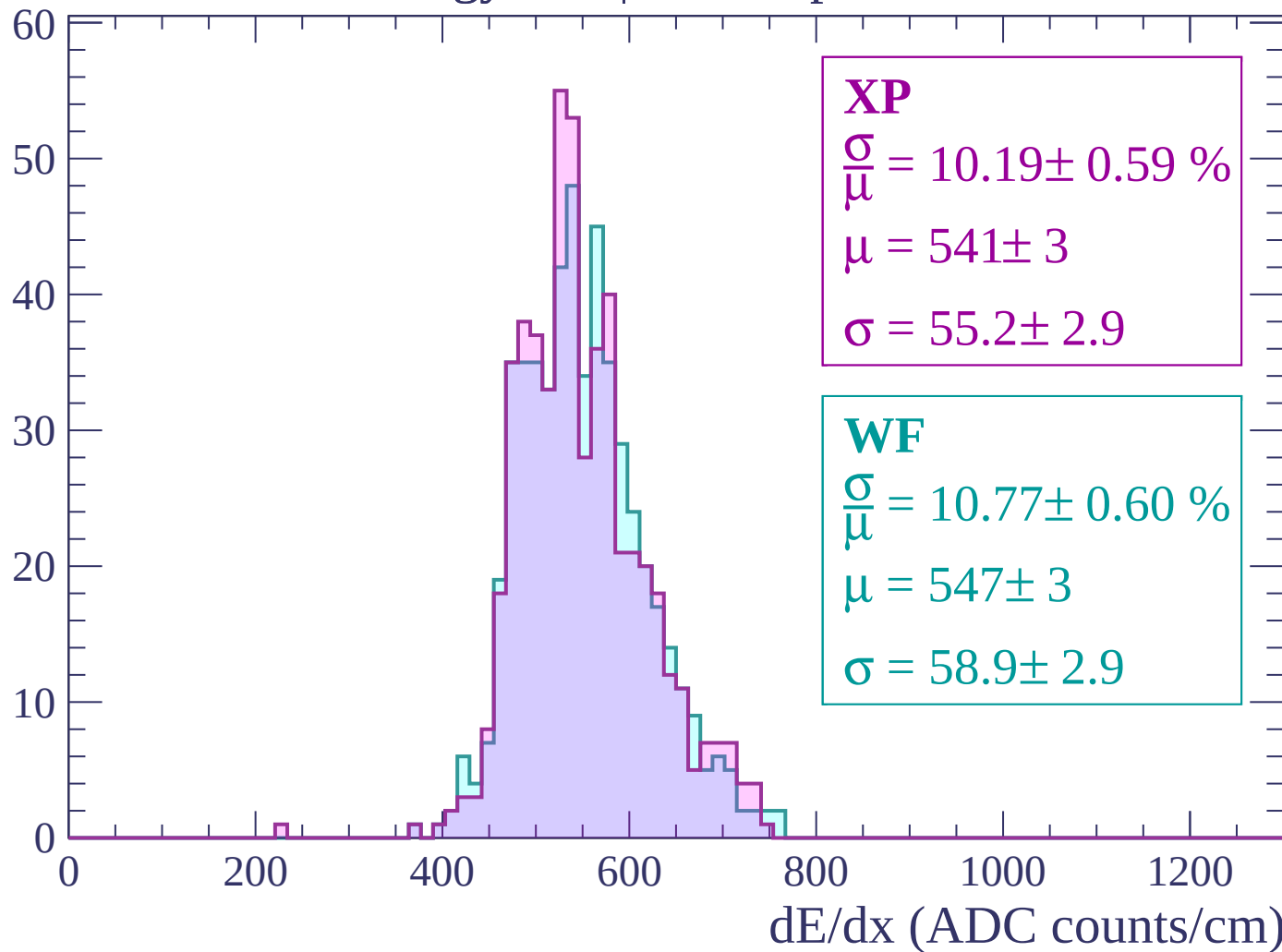
Energy loss | $2880 < p < 2940$

Count



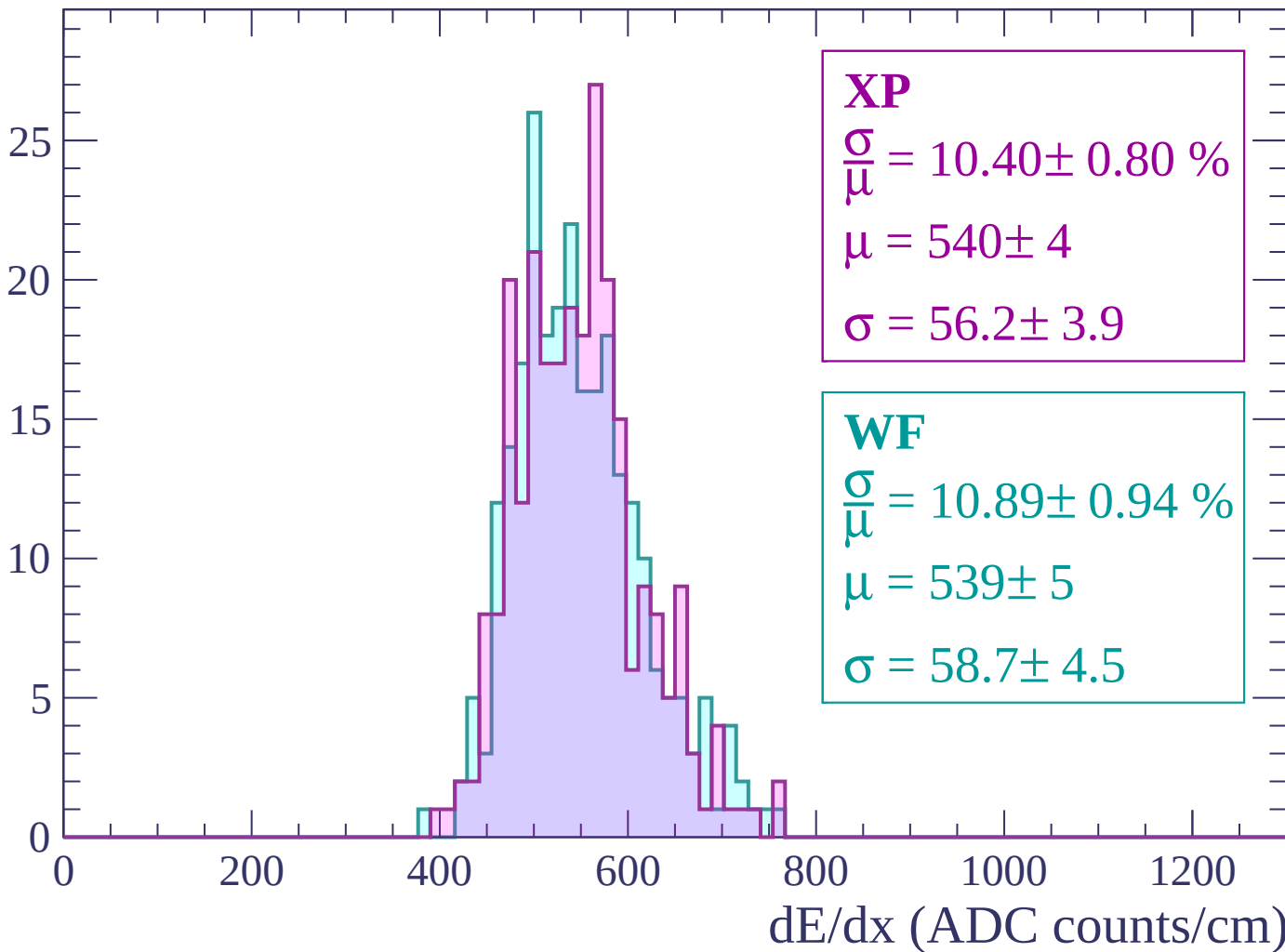
Energy loss | $2940 < p < 3000$

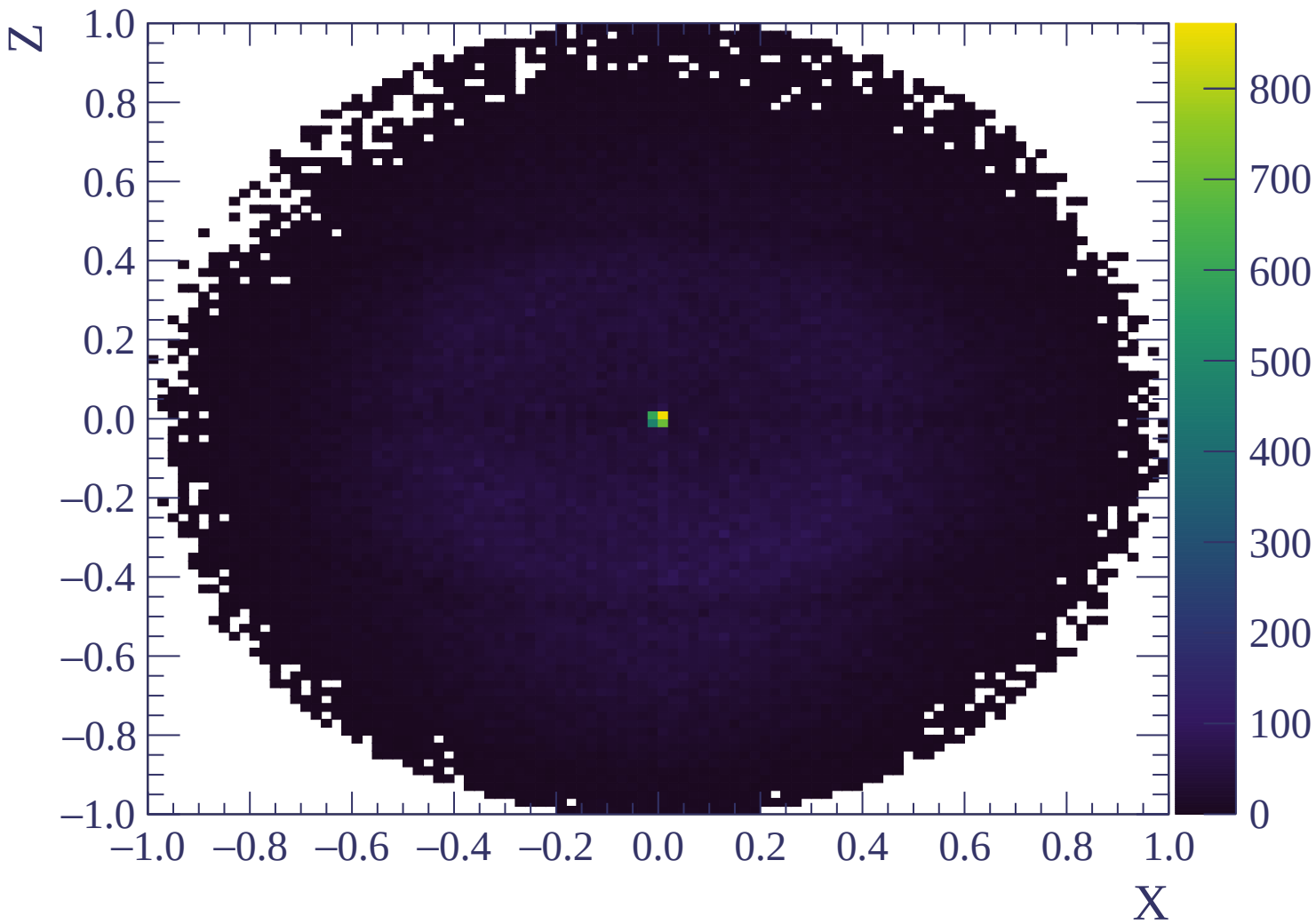
Count

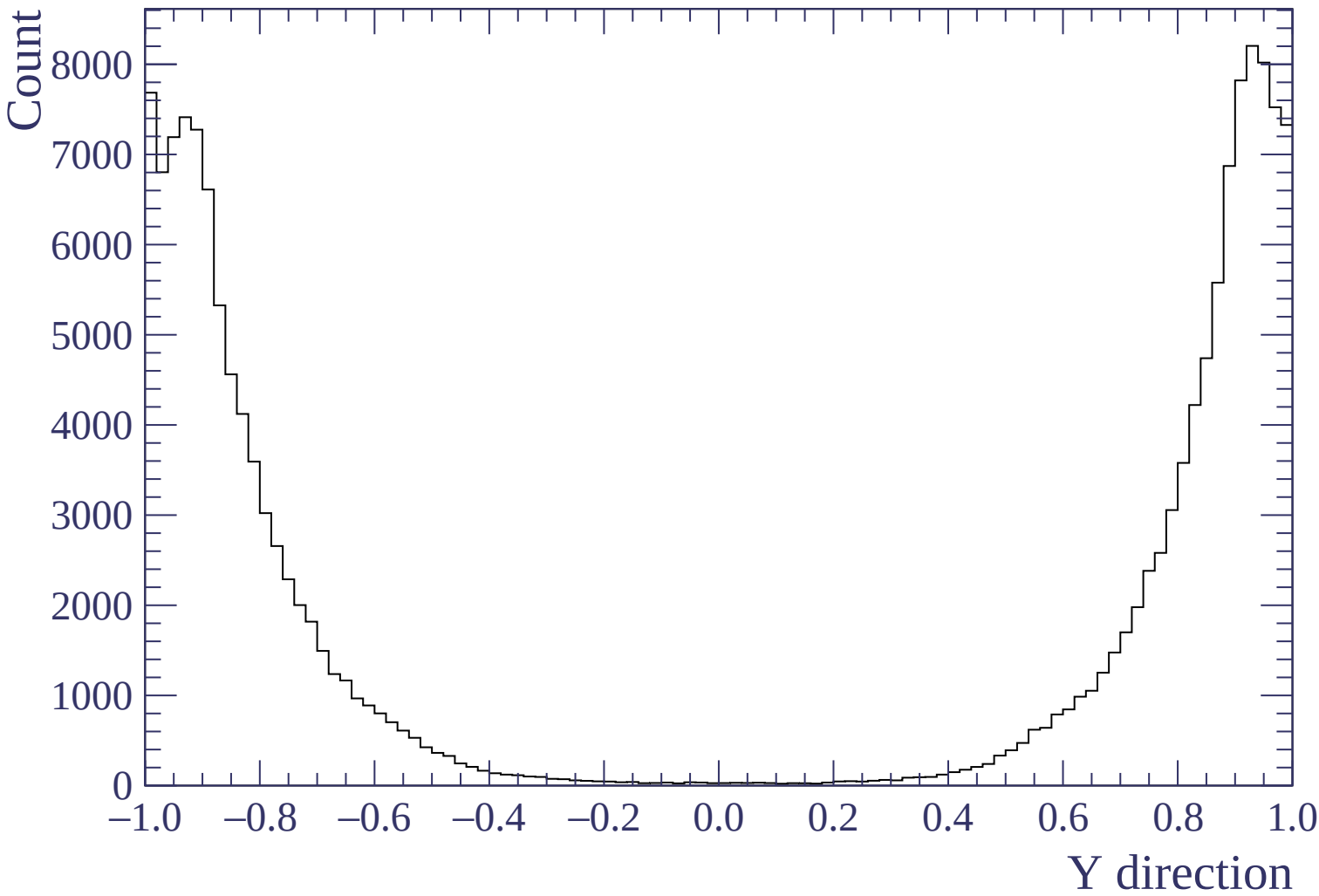


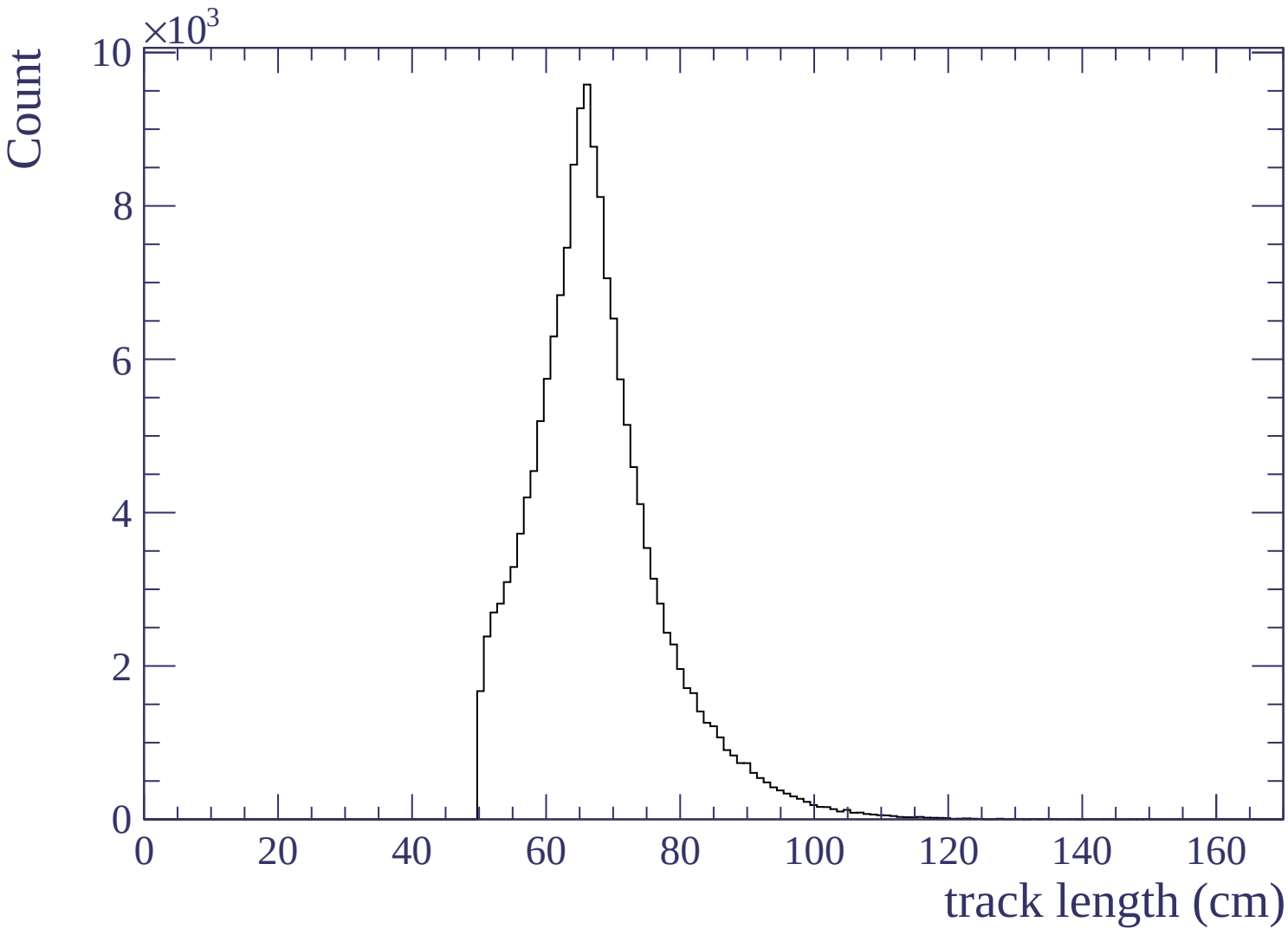
Energy loss | $3000 < p < 3060$

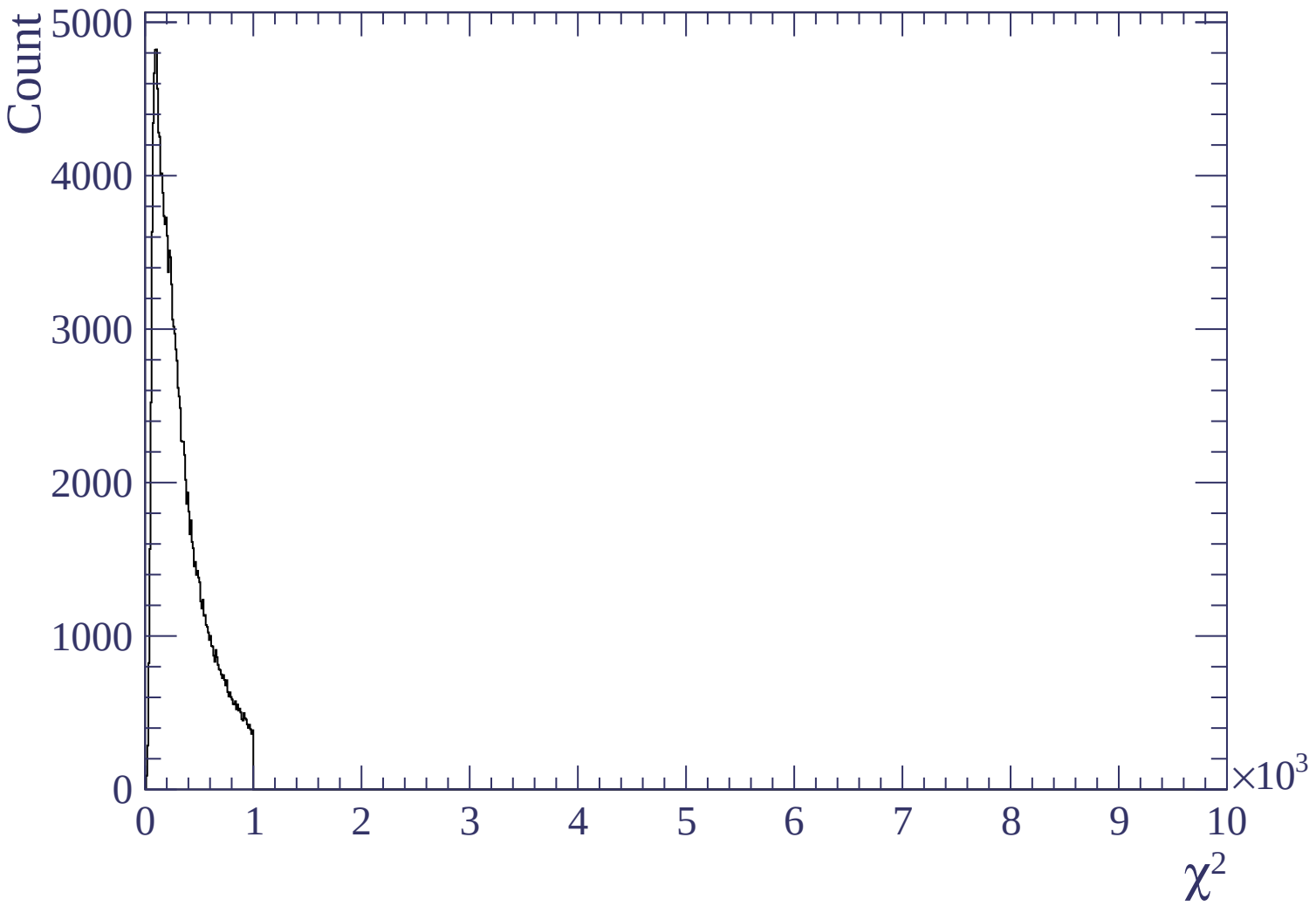
Count

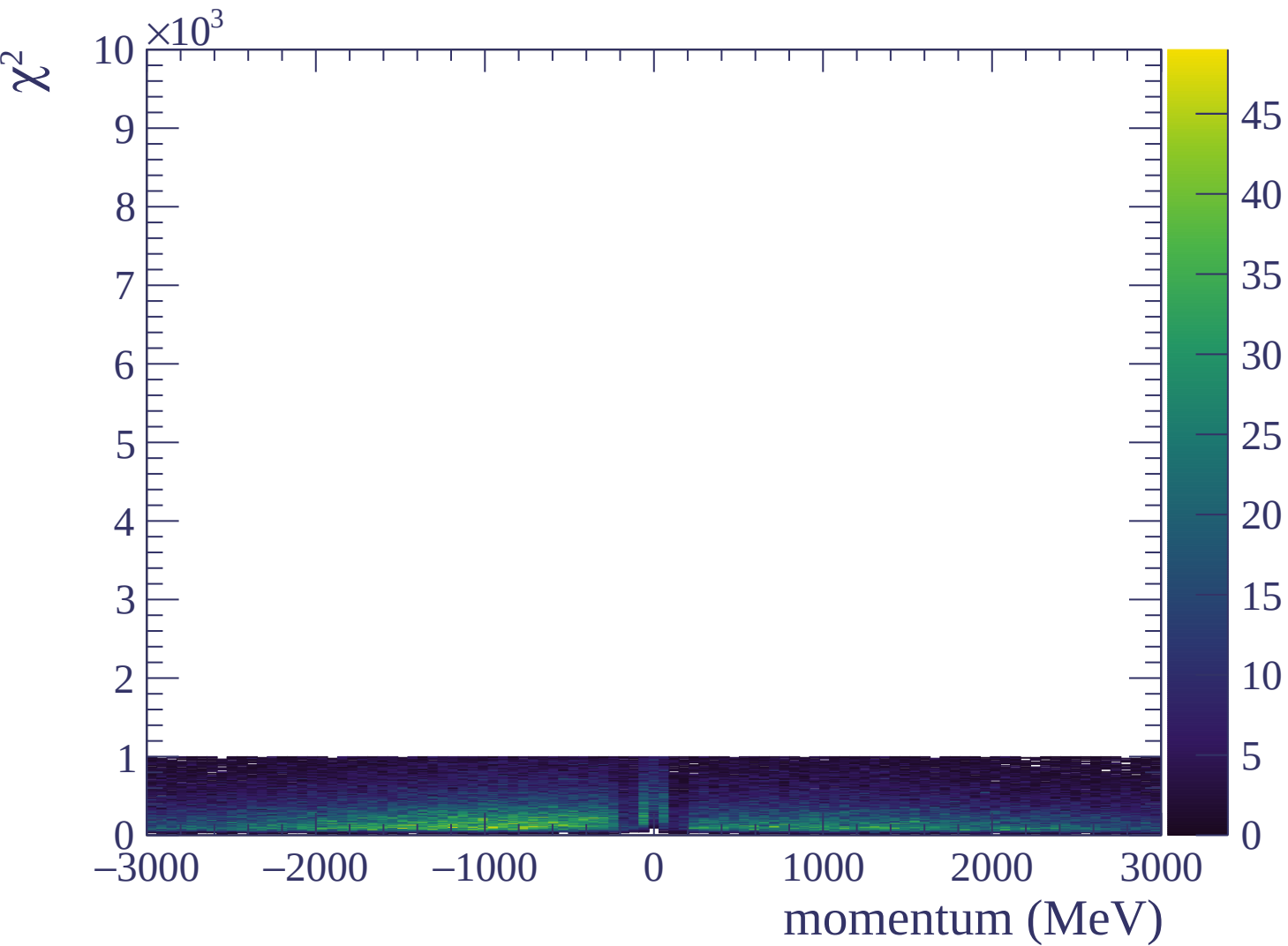












Count

